THE COMPLETE INTRINSIC CONGRUENCE SYSTEM FOR THE ERDŐS–STRAUS CONJECTURE: EXACT STRUCTURE AND PROVISIONAL EXCEPTIONAL-SET BOUNDS FOR GENERAL NUMERATORS

ANONYMOUS

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ABSTRACT. Subject to external verification, we obtain the unconditional bound

$$E_{\mathrm{all}}(N) \ll N \exp\{-c(\log N)^{3/4}\}$$

for Erdős–Straus exceptional denominators. More generally, if $E_m(N)$ counts $n \le N$ for which m/n is not a sum of three positive unit fractions, then, for every fixed $\varepsilon > 0$, uniformly for $3 \le m \le (\log N)^{3-\varepsilon}$, we obtain the second headline

$$E_m(N) \ll_\varepsilon N \exp\left\{-c_\varepsilon \left(\frac{\eta_2(m)(\log N)^3}{\varphi(m)}\right)^{1/4}\right\}, \quad \eta_2(m) = \prod_{p\mid m} \frac{p^2}{p^2+p-1}.$$

Both 3/4 theorems are explicitly CLAIMED/PROVISIONAL; their load-bearing pruned cubic prime-slice input, Theorem 9.4 (the analogue of Theorem 34.8 in the source notes), is likewise CLAIMED/PROVISIONAL, and the second headline inherits the three qualifications of the source §39.7. No assertion in this draft should be cited as an established improvement before independent expert review. *Priority note (November 2026):* the exponent 3/4 for $E(N)$, and for $E_m(N)$ at each fixed $m \ge 4$ with constants depending on m, was obtained earlier by Kandukoori, Ramesh and Kumar [16] (unrefereed preprint dated 29 July 2026, with a Lean formalisation); no priority is claimed for the exponent, and any "record" wording below is historical. A separate, completely written fixed-m layer gives $E_m(N) \ll_m N \exp\{-c(\eta_1(m)/\varphi(m))^{1/3}(\log N)^{2/3}(\log\log N)^{1/3}\}$. Because this too would improve the published general-numerator benchmark (though not the fixed-m preprint bound of [16]), it remains CLAIMED/PROVISIONAL pending external priority and referee checks, although it assumes no unproved hypothesis. We further blind-adjudicate an exact, record-adjacent CLAIMED/PROVISIONAL effectivity perimeter, prove a two-scale tail for the minimal Type-II witness modulus W (cubic variant CLAIMED/PROVISIONAL), and prove fixed-slice and stacked Type-I almost-all theorems. Effectively, (50.7) fails on at most an $\exp\{-c_A(\log\log X)^{3/2}/\sqrt{\log\log\log X}\}$ proportion of hard primes for fixed $0 < A < 1$. In the opposite direction, we prove effectively that $W(p) > c\log p$ and $ck_{\min}(p) > c\log p$ infinitely often; hence both $H_{\mathrm{MOD}}(A)$ and $H_{\mathrm{SPF}}(A)$ are false for every $0 < A < 1$, leaving exactly $A \ge 1$ open in the two pointwise frames. Complete congruence-class certificates through depth T satisfy the effective lower bounds $\log Q \ge (1/2 + o(1))T$ in Type II and $\log Q \ge T + o(T)$ in Type I; these constrain only the certificate-plus-least-prime architecture. A **proved-conditional** reduction would settle all sufficiently large primes if the cubic depth window reached any $\theta > 1/3$, or the square-log window reached $\theta \ge 1/2$. Both window hypotheses are UNPROVED; the currently realized common exponent is only 1/4, with every provisional inheritance retained. We also prove the square-escape theorem

$$W(m^2) = +\infty \quad (m \ge 1), \qquad T < L_{\mathrm{int}}(T) \le T + 2\sqrt{T} + 1,$$

under the moving inverse convention. Thus all-integer tails always contain square survivors, making count below one a prime-essential route; more exactly, the all-integer cubic window fails beyond $\theta = 1/3$, with endpoint obstruction $cb^3 > 1/2$. The all-prime inverse is proved only to satisfy $L_p(T) \le \exp\{(5.2 + o(1))T\}$; no subexponential prime-survivor theorem is claimed. We further classify uniform polynomial escapes completely through degree two: they are exactly the polynomial squares, so none takes a positive prime value. In arbitrary degree we prove only a qualitative, ineffective root-field/cyclotomic obstruction, and the higher- degree classification C_{POLY} remains open. We then prove a finite normal form for every witness datum at fixed n, with $a, u \le 2\lfloor (n+1)/3\rfloor$ and $M \le 8\lfloor (n+1)/3\rfloor^2 - 1$; thus $W(n) = +\infty$ is decidable by finite computation. The resulting computer-assisted theorem proves $W(288) = W(336) = W(4545) = +\infty$, refuting the former complete-square conjecture C_{SQ}. Its open repair C'_{SQ} adds exactly these three composite sporadics to the squares and would still imply Erdős–Straus. The sporadics escape this witness statistic, but explicit unit-fraction representations show that they are not Erdős–Straus counterexamples. A simultaneous forward normal form gives the complete computer-assisted census through 200,000, and an optional full replay through 10^6: the $W = +\infty$ set is exactly the squares together with $288, 336, 4545$ in both ranges. This is finite stress evidence for C'_{SQ}, not a proof. For every prime $p \equiv 1 \pmod{24}$ we further prove the exact finite criterion

$$W(p) < +\infty \iff \exists\, a \equiv 3 \pmod 4, \quad 1 \le a \le 2\lfloor (p+1)/3\rfloor, \quad \exists\, D \mid ((p+a)/4)^2 : D \equiv -(p+a)/4 \pmod a.$$

It is the distilled finite form of the Type-II pointwise problem, not a proof that the displayed divisor set is nonempty for every prime. We further rewrite this criterion as the exact condition that -1 lie in a finite divisor-ratio spectrum, prove exact local laws for $a = 3, 7, 11, 15, 19, 23$, and isolate the resulting finite conspiracy. This is a change of coordinates and a collection of local tests, not a supply theorem, tail bound, pointwise result, or Erdős–Straus progress. The dual taxonomy also proves effective $W(p) \le \sqrt{p+4}$ coverage outside $O(N/(\log N)^{3/2})$ hard primes; this is weaker than Section 51 on both axes. In the same a-frame, for every fixed prime $a \equiv 3 \pmod 4$ we prove

$$\#\{h \le H : (h, a) = 1,\ -1 \notin \mathrm{Rat}_a(h)\} = (C_a + o_a(1))H/\sqrt{\log H},$$

with fixed-progression and shifted-prime upper laws. This proves that no fixed compatible congruence class forces failure at fixed a, but gives no uniformity for a growing family of coefficients. Shifted-form stacking is unconditional only for fixed J. The moving-J tail is conditional on the two named open hypotheses H_{FAIL} and H_{STACK}; no unconditional result on this route beats the standing literature. The Computational a_1 census through all 719781 hard primes below 10^8 has maximum 107, uniquely at 8803369. Its frequency, dependence, conditioning, and budget panels are finite INFO only and imply no tail, independence, or pointwise theorem. An exact gcd theorem localizes post-quotient eligibility failure, but the shifted laws

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1. Introduction and statement map

The Erdős–Straus conjecture asks whether, for every integer $n \geq 2$, there are positive integers x, y, z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

It remains open. The solution parametrizations and average counts of Elsholtz–Tao [8], the divisor criterion presented by Bloom–Elsholtz [2], and the exceptional-set argument reconstructed by Pomerance–Weingartner [20] form the nearest literature context. Yamamoto's reciprocity analysis [27], Salez's computation [22], and the

later counting work of Elsholtz–Planitzer [7] provide additional context; none is used to promote a conditional or provisional statement below.

The two headline assertions of this draft are

$$\boxed{E_{\mathrm{all}}(N) \ll N \exp\{-c(\log N)^{3/4}\}}, \qquad \boxed{E_m(N) \ll_\varepsilon N \exp\left\{-c_\varepsilon \left(\frac{\eta_2(m)(\log N)^3}{\varphi(m)}\right)^{1/4}\right\}}$$

where the second bound is uniform, for every fixed $\varepsilon > 0$, in $3 \le m \le (\log N)^{3-\varepsilon}$; both are CLAIMED/PROVISIONAL. Here $E_m(N)$ has the meaning given in the abstract. The assertions are unconditional in the narrow sense that no unproved hypothesis is assumed, but they are subject to external verification. Their load-bearing prime-slice input, Theorem 9.4, has the same provisional status, and the general form inherits the source §39.7 qualifications. Its exact statement is Theorem 11.8; Theorem 11.7 records the weaker linear-thinning replay. The first headline, and the second at each fixed $m \ge 4$ (with m-dependent constants), were obtained earlier by Kandukoori, Ramesh and Kumar [16, Thms. 1–2]; no priority is claimed for them. What may be new in the second headline is only its uniformity in m; this has not been compared in detail with [16]. Beyond the records, Theorems 14.2 and 14.6 give the exact W windows and effectivity perimeter, while Theorem 15.1 and Corollary 16.5 give the fixed-slice and stacked almost-all frontiers. These upper tails are not pointwise. Theorems 17.1 and 17.3 supply the opposite, pointwise lower tail effectively. Theorems 18.1 and 18.2 then give the exact proved exponential modulus costs for complete-class certificates, while Corollary 18.3 records the scoped least-prime consequence. Theorem 19.3 is the proved-conditional count-below-one reduction: its cubic and square-log window thresholds are respectively $\theta > 1/3$ and $\theta \ge 1/2$, but both named hypotheses are unproved and the common realized window is only $\theta = 1/4$. Theorem 20.1 proves $W(m^2) = +\infty$ for every m, while Corollary 20.4 shows why this reduction must be prime-restricted. Theorem 21.4 then classifies all linear and quadratic uniform polynomial escapes as exactly the squares; Theorem 21.3 gives the qualitative, ineffective necessary root-field condition in every degree. The higher-degree C_{POLY} statement remains a conjecture, not a pointwise conclusion. Theorem 22.1 then gives an exact finite normal form and terminating decision procedure for $W(n)$, without assuming an $a \leftrightarrow M$ swap. The computer-assisted Theorem 22.3 proves that 288, 336, and 4545 have $W = +\infty$, refutes the former C_{SQ}, and motivates the open repair C'_{SQ}. These three values are composite statistic-escapes, not Erdős–Straus counterexamples. Lemma 23.1 then reorders the finite normal form into a simultaneous census: the complete $W = +\infty$ set through 200,000, and through 10^6 in the optional full replay, is exactly the squares and those three sporadics. Theorem 23.2 retains every cancellation branch for sm^2, while Theorem 23.3 proves the exact first residue layers for $2m^2$. Theorem 23.4 is the standalone finite divisor criterion for $p \equiv 1 \pmod{24}$. Its gcd simplification is complete, including even divisors, but it does not prove that a criterion row always exists. Theorem 24.1 reexpresses that criterion by an exact divisor-ratio spectrum, and Theorem 24.2 proves the six explicit small-a laws; neither supplies pointwise nonemptiness. Theorems 32.5, 32.7, and 32.11 then give the exact fixed-a failure law, lower-order budget term, and absence of a fixed congruence wall. These theorems have no growing-a uniformity. Theorem 33.3 is unconditional for every fixed J, while Theorem 33.8 is conditional on the open Hypotheses 33.6 and 33.7; its conditional tail is not an unconditional record. Section 34 is wholly Computational/INFO

and gives no asymptotic or pointwise conclusion. Theorem 25.4 unifies the dual coordinate and Corollary 25.8 gives effective, but weaker, square-root coverage. Theorems 26.1 and 26.2 localize composite cancellation and give necessary shifted laws without a classification. The Computational/INFO census in Section 27 does not advance H_{MOD}. Theorems 28.1 and 28.3 prove finite-depth parameter cylinders and unbounded depth in the three sporadic families without producing a new $W = +\infty$ member. Proposition 28.5 is the anchor-containing obstruction, and Theorem 31.4 closes its one-sided gap for one polynomial datum on each anchored ray, in both divisibility regimes and with no degree, period, or tail-start restriction. Arbitrary non-polynomial witness functions remain untouched. Section 29 gives finite stall anatomy and an explicitly no-evidence toy model, and restates the eventual frontier as every $A \geq 1$ open. Section 30 audits the current Pomerance–Weingartner source and its bibliography without promoting any provisional theorem. Theorem 35.2 records the status-preserving general-m truncation.

The paper has five layers.

(i) Section 3 derives a complete two-case divisor criterion for prime denominators and the identity families that lead to the intrinsic congruence system.

(ii) Sections 4–5 determine that system exactly. The distinct-modulus class mass is cubic, while the full prime slice and its avoidance exponent are quadratic. Squares survive every modulus, and a multi-shift certificate gives a subcubic-cost lower bound for a growing part of the composite system.

(iii) Sections 6–7 state the literal cubic-mass majorant hypothesis H_{PF}, prove its formerly desired conditional consequence, and then refute the hypothesis with all quantifiers visible. The analytic inputs are upper mean-value estimates of Nair–Tenenbaum [18] and Henriot [13]; Shiu's theorem [24] is used in the prime-harvesting arguments.

(iv) Section 8 separates that all-range refutation from the critical window and records the cylinder, tensor, finite-transfer, and factorial-moment walls.

(v) Section 9 states the provisional record inputs. Section 10 then gives the complete fixed-c-free assembly leading, provisionally, to the 3/4 exponent, followed by an explicit internal-verification pedigree. Section 11 proves the m-uniform identity and the two general-numerator layers, including the attested-blind thinned-window strengthening, its fixed-gap range, and the post-adjudication two-regime comparison with Pomerance–Weingartner. Section 12 records the moving genus obstruction, the exact fixed-divisor progression law and its residue-one escape, while keeping pointwise Type-I existence open. Section 13 then proves the one-prime-factor criterion, states the bespoke unproved slice-factor hypothesis, and audits why the named standard hypotheses do not supply its pointwise quantifier. Sections 14– 16 add the witness tail, effectivity/priority register, fixed-slice sieve, and effective stacked almost-all theorem. Section 17 then proves the logarithmic lower tail and sharpens both pointwise exponent frontiers. Section 18 proves the complete-class certificate ceilings, states their exact architecture-only perimeter, and extends all three finite depth censuses. Section 19 then isolates the exact conditional window thresholds, the mass-invariant gap within its formalized absolute-remainder architecture, and the open crossing hypotheses. Section 20 proves square escape, separates the all-integer and prime inverse scales, and audits the Jacobsthal and prime-survivor walls. Section 21 gives the exact local escape

criterion, the ineffective cyclotomic obstruction, the complete degree-two classification, the all-integer square-crossing obstruction, and the historical C_{SQ} proposal with its still-valid census and reductions. Section 22 supersedes that proposal: it proves the finite decision normal form, closes the three-value survivor frontier, refutes C_{SQ}, and states open C'_{SQ}. Section 23 runs that form forward to obtain the complete bounded census, proves the twisted-square first-layer laws, and distills the hard-prime pointwise problem to one finite divisor-class test, without asserting that the test always succeeds. Section 24 turns that test into the exact divisor-ratio spectrum, proves six local laws, and records the finite-conspiracy and bounded fourth-sporadic stress without promoting either to a supply or classification theorem. Sections 25–27 give the dual taxonomy and weaker square-root law, the exact gcd mechanism and coordinate-box hunts, and the independently replayed Computational/INFO census, without a classification or growth law. Section 28 proves finite-depth class decidability, three unbounded limsups, a family finite-certificate ceiling, and the anchor-containing identity obstruction; its projected classes and censuses remain bounded computation. Section 29 separates two finite local diagnostics from a deliberately fragile no-evidence spacing model and restores the exact open exponent frontier. Section 30 records the exact Pomerance–Weingartner statements, range algebra, source-bounded priority verdicts, and Vaughan secondary access record. Section 31 proves that integer-wise polynomial divisibility supplies integer-valued polynomial cofactors and propagates tail positivity back to each of the three anchors. It rules out one polynomial witness family on every anchored forward ray, not non-polynomial or nonuniform supply, and does not support C'_{SQ}. Section 32 proves the fixed-modulus subgroup, confinement, budget, shifted-prime, composite-kernel, and congruence-wall laws, with all constants and errors allowed to depend on fixed a. Section 33 quotes the three mean-value inputs, proves only fixed-J stacking unconditionally, computes the exact local collision factor, and isolates the open H_{FAIL} and H_{STACK} requirements for moving J. Section 34 records the Computational third-decade a_1 census and finite dependence anatomy; its INFO panels establish no uniformity or decorrelation. Section 35 gives the status-preserving general-numerator tails and their exact effectivity perimeter. Finally, Section 36 gives the endpoint sparsity, Kloosterman-fraction audit, widened prefix and large-c wedge, and exact open collision remainder. The composite frontier records both successive refutations: the divisor cube kills the raw hierarchy, and the maximal semiprime grid kills its implication-antichain repair. Scott–Sokal's partition-function correspondence [23] does not itself provide the required pointwise minorant.

Labels are part of the assertions. "Proved" means proved internally in this draft. "Open" means no proof is claimed. The record results in Sections 9–11, the cubic witness tail, Theorem 14.6, and the cubic row of Theorem 35.2 remain CLAIMED/PROVISIONAL as labelled. The logarithmic lower-tail, congruence-certificate, square-escape, local polynomial-escape, and degree-at-most-two classification theorems are proved. The cyclotomic obstruction is proved but qualitatively ineffective; C_{POLY} and C'_{SQ} remain open, while C_{SQ} is refuted and the three former finite-census survivors are proved to have $W = +\infty$. Only the right-to-left inclusion in C'_{SQ} is proved; the complete bounded census is Computational stress evidence,

not an unbounded classification. The forward, ratio, dual-taxonomy, square-root, gcd-cover, shifted-support, and displayed exclusion laws are proved, but do not force prime nonemptiness. The fixed-a failure and congruence-wall laws are proved, but their constants and errors are not uniform in growing a. The shifted-indicator theorem is unconditional only at fixed J; every moving-J conclusion assumes both explicitly open hypotheses H_{FAIL} and H_{STACK}, and no unconditional result in that section improves the standing literature. The a_1 data, ledgers, hunts, hard-prime census, depth-1000 projection, and multiplier censuses are Computational in their exact ranges; normalized trajectories are INFO only. The escape-cylinder limsups are proved but do not produce a new fixed $W = +\infty$ input, and the harvested ceiling concerns only finite fixed laws. The forward-ray theorem closes the former positivity/eligibility leak only for a single polynomial datum on one of the three anchored rays; it covers integer-wise and polynomial divisibility at every degree, period, and tail start, but no arbitrary non-polynomial witness function or classification. It is not evidence for C'_{SQ}. Purity depth is a separate diagnostic and the record-spacing model has no evidential force. Square-root coverage is weaker than Section 51 on both axes, nonsquare finiteness and C'_{SQ} remain open with only its right-to-left inclusion proved, every eventual exponent $A \geq 1$ remains open, and H_{MOD} is unchanged. The Pomerance–Weingartner verdicts are source-bounded and preserve every CLAIMED/PROVISIONAL label. These conclusions concern W, not all Egyptian-fraction decompositions. The count-below-one and crossing implications are proved conditional on their explicitly UNPROVED hypotheses; the mass-invariant proposition is only an architecture theorem. The normalized census panels are informational, the GRH least-prime remark is conditional, and the congruence and prime-survivor walls are Assessments rather than universal pointwise theorems. Writing out internal arguments and reviews for the provisional results does not replace independent external priority and referee checks. In particular, no result of this paper proves the Erdős–Straus conjecture.

Related formalization and frontier. A publicly available Lean development titled "auro-zera" formalizes a sound algebraic identity and finite case-split layer, then assumes one axiom, `good_divisor_exists`, for the remaining hard classes. That axiom is the open divisor-existence core rather than a proof of it; local compilation of the file was not checked. Its cited support is Dyachenko [6], whose own Conclusion says that the infinite general case is not proved constructively and that the relevant finite-covering argument is conditional. The development is nevertheless useful as a formal isolation of the same frontier: Sections 14– 20 quantify the almost-all tail, effective lower tail, complete-class certificate perimeter, conditional count-below-one window, and inverse scales surrounding the witness existence that its axiom assumes. No result here depends on that file or on Dyachenko's conditional existence claim.

The historical comparison is also provisional. Vaughan's 1970 bound [25] is the record identified by the search. The full source-bounded audit in Section 30 checks all 25 pages and the bibliography of Pomerance–Weingartner v2 [20]. That source quotes Vaughan's 2/3 exponent, uses his large-sieve route, and states no stronger $m = 4$ upper bound. It therefore supplies no threat from that PDF to the would-be record, but this is not a complete literature guarantee. Access to Vaughan's primary paper remains blocked, although its DOI is confirmed. A later audit (November 2026) found that the exponent 3/4 for $m = 4$ was obtained earlier by Kandukoori, Ramesh and Kumar [16], so the $m = 4$ theorem is not claimed as a first improvement

since 1970. The general-m comparison is only with exact Theorem 1.3 as displayed in (68.1) and (68.6); all record language remains CLAIMED/PROVISIONAL.

2. Notation and elementary tools

All logarithms are natural. We write $\tau_k(n)$ for the ordered k-fold divisor function, $\tau = \tau_2$, $\omega(n)$ for the number of distinct prime factors, $P^-(n)$ for the least prime factor, and φ for Euler's function. Constants implicit in $O, \ll, \asymp$ are absolute unless a subscript says otherwise.

For $X \geq 3$, put $L = \log X$. Once the intrinsic sets have been defined, their full avoider set is

$$\mathrm{Av}_X(N) = \{n \leq N : n \bmod M \notin \mathscr{R}(M) \text{ for every } M \leq X,\ M \equiv 3 \pmod 4\}. \quad (2)$$

For fixed X this predicate is periodic, so its natural density δ_X exists. The superscript "prime" restricts the moduli to primes $\ell \equiv 3 \pmod 4$.

We use three standard inputs: Bombieri–Vinogradov in a fixed power-saving form, Brun–Titchmarsh, and the quantitative Lovász local lemma. The latter says that events A with a dependency graph have positive joint avoidance probability at least $\prod_A (1 - x_A)$ if

$$\Pr(A) \leq x_A \prod_{B \sim A} (1 - x_B), \qquad 0 \leq x_A < 1.$$

The product-space form and its relation to the independent-set polynomial are discussed in [23].

3. Criterion and identity families

3.1. A two-term lemma and the complete prime criterion.

Lemma 3.1 (two-term criterion)**.** *Let a, b be coprime positive integers. The equation $a/b = 1/y + 1/z$ has a solution in positive integers if and only if b^2 has a divisor $d \equiv -b \pmod a$. Its solutions are*

$$y = \frac{b+d}{a}, \qquad z = \frac{b + b^2/d}{a}.$$

Proof. If $d \mid b^2$ and $d \equiv -b \pmod a$, then $d' = b^2/d$ also satisfies $d' \equiv -b \pmod a$, since b is invertible modulo a. The displayed y, z are integral and

$$(d+b)(d'+b) = b(d + d' + 2b),$$

which gives $1/y + 1/z = a/b$. Conversely, from $ayz = b(y+z)$ put $d = ay - b$ and $d' = az - b$. Positivity follows from $1/y < a/b$ and $1/z < a/b$, and $dd' = a^2yz - ab(y+z) + b^2 = b^2$. Thus $d \mid b^2$ and $d \equiv -b \pmod a$. □

If p is an odd prime in a solution of (1), at least one denominator is divisible by p, because $4xyz = p(xy + yz + zx)$, and not all three are, since after division their reciprocal sum would be at most 3. Thus exactly one or exactly two denominators are divisible by p.

Theorem 3.2 (complete divisor criterion)**.** *Let p be an odd prime. Equation (1) is soluble if and only if at least one of the following holds.*

(B): *There is $q > 0$ with $q \equiv -p \pmod 4$ and a divisor $d \mid x_0^2$, where $x_0 = (p+q)/4$, such that $q \mid d + x_0$.*

(A):

There is $m > 0$ with $m \equiv 3p \pmod 4$ and a divisor $d \mid z_0^2$, where $z_0 = (pm+1)/4$, such that $m \mid d + z_0$.

The corresponding solutions are respectively

$$\left(x_0, \frac{p(x_0+d)}{q}, \frac{p(x_0+x_0^2/d)}{q}\right), \qquad \left(\frac{z_0+d}{m}, \frac{z_0+z_0^2/d}{m}, pz_0\right).$$

Proof. If two denominators are divisible by p, write them py', pz'. Then

$$\frac{4}{p} - \frac{1}{x_0} = \frac{1}{p}\left(\frac{1}{y'} + \frac{1}{z'}\right), \qquad q := 4x_0 - p > 0,$$

so $q/x_0 = 1/y' + 1/z'$. Also $(q, x_0) = (p, x_0) = 1$, since $p \mid x_0$ would put p in all three denominators. Lemma 3.1 gives (B), both formulas, and the converse.

If exactly one denominator is divisible by p, call it pz_0. The equation becomes

$$xy(4z_0 - 1) = pz_0(x + y).$$

Since $p \nmid xy$, write $4z_0 - 1 = pm$. Then $m/z_0 = 1/x + 1/y$ and $(m, z_0) = 1$. Lemma 3.1 gives (A), while $m \equiv 3p \pmod 4$ is exactly the integrality condition for z_0. Reversing the calculation proves the converse. □

This is an independent derivation of the criterion whose shape agrees with Theorem 1 of Bloom–Elsholtz [2].

3.2. **Forced identities.** For $D = \prod r^{e_r}$ define

$$R_0(D) = \prod_r r^{\lceil e_r/2 \rceil}. \tag{3}$$

Thus $R_0(D)$ is the least positive integer R for which $R \mid x$ forces $D \mid x^2$.

Fix q, d in case (B) and an R divisible by $R_0(d)$. The simultaneous conditions

$$x_0 \equiv 0 \pmod R, \qquad x_0 \equiv -d \pmod q, \qquad p = 4x_0 - q$$

form, when consistent, one progression modulo $4\,\mathrm{lcm}(R, q)$, and every member is represented by Theorem 3.2. The case-(A) construction is identical with m in place of q. Two classical-looking forms are also immediate.

Proposition 3.3 (elementary identity families)**.** *For positive integers g, u, v,*

$$p = 4guv - u - v \quad \Longrightarrow \quad \frac{4}{p} = \frac{1}{guv} + \frac{1}{pgu} + \frac{1}{pgv}.$$

For $j \geq 1$, the choice $q = (x_0+1)/j$ and $d = 1$ in case (B) gives

$$jp = (4j-1)x_0 - 1,$$

and hence covers exactly the class $p \equiv -4 \pmod{4j-1}$.

Proof. The first right side has common denominator $pguv$ and numerator $p+u+v = 4guv$. For the second, substitute $q = (x_0+1)/j$ into $p = 4x_0 - q$. Since $4j \equiv 1 \pmod{4j-1}$, integrality is equivalent to $p \equiv -4 \pmod{4j-1}$; then $q \mid x_0+1 = d+x_0$ and Theorem 3.2 applies. □

Dyachenko's ED2 identity family [6] is an algebraic cousin of the identities in Proposition 3.3 and of the multiplier supply below. We cite it only at the identity level: the paper's own Conclusion characterizes its infinite general case as not constructively proved and the relevant finite-covering steps as conditional.

The identity from which the complete intrinsic system arises is the following multiplier form.

Lemma 3.4 (intrinsic multiplier identity)**.** *Let $M \equiv 3 \pmod 4$, $A = (M+1)/4$, and $A = uvw$ a factorization into positive integers. If the positive integer n satisfies $nv \equiv -u \pmod M$ and $s = (nv+u)/M$, then*

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw}. \tag{4}$$

In particular $n \equiv -uv^{-1} \pmod M$ is a forced-solubility class.

Proof. Since $(A, M) = 1$, the factor v is invertible modulo M, and s is a positive integer. On the common denominator $nsuvw$, the numerator on the right is $nv + u + s = sM + s = s(M+1) = 4suvw$. □

4. The complete intrinsic system

For $M \equiv 3 \pmod 4$, define

$$\mathscr{R}(M) = \{-uv^{-1} \pmod M : uvw = (M+1)/4\}, \qquad F(M) = |\mathscr{R}(M)|.$$

"Complete" always means complete for the multiplier identity of Lemma 3.4; it does not assert that all possible representations of (1) arise from one fixed M.

Lemma 4.1 (exact structure and deduplication)**.** *Put $A = (M+1)/4$. Then*

$$\mathscr{R}(M) = \{-4D \pmod M : D \mid A^2\}, \qquad \frac{\tau(A^2)-1}{2} \le F(M) \le \tau(A^2). \tag{5}$$

The lower-bound classes may be taken from the divisors $D < A$, and they are pairwise distinct.

Proof. A factorization $A = uvw$ gives $D = u^2w \mid A^2$, and $4A \equiv 1 \pmod M$ gives

$$-uv^{-1} \equiv -u^2wA^{-1} \equiv -4u^2w = -4D \pmod M.$$

Conversely, write $A = \prod p^{e_p}$ and $D = \prod p^{d_p}$ with $0 \le d_p \le 2e_p$. Set

$$v_p(u) = \lfloor d_p/2 \rfloor, \quad v_p(w) = d_p - 2\lfloor d_p/2 \rfloor, \quad v_p(v) = e_p - v_p(u) - v_p(w).$$

The last number is nonnegative; hence $uvw = A$ and $u^2w = D$. This proves the set identity and upper bound. The involution $D \mapsto A^2/D$ has unique fixed point A, so exactly $(\tau(A^2)-1)/2$ divisors have $D < A$. For them $0 < 4D < M$, making the residues $-4D$ distinct. □

The set depends on M, not on a factorization $M = k\ell$. Thus a sum over pairs (k, ℓ) must be quotiented by their product before it represents distinct intrinsic moduli.

Theorem 4.2 (exact cubic supply)**.** *As $Q \to \infty$,*

$$\sum_{\substack{M \le Q \\ M \equiv 3 \ (4)}} \frac{F(M)}{M} \asymp (\log Q)^3. \tag{6}$$

More precisely,

$$\left(\frac{1}{48\zeta(2)}+o(1)\right)(\log Q)^3 \le \sum_{\substack{M\le Q\\ M\equiv 3 \pmod 4}} \frac{F(M)}{M} \le \left(\frac{1}{24\zeta(2)}+o(1)\right)(\log Q)^3.$$

Proof. At each prime,

$$\sum_{e\ge 0}(2e+1)z^e = \frac{1-z^2}{(1-z)^3},$$

so

$$\sum_{n\ge 1}\frac{\tau(n^2)}{n^s} = \frac{\zeta(s)^3}{\zeta(2s)}, \qquad \tau(n^2)=\sum_{d^2\mid n}\mu(d)\tau_3(n/d^2).$$

The elementary three-dimensional hyperbola argument gives

$$\sum_{m\le x}\tau_3(m) = \tfrac{1}{2}x(\log x)^2 + O(x\log x).$$

Substitution in the coefficient identity, using absolute convergence of the d^{-2} sum, yields

$$\sum_{n\le x}\tau(n^2) = \frac{x(\log x)^2}{2\zeta(2)} + O(x\log x).$$

Partial summation gives

$$\sum_{n\le x}\frac{\tau(n^2)}{n} = \frac{(\log x)^3}{6\zeta(2)} + O((\log x)^2). \tag{7}$$

Write $M = 4A-1$. Since $(4A-1)^{-1} = (4A)^{-1} + O(A^{-2})$ and $\tau(A^2) \ll_\varepsilon A^\varepsilon$, replacing the denominator costs $O(1)$. Apply the two halves of Lemma 4.1 in (7); the subtracted 1 in the lower bound costs only $O(\log Q)$. □

A mass-only Rankin sieve with auxiliary scale $X = \mathrm{e}^t$, class mass $O(t^B)$, and product-tail budget $t^{B+1} \ll \log N$ can save at most $O((\log N)^{B/(B+1)})$. Thus the intrinsic supply has the formal ceiling 3/4. This is only a ceiling inside that mass/product-tail model; it is not a nonexistence theorem for other uses of the class geometry.

4.1. **Shifted-divisor form.**

Lemma 4.3 (exact reorganization)**.** *For $A = (M+1)/4$,*

$$D \mid A^2 \iff R_0(D) \mid A \iff M \equiv -1 \pmod{4R_0(D)}.$$

Consequently $n \notin \mathrm{Av}_X$ if and only if there are $D, M \ge 1$ such that

$$M \mid n+4D, \qquad M \le X, \qquad M \equiv -1 \pmod{4R_0(D)}. \tag{8}$$

If $R = \prod p^{a_p}$, then

$$R_0(D) = R \iff D = R^2/s \quad \text{for some } s \mid \mathrm{rad}(R). \tag{9}$$

Proof. At a prime, $D \mid A^2$ says $e_p(D) \le 2v_p(A)$, exactly the assertion $\lceil e_p(D)/2\rceil \le v_p(A)$. This proves the first line. Lemma 4.1 changes $n \equiv -4D \pmod M$ into (8), in both directions. Finally, if $v_p(R) = a$, the condition $\lceil v_p(D)/2\rceil = a$ allows $v_p(D) = 2a$ or $2a-1$, independently at each $p \mid R$, which is (9). □

The congruence in (8) implies $R_0(D) \le (X+1)/4$ and hence $D \le (X+1)^2/16$; no D-tail is omitted.

5. Squares and the prime slice

5.1. The square-class phenomenon.

Lemma 5.1 (quadratic sign). *If $M = 4A - 1$ and $D \mid A^2$, then*

$$\left(\frac{D}{M}\right) = 1, \qquad \left(\frac{-4D}{M}\right) = -1.$$

Thus every unit class in $\mathscr{R}(M)$ has Jacobi sign -1. Every n with $(n/M) \in \{0, 1\}$ avoids $\mathscr{R}(M)$; in particular, every perfect square avoids the complete intrinsic system for every X.

Proof. For an odd prime $p \mid A$, quadratic reciprocity and $M \equiv -1 \pmod p$, $M \equiv 3 \pmod 4$ give

$$\left(\frac{p}{M}\right) = \left(\frac{M}{p}\right)(-1)^{(p-1)(M-1)/4} = \left(\frac{-1}{p}\right)(-1)^{(p-1)/2} = 1.$$

If $2 \mid A$, then $M \equiv 7 \pmod 8$ and $(2/M) = 1$. Multiplicativity gives $(D/M) = 1$. Since $(-1/M) = -1$ and $(4/M) = 1$, the second identity follows. Also $(A, M) = 1$, so all intrinsic classes are units. □

In particular $|\operatorname{Av}_X(N)| \geq \lfloor\sqrt{N}\rfloor$. Unit squares modulo the odd period $P_X = \operatorname{lcm}\{M \leq X : M \equiv 3 \pmod 4\}$ also give

$$\delta_X \geq \frac{\varphi(P_X)}{2^{\omega(P_X)}P_X} \geq \exp\{-O(X/\log X)\}.$$

These bounds are too small to test a cubic exponent when $\log N \asymp L^4$.

5.2. Harvesting prime classes.

We record the fixed-multiplier harvesting input in a self-contained form.

Lemma 5.2 (prime harvest). *For primes $\ell \equiv 3 \pmod 4$, there are subsets $\mathcal{C}_\ell \subseteq \mathscr{R}(\ell)$ such that*

$$\sum_{X^{1/2} < \ell \leq X} \frac{|\mathcal{C}_\ell|}{\ell} \asymp L^2. \tag{10}$$

Proof. In a dyadic interval $x < \ell \leq 2x$, put $z = x^{1/6}$ and take coprime $u, v \leq z$ with $\omega(uv) \leq D \log\log X$. The progression $\ell \equiv -1 \pmod{4uv}$ makes $uv \mid (\ell + 1)/4$, so Lemma 3.4 supplies the residue $-uv^{-1} \pmod \ell$. Distinct reduced pairs give distinct residues: a collision implies $\ell \mid uv' - u'v$, while $|uv' - u'v| < \ell^{2/3} < \ell$, and hence $u/v = u'/v'$.

The elementary harmonic estimates

$$\sum_{\substack{u,v \leq z \\ (u,v)=1}} \frac{1}{uv} \asymp (\log z)^2, \qquad \sum_{u,v \leq z} \frac{1}{\varphi(u)\varphi(v)} \ll (\log z)^2$$

follow respectively by Möbius inversion and from $\sum_{n \leq z} 1/\varphi(n) \ll \log z$. Moreover, for fixed $1 < t < 2$,

$$\sum_{u,v \leq z} \frac{t^{\omega(u)+\omega(v)}}{uv} \ll_t (\log z)^{2t}.$$

Rankin's inequality therefore shows that a fixed D leaves a positive proportion of the first harmonic sum. A modulus $4uv$ is then generated at most $2^{\omega(uv)} \leq (\log X)^{D \log 2}$

times. Bombieri–Vinogradov applies since $4uv \le 4x^{1/3}$ and absorbs this fixed logarithmic multiplicity. It gives $\gg x \log x$ retained prime incidences in the block; division by $\ell \asymp x$ gives $\gg \log x$. There are $\asymp L$ dyadic blocks between $X^{1/2}$ and X, proving the lower bound in (10). Brun–Titchmarsh and the second harmonic estimate give $O(x \log x)$ incidences per block and hence the matching upper bound. □

Lemma 5.3 (full prime mass)**.** *There are $c, C > 0$ such that*

$$cL^2 \le \sum_{\substack{\ell \le X \\ \ell \equiv 3 \ (4)}} \frac{F(\ell)}{\ell} \le CL^2, \tag{11}$$

where ℓ is prime.

Proof. The lower bound follows from Lemma 5.2. For the upper bound put $A = (\ell+1)/4$. Lemma 4.1 and $2e+1 \le (e+1)(e+2)/2$ give $F(\ell) \le \tau(A^2) \le \tau_3(A)$. On $A \asymp Y$, write $A = abc$ and designate a largest coordinate as c, at a factor-three cost; then $ab \ll Y^{2/3}$. For fixed a, b, Brun–Titchmarsh for the prime $4abc - 1$ modulo $4ab$ gives

$$\#\{c : A \asymp Y,\ 4abc - 1 \text{ prime}\} \ll \frac{Y}{\varphi(4ab) \log Y}.$$

Since $\varphi(4ab) \ge \varphi(a)\varphi(b)$ and $\sum_{m \le y} 1/\varphi(m) \ll \log(2y)$,

$$\sum_{\substack{A \asymp Y \\ 4A-1 \text{ prime}}} \tau(A^2) \ll Y \log Y.$$

After division by $\ell \asymp Y$, each dyadic block costs $O(\log Y)$, and summing proves the upper bound. □

Theorem 5.4 (two-sided prime-slice avoidance)**.** *There are $c_1, C_2, C_3 > 0$ such that, for all large X and $\log N \ge C_3 L^3$,*

$$N\mathrm{e}^{-C_2 L^2} \le |\operatorname{Av}_X^{\mathrm{prime}}(N)| \le N\mathrm{e}^{-c_1 L^2}. \tag{12}$$

Its natural density is $\mathrm{e}^{-\Theta(L^2)}$.

Proof. Chinese remaindering over distinct prime moduli gives exact density

$$V_X = \prod_{\substack{\ell \le X \\ \ell \equiv 3 \ (4)}} \left(1 - \frac{F(\ell)}{\ell}\right).$$

Lemma 5.1 gives $F(\ell)/\ell < 1/2$. Lemma 5.3 and $-2u \le \log(1-u) \le -u$ imply $V_X = \mathrm{e}^{-\Theta(L^2)}$.

For the finite interval put $a_\ell = F(\ell)/\ell$ and $\mu = \sum a_\ell = \Theta(L^2)$. Odd and even Bonferroni truncations at degrees $r = O(\mu)$ bracket the avoider indicator. Their omitted CRT main tails are bounded by $\sum_{j>r} \mu^j/j!$; taking the constant in r large makes each tail at most $V_X/4$. An intersection indexed by a product d of j primes occupies $\prod_{\ell \mid d} F(\ell)$ classes modulo d, so its finite count is

$$N \prod_{\ell \mid d} a_\ell + O\left(\prod_{\ell \mid d} F(\ell)\right).$$

The preceding proof of Lemma 5.3, before division by ℓ, gives $W := \sum_{\ell \le X} F(\ell) \ll XL$. Hence all rounding errors through degree r total at most $\sum_{j \le r} W^j/j! = \exp\{O(L^3)\}$. If C_3 is large this is at most $NV_X/4$. The two Bonferroni bounds now give (12). □

Since the full avoiders are contained in the prime-slice avoiders, we obtain the promised unconditional full-system upper bound.

Corollary 5.5. *If $\log N \ge CL^3$, then*

$$|\operatorname{Av}_X(N)| \ll N \exp\{-cL^2\}.$$

The same upper bound holds for δ_X without a condition on N.

5.3. **A growing multi-shift certificate.** For $Y \ge 1$, let $\operatorname{Av}_X^{(R \le Y)}$ denote avoidance of every pair in (8) with $R_0(D) \le Y$.

Theorem 5.6 (multi-shift prime certificate)**.** *Fix $\varepsilon > 0$. For all sufficiently large X, uniformly for $1 \le Y \le L^{3-\varepsilon}$ and $\log N \ge L^4$,*

$$|\operatorname{Av}_X^{(R \le Y)}(N)| \ge N \exp\{-C_\varepsilon Y \log(2Y) \log L\}.$$

The corresponding natural density has the same lower bound. At $Y = L^{3-\varepsilon}$ the exponent is $o(L^3)$.

Proof. Let

$$\mathcal{D}_Y = \{(R, D) : R \le Y,\ D = R^2/s,\ s \mid \operatorname{rad}(R)\}, \qquad H = |\mathcal{D}_Y|.$$

The standard Euler-product estimate

$$\sum_{R \le Y} 2^{\omega(R)} \ll Y \log(2Y)$$

gives $H \ll Y \log(2Y)$. For each prime $p \le X$, $p \equiv 3 \pmod 4$, forbid

$$B_p = \{-4D \pmod p : (R, D) \in \mathcal{D}_Y,\ p \nmid R\}.$$

Every B_p is nonzero and has size $g(p) \le \min(H, p-1)$. Avoiding all B_p certifies the desired avoidance: a hit in (8) has $(M, R) = 1$, and some prime $p \equiv 3 \pmod 4$ divides M to odd exponent; then $p \le X$, $p \nmid R$, and $n \equiv -4D \pmod p$, a contradiction.

For $p \le 2H$, choose the allowed residue $n \equiv 0 \pmod p$, and let Q_0 be their product. Then $\log Q_0 = O(H)$. In $n = Q_0 m$, every remaining forbidden set has $g(p) \le H < p/2$, and

$$\mu_Y = \sum_{2H < p \le X} \frac{g(p)}{p} \ll H \log L.$$

Odd Bonferroni at degree $r \asymp \mu_Y$ retains a fixed fraction of the Euler product, at least $\exp(-2\mu_Y)$. Since $W_Y = \sum g(p) \le H\pi(X)$, the total rounding error is

$$\exp\{O(r \log W_Y)\} = \exp\{O(HL \log L)\}.$$

For $Y \le L^{3-\varepsilon}$ this is $\exp\{o(L^4)\}$ and is negligible under $\log N \ge L^4$. Restoring Q_0 proves the claimed bound. On complete periods there is no rounding error, proving the density statement. □

6. The cubic-mass hypothesis and its conditional consequence

Put

$$\mu_X = \sum_{\substack{M\le X\\ M\equiv 3\ (4)}} \frac{F(M)}{M} \asymp L^3.$$

The following is the literal hypothesis proposed for partition-free assembly. Its all-range quantifier in N is essential.

Hypothesis 6.1 (the literal cubic-mass majorant H_{PF})**.** *There are absolute $c, C > 0$ such that, for every sufficiently large X, every $J \ge C\mu_X$, and every N satisfying*

$$J \log X \le \tfrac{1}{2} \log N,$$

there is a nonnegative arithmetic function ν on $[1, N]$ which is at least 1 on every member of $\mathrm{Av}_X(N)$ and satisfies

$$\frac{1}{N} \sum_{n\le N} \nu(n) \ll \exp(-c\mu_X).$$

Its evaluation uses exact congruence counts modulo lcms of at most J participating moduli; every such lcm is at most $X^J \le N^{1/2}$. The coefficient sum and number of congruence-count terms are at most $\exp(O(J))$, so aggregate rounding errors are required to be negligible under the displayed degree budget.

This hypothesis concerns the complete system, not arbitrary subfamilies. The latter formulation is already false: all $D = 1$ classes with $3 \mid M$ lie in $n \equiv 2 \pmod 3$, while the complement has density $2/3$ despite growing class mass.

Theorem 6.2 (formal conditional 3/4 bound)**.** *If H_{PF} holds, then for some $c > 0$ the number $E_{\mathrm{all}}(N)$ of exceptional denominators $n \le N$ satisfies*

$$E_{\mathrm{all}}(N) \le N \exp\{-c(\log N)^{3/4}\}.$$

Proof. Put $t = \alpha(\log N)^{1/4}$, $X = \mathrm{e}^t$. By Theorem 4.2, $\mu_X \asymp t^3$. Take $J = C\mu_X$. Then $J \log X \ll \alpha^4 \log N$, so small fixed α meets the degree condition. Hypothesis 6.1 bounds the avoiders by $N \exp(-c\mu_X)$. Every exceptional denominator is an avoider, because every intrinsic class is forced soluble by Lemma 3.4. □

The implication is valid, but its antecedent will now be disproved.

7. Rough moduli and refutation of H_{PF}

For primes q define

$$b_q = \frac{L^3}{q \log z}, \qquad W_{\boldsymbol{a}}(M) = \mathbf{1}_{P^-(M)>z} \exp\left(2 \sum_{p\mid M} a_p\right).$$

The sum in the exponential is over distinct prime factors. An *atom* is a pair (M, r) with $M \le X$, $M \equiv 3 \pmod 4$, and $r \in \mathscr{R}(M)$; it is the event $n \equiv r \pmod M$. Thus a modulus M contributes exactly $F(M)$ atoms.

Theorem 7.1 (weighted rough-modulus mean)**.** *Uniformly for $3 \le z \le X^{1/10}$ and charges $0 \le a_p \le 1$, put $E = \sum_{z<p\le X} a_p/p$. Then*

$$\sum_{\substack{M\le X\\ M\equiv 3 \ (4)}} \frac{F(M)}{M} W_{\boldsymbol{a}}(M) \ll \mathrm{e}^{CE} \frac{L^3}{\log z}. \tag{13}$$

In particular, if $a_p = O(L^3/(p\log z))$ and $z = L^3/\sqrt{\log L}$, then $E = o(1)$ and the left side is $o(L^3)$.

Proof. Write $M = 4m-1$ and define the multiplicative function

$$w(p^\nu) = \begin{cases} 0, & p \le z, \\ \mathrm{e}^{2a_p}, & p > z, \end{cases} \qquad w(1) = 1.$$

Lemma 4.1 gives $F(4m-1)W_{\boldsymbol{a}}(4m-1) \le w(4m-1)\tau_3(m)$. Apply the Nair–Tenenbaum estimate in Henriot's discriminant-uniform form [18, 13] to the determinant-one linear forms $Q_1(t) = 4t-1$, $Q_2(t) = t$ and the multiplicative function $\mathcal{F}(u,v) = w(u)\tau_3(v)$. The class parameters are uniform in the charges. On a dyadic interval of length V, the local factors give

$$V \prod_{p\ll V} (1-2/p)(1+w(p)/p)(1+3/p) \ll V \frac{\log^2(2V)}{\log z} \exp\left(C \sum_{z<p\ll V} \frac{a_p}{p} \right).$$

Indeed the coefficient of $1/p$ in the logarithm is 1 for $p \le z$ and $2 + O(a_p)$ for $p > z$. The first possible rough value exceeds z, so the lowest nonempty blocks obey the same estimate after changing the constant. Dyadic summation and partial summation yield

$$\sum \frac{F(M)W_{\boldsymbol{a}}(M)}{M} \ll \frac{\mathrm{e}^{CE}}{\log z} \left(L^2 + \int_z^X \frac{(\log t)^2}{t}\, dt \right),$$

which proves (13). Finally $E \ll L^3/(z\log z) = o(1)$ for the stated charges and cutoff. □

The pointwise neighborhood estimate needs a small maximal-order correction. Fix a sufficiently large absolute C_d and put

$$u_q = \frac{\log(2q)}{q} \exp\left(\frac{C_d \log(2q)}{\log\log(3q)} \right).$$

For an atom modulus divisible by q, set

$$T_q(z;\boldsymbol{a}) = \sum_{\substack{M\le X,\ M\equiv 3 \ (4)\\ P^-(M)>z,\ q\mid M}} \frac{F(M)}{M} \exp\left(2\sum_{p\mid M} a_p \right).$$

Theorem 7.2 (softened neighborhood bound)**.** *Let*

$$z = \frac{L^3 \log\log L}{\log L}. \tag{14}$$

For all large X there is an absolute $A > 0$ such that $a_q = A(b_q + u_q)$ for primes $z < q \le X$ satisfies

$$T_q(z; \boldsymbol{a}) \le a_q, \quad \max_{q>z} a_q = o(1), \quad \sum_{q>z} \frac{a_q}{q} = o(1),$$

and

$$\sum_{\substack{M \le X \\ M \equiv 3 \ (4)}} \frac{F(M)}{M} W_{\boldsymbol{a}}(M) \ll \frac{L^3}{\log z} = o(L^3). \tag{15}$$

The same conclusions hold for $z = L^3 g(L)/\log L$ if $g(L) \to \infty$ and $g(L) = o(\log L)$.

Proof. Use the maximal-order bound

$$\tau_3(n) \le \exp\left(\frac{C_d' \log(2n)}{\log\log(3n)} \right). \tag{16}$$

Fix $q > z$, write $M = qk$, and choose $r \in \{1, 3\}$ with $qr \equiv 3 \pmod 4$. With $k = 4t + r$,

$$\frac{qk+1}{4} = qt + c, \qquad c = \frac{qr+1}{4}, \qquad 4c - qr = 1.$$

The cofactor and shifted factor are determinant-one linear forms, and the coefficient norm of their product is at most $13q$. Also $W_{\boldsymbol{a}}(qk) \le \mathrm{e}^{2a_q} w(k)$ and $F(qk) \le \tau_3((qk+1)/4)$.

Split at $k = C_0 q^{1/2}$, where C_0 is large enough for Henriot's coefficient-size condition. In the lower range, (16), roughness, and $\sum_{k \le y} 1/k \ll \log(2y)$ give

$$\frac{1}{q} \sum_{k < C_0 q^{1/2}} \frac{F(qk) W_{\boldsymbol{a}}(qk)}{k} \ll u_q.$$

Here $a_p = O(1/\log\log L + p^{-1+o(1)})$, and a rough qk has at most $\log(qk)/\log z$ prime factors, so its exponential weight is absorbed by enlarging the constant in (16). This includes $k = 1$.

Decompose the high range dyadically in t. Henriot's theorem applies to the two determinant-one forms and $w(4t + r)\tau_3(qt + c)$, giving

$$\sum_{V < k \le 2V} w(k) \tau_3((qk+1)/4) \ll V \frac{(\log(2V))^2}{\log z}.$$

After division by qk and summation over blocks, this is $O(L^3/(q \log z)) = O(b_q)$. Thus $T_q \ll b_q + u_q$ uniformly once $\max a_p \le 1$ and $\sum a_p/p \le 1$. Choose the fixed A larger than this implicit constant; the estimates below verify those two conditions for all sufficiently large X, closing the argument without circularity.

Indeed

$$\sum_{q>z} \frac{b_q}{q} \ll \frac{L^3}{z \log z} = o(1).$$

For large q, the exponential in u_q is at most $q^{1/4}$, whence

$$\sum_{q>z} \frac{u_q}{q} \ll \sum_{n>z} \frac{\log(2n)}{n^{7/4}} = o(1).$$

The same estimates give $\max a_q = o(1)$. Theorem 7.1 proves (15). Replacing $\log\log L$ by $g(L)$ changes the first displayed bound to $O(1/g(L))$ and nothing else. □

Theorem 7.3 (natural-density lower bound)**.** *With* (14),

$$\delta_X \geq \exp\left\{-O\left(\frac{L^3 \log\log L}{\log L}\right)\right\} = \exp\{-o(L^3)\}. \tag{17}$$

More generally, the last cutoff in Theorem 7.2 gives $\delta_X \geq \exp\{-O(L^3 g(L)/\log L)\}$.

Proof. First impose $n \equiv 0 \pmod{p}$ for every odd prime $p \leq z$. Work on the finite uniform CRT space modulo the lcm of their product and all participating moduli. The conditioning costs $\exp\{-\vartheta(z) + O(1)\} = \exp\{-O(z)\}$. If an intrinsic atom has a modulus divisible by such a p, it is now impossible: its residue $-4D$ is a unit modulo M, since $(D, M) = 1$. Every surviving distinct atom has a z-rough modulus M, remains uniform, and has probability exactly $1/M$.

Join two atoms when their moduli share a prime and set

$$x_A = \frac{1}{M} \exp\left(2 \sum_{q \mid M} a_q\right)$$

for an atom of modulus M. Roughness and $\max a_q = o(1)$ give $x_A < 1/2$. Theorem 7.2 bounds the total activity of the neighbors of A by $\sum_{q \mid M} a_q$. Therefore

$$\prod_{B \sim A} (1 - x_B) \geq \exp\left(-2 \sum_{B \sim A} x_B\right) \geq \exp\left(-2 \sum_{q \mid M} a_q\right),$$

so $\Pr(A) \leq x_A \prod_{B \sim A}(1 - x_B)$. The quantitative local lemma gives

$$\begin{aligned}\Pr(\text{no surviving atom} \mid \text{quarantine}) &\geq \prod_A (1 - x_A) \\ &\geq \exp\left(-2 \sum_A x_A\right) \geq \exp\{-O(L^3/\log z)\}.\end{aligned}$$

where the last inequality is (15). Multiplying by the quarantine cost proves (17). □

Corollary 7.4 (refutation of the literal hypothesis)**.** *Hypothesis* H_{PF} *as stated in Hypothesis 6.1 is false.*

Proof. Theorem 4.2 gives $\mu_X \geq c_0 L^3$. If H_{PF} held with constants c, C, choose $J \geq C\mu_X$. For each fixed sufficiently large X, take arbitrarily large multiples $N = mP_X$ of the full period satisfying $J \log X \leq \frac{1}{2} \log N$. Periodicity gives $|\operatorname{Av}_X(N)|/N = \delta_X$. Since the majorant is at least one on every avoider, H_{PF} would imply

$$\delta_X \ll \mathrm{e}^{-c\mu_X} \leq \mathrm{e}^{-cc_0 L^3}.$$

This contradicts Theorem 7.3 for all sufficiently large X. □

The scope is exact: this refutes the all-range quantifiers of H_{PF}, not the conclusion of Theorem 6.2 by every possible route and not the critical-window statement below.

8. The critical window and transfer walls

Hypothesis 8.1 (critical-window variant H'_{PF}; open)**.** *There are* $c, C > 0$ *and* $0 < c_- < c_+$*, with* c_- *large enough for the degree budget, such that the majorant in Hypothesis 6.1 exists whenever*

$$c_- L^4 \leq \log N \leq c_+ L^4, \qquad J \geq C\mu_X, \qquad J \log X \leq \tfrac{1}{2} \log N.$$

No assertion is made outside this window.

This remains open and would still imply Theorem 6.2. Corollary 7.4 uses arbitrarily large complete-period multiples at fixed X, which need not lie in this window.

8.1. **Cylinder and tensor obstructions.**

Theorem 8.2 (all-avoider cylinder wall)**.** *If a nonempty union $\mathcal{U}$ of complete classes modulo Q is contained in the full intrinsic avoider set, then*

$$\prod_{\substack{\ell \le X,\ \ell \equiv 3 \pmod 4 \\ \ell \text{ prime}}} \ell \mid Q, \qquad \log Q \ge (1/2 + o(1))X.$$

Proof. Choose one class $a \pmod Q$ in $\mathcal{U}$. For each prime $\ell \equiv 3 \pmod 4$, the divisor $D = 1$ gives the forbidden class $-4 \pmod \ell$. If $\ell \nmid Q$, the Chinese remainder theorem produces an integer lying both in $a \pmod Q$ and in that forbidden class, a contradiction. Thus every such prime divides Q, and the prime number theorem in the progression $3 \pmod 4$ gives the estimate. □

The CRT space is a product over prime powers, but the avoider indicator is not a tensor product: its atomic factors overlap.

Lemma 8.3 (explicit non-tensor obstruction)**.** *Already at $X = 15$, the complete avoider indicator modulo $1155 = 3 \cdot 5 \cdot 7 \cdot 11$ does not factor into functions of the four prime coordinates.*

Proof. The residues

n	$n \bmod 3$	$n \bmod 5$	$n \bmod 7$	$n \bmod 11$	$\mathbf{1}_{\mathrm{Av}_{15}}(n)$
231	0	1	0	0	1
616	1	1	0	0	1
693	0	3	0	0	1
1078	1	3	0	0	0

form a two-coordinate rectangle with the other coordinates fixed. A tensor product containing the first three corners contains the fourth. To check the final column without an enumeration, Lemma 4.1 gives

$$\mathscr{R}(3) = \{2\}, \quad \mathscr{R}(7) = \{3, 5, 6\}, \quad \mathscr{R}(11) = \{7, 8, 10\}, \quad \mathscr{R}(15) = \{7, 11, 13, 14\}.$$

The first three rows miss these four sets. The last row is $13 \pmod{15}$ and is killed by $D = 8 \mid 4^2$. □

Thus Fourier transformation produces a hypergraph convolution, not a product of local Fourier transforms.

8.2. **The corrected finite-transfer criterion.** After the odd-prime quarantine, write $n = P_z m$. For $U \subseteq \mathbb{Z}/d\mathbb{Z}$, put $1_{U,d}(m) = \mathbf{1}_{m \bmod d \in U}$ and $\rho(U) = |U|$.

Lemma 8.4 (residue-set transfer)**.** *Let $\mathcal{F}_X$ be a specified family of surviving atoms and I_X a finite index set. Suppose*

$$B_X(m) = \sum_{\alpha \in I_X} c_\alpha 1_{U_\alpha, d_\alpha}(m) \le \mathbf{1}_{\{\text{no atom of } \mathcal{F}_X \text{ at } m\}}$$

pointwise, where each d_α divides the surviving CRT period and $(d_\alpha, P_z) = 1$. If

$$\mathbb{E}_{\mathrm{CRT}} B_X \geq \mathrm{e}^{-o(L^3)}, \qquad \log \max_\alpha d_\alpha = o(L^4),$$
$$\log \mathcal{R}_X = o(L^4), \qquad \mathcal{R}_X = \sum_\alpha |c_\alpha| \rho(U_\alpha).$$

then, uniformly for fixed $0 < c_- < c_+$ and $c_- L^4 \leq \log N \leq c_+ L^4$,

$$|\operatorname{Av}(\mathcal{F}_X; N)| \geq N \mathrm{e}^{-o(L^3)}.$$

Proof. Put $H = \lfloor N/P_z \rfloor$. A set of $\rho(U_\alpha)$ classes modulo d_α has count

$$H \frac{\rho(U_\alpha)}{d_\alpha} + O(\rho(U_\alpha)).$$

Hence $\sum_{m \leq H} B_X(m) = H \mathbb{E}_{\mathrm{CRT}} B_X + O(\mathcal{R}_X)$. The main term has logarithm $\log N - O(z) - o(L^3) = \log N - o(L^3)$, while the error has logarithm $o(L^4)$ and is negligible because $\log N \geq c_- L^4$. The pointwise inequality and $P_z = \exp\{O(z)\} = \exp\{o(L^3)\}$ finish the proof. □

For an atom monomial, compatibility gives one residue class and incompatibility gives none, so $\rho \leq 1$. The following is the isolated low-degree criterion; the factor $\rho(U)$ in the general ledger is essential.

Corollary 8.5 (monomial finite-transfer criterion)**.** *Suppose real coefficients c_S, supported on sets of at most $r = o(L^3)$ surviving atoms, make*

$$B_X(n) = \sum_S c_S \prod_{A \in S} \mathbf{1}_A(n) \leq \mathbf{1}_{\{\text{no surviving atom at } n\}}$$

pointwise. If its exact CRT mean is at least $\mathrm{e}^{-o(L^3)}$ and $\log \sum_S |c_S| = o(L^4)$, then $|\operatorname{Av}_X(N)| \geq N \mathrm{e}^{-o(L^3)}$ uniformly for $\log N \asymp L^4$.

Proof. Apply Lemma 8.4. Compatible monomials are single classes modulo lcms $d_S \leq X^r = \exp\{o(L^4)\}$, incompatible monomials vanish, and the weighted ledger is at most $\sum_S |c_S|$. □

8.3. **The prime side transfers.** Let $S_\ell = P_z^{-1} \mathscr{R}(\ell) \pmod{\ell}$. Choose $h(L) \to \infty$ slowly and put $Y = L^4/h(L)$, so $z < Y < X$. Define

$$d_0 = \prod_{\substack{z < \ell \leq Y \\ \ell \equiv 3 \ (4)}} \ell, \qquad U_0 = \prod_{\substack{z < \ell \leq Y \\ \ell \equiv 3 \ (4)}} ((\mathbb{Z}/\ell\mathbb{Z}) \setminus S_\ell), \qquad T_0 = 1_{U_0, d_0}.$$

Here and below ℓ is prime. Then $\log d_0 = O(Y) = o(L^4)$. Let H_P count violated prime atoms with $Y < \ell \leq X$, put $\mu_P = \mathbb{E} H_P = \Theta(L^2)$, let r be the least odd integer at least $8\mu_P$, and define

$$Q_r(H_P) = \sum_{j=0}^{r} (-1)^j \binom{H_P}{j}, \qquad B_P = T_0 Q_r(H_P).$$

Theorem 8.6 (prime-side two-level minorant)**.** *The function B_P is pointwise at most the surviving prime-system avoider indicator. Its tail degree is $O(L^2)$, every term has modulus logarithm $o(L^4)$,*

$$\mathbb{E}_{\mathrm{CRT}} B_P \geq \mathrm{e}^{-O(L^2)}, \qquad \log \sum_\alpha |c_\alpha| \rho(U_\alpha) = o(L^4).$$

Thus Lemma 8.4 transfers a lower bound $N\mathrm{e}^{-o(L^3)}$ in the critical window. The sharper $N\mathrm{e}^{-O(L^2)}$ bound is already Theorem 5.4.

Proof. If $T_0 = 0$ the pointwise assertion is immediate; if $T_0 = 1$, it is odd Bonferroni. For $h \geq 1$,

$$\sum_{j=0}^{r} (-1)^j \binom{h}{j} = -\binom{h-1}{r}.$$

Distinct prime coordinates are independent and atoms at one coordinate are disjoint, so the loss from the exact tail void is at most

$$\mathbb{E}\binom{H_P}{r+1} \leq \frac{\mu_P^{r+1}}{(r+1)!} \leq \left(\frac{\mathrm{e}\mu_P}{r+1}\right)^{r+1}.$$

The tail Euler product is at least $\mathrm{e}^{-2\mu_P}$; the last display is a small fraction of it. This proves the mean bound, including the independent low tensor.

Expand only the tail factor. If $W_P = \sum_{Y<\ell\leq X} F(\ell)$, the maximal-order divisor bound gives $W_P = \exp\{O(L)\}$. Each term has at most $\rho(U_0)$ residue classes, so

$$\mathcal{R}_X \leq \rho(U_0) \sum_{j\leq r} \frac{W_P^j}{j!}, \qquad \log \mathcal{R}_X \leq o(L^4) + O(rL) = o(L^4).$$

Its modulus divides $d_0 X^r$, giving the other budget. Apply Lemma 8.4; the quarantine costs $\mathrm{e}^{-O(z)} = \mathrm{e}^{-o(L^3)}$. □

8.4. **The composite frontier.** Delete a composite atom if one of its prime projections is already a prime intrinsic atom; retain the residual family $\mathcal{C}^\circ$. Condition on $T_0 = 1$, and let H count the tail prime atoms together with $\mathcal{C}^\circ$. Put

$$\Lambda = \frac{L^3}{\log L}.$$

For $A \in \mathcal{C}^\circ$, conditioning on $T_0 = 1$ gives

$$\Pr(A \mid T_0 = 1) = \frac{1}{M_A} \prod_{\substack{p \mid M_A \\ z<p\leq Y}} \frac{p}{p - F(p)} = \frac{1 + o(1)}{M_A}$$

uniformly: $F(p) = p^{o(1)}$, $p > z = L^{3+o(1)}$, and $\omega(M_A) \leq L/\log z$. The rough mean therefore gives $\mathbb{E}H = O(\Lambda)$. The conditional full void after the quarantine is at least $\mathrm{e}^{-O(\Lambda)}$ by the proof of Theorem 7.3; dividing by $\Pr(T_0 = 1) \leq 1$ gives $\Pr(H = 0 \mid T_0 = 1) \geq \mathrm{e}^{-O(\Lambda)}$.

Proposition 8.7 (factorial-moment reduction)**.** *Suppose fixed $C, D > 0$, with D sufficiently large in terms of C and of the constant in the conditioned void lower bound (37.18) of the source notes, and an even $m \in [D\Lambda, D\Lambda + 2]$ satisfy*

$$\mathbb{E}(H)_m \leq (C\Lambda)^m. \tag{18}$$

Then a two-level pointwise minorant satisfies Lemma 8.4 and proves $|\mathrm{Av}_X(N)| \geq N\mathrm{e}^{-o(L^3)}$ uniformly for $\log N \asymp L^4$.

Proof. Use $B = T_0 Q_{m-1}(H)$. Since $m - 1$ is odd, Bonferroni makes it a pointwise minorant. The preceding binomial identity and $\binom{h-1}{m-1} \leq \binom{h}{m}$ give

$$\mathbb{E}(Q_{m-1}(H) \mid T_0 = 1) \geq \Pr(H = 0 \mid T_0 = 1) - \frac{(C\Lambda)^m}{m!}.$$

By Stirling, the second term is at most $(\mathrm{e}C/D)^m$, a small fraction of the first for large D. There are $\exp\{O(L)\}$ atoms. Thus every term, after multiplication by T_0, is a residue set and

$$\begin{aligned}\deg B &= O(\Lambda) = o(L^3),\\ \log(\text{modulus}) &\le Y + O(\Lambda L) = o(L^4),\\ \log(\text{ledger}) &\le Y + O(\Lambda L) = o(L^4).\end{aligned}$$

Lemma 8.4 applies. □

Proposition 8.7 is a valid sufficient reduction, but Subsection 8.5 below shows that (18) is false for this unreduced count H. Its pair-dispersion subproblem remains meaningful. For a high prime p and $a \pmod p$, set

$$u_{p,a} = \sum_{\substack{A\in\mathcal{C}^\circ\\ p\mid M_A,\ r_A\equiv a\ (p)}} \Pr(A \mid T_0 = 1), \qquad t_p = \sum_a u_{p,a}.$$

For squarefree pairs whose gcd is p, their pair contribution is bounded by a sub-sum of $p\sum_a u_{p,a}^2$. Even granting the ideal incidence estimate $t_p \ll \Lambda/p$, the available ℓ^1 information yields only

$$\sum_{z<p\le X} p\sum_a u_{p,a}^2 \le \sum_{z<p\le X} p t_p^2 \ll \Lambda^2 \sum_{z<p\le X}\frac{1}{p} \asymp \Lambda^2 \log L. \tag{19}$$

The live pair-dispersion frontier is

$$\sum_{z<p\le X} p\sum_a u_{p,a}^2 = O(\Lambda^2). \tag{20}$$

Equation (19) is not a counterexample to (20); it identifies the factor lost by present information. Scott–Sokal's cluster expansion [23] controls a numerical partition function, not the needed pointwise polynomial. Indeed the literal independent-set polynomial in violated-event indicators fails pointwise: on a five-vertex path its alternating sum is $1-5+6-1=1$ although the void indicator is zero.

8.5. **Nested implication and the two hierarchy refutations.** The more basic failure is logical redundancy among composite atoms. For a compatible nonempty set S, put $Q_S = \operatorname{lcm}_{A\in S} M_A$ and let $\mathcal{L}(S)$ be the sum, over compatible $B \notin S$, of the conditional extension ratios

$$\frac{\Pr(\bigcap_{A\in S\cup\{B\}} A \mid T_0 = 1)}{\Pr(\bigcap_{A\in S} A \mid T_0 = 1)}.$$

The literal source target (40.28) asserted $\mathcal{L}(S) \ll \Lambda$ uniformly for $|S| < m \asymp \Lambda$.

Theorem 8.8 (nested retained-atom obstruction; source Theorem 47.2)**.** *For all sufficiently large X, the family counted by the raw H contains a squarefree atom A such that*

$$\mathcal{L}(\{A\}) \ge \exp\{cL/\log L\}.$$

Moreover, for $m \asymp \Lambda$ and every fixed $C > 0$,

$$\mathbb{E}((H)_m \mid T_0 = 1) > (C\Lambda)^m$$

for all sufficiently large X.

Mechanism sketch. Choose $q \in (Y, 2Y)$ with $q \equiv 3 \pmod 8$ and distinct $p_1, \dots, p_J \in (Y, 2Y)$ with $p_i \equiv 5 \pmod 8$, where J is the largest odd integer at most $L/(3\log(2Y))$; the prime number theorem in fixed progressions supplies them, and $M = q\prod_i p_i < X$. For every odd $I \subseteq \{1, \dots, J\}$, the squarefree $M_I = q\prod_{i\in I} p_i$ is $7 \pmod 8$, so $D = 2 \mid ((M_I + 1)/4)^2$ supplies the atom $n \equiv -8 \pmod{M_I}$. It survives prime-implied deletion because $(-8/q) = 1$, whereas every intrinsic unit class modulo q has Legendre sign -1; the p_i have no prime atom, and $D = 2$ is the least representative. The top event modulo M implies all $2^{J-1} - 1$ proper odd-subset events, which proves the codegree bound. On that event $H \geq 2^{J-1}$ and its conditional probability is $1/M$ (all its primes exceed Y), so $\mathbb{E}(H)_m \geq (2^{J-1})_m/M > (C\Lambda)^m$ for $m \asymp \Lambda$. □

Thus the literal one-step hierarchy (40.28) and the raw-H moment target (37.19), namely (18), are REFUTED. These were internal sufficient mechanisms: Theorem 8.8 does not refute H'_{PF}, (20), or the Erdős–Straus conjecture.

The proposed repair was to pass to the implication antichain $\mathcal{A}^*$: delete an atom A whenever another retained atom B has $M_B \mid M_A$ and $r_A \equiv r_B \pmod{M_B}$. Then $A \subseteq B$, so the union and its void are unchanged. For a candidate B put

$$w_B = \Pr(B \mid T_0 = 1),$$

$$\Gamma_S(B) = \prod_{p\mid(M_B,Q_S)} p^{\min(v_p(M_B),v_p(Q_S))} \prod_{\substack{p\mid(M_B,Q_S)\\ z<p\le Y,\, p\equiv 3(4)}} \frac{p - F(p)}{p}.$$

The exact proposed wall (source (47.16)) asked for fixed $C, D > 0$ such that, for every compatible $S \subset \mathcal{A}^*$ with $1 \leq |S| < m$ and even $m \in [D\Lambda, D\Lambda + 2]$,

$$\sum_{\substack{B\in\mathcal{A}^*\setminus S,\ B\sim S\\ (M_B,Q_S)>1}} w_B\Gamma_S(B) \leq C\Lambda. \tag{47.16}$$

The coprime part is already $O(\Lambda)$; hence (47.16) would recover the factorial-moment reduction for the reduced count. It was the proposed repair; the following exact structure and grid show that this repair also fails.

8.6. **The implication antichain and the semiprime grid.** Let $\mathcal{A}^\circ_X$ be the finite retained family counted by H: tail-prime atoms and composite atoms after prime-implied deletion, with classes deduplicated. For an intrinsic class $r \pmod M$, put

$$\delta_M(r) = \min\{D : D \mid A_M^2,\ -4D \equiv r \pmod M\}, \qquad A_M = \frac{M+1}{4}. \tag{49.1}$$

Theorem 8.9 (exact implication criterion; source Theorem 49.1)**.** *For two distinct retained atoms,*

$$\begin{aligned} E_{M,r} \subseteq E_{m,s} &\iff m \mid M \ \text{ and } \ r \equiv s \pmod m \\ &\iff m \mid M \ \text{ and } \ \delta_M(r) \equiv \delta_m(s) \pmod m. \end{aligned} \tag{49.2}$$

If $M = tm$, the intrinsic conditions unpack exactly as

$$t \equiv 1 \pmod 4, \qquad A_M = tA_m - \frac{t-1}{4}, \qquad D = D' + jm \quad (j \in \mathbb{Z}), \tag{49.3}$$

where $D \mid A_M^2$, $D' \mid A_m^2$, and both corresponding classes survive the deletions. There is no further division-by-four condition. At one modulus, distinct retained classes are disjoint and cannot nest.

Proof. A full cylinder $E_{M,r}$ lies in $E_{m,s}$ exactly when reduction modulo m is defined and equals s, giving the first equivalence. Since every participating modulus is odd, multiplication by -4^{-1} gives the second. If $M = tm$, reducing modulo four and comparing $4A_M = M+1$ with $4A_m = m+1$ gives (49.3). Equal-modulus nesting is excluded by class deduplication. □

Consequently the antichain, now denoted $\mathcal{A}_X^\dagger$, is exactly

$$\mathcal{A}_X^\dagger = \{(M,r) \in \mathcal{A}_X^\circ : \nexists (m,s) \in \mathcal{A}_X^\circ,\ m < M,\ m \mid M,\ s \equiv r \pmod m\}. \tag{49.4}$$

It keeps the inclusion-largest events, equivalently the divisibility-minimal compatible moduli.

Corollary 8.10 (pointwise void identity; source Corollary 49.2)**.** *If $H^\dagger$ counts $\mathcal{A}_X^\dagger$, then, on every integer and also pointwise inside the $T_0 = 1$ fibre,*

$$H = 0 \quad \Longleftrightarrow \quad H^\dagger = 0. \tag{49.5}$$

Proof. A deleted cylinder lies in a retained cylinder of smaller modulus; finite descent terminates in $\mathcal{A}_X^\dagger$. Thus the two unions agree. □

Writing $w_A = \Pr(A \mid T_0 = 1)$, $\mu = \sum_{\mathcal{A}_X^\circ} w_A$, and $\mu^\dagger = \sum_{\mathcal{A}_X^\dagger} w_A$, only

$$\Theta(L^2) = \mu_P \le \mu^\dagger \le \mu = O(\Lambda) \tag{49.6}$$

is proved. No $\Theta(\Lambda)$ first moment and no constant-factor lower bound $\mu^\dagger \gg \mu$ is available. The void identity and all upper charge, degree, modulus, and ledger estimates survive reduction.

For a divisor g and a reduced class $a \pmod g$, define

$$W_{g,a}^\dagger = \sum_{\substack{A \in \mathcal{A}_X^\dagger \\ g \mid M_A,\ r_A \equiv a \ (g)}} w_A, \qquad W_g^\dagger = \sum_{a \ (g)} W_{g,a}^\dagger. \tag{49.8}$$

Theorem 8.11 (uniform residue profile is false; source Theorem 49.4)**.** *There is no absolute C for which*

$$W_{g,a}^\dagger \le \frac{C}{\varphi(g)} W_g^\dagger \tag{49.9}$$

holds uniformly in X, g, a. In fact, there are $g \asymp X$ and reduced $a \pmod g$ for which the ratio between the two sides without C is $g^{1-o(1)}$.

Proof. Choose primes $q \equiv 3 \pmod 8$ and $p \equiv 5 \pmod 8$, both in $(3X^{1/2}/4, 4X^{1/2}/5)$, and put $M = qp$. Then $X/2 < M < X$ and, for large X, both primes exceed z and Y. Since $M \equiv 7 \pmod 8$, the intrinsic $D = 2$ atom is

$$n \equiv -8 \pmod M. \tag{49.10}$$

Its projection modulo q has Legendre sign $(-8/q) = 1$, opposite to every intrinsic prime-modulus class; the other prime is $1 \pmod 4$. Thus the atom survives prime deletion. Its proper modulus divisors $1, q, p$ carry no retained atom in this class, so Theorem 8.9 puts it in $\mathcal{A}_X^\dagger$. Take $g = M$, $a = -8$. Since $2M > X$, every atom in $W_g^\dagger$ has modulus M and weight $1/M$. Their number is at most $F(M) \le \tau(A_M^2) = M^{o(1)}$, while the displayed bin is nonempty. Hence

$$\frac{\varphi(g) W_{g,a}^\dagger}{W_g^\dagger} \ge \frac{\varphi(M)}{F(M)} = M^{1-o(1)}. \tag{49.11}$$

□

For fixed $q \equiv 3 \pmod 8$, every surviving semiprime atom with $p \equiv 5 \pmod 8$ still projects to the one class $-8 \pmod q$. Thus the $D = 2$ slice remains residue-concentrated after maximal reduction and cannot support a literal transplant of the $1/\varphi(g)$ cancellation used in the provisional record proof. This does not refute a profile averaged over g or restricted to a sufficiently small range.

Choose distinct primes

$$q_1, \dots, q_t \equiv 3 \pmod 8, \qquad p_1, \dots, p_t \equiv 5 \pmod 8, \tag{49.12}$$

all greater than z and with every $q_i p_j < X$. By the preceding proof, every edge

$$A_{ij} : \qquad n \equiv -8 \pmod{q_i p_j} \tag{49.13}$$

is a retained maximal atom, giving a complete bipartite grid in $\mathcal{A}_X^\dagger$. Let $\mathcal{L}^\dagger(S)$ denote the same conditional extension load as $\mathcal{L}(S)$, but within this antichain.

Theorem 8.12 (collective-closure obstruction; source Theorem 49.5)**.** *For every fixed $C, D > 0$, every sufficiently large X, and* every *even integer $m \in [D\Lambda, D\Lambda + 2]$, there is a compatible selected set $S \subset \mathcal{A}_X^\dagger$, with pairwise-coprime squarefree moduli and $|S| < m$, such that*

$$\mathcal{L}^\dagger(S) > C\Lambda. \tag{49.14}$$

For the same fixed C, D and every such m, the factorial-moment input also fails for all sufficiently large X:

$$\mathbb{E}((H^\dagger)_m \mid T_0 = 1) > (C\Lambda)^m. \tag{49.15}$$

Proof. For the hierarchy assertion take the diagonal matching $S = \{A_{ii} : 1 \le i \le t\}$. Its merged class is -8 modulo

$$Q_S = \prod_{i=1}^{t} q_i p_i. \tag{49.16}$$

For $i \neq j$, $q_i p_j \mid Q_S$ and the merged class projects to A_{ij}. Thus the intersection of the diagonal events implies every cross edge, and its exact conditional extension ratio is

$$\frac{\Pr(\bigcap_{A \in S} A \cap A_{ij} \mid T_0 = 1)}{\Pr(\bigcap_{A \in S} A \mid T_0 = 1)} = 1. \tag{49.17}$$

Set $t = m - 1$. For either required class $a = 3, 5 \pmod 8$, and any fixed $B_D > 12D$, the prime number theorem gives

$$\#\{z < r \le B_D L^3 : r \equiv a \pmod 8\} = \left(\frac{B_D}{12} + o(1)\right)\Lambda.$$

This supplies the grid for every admissible m; its products are $O_D(L^6) < X$. The diagonal moduli are pairwise coprime and squarefree, and the $t(t-1)$ forced cross edges prove (49.14).

For the moment assertion instead put

$$t = \left\lfloor \frac{m}{2\log(2z)} \right\rfloor. \tag{49.18}$$

There are enough primes of each class in $(z, 2z)$, since their number is asymptotic to $z/(4\log z)$ and exceeds t by a factor tending to infinity. Put $R = \prod_i q_i p_i$ and let

C_R be $n \equiv -8 \pmod{R}$. At each conditioned q_i, this class is allowed, so

$$\Pr(C_R \mid T_0 = 1) = \frac{1}{R} \prod_{q_i \le Y} \frac{q_i}{q_i - F(q_i)} \ge \frac{1}{R}. \tag{49.19}$$

On C_R all t^2 grid atoms occur. Since $R \le (2z)^{2t}$, $t^2 \ge 2m$ for large X, and $(u)_m \ge (u/2)^m$ for $u \ge 2m$,

$$\mathbb{E}((H^\dagger)_m \mid T_0 = 1) \ge \frac{(t^2)_m}{R} \ge \left(\frac{t^2}{2}\right)^m (2z)^{-2t}. \tag{49.20}$$

After division by $(C\Lambda)^m$, the logarithm is at least

$$m \log\left(\frac{t^2}{2C\Lambda}\right) - 2t \log(2z). \tag{49.21}$$

The second term is at most m, whereas $t^2/\Lambda \gg_D \Lambda/(\log z)^2 \to \infty$, proving (49.15). □

The point is collective: no diagonal edge contains an off-diagonal edge, but their intersection implies all cross edges. Proposition 47.3's prime-only rung survives. Bounded $\omega(M)$ does not repair these formulations because every grid atom has $\omega(M) = 2$.

The scope is exact. Theorems 8.9 and 8.12 do not rule out a weighted count, a redesigned conditioning selector, a different pointwise minorant, or a further collective void-preserving reduction. Blindly deleting grid atoms can enlarge the void and supplies no pointwise minorant. The endpoint pair estimate (40.19), the corresponding pair target (37.27), the existence of some other minorant proving (33.16), (33.16) itself, and H'_{PF} all remain OPEN. Nothing here proves or refutes the Erdős–Straus conjecture.

Computational 49.7 (exact finite scope). Exact divisor-enumeration diagnostics reduce 1042 retained atoms to 970 at $(X, z, Y) = (1000, 2, 1000)$, retaining conditional first-moment share $0.939936\ldots$; the maximum normalized profile there is 220 at $g = 943$. The exact square $(q_1, q_2; p_1, p_2) = (3, 11; 5, 13)$ verifies that selected moduli $15, 143$ force the cross atoms at $39, 55$ with extension ratio one. These finite fractions are computational diagnostics, not asymptotic evidence.

Assessment 49.8 (wave-17 verdict). The uniform profile, (47.16), and the ordinary factorial-moment target for $H^\dagger$ are REFUTED; the cube killed nesting and the grid kills antichains. The complete-system moment axis is therefore DORMANT: its original stakes have largely collapsed since the CLAIMED/PROVISIONAL unconditional record of Theorem 10.7, and any revival needs a genuinely new pointwise formulation.

9. Provisional record context

This section is deliberately segregated. Both records are CLAIMED/PROVISIONAL; neither is used in the refutation of H_{PF}. "Record" is historical wording of this draft: neither statement is now claimed as a record, since the stronger exponent 3/4 was obtained earlier by [16] (see Remark 9.2).

9.1. **The provisional log-log exceptional-set bound.** Let $E(N)$ count exceptional prime denominators $p \le N$.

Theorem 9.1 (CLAIMED/PROVISIONAL)**.** *There is an absolute $c > 0$ such that*

$$E(N) \ll N \exp\{-c(\log N)^{2/3}(\log\log N)^{1/3}\}.$$

Internal argument; provisional status retained. Put

$$\mathcal{K}(K) = \{k \le K : k \equiv 1 \pmod 4\}, \qquad L_K = \operatorname{lcm}_{k\in\mathcal{K}(K)} k,$$
$$h(\mathcal{J}) = \sum_{k\in\mathcal{J}} \frac{\varphi(k)}{k^2}.$$

The multiplier form of Lemma 3.4, with $M = k\ell$, has the following class-mass consequence. Uniformly for $K \le (\log X)^A$, every subfamily $\mathcal{J} \subseteq \mathcal{K}(K)$ containing 1, and every $c \pmod{24L_{\mathcal{J}}}$ with $(c, 24L_{\mathcal{J}}) = 1$, there are distinct forced classes modulo primes $X^{1/2} < \ell \le X$ whose counts $f_c(\ell)$ satisfy

$$\sum_{X^{1/2}<\ell\le X} \frac{f_c(\ell)}{\ell} \asymp (\log X)^2 h(\mathcal{J}). \tag{21}$$

Here the classes arise from

$$H < u, v \le \ell^{1/3}, \quad (u,v) = (uv,k) = 1, \quad uv \mid (k\ell+1)/4, \quad k \mid u + cv,$$

with $H = K^{10}$ and $\omega(uv) \le D\log\log X$.

For completeness, the harmonic lattice input behind (21) is obtained by partitioning $(H, z]$ into fixed-ratio boxes. Möbius inversion of $(u,v) = 1$ gives, in $I \times J$ with lengths $\eta U, \eta V$,

$$\#\{(u,v) \in I\times J : (u,v) = (uv,k) = 1,\ k \mid u + cv\} = \eta^2 UV \frac{\varphi(k)}{k^2} \prod_{p\nmid k}(1-p^{-2}) + O((U+V)\tau(k)\log(2UV)).$$

After division by UV and summation over boxes, the boundary/main ratio is $O(K^2/H)$, uniformly in k, c. Shiu's theorem [24], applied successively in the two variables to $t^{\omega(n)}$ and $n/\varphi(n)$, gives

$$\sum^{*} \frac{t^{\omega(u)+\omega(v)}}{uv} \ll_t \frac{\varphi(k)}{k^2}(\log z)^2(\log z)^{2(t-1)}, \qquad \sum^{*} \frac{1}{\varphi(u)\varphi(v)} \ll \frac{\varphi(k)}{k^2}(\log z)^2.$$

Rankin's inequality removes the high-ω tail after fixed large D. Summing over k gives the factor $h(\mathcal{J})$.

In a dyadic prime interval, the divisor condition is the progression $\ell \equiv -k^{-1} \pmod{4uv}$. Colliding classes imply $\ell \mid uv' - u'v$ and hence equality of reduced pairs; different k, k' would force the common uv to divide $(k-k')/4$, impossible because $uv > H^2 > K$. A fixed modulus has multiplicity at most $K2^{\omega(uv)} = (\log X)^{O(1)}$. Bombieri–Vinogradov absorbs it and gives the lower half of (21); Brun–Titchmarsh and the second weighted lattice estimate give the upper half. This proves (21).

Now choose $K = \lfloor \delta \log N \rfloor$ and partition primes by reduced classes modulo $M_0 = 24L_K$. Since $\log M_0 \le (1+o(1))K$, each subsequence has length $Y \ge N^{1-2\delta}$. Put

$$X = \exp\{\alpha(\log N/\log\log N)^{1/3}\}.$$

For a fixed subsequence, the larger-sieve denominator is

$$S_c = \sum_{\substack{s \le Y^{1/2} \\ s \mid P}} \mu^2(s) \prod_{\ell \mid s} \frac{f_c(\ell)}{\ell - f_c(\ell)}, \quad P = \prod_{X^{1/2} < \ell \le X} \ell.$$

Here $f_c(\ell) \le K\ell^{2/3} < \ell/2$ for large N. Writing G_c for the untruncated Euler product, Rankin's trick with $v = 1/\log X$ gives

$$\frac{G_c - S_c}{G_c} \le \exp\{-\log Y/(2\log X) + C(\log X)^2 \log K\}.$$

Small α makes this at most $1/2$. Since $h(\mathcal{K}(K)) \asymp \log K$, (21) gives $S_c \ge \frac{1}{2}\exp\{a(\log X)^2 \log K\}$. Summing the uniform bound $O(Y/S_c)$ over the reduced subsequences proves the theorem. □

Remark 9.2 (priority and verification status)**.** *Theorem 9.1 remains* CLAIMED/PROVISIONAL *pending external checking and primary-literature verification. Vaughan's primary paper* [25] *was located by its confirmed DOI but its PDF was access-blocked. Pomerance–Weingartner's 2025 reconstruction* [20, §4] *cites Vaughan as the state of the art, and the fresh search found no post-1970 exponent improvement. The draft therefore does not claim a direct verification of Vaughan's argument or a settled record; the search result is not a literature guarantee. A later audit (November 2026) found the preprint* [16] *with the stronger exponent* 3/4*; Theorem 9.1 is therefore not a record in any sense.*

9.2. **The pruned cubic prime slice.** The current harvesting frontier uses the incidence

$$r_{\mathcal{J}}(u, v; c) = \#\{k \in \mathcal{J} : k \mid u + cv\}.$$

Lemma 9.3 (incidence second moment; provisional context)**.** *For $K \ge K_0$, $H = K^{10}$, and $z > H^2$,*

$$\sum_{\substack{H < u,v \le z \\ (u,v)=1}} \frac{r_{\mathcal{J}}(u, v; c)^2}{uv} \ll (\log z)^2 (1 + \log K)^3,$$

uniformly for $\mathcal{J} \subseteq \mathcal{K}(K)$ and $c \pmod{24L_{\mathcal{J}}}$ with $(c, 24L_{\mathcal{J}}) = 1$.

Proof. Expand the square. For $k, k' \in \mathcal{J}$, both divisibilities are the congruence $[k, k'] \mid u + cv$. The preceding box calculation, now with $d = [k, k'] \le K^2$, gives

$$\sum\nolimits^{*} \frac{1}{uv} \ll \frac{\varphi(d)}{d^2}(\log z)^2 + \frac{\tau(d)(\log z)^2}{H}.$$

Furthermore

$$\sum_{k,k' \le K} \frac{\varphi([k, k'])}{[k, k']^2} \le \sum_{k,k' \le K} \frac{(k, k')}{kk'} \ll (1 + \log K)^3,$$

by writing $k = da, k' = db$. Since $\tau([k, k']) \le 2K$, all boundary terms total $O(K^3(\log z)^2/H)$, proving the result. □

Theorem 9.4 (unconditional pruned cubic prime slice; CLAIMED/PROVISIONAL)**.** *Fix $0 < \kappa < 1/240$ and let $K = X^{\kappa}$. For every $\mathcal{J} \subseteq \mathcal{K}(K)$ containing 1 and every $c \pmod{24L_{\mathcal{J}}}$ with $(c, 24L_{\mathcal{J}}) = 1$, retain only triples in (21) satisfying $r_{\mathcal{J}}(u, v; c) \le (\log X)^4$. Then (i) the restriction of the H_{kBV} error sum to these*

retained triples is $o(x \log x\, h(\mathcal{J}))$ unconditionally, uniformly in all the quantified data; and (ii) their distinct class counts $f_c^{\mathrm{good}}(\ell)$ satisfy, uniformly in these data,

$$\sum_{X^{1/2}<\ell\le X} \frac{f_c^{\mathrm{good}}(\ell)}{\ell} \asymp (\log X)^2 h(\mathcal{J}).$$

For the full $\mathcal{J} = \mathcal{K}(X^\kappa)$, this is $\asymp (\log X)^3$.

Internal argument; provisional status retained. The harmonic lattice proof above remains uniform whenever $z > K^{20}$: its box error is $O(K^2/H)$, and Shiu applies because every variable exceeds $H = K^{10}$. At the lowest dyadic interval, $z \ge X^{1/12} > K^{20}$ follows from $\kappa < 1/240$. Lemma 9.3 and Markov's inequality bound the harmonic mass of discarded triples by

$$\frac{(\log z)^2(1+\log K)^3}{(\log X)^4} = o((\log z)^2 h(\mathcal{J})).$$

Thus the retained first moment has full order. For fixed $4uv$, coprimality permits at most $2^{\omega(uv)}$ allocations to (u,v), and the pruning permits at most $(\log X)^4$ choices of k. The retained progression multiplicity is therefore $(\log X)^{D\log 2+4}$. Ordinary Bombieri–Vinogradov absorbs it. Distinctness is unchanged, and Brun–Titchmarsh gives the matching upper bound. Finally $h(\mathcal{K}(X^\kappa)) \asymp \log X$. □

Theorem 9.4 does not prove the literal weighted error-sum hypothesis from which it originated, does not assemble composite moduli, and does not improve Theorem 9.1 unconditionally. Its soft spots are the power-range extension of the Shiu/box estimates and the uniform second-moment boundary sum; these require external review.

10. The fixed-c-free critical-window assembly

> Claimed/provisional. Every result in this section which feeds Theorem 10.7 is subject to hostile external checking. In particular, the load-bearing prime-slice input Theorem 9.4 is likewise provisional. The argument is not presented as an established record.

This section attacks the restricted critical-window assembly. It does *not* prove the literal coefficient-sum clause of the source-notes hypothesis $H'_{\mathrm{PF}}(k\ell;\mathrm{good})$. Instead it removes the moving c-family, uses one fixed unpruned atom family, and proves the weaker ledger $\exp\{O(J\log X)\}$. That ledger is of the same order as the lcm budget and is still absorbed when $\log N \ge C(\log X)^4$. Subject to the caveats in Section 10.7, this gives the claimed/provisional unconditional 3/4 exponent.

Put

$$t = \log X, \qquad K = \lfloor X^\kappa \rfloor, \qquad H = K^{10}, \qquad 0 < \kappa < 1/240, \tag{39.1}$$

and partition $(X^{1/2}, X]$ into disjoint dyadic blocks $(x, 2x]$, truncating the two endpoint blocks. In a block put $z = x^{1/6}$. Let $\mathcal{A}_X$ be the following fixed *c-free* family. An atom is

$$A = (k,\ell,u,v), \quad k \le K,\ k \equiv 1 \pmod 4, \quad \ell \in (x,2x] \text{ prime}, \quad H < u,v \le z, \tag{39.2}$$

with

$$(u,v) = (uv,k) = 1, \qquad \omega(uv) \le D\log\log X, \qquad 4uv \mid k\ell + 1. \tag{39.3}$$

Here D is the fixed low-ω cutoff constant used in the source Lemma 16.2 and in the proof of Theorem 9.4. The atom denotes the cylinder

$$E_A = \{n : n \equiv -uv^{-1} \pmod{k\ell}\}. \tag{39.4}$$

The dyadic cutoff is only a canonical subfamily of the multiplier-prime family. The lower proof of Theorem 9.4, specifically the source-notes estimates (34.20)–(34.21), furnishes its retained main mass inside these boxes.

10.1. **Removing the moving fibre and deduplicating the atoms.**

Lemma 10.1 (source Lemma 39.1: c-free reduction and distinct large coordinates; proved, provisionally checked)**.** *The following hold.*

(1) If n is exceptional, then it avoids every atom in $\mathcal{A}_X$. If additionally $c \equiv n \pmod{24L_K}$, then every atom hit by n automatically obeys $k \mid u + cv$. Thus the coupling condition used to define the source family $\mathscr{G}_{X,c}$ is implied by the atom itself; it need not be built into the polynomial.

(2) Distinct atoms which have a compatible intersection have distinct primes ℓ. At a fixed ℓ, all atom classes modulo ℓ are distinct. At a fixed modulus $k\ell$, no two ordered pairs (u, v) produce the same class.

(3) Consequently, for a compatible ordered set $A_1, \dots, A_j$,

$$\Pr\Big(\bigcap_{i\le j} E_{A_i}\Big) = \frac{\mathbf{1}_{\text{compatible}}}{\operatorname{lcm}(k_1,\dots,k_j)\prod_{i\le j}\ell_i}. \tag{39.5}$$

Proof. Lemma 3.4 proves the first assertion. Modulo k, (39.4) says $u + nv \equiv 0$; since $c \equiv n \pmod{k}$, it also says $u + cv \equiv 0$.

Suppose two atoms at the same ℓ are compatible. Their projections modulo ℓ agree, so

$$\ell \mid uv' - u'v.$$

In the same block the absolute value is at most $z^2 = x^{1/3} < \ell$. The same conclusion holds if two admissible cutoffs are compared across blocks, since every $u, v \le X^{1/6}$, while $\ell > X^{1/2} > X^{1/3}$. Hence $uv' = u'v$, and reducedness gives $(u, v) = (u', v')$. The apparent swap can agree only when $u = v$, which is impossible here because $(u, v) = 1$ and $u, v > H$. Finally $uv \mid (k\ell + 1)/4$ is exactly

$$k\ell \equiv -1 \pmod{4uv}.$$

Since $(\ell, 4uv) = 1$, this determines $k \pmod{4uv}$, and $4uv > 4H^2 > K$ makes the admissible k unique. This proves both forms of deduplication and the distinct-ℓ assertion. The Chinese remainder theorem now gives (39.5). □

The logical point is as important as the geometry. The majorant below covers the fixed predicate $\mathbf{1}_{(n,P_y)=1}\mathbf{1}_{H_X(n)=0}$, where $H_X = \sum_{A\in\mathcal{A}_X} \mathbf{1}_{E_A}$. It does not majorize every avoider of the smaller, c-dependently pruned source family. It does majorize every exceptional prime, which is all the exceptional-set implication needs.

10.2. **Exact mass profiles.** For $g \mid k$ and $a \in (\mathbb{Z}/g\mathbb{Z})^\times$, write

$$W_{k,a}(g) = \sum_{\substack{A\in\mathcal{A}_X \text{ with multiplier } k \\ -uv^{-1}\equiv a \ (g)}} \frac{1}{k\ell}, \qquad W_k = \sum_{A : k_A = k} \frac{1}{k\ell}. \tag{39.6}$$

Lemma 10.2 (source Lemma 39.2: multiplier and residue profiles; proved, provisional)**.** *Uniformly for $k \le K$, $k \equiv 1 \pmod 4$, $g \mid k$, and reduced $a \pmod g$,*

$$W_{k,a}(g) \ll \frac{t^2}{\varphi(g)} \frac{\varphi(k)^2}{k^3}, \qquad W_k \asymp t^2 \frac{\varphi(k)^2}{k^3}. \tag{39.7}$$

Consequently

$$\mu_X := \sum_{A \in \mathcal{A}_X} \frac{1}{k_A \ell_A} \asymp t^2 \sum_{\substack{k \le K \\ k \equiv 1(4)}} \frac{\varphi(k)^2}{k^3} \asymp t^2 \log K \asymp t^3. \tag{39.8}$$

For each fixed odd prime p, the share of the proxy Euler weight $b(k)/k = \varphi(k)^2/k^3$ carried by multipliers divisible by p is exactly

$$\alpha_p = \frac{p-1}{p^2+p-1} \sim \frac{1}{p}. \tag{39.9}$$

Because (39.7) determines W_k only up to constants, (39.9) is a statement about the proxy weight, not an exact share of the atom mass; the moment proof below never uses α_p.

Proof. We give the review-inserted derivation rather than appeal to an average over c. Fix the block and k, and put $F(n) = n/\varphi(n)$ and

$$q_0 = \operatorname{lcm}(g, \operatorname{rad} k) \le k.$$

Work in multiplicative boxes $u \asymp U$, $v \asymp V$. The conditions $(uv, k) = 1$ and $-uv^{-1} \equiv a \pmod g$ restrict $(u \bmod q_0, v \bmod q_0)$ to reduced pairs with a fixed ratio modulo g: for each of the $\varphi(q_0)$ choices of v-class, the u-class is determined modulo g and free at the remaining part, so exactly $\varphi(q_0)^2/\varphi(g)$ ordered class pairs are compatible. Shiu's theorem for the multiplicative function F, on the fixed-relative-length interval $u \asymp U$ in one reduced class modulo q_0, is applicable since $q_0 \le k \le K$ and $U > H = K^{10}$, and gives

$$\sum_{\substack{u \asymp U \\ u \equiv r \ (q_0)}} F(u) \ll \frac{U}{\varphi(q_0)} \exp\Big\{ -\sum_{p \mid k} \frac{1}{p-1} \Big\} \asymp \frac{U}{\varphi(q_0)} \frac{\varphi(k)}{k}.$$

The last comparison holds because the logarithm of the quotient is $O(\sum_p p^{-2})$. Multiplying the u- and v-class sums by the number of compatible class pairs and dividing by UV yields, per box pair,

$$\sum_{\text{box}} \frac{1}{\varphi(u)\varphi(v)} \ll \frac{1}{\varphi(g)} \left(\frac{\varphi(k)}{k} \right)^2.$$

Summing the $O((\log(z/H))^2) = O(t^2)$ box pairs gives

$$\sum \frac{1}{\varphi(u)\varphi(v)} \ll \frac{(\log z)^2}{\varphi(g)} \frac{\varphi(k)^2}{k^2}. \tag{39.11}$$

For fixed (u, v), Brun–Titchmarsh in the progression $\ell \equiv -k^{-1} \pmod{4uv}$, whose modulus is $4uv \le 4x^{1/3}$, gives

$$\sum_{\substack{x < \ell \le 2x \\ \ell \equiv -k^{-1} \ (4uv)}} \frac{1}{\ell} \ll \frac{1}{\varphi(4uv) \log x}, \qquad \frac{1}{\varphi(4uv)} \ll \frac{1}{\varphi(u)\varphi(v)},$$

where the last inequality uses $(u,v)=1$. Combining this with (39.11) over the boxes proves

$$\sum_{\substack{A \text{ in the block}\\ k_A=k,\ -uv^{-1}\equiv a\ (g)}} \frac{1}{\ell} \ll \frac{\log x}{\varphi(g)}\frac{\varphi(k)^2}{k^2}, \tag{39.10}$$

with an absolute constant, uniformly in $k\le K$, $g\mid k$, and reduced a: no k^{ε}, divisor, or logarithmic loss enters. After the $O(t)$ blocks and division by k, this is the upper half of (39.7).

For the lower half drop the ratio condition. For one fixed k, a modulus $4uv$ has at most $2^{\omega(uv)}=(\log X)^{O(1)}$ ordered allocations. Thus ordinary Bombieri–Vinogradov, with a fixed sufficiently large logarithmic saving, evaluates the prime progressions without the factor K: k is fixed on this line. The low-ω Rankin argument from the source proof retains a fixed proportion uniformly. At a prime $p\mid k$ the exact coprime-pair local factor differs from $(1-1/p)^2$ by $1+O(p^{-2})$, so the unrestricted harmonic pair mass is $\asymp(\log(z/H))^2(\varphi(k)/k)^2$ uniformly. This gives the matching lower bound block by block and proves (39.7).

Finally $b(k)=(\varphi(k)/k)^2$ is multiplicative and $\sum b(k)/k\asymp\log K$, also after selecting $k\equiv 1 \pmod 4$, by the principal/χ_4 decomposition. At an odd prime its local factor is

$$1+\sum_{e\ge 1}\frac{(1-1/p)^2}{p^e}=1+\frac{p-1}{p^2}, \tag{39.12}$$

so the part with $p\mid k$, divided by the full factor, is exactly (39.9). □

Given a reduced fibre c, an atom with multiplier k is active for one of the $\varphi(k)$ unit residues and then has prime-coordinate probability $1/\ell$. Averaging those conditional masses produces the factor $1/\varphi(k)$, whereas the unconditioned cylinder has probability $1/(k\ell)$. Formula (39.7), not the conditional mass profile used in the earlier moving-c family, is the correct profile for factorial moments in the c-free CRT space.

10.3. **Collapse versus consistency.** The rough collapse bound which discards residue consistency would charge a shared prime p by $p\alpha_p^2\asymp 1/p$, producing a false $\log\log K$ loss. The required cancellation is exact at the level of a new compatible atom.

For a cutoff y, put

$$q_y(k)=\prod_{\substack{p\mid k\\ p\le y}}\frac{p}{p-1}, \qquad b_y(g)=\prod_{\substack{p^e\parallel g\\ p\le y}}\varphi(p^e)\prod_{\substack{p^e\parallel g\\ p>y}}p^e. \tag{39.13}$$

Condition the CRT space by $(n,P_y)=1$, where $P_y=\prod_{p\le y}p$. An atom then has probability $q_y(k)/(k\ell)$. If a compatible previously selected set has multiplier lcm R, adding an atom of multiplier k, with $g=(k,R)$, multiplies its intersection probability by

$$\frac{q_y(k)}{k\ell}\,b_y(g). \tag{39.14}$$

Compatibility fixes one unit residue modulo g. Lemma 10.2 says that the atom mass in that residue is at most $1/\varphi(g)$ of its multiplier mass. Thus the collapse left

after summing the compatible new atoms is

$$\frac{b_y(g)}{\varphi(g)} = \prod_{\substack{p\mid g\\ p>y}} \frac{p}{p-1}; \tag{39.15}$$

small conditioned primes cancel exactly, and an unconditioned shared prime costs only $p/(p-1)$, not p.

Lemma 10.3 (source Lemma 39.3: uniform multiplier Euler bound; proved)**.** *Uniformly in $y \ge 2$, every integer R, and $K \ge 3$,*

$$\sum_{\substack{k\le K\\ k\equiv 1(4)}} \frac{\varphi(k)^2}{k^3} q_y(k) \prod_{\substack{p\mid (k,R)\\ p>y}} \frac{p}{p-1} \ll \log K. \tag{39.16}$$

Proof. It is enough to sum over all odd k. The unmodified local nonconstant mass is $(p-1)/p^2$. Multiplication either by q_y for $p \le y$, or by $p/(p-1)$ when $p > y$ and $p \mid R$, changes this to $1/p$. The quotient

$$\frac{1+1/p}{1+(p-1)/p^2} = 1 + O(p^{-2}) \tag{39.17}$$

has a convergent product, uniformly in the chosen set of modified primes. The remaining harmonic Euler product is $O(\log K)$. □

Theorem 10.4 (source Theorem 39.4: all required upper factorial moments; proved, provisional)**.** *In the CRT space conditioned by $(n, P_y) = 1$, for every integer $m \ge 1$,*

$$\mathbb{E}(H_X)_m \le (Ct^3)^m, \tag{39.18}$$

with C depending at most on κ, uniformly in $2 \le y < X^{1/2}$. The upper restriction is necessary: for $y \ge \ell$ the coprimality conditioning would renormalize the ℓ-coordinate itself by $\ell/(\ell-1)$, which (39.14) does not include. The application below has $y = Bt^3 < X^{1/2}$ for all sufficiently large X. In particular the bound holds through every order $m = D_B t^3$.

Proof. Expand the ordered factorial moment into distinct compatible atoms. By Lemma 10.1, every admissible next atom has a new ℓ. For a fixed preceding compatible set, (39.14), the residue-profile upper bound (39.7), and (39.15) bound the sum over all possible next atoms by

$$Ct^2 \sum_{\substack{k\le K\\ k\equiv 1(4)}} \frac{\varphi(k)^2}{k^3} q_y(k) \prod_{\substack{p\mid (k,R)\\ p>y}} \frac{p}{p-1} \le Ct^2 \log K \le Ct^3. \tag{39.19}$$

Removing the forbidden old ℓ's only decreases this sum. Iteration proves (39.18). □

At pair level, for fixed k, k' and $g = (k, k')$,

$$\sum_{A:k_A=k} \sum_{\substack{B:k_B=k'\\ A\sim B}} \Pr(E_A \cap E_B) \ll \frac{g}{\varphi(g)} W_k W_{k'}. \tag{39.20}$$

Averaging (39.20) uses $\alpha_p^2(p/(p-1)-1) = O(p^{-3})$, not $p\alpha_p^2 \asymp 1/p$. Hence the pair excess is $O(\mu_X^2)$ with an absolute constant. It is *not* $o(\mu_X)$: a fixed small prime already contributes a constant multiple of μ_X^2 to the usual Janson dependency sum. Therefore the proposed direct condition $\Delta = o(\mu)$ and the naive Janson proof fail

quantitatively. The factorial-moment theorem survives because an absolute base C^m, rather than Poisson relative error, is enough after the separate void estimate below.

10.4. **The CRT void without partitioning by all of M_0.** A completely unconditioned c-free void still has a quarantine floor: taking $n \equiv 0 \pmod p$ for many small p disables every atom whose multiplier contains one of them. Thus $Q_r(H_X)$ alone cannot have cubic mean. Take

$$y = Bt^3 \tag{39.21}$$

with a sufficiently large fixed B, and multiply by the exact small-prime coprimality selector

$$S_y(n) = \mathbf{1}_{(n,P_y)=1} = \sum_{d \mid P_y} \mu(d)\mathbf{1}_{d\mid n}. \tag{39.22}$$

Every exceptional prime greater than y has $S_y = 1$. Notice that (39.22) has degree $\pi(y) = O(t^3/\log t)$, not the impossible degree $\pi(K)$.

Lemma 10.5 (source Lemma 39.5: conditioned c-free void; proved, provisional)**.** *In the exact CRT space,*

$$\Pr(H_X = 0 \mid S_y = 1) \le e^{-ct^3}. \tag{39.23}$$

Proof. Reveal all multiplier coordinates. Let c denote the resulting class modulo $24L_K$, where $L_K = \operatorname{lcm}\{k \le K : k \equiv 1 \pmod 4\}$, and put

$$\mathcal{J}_c = \{k \le K : k \equiv 1 \pmod 4,\ (k,c) = 1\}, \qquad Z(c) = \sum_{\substack{y < p \le K,\ p \mid L_K \\ p \mid c}} \frac{1}{p}. \tag{39.24}$$

The restriction $p \mid L_K$ makes Z well defined: a prime $p \equiv 3 \pmod 4$ with $3p > K$ divides no admissible multiplier, so it has no CRT coordinate here and can be omitted; it removes nothing from $\mathcal{J}_c$. The coordinates for $p > y$, $p \mid L_K$, remain independent and $\Pr(p \mid c) = 1/p$. With exponential parameter y,

$$\mathbb{E}e^{yZ} \le \exp\left\{Cy \sum_{p>y} p^{-2}\right\} = e^{O(1)}, \qquad \Pr(Z > \eta) \le e^{-\eta y + O(1)}. \tag{39.25}$$

Write $h(\mathcal{J}) = \sum_{k\in\mathcal{J}} \varphi(k)/k^2$. The elementary bounds

$$h(\mathcal{K}(K)) \asymp \log K, \qquad \sum_{\substack{k \le K \\ p \mid k}} \frac{\varphi(k)}{k^2} \ll \frac{\log K}{p} \tag{39.26}$$

show by the union bound that, for a sufficiently small fixed η, $Z(c) \le \eta$ implies $h(\mathcal{J}_c) \ge c_1 \log K$. Also $S_y = 1$ makes c coprime to 24, and by definition it is coprime to $L_{\mathcal{J}_c}$.

Apply Theorem 9.4, the source-notes Theorem 34.8, to this arbitrary subfamily $\mathcal{J}_c$. Its lower proof lies in the canonical boxes (39.2), and its good triples are contained in the c-free unpruned family. Conditional on the multiplier coordinates, the distinct ℓ-coordinates are independent by Lemma 10.1. Therefore, when $Z(c) \le \eta$,

$$\Pr(H_X = 0 \mid c) \le \prod_{\ell} \left(1 - \frac{f_c^{\mathrm{good}}(\ell)}{\ell}\right) \le \exp\{-c_2 t^2 h(\mathcal{J}_c)\} \le e^{-c_3 t^3}. \tag{39.27}$$

The bad multiplier coordinates cost at most $e^{-\eta Bt^3+O(1)}$ by (39.25). Increasing B if necessary and averaging (39.27) proves (39.23). □

This is the load-bearing use of Theorem 9.4. It replaces Janson by supplying cubic active prime-coordinate mass uniformly for every reduced surviving multiplier fibre. The only fibres not reduced are those which set too much large-prime multiplier mass to zero, and (39.25) makes those fibres exponentially rare after the polynomial-sized coprimality cutoff. The dependency on that provisional theorem is therefore explicit and unavoidable in the present argument.

10.5. **A low-degree majorant with the honest ledger.** For even r, put

$$Q_r(h) = \sum_{j=0}^{r} (-1)^j \binom{h}{j}.$$

The exact identity

$$Q_r(0) = 1, \qquad Q_r(h) = \binom{h-1}{r} \geq 0 \quad (h \geq 1) \tag{39.28}$$

shows that

$$\nu_X(n) = S_y(n) Q_r(H_X(n)) \tag{39.29}$$

is nonnegative and is at least one on every exceptional prime $n > \max(K, y)$.

Theorem 10.6 (source Theorem 39.6: modified c-free critical-window assembly; proved, CLAIMED/PROVISIONAL)**.** *There are constants $c, C_0, D_B > 0$, depending at most on κ, such that if r is the least even integer at least $D_B t^3$, then (39.29) has exact CRT mean*

$$\mathbb{E}_{\mathrm{CRT}} \nu_X \leq e^{-ct^3}. \tag{39.30}$$

Here and below D_B is the Bonferroni-degree constant, distinct from the source Lemma 16.2's ω-cutoff constant D in (39.3). Its congruence expansion has the explicit bounds

$$\begin{aligned} \textit{degree} &\leq r + \pi(y) = O(t^3), \\ \log d_{\mathrm{term}} &\leq O(y) + r(t + \log K) = O(t^4), \\ \log \sum_{\text{terms}} |c_{\mathrm{term}}| &= O(rt) = O(t^4). \end{aligned} \tag{39.31}$$

Every nonzero term is one plain congruence class. Consequently, whenever $\log N \geq C_0 t^4$,

$$\sum_{n \leq N} \nu_X(n) \ll N e^{-ct^3}. \tag{39.32}$$

Proof. From (39.28), for $h \geq 1$, $\binom{h-1}{r} \leq \binom{h}{r+1}$. Lemma 10.5 and Theorem 10.4 therefore give

$$\begin{aligned} \mathbb{E}(Q_r(H_X) \mid S_y = 1) &\leq \Pr(H_X = 0 \mid S_y = 1) + \frac{\mathbb{E}((H_X)_{r+1} \mid S_y = 1)}{(r+1)!} \\ &\leq e^{-c_1 t^3} + \frac{(Ct^3)^{r+1}}{(r+1)!} \leq e^{-ct^3}, \end{aligned} \tag{39.33}$$

when D_B is sufficiently large. Multiplication by $\Pr(S_y = 1) \leq 1$ proves (39.30).

Expand (39.22) and the elementary symmetric polynomials in Q_r. Lemma 10.1 makes every compatible atom set use distinct ℓ's; CRT then makes its further

intersection with $d \mid P_y$ either empty or one class. If $A_X = |\mathcal{A}_X|$, the crude choice count from (39.2) gives $\log A_X = O(t)$, and hence

$$2^{\pi(y)} \sum_{j \le r} \binom{A_X}{j} = \exp\{O(rt)\}. \tag{39.34}$$

The modulus is at most $P_y(KX)^r$, proving (39.31). On $[1, N]$, each class count is its CRT mean times N, plus $O(1)$. Choosing C_0 larger than the constants in (39.31) makes the aggregate rounding error $\exp\{O(rt)\}$ at most, say, $N^{1/2}$, which is negligible beside Ne^{-ct^3}. This proves (39.32). □

The literal source hypothesis $H'_{\mathrm{PF}}(k\ell; \mathrm{good})$ demanded coefficient sum $e^{O(J)}$, with $J \asymp t^3$, and a majorant for its larger aggregate avoider predicate. Theorem 10.6 proves *neither* clause: its direct even-Bonferroni expansion has ledger $e^{O(Jt)}$, and it only covers the c-free avoider, hence all exceptional primes. These are deliberate modifications, not suppressed losses. The ledger still closes because the critical window itself has logarithmic size $\asymp Jt$. Thus the literal $e^{O(J)}$ clause remains OPEN as stated, but is no longer needed for the exceptional set if Theorem 10.6 survives review.

10.6. **Exceptional denominators.**

Theorem 10.7 (source Theorem 39.7: unconditional 3/4 exceptional-set bound; CLAIMED/PROVISIONAL)**.** *There is a constant $c > 0$ such that*

$$E_{\mathrm{all}}(N) \ll N \exp\{-c(\log N)^{3/4}\}. \tag{39.35}$$

The same bound holds for exceptional primes.

Internal argument; provisional status retained. Put $L = \log N$, take $t = \alpha L^{1/4}$, and set $X = e^t$. Choose the fixed $\alpha > 0$ small enough that $L \ge C_0 t^4$. Lemma 10.1 and (39.29) majorize every exceptional prime larger than $\max(K, y)$; that omitted range is negligible. Equations (39.32) and $t^3 = \alpha^3 L^{3/4}$ give the prime bound.

For completeness, we spell out the semigroup transfer used for all denominators. Every prime factor of an exceptional integer $n > 1$ is itself exceptional: if $p \mid n$ and $4/p = 1/x_1 + 1/x_2 + 1/x_3$, scaling the three denominators by n/p represents $4/n$. Thus the exceptional integers are contained in the multiplicative semigroup generated by exceptional primes, including the empty product. The integer 1 contributes only $O(1)$.

Write $g(u) = u^{3/4}$ and $x = e^L$. Fix x_0 large enough that the prime bound gives, for every $x \ge x_0$,

$$E_{\mathrm{prime}}(x) \ll x \exp\{-c_0 g(\log x)\}.$$

Fix $0 < \eta < c_0$ and, for large L, put $\delta = \eta g(L)/L$, so $0 < \delta < 1/4$. Rankin's inequality over the semigroup gives

$$E_{\mathrm{all}}(x) \le x^{1-\delta} \prod_{\substack{p \le x \\ p \text{ exceptional}}} (1 - p^{-1+\delta})^{-1}.$$

The finitely many exceptional primes $p \le x_0$ contribute $O(1)$, uniformly in δ. Partial summation and the endpoint estimate $\delta L - c_0 g(L) = -(c_0 - \eta) g(L) < 0$ give

$$\sum_{\substack{p \le x \\ p \text{ exceptional}}} p^{-1+\delta} \ll 1 + \int_{\log x_0}^{L} \exp\{\delta u - c_0 g(u)\}\, du.$$

Because $g(u)/u = u^{-1/4}$ is decreasing, for $u \le L$ one has $\delta u \le \eta g(u)$. The last integral is therefore bounded by $\int_{\log x_0}^{\infty} e^{-(c_0-\eta)g(u)}\,du < \infty$. The prime-power terms in the logarithm of the Euler product also satisfy

$$\sum_{p \le x} \sum_{j \ge 2} \frac{p^{-j(1-\delta)}}{j} \le \sum_{p} \sum_{j \ge 2} \frac{p^{-3j/4}}{j} = O(1).$$

Hence the Euler product is $O(1)$ and $E_{\mathrm{all}}(x) \ll x^{1-\delta} = x \exp\{-\eta g(\log x)\}$, proving (39.35). □

> Theorem 10.7 is CLAIMED/PROVISIONAL. It uses the likewise provisional Theorem 9.4. It does not use the source hypotheses $H_{k\mathrm{BV}}$, H'_{PF}, or Elliott–Halberstam, but "unconditional" does not remove the need for external verification.

10.7. **Remarks on verification status.** The three weakest steps identified by the maximum-severity self-review are reproduced faithfully here.

(1) The new residue-resolved upper bound (39.11) must be rederived by a referee with every local factor present. It needs simultaneous uniformity in $k \le X^{\kappa}$, every divisor $g \mid k$, and every unit residue; the proof uses Brun–Titchmarsh for the upper bound, so it does not hide a polynomially small Bombieri–Vinogradov main term. This is the most load-bearing new analytic lemma.

(2) Lemma 10.5 uses Theorem 9.4 for the data-dependent subfamily $\mathcal{J}_c$ and uses the canonical lower boxes, not merely the larger family in the statement of the source formula (34.18). The quantifiers of Theorem 9.4 do allow every subfamily containing 1, and its proof furnishes the lower mass in those boxes, but this inheritance is new and needs an independent line-by-line check together with Theorem 9.4 itself.

(3) The ordered-moment induction (39.14)–(39.19) is where lcm collapse and residue consistency cancel at every prime power. A missed factor there would be exponentiated through $r \asymp t^3$. The displayed conditional factor is $\varphi(p^e)$ for $p \le y$ and p^e for $p > y$; after the $1/\varphi(g)$ residue share, only $p/(p-1)$ remains at an unconditioned shared prime. This exact replay, including unequal prime-power exponents, is mandatory in external review.

Internally, the quantifier chain is complete: Theorem 9.4 is uniform in $\mathcal{J}_c$; (39.25) removes the nonreduced fibres; (39.18) controls the even Bonferroni tail; and (39.31)–(39.34) pay the full $e^{O(t^4)}$ coefficient ledger before selecting $t = \alpha(\log N)^{1/4}$. The direct Janson proposal is rejected because its dependency sum is $\Theta(\mu^2)$, while the original $e^{O(J)}$ formulation remains open. Theorem 10.7 should not be cited as established outside this campaign until an independent expert has checked the three steps above. No priority is claimed for it: the same exponent was obtained earlier by Kandukoori, Ramesh and Kumar [16, Thm. 1] (unrefereed preprint dated 29 July 2026, with a Lean formalisation).

The search at the time of this draft found no post-1970 improvement of Vaughan's $N/\exp(c(\log N)^{2/3})$ exceptional-set bound. Pomerance–Weingartner's 2025 preprint [20, §4] explicitly cites Vaughan 1970 as the state of the art and reproduces the same 2/3 exponent when its general theorem is specialized to $m = 4$. Access to Vaughan's paper was blocked, but DOI 10.1112/S0025579300002886 was confirmed. That search missed [16], located by a later audit (November 2026), so Theorem 10.7

is not a first exponent improvement since 1970; the CLAIMED/PROVISIONAL label stands.

10.8. **Internal verification pedigree.** The counts in this subsection concern internal checks, not independent external validation. Theorem 9.4 has had three independent internal checks. In the third, a fresh reviewer rederived the power-sized extension of the box argument, expanded the second incidence moment, paid all boundary errors, and rebuilt the low-congestion Bombieri–Vinogradov assembly. The reviewer checked Shiu's 1980 paper [24] directly: the interval and modulus inequalities hold with Shiu's parameters $\alpha = \beta = 1/4$, the residue classes are reduced, and both multiplicative-function hypotheses apply in the fixed-relative-length boxes with $H = K^{10}$. The use of $1 \in \mathcal{J}$, same-prime distinctness, the canonical lower boxes, and the data-dependent family $\mathcal{J}_c$ in Lemma 10.5 were separately replayed. That check was CONFIRMED-AFTER-REPAIRS; the provisional register did not change.

The downstream fixed-c-free chain has, in addition to its original construction, one attested-blind parallel construction and two paper fidelity passes. For the blind construction, the unit read only the task brief and the source-note analogues of Sections 16 and 34 at repository commit `c489912b`, and froze its complete S1–S7 derivation at commit `347a570` before opening the source analogue of Section 39. It did not inspect the earlier verification program, although the mandatory baseline run printed that program's summary line. Git verifies the snapshots, texts, and commit order; it cannot verify what was visible on screen or enforce read isolation. The blindness claim is therefore an explicit self-attestation, not an independently audited fact. Reconciliation classified all seven steps as CONFIRMED-SAME-METHOD; the variants (modulus k versus $\operatorname{lcm}(g, \operatorname{rad} k)$ in Shiu, a pointwise Euler bound, and direct weighted-fibre averaging) made no quantifier or range change. Its separate exact-rational stress test checked moments through order six, 14,210 atoms, and 1,128,378 compatible pairs. Those finite checks are bug hunts, not asymptotic evidence, and the blind unit treated Theorem 9.4 as an allowed input.

Wave 15 then ran three separate maximum-severity hostile reviews, one each for the general-numerator transfer, the slice-conspiracy analysis, and the Kloosterman endpoint. Each was repaired before inclusion here. The slice review repaired the raw/primitive boundary. The endpoint review checked the actual DFI and Bettin–Chandee statements and discovered Theorem 36.2, which was then independently verified internally.

Wave 16 added an attested-blind general-m construction. The unit read only the prescribed source §§16, 34, 39 and the archived Pomerance–Weingartner text; `blind43.md` logs its reads, and git records its S1–S7 freeze at `a0bd2940c322` before source §43 was opened. This is a self-attestation, not an audit. The identity, aggregate supply, atoms, and profiles converged; the cutoff required repair, while the moment window and assembly diverged and supplied (43.36). A separate maximum-severity adjudication rechecked the quarantine, moments, ledger, range, semigroup transfer, and crossover and returned STRENGTHENING CONFIRMED. It retained Theorem 9.4, the source §39.7 qualifications, and the new headline as CLAIMED/PROVISIONAL; none of these passes was external refereeing or priority certification.

Wave 17 added maximum-severity hostile reviews of the implication-antichain analysis and the conditional slice frontier. The first returned CONFIRMED-AFTER-REPAIRS: it checked the exact inclusion direction, semiprime-atom retention, every

quantifier in the grid, the extension ratio one, and the direct reduced-moment lower bound. The second returned SOUND-AFTER-REPAIRS: it independently rederived the prime-factor criterion and checked the genus, raw/primitive, admissibility, standard- hypothesis, and streamed-census boundaries. These are internal reviews and computational bug hunts, not external validation of either provisional record.

Wave 18's separate hostile reviews of the witness/effectivity and per-slice sections both returned SOUND-AFTER-REPAIRS: they repaired prime scope and the induced-character/dyadic-conductor ledger, and replayed the three local classes, exact exponent, elementary remainder, fixed-refinement corollary, and ineffective uniform scope. Wave 19's stacked-sieve review returned the same verdict after repairs to the overlap threshold, empty Page branch, growing-dimensional sieve ledger, and varying-cutoff inclusion; it confirmed Theorem 53.3 as proved and effective. The attested-blind protocol in Section 14 was adjudicated CONVERGED-WITH-DIVERGENCES: exact tails and effectivity converged, but its ancillary-$o(1)$ overclaim was rejected and no label changed. Blocks `(ax)`–`(az)` check respectively the W tail, the fixed-slice sieve, and stacked geometry. These three hostile reviews, the adjudication, and all finite checks remain internal, not external validation.

Wave 20 added hostile reviews of the logarithmic lower tail and the m-uniform effectivity perimeter. The first returned CONFIRMED-AFTER-REPAIR; the repair enlarged verifier coverage but did not change Section 17. The second returned SOUND-AFTER-REPAIRS; it narrowed the favorable-sign claim to the full Layer-1 family and the structured c-free fibres, inserted the worst-case fibre subtraction, and kept every inherited provisional label. Block `(ba)` checks the two residue-one lower-tail constructions, and block `(bb)` checks the general-m finite harvest and favorable-sign mass. A wave-21 hostile review of the certificate ceilings and extremal census returned SOUND-AFTER-MINIMAL-SCOPE-REPAIRS. It preserved both constants and proofs while making explicit that a generic least-prime bound must select a prime beyond an eventual certificate's finite exceptional prefix, correcting the census-memory description to a linear resident-state bound, and restoring the architecture-only pointwise pointer. Block `(bc)` replays the certificate data, record tables, natural-log normalizations, and the memory-bounded full scans.

Wave 22 added separate hostile reviews of the supercritical-window and inverse-census sections; both returned SOUND-AFTER-REPAIRS. The first independently rederived both threshold exponents, made the full $W = +\infty$ containment and dyadic quantifiers explicit, corrected the $cb^3 = 1$, $C < 1$ endpoint, retained the UNPROVED window hypotheses and every inherited Section 10 and CLAIMED/PROVISIONAL label, restored the source §18 and Lemma-33.3 antecedents, made (57.11) sufficient rather than necessary, scoped every modulus/cylinder claim to its proved architecture, fixed the crossing families before quantification, and repaired the factor-two transfer proof. Block `(bd)` uses the shared exact harvest, reproduces the integer and prime tails and exact displayed logarithms, and checks only finite toy instances of the crossing geometry.

The inverse review confirmed the square theorem only after expanding the nonunit, common-factor, 2-adic, prime-power, and local-obstruction cases in its full proof. It changed the dichotomy to the moving count $T < n \le N$, retained square density, restricted $(k \log k)^2$ to $k \ge 2$, and made the secondary-source provenance and both directions of the liminf duality explicit. Block `(be)` compares the original and divisor harvests, tests every harvested divisor and every $m^2 \le 10^6$ for each eligible

$M \le 3000$, checks leastness of each inverse row, and chunks prime generation; optional rows remain behind `ES_FULL_SCAN=1`. Its independent replay covered 750 moduli, 15,754 divisor data, 14,745 distinct classes, and 750,000 square residues (15,754,000 (M, D, m) incidences). These finite checks are regressions, not proofs. Costello–Watts page 1 and Ford–Green–Konyagin–Maynard–Tao pages 2–3 were checked as complete archived secondary sources; Iwaniec's primary 1978 PDF remains uninspected. Wave 23's maximum-severity hostile review of the polynomial-escape section returned SOUND-AFTER-REPAIRS. It rederived the exact local criterion, qualitative cyclotomic obstruction, complete linear/quadratic classification, and square-count tail obstruction. The degree-two repair made the leading-coefficient choice, negative-leading sign, zero Discriminant, and every real/imaginary discriminant case explicit; three nonsquare quadratics passing the whole $D = 1$ layer now have exact later-shift hits. The tail proof chooses an interior $t = \lambda L^{1/3}$ even when the nominal power window extends beyond $T = N$. The review also kept Chebotarev qualitative and ineffective and, at the source-Section-59 stage, kept both conjectures and all three constants open while recording only the implication $C_{\mathrm{SQ}} \Rightarrow$ Erdős–Straus, not a converse. That historical frontier is superseded by Theorem 22.3.

Block `(bf)` independently uses Python-integer divisor products, rejects the false fixed-width-overflow witnesses, proves the default $3 \cdot 10^6$ finite frontier, and records the completed $1.5 \cdot 10^8$ streamed run. It also checks the exact Jacobi, forbidden-divisor, and square-scaling reductions and all 180 twisted-square controls; 177 are hit by $M \le 131$. The deep run remained below 170 MB. At the Section-59 stage these finite checks were regressions and stress tests, not evidence for C_{SQ} or proofs of infinite escape.

Wave 24's maximum-severity review of the witness duality returned SOUND-AFTER-REPAIRS. It independently rederived the normal form without an $a \leftrightarrow M$ assumption, checked the 2-adic and gcd cancellation branches and the partial swap criterion, and independently reimplemented the original M-harvest to all three exact ceilings. The largest replay streamed 409,000,770 divisor incidences. Block `(bg)` checks both finite normal-form parameterizations, near misses, coverage union, and bounded duality replay.

Wave 25's maximum-severity review of the decidable census, twisted-square laws, and prime criterion returned SOUND-AFTER-REPAIRS. It independently rederived the simultaneous forward bounds, reproduced the complete original-coordinate classification through $n = 5000$, replayed the twisted-square box, and checked necessity and sufficiency of the prime criterion without any odd-D assumption. The repairs added an original/duality comparison for every one of the 3202 hard primes through 300000 and made the even-D argument explicit; 1941 selected minimum rows have even D. Seeded samples of 2000 integers through 200000 and 500 through 10^6, and the optional complete million-point run, supplied separate finite replications. Block `(bh)` checks the forward census, two-method overlap, twisted-square laws and box, all-hard-prime comparison, and later prime replays. These are internal reviews and bug hunts, not external validation.

Wave 26's maximum-severity review of the divisor-ratio spectrum returned SOUND-AFTER-REPAIRS. It independently rederived the ratio bijection and pairing, all six local laws (including the ten minimal mod-11 budgets and the $C_4 \times C_2$ law at 15), the four record-prime rows, and the staged-hunt maximum. Fresh divisor-spectrum

enumeration for every unit $h \le 10^5$ reproduced the six failure counts; an independent original-coordinate harvest rechecked all 146016 members of the stated (s, m)-box and found the same unique maximum $W(9028800) = 3359$. The review retained the box-subset scope, the right-to-left-only proved inclusion for C'_{SQ}, the exhibited-mechanisms-only assessment, and every Heuristic and no-pointwise caveat. Block `(bi)` checks the ratio pairing, local laws, 1181 hard primes through 10^5, record and sporadic ledgers, and the gated staged hunt with Python integers and bounded live factor/divisor state. These checks and the verdict are internal validation, not external refereeing.

Wave 27's three maximum-severity reviews returned SOUND-AFTER-REPAIRS. The taxonomy review rederived DIV/D1, fixed-D, CRT coupling, and the dimension-3/2 sieve, and reproduced the 9732 default and 82887 gated census scopes; block `(bj)` pins these laws, 7146 fixed-D rows, and the square-root and a_1 censuses. The gcd review rederived the valuation and shifted/unit laws and independently replayed all 242837 new pairs, the 146016/242837/388853 counts, ledgers, and $W(3201660) = 5303$; block `(bk)` gates the full box and thin slice. Because the original full census runner was not preserved, the §65 review committed independent `scripts/review65_independent.py`; without `verify.py` or SymPy it replayed all 719781 hard primes below 10^8 and asserted no survivor at $M = 30000$. Block `(bl)` instead replays records only, even when gated. Coverage runs through `(bl)`; all three reviews remain internal validation. A separate full-source review of the auro-zera Lean file replayed its identities but did not compile it; its open-core characterization and the source paper's own conditional Conclusion are related-work checks, not validation of this paper.

Wave 28 added three maximum-severity reviews. The escape-cylinder review returned SOUND-AFTER-MINOR-REPAIRS: it independently rederived the 2/3 period constant and all class quantifiers, scoped the all-degree identity obstruction to rays containing the anchor, and replayed all 54,990 candidates, 65 projected classes, and three censuses. Block `(bm)` pins the prime-power criterion, cylinders, projection, digest, and bounded hunt. The record-stall review returned SOUND-AFTER-REPAIRS: it made the purity depth an integer supremum allowing infinity, exposed every toy-model input and removed evidential language, and restored the exact frontier that every $A \ge 1$ is open. Block `(bn)` checks the full displayed anatomy, all 15 records, residue-one rows, exponent table, and calibrated toy arithmetic, with the complete ranking gated and memory-bounded. The Pomerance–Weingartner v2 audit returned CONFIRMED AFTER REPAIRS, SOURCE-BOUNDED ONLY; it read all 25 pages and the bibliography, repaired the Section 55 range comparison, and found no threat in that source to any provisional priority position. Block `(bo)` pins the crossover and all three range regimes symbolically. These attestations and computations remain internal validation and source-bounded literature review, not external refereeing or complete priority certification.

Wave 29's refutation-class review of the forward-ray theorem returned SOUND-AFTER-REPAIRS. It independently rederived the integer-wise collapse, the complete endpoint-shadow alternatives, and the two-sign backward induction for every degree, period, and tail start. Its fresh finite replay reproduced the bound-500 ledger $(56{,}226; 1{,}240; 54{,}986; 0)$, the bound-5,000 ledger $(5{,}792{,}112; 21{,}390; 5{,}770{,}722; 0)$, and the 65-class fixed-data counts $(445, 1{,}709, 0)$. Block `(bp)` pins these ledgers, the integer-valued quotient and resultant tests, and the shadow blockage. The

review and bounded zeros are internal checks; the proof, not the hunts, supplies the all-degree anchored conclusion, and no non-polynomial or classification claim follows. At that stage coverage ran through `(bp)`.

Wave 30's three maximum-severity reviews were repaired before the present absorption. The fixed-a review returned SOUND-AFTER-REPAIRS; it added the required character-twist hypothesis to the progression asymptotic, repaired the primitive two-form sieve scope, and rejected the false quadratic-character-only composite classification using the $a = 15$ example. Block `(bq)` checks the subgroup laws, finite spectra, progressions, and independent divisor comparisons. The fixed-J review returned SOUND-AFTER-REPAIRS; it audited the verbatim Shiu, Nair–Tenenbaum, and Henriot statements and restored Henriot's prime-density factor while retaining fixed-J scope and both named open hypotheses. Block `(br)` checks the discriminant factors, exponent sums, and exact local collisions. The census review returned CONFIRMED-AFTER-REPAIRS; its independent replay covered all 719781 hard primes below 10^8 and reproduced the complete a_1 histogram and finite anatomy. Block `(bs)` carries the bounded and gated census checks. These are internal reviews and finite replications, not external validation, asymptotic evidence, or pointwise progress. Coverage now runs through `(bs)`.

11. General numerators

For $m \geq 3$, define

$$E_m(N) = \#\{n \leq N : m/n \text{ is not a sum of three positive unit fractions}\}. \quad (43.1)$$

This concerns the rational number: reducing a common factor does not change representability. We nevertheless count the unreduced pairs, as does Pomerance–Weingartner [20].

11.1. The identity and the first exceptional-set layer.

Lemma 11.1 (general-numerator multiplier identity)**.** *Let $k, \ell \geq 1$, suppose $k\ell \equiv -1 \pmod m$, and let $A = (k\ell+1)/m = uvw$ be a factorization into positive integers. If $n > 0$, $nv \equiv -u \pmod{k\ell}$, and $s = (nv+u)/(k\ell)$, then*

$$\frac{m}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw}. \quad (43.2)$$

Moreover $(A, k\ell) = 1$, so v is invertible modulo $k\ell$.

Proof. The congruence makes s a positive integer. The numerator over the common denominator $nsuvw$ is $nv + u + s = s(k\ell+1) = smuvw$. A common divisor of A and $k\ell$ divides $mA - k\ell = 1$. □

No parity, primality, or coprimality condition on m and n occurs here. For a prime $\ell \nmid m$, the old condition $k \equiv 1 \pmod 4$ is replaced by $k \equiv -\ell^{-1} \pmod m$.

Put

$$\eta_1(m) = \prod_{p\mid m} \frac{p}{p+1}, \qquad \eta_2(m) = \prod_{p\mid m} \frac{p^2}{p^2+p-1}, \qquad C_2 = \prod_p (1-p^{-1})\left(1 + \frac{p-1}{p^2}\right). \quad (43.3)$$

Lemma 11.2 (exact thinning factors)**.** *For fixed m and every reduced $r \pmod m$,*

$$\sum_{\substack{k\le K\\ k\equiv r\ (m)}} \frac{\varphi(k)}{k^2} \sim \frac{\eta_1(m)}{\varphi(m)\zeta(2)}\log K,$$
$$\sum_{\substack{k\le K\\ k\equiv r\ (m)}} \frac{\varphi(k)^2}{k^3} \sim \frac{C_2\eta_2(m)}{\varphi(m)}\log K. \tag{43.4}$$

If $\log m = O(\log K)$, the aggregate sums over $(k,m)=1$ are, up to constants depending only on that O-constant, $\eta_1(m)\log K$ and $\eta_2(m)\log K$. Also

$$\frac{\varphi(m)}{m} \le \eta_1(m) \le \zeta(2)\frac{\varphi(m)}{m}, \qquad \eta_2(m) \asymp \frac{\varphi(m)}{m}. \tag{43.5}$$

Proof. The Euler series for $\varphi(k)/k$ is $\zeta(s+1)/\zeta(s+2)$; removing multiples of $p \mid m$ deletes the local factor $1+1/p$. For $b(k)=(\varphi(k)/k)^2$, the local harmonic factor is

$$1+\sum_{e\ge1}\frac{(1-p^{-1})^2}{p^e} = 1+\frac{p-1}{p^2}.$$

Character orthogonality divides the principal pole by $\varphi(m)$ and, for fixed m, each nonprincipal twist is regular there. The same convolution argument without characters proves the uniform aggregate bounds. Finally, divide each product in (43.3) by $\varphi(m)/m$; the resulting Euler products converge absolutely. □

The fixed-class asymptotics in (43.4) are deliberately not asserted uniformly for growing m. In particular, no Siegel-uniform estimate and no equidistribution among short residue classes is used below. Only the aggregate nonnegative family $(k,m)=1$ is used.

Define the multiplicative function

$$F_m(p^a) = \begin{cases} 1, & p \mid m,\\ p/(p-1), & p \nmid m. \end{cases}$$

For $(u,v)=1$ one has the exact identity

$$\frac{1}{\varphi(muv)} = \frac{F_m(u)F_m(v)}{\varphi(m)uv}. \tag{43.6}$$

This avoids both a false extra factor $1/m$ and the false restriction $(uv,m)=1$.

Lemma 11.3 (general-m class mass)**.** *Fix $B>0$. Let $K_0 \le K \le (\log X)^B$, $H = K^{10}$, $m \le (\log X)^B$, and $\mathcal{J} \subseteq \{k \le K : (k,m)=1\}$ contain 1. Put $h(\mathcal{J}) = \sum_{k\in\mathcal{J}}\varphi(k)/k^2$ and suppose $(c, L_{\mathcal{J}})=1$. For primes $X^{1/2} < \ell \le X$, let $f_{m,c}(\ell)$ count the distinct residues $-uv^{-1} \pmod\ell$ furnished by*

$$k \in \mathcal{J},$$
$$H < u,v \le \ell^{1/3}, \quad (u,v)=(uv,k)=1, \quad \omega(uv) \le D\log\log X, \tag{43.7}$$
$$muv \mid k\ell+1, \qquad k \mid u+cv.$$

Then, with constants depending only on B,

$$\sum_{X^{1/2}<\ell\le X}\frac{f_{m,c}(\ell)}{\ell} \asymp_B \frac{(\log X)^2 h(\mathcal{J})}{\varphi(m)}. \tag{43.8}$$

Each counted residue, combined with the congruence modulo k, is forced by Lemma 11.1.

Proof. For fixed k, the lattice congruence is still $u + cv \equiv 0 \pmod{k}$. Thus the box count, boundary ratio $O(K^2/H)$, low-ω Rankin tail, and Shiu modulus condition $k < U^{1/2}, V^{1/2}$ are unchanged. The condition $k \equiv -\ell^{-1} \pmod{m}$ is not a second lattice condition: after (k, u, v) is fixed, all congruences are carried by the one reduced prime progression

$$\ell \equiv -k^{-1} \pmod{muv}. \tag{43.9}$$

The modulus is the product muv, not $\operatorname{lcm}(m, uv)$; common prime powers, including powers of 2, are already present with their summed exponent. In the canonical boxes $uv \le x^{1/3}$, so a fixed log-power m remains below the Bombieri–Vinogradov level. Each modulus has at most $K2^{\omega(uv)}$ descriptions, absorbed by choosing the logarithmic saving in terms of B. Brun–Titchmarsh gives the upper half. Identity (43.6) and the same two-variable harmonic calculation give exactly the factor $1/\varphi(m)$ on both sides. Shiu's auxiliary modulus remains k; no condition $(uv, m) = 1$ is inserted.

For deduplication, a collision at one ℓ gives $\ell \mid uv' - u'v$ and $|uv' - u'v| < \ell^{2/3} < \ell$, hence reducedness gives $(u, v) = (u', v')$. The two atoms then fix $k \pmod{muv}$, and $muv > mH^2 > K$ gives $k = k'$. Finally $n \equiv c \pmod{L_{\mathcal{J}}}$ and a hit modulo ℓ combine to $nv \equiv -u \pmod{k\ell}$. □

Theorem 11.4 (first general-numerator layer; CLAIMED/PROVISIONAL)**.** *For every fixed $m \ge 3$ there is an absolute $c_0 > 0$ such that, for all sufficiently large N,*

$$E_m(N) \ll_m N \exp\left\{-c_0 \left(\frac{\eta_1(m)}{\varphi(m)}\right)^{1/3} (\log N)^{2/3} (\log\log N)^{1/3}\right\}. \tag{43.10}$$

The prime-denominator and all-denominator forms are both uniform, for every fixed $\varepsilon > 0$, when $3 \le m \le (\log N)^{2-\varepsilon}$, after an ε-dependent lower threshold. This theorem has a complete internal proof but remains provisional externally because it would improve the published general-m benchmark.

Internal argument; provisional status retained. Take $K = \lfloor \delta \log N \rfloor$, all $k \le K$ coprime to m, and $M_0 = \operatorname{lcm}_{k \le K, (k,m)=1} k$. Since $\log M_0 \le (1 + o(1))K$, each reduced subsequence has length $N^{1-O(\delta)}$. Apart from the $O(\log m)$ primes dividing m, every prime denominator above K occurs in one. Lemmas 11.2 and 11.3 give uniformly in its residue c the prime class mass

$$\asymp \lambda_m (\log X)^2 \log K, \qquad \lambda_m = \frac{\eta_1(m)}{\varphi(m)}.$$

The upper mass has the same factor, so the larger-sieve Rankin truncation used in Theorem 9.1 closes with

$$\log X = \alpha \left(\frac{\log N}{\lambda_m \log\log N}\right)^{1/3}.$$

This gives (43.10) for primes. In the stated range, $\log X = o(\log N)$ and K, m are fixed powers of $\log X$; the sieve length and BV/Shiu levels follow with a saving chosen in terms of ε, not m.

If $a \mid n$ and m/a is representable, scaling its denominators by n/a represents m/n. Thus every divisor, in particular every prime factor, of an exceptional n is exceptional. The semigroup argument in Theorem 10.7 applies with $g_m(u) = \lambda_m^{1/3} u^{2/3} (\log u)^{1/3}$.

For the uniform transfer put $\Theta_1 = \varphi(m)/\eta_1(m) \asymp m$, fix the final margin $0 < \varepsilon_0 < 2$, choose $0 < \varepsilon' < \varepsilon_0/(3-\varepsilon_0)$, and start the prime estimate at $\log x_0 \asymp \Theta_1^{1/(2-\varepsilon')}$. Its Rankin parameter is $\delta \asymp \Theta_1^{-1/3}(\log N)^{-1/3}(\log\log N)^{1/3}$, and for $\Theta_1 \ll (\log N)^{2-\varepsilon_0}$,

$$\delta \log x_0 \ll (\log N)^{-\gamma}(\log\log N)^{1/3} = o(1), \qquad \gamma = \frac{\varepsilon_0(1+\varepsilon') - 3\varepsilon'}{3(2-\varepsilon')} > 0.$$

Declaring every smaller prime exceptional costs $\exp\{O(\log \Theta_1)\}$, absorbed by the displayed saving. Above x_0, $g_m(u)/u$ decreases and $g_m(\log x_0)$ is a positive power of Θ_1, so partial summation is uniform. This gives the same $3 \le m \le (\log N)^{2-\varepsilon_0}$ range for all denominators. □

11.2. **The provisional cubic transfer.** Fix $0 < \kappa < 1/240$, put $t = \log X$, $K = \lfloor X^{\kappa} \rfloor$, $H = K^{10}$, and use the dyadic blocks $x < \ell \le 2x$ in $(X^{1/2}, X]$. The general-m c-free atoms are

$$\begin{aligned} &A = (k, \ell, u, v), \quad k \le K, \quad (k, m) = 1, \quad H < u, v \le x^{1/6}, \\ &(u, v) = (uv, k) = 1, \quad \omega(uv) \le D \log\log X, \quad muv \mid k\ell + 1, \\ &\qquad E_A = \{n : n \equiv -uv^{-1} \pmod{k\ell}\}. \end{aligned} \tag{43.11}$$

The proof of Lemma 10.1, with $4uv$ replaced by muv, shows that the coupling $k \mid u + cv$ is automatic after revealing $c \equiv n \pmod{L_{m,K}}$, distinct compatible atoms have distinct ℓ's, and

$$\Pr\left(\bigcap_{i \le j} E_{A_i}\right) = \frac{\mathbf{1}_{\text{compatible}}}{\operatorname{lcm}(k_1, \dots, k_j) \prod_i \ell_i}. \tag{43.12}$$

The two relevant inequalities only improve: $x^{1/3} < \ell$ makes (u, v) unique, and $muv > K$ then makes k unique.

Lemma 11.5 (general-m mass profiles)**.** *Uniformly for $m \le t^B$, $g \mid k$, and reduced $a \pmod g$,*

$$W_{k,a}(g) \ll_B \frac{t^2}{\varphi(m)\varphi(g)} \frac{\varphi(k)^2}{k^3}, \qquad W_k \asymp_B \frac{t^2}{\varphi(m)} \frac{\varphi(k)^2}{k^3}. \tag{43.13}$$

Consequently

$$\mu_{m,X} \asymp_B \frac{\eta_2(m)}{\varphi(m)} t^3. \tag{43.14}$$

Proof. Identity (43.6) extracts $1/\varphi(m)$. In boxes put $q_0 = \operatorname{lcm}(g, \operatorname{rad} k)$. Since $(k, m) = 1$, there are still $\varphi(q_0)^2/\varphi(g)$ compatible reduced class pairs. Shiu applies to $F_m \le n/\varphi(n)$ because $q_0 \le k < U^{1/10}, V^{1/10}$, and gives

$$\sum_{\text{boxes}} \frac{F_m(u) F_m(v)}{uv} \ll \frac{t^2}{\varphi(g)} \left(\frac{\varphi(k)}{k}\right)^2.$$

Brun–Titchmarsh on modulus muv proves the upper profile. For the BV lower profile, replacing $n/\varphi(n)$ by $F_m(n)$ changes the coprime-pair factor at $p \mid m$ by $(p^2-1)/(p^2+1)$; the product is bounded away from zero. The Rankin truncation and allocation multiplicity are unchanged. Summation and the second aggregate formula in Lemma 11.2 prove (43.14). □

Theorem 11.6 (general-m pruned cubic slice; CLAIMED/PROVISIONAL)**.** *For fixed B, uniformly for $m \le t^B$, every $\mathcal{J} \subseteq \{k \le K : (k,m) = 1\}$ containing 1, and every reduced fibre $(c, L_{\mathcal{J}}) = 1$, the restriction $r_{\mathcal{J}}(u, v; c) \le t^4$ leaves*

$$\sum_{X^{1/2} < \ell \le X} \frac{f_{m,c}^{\mathrm{good}}(\ell)}{\ell} \asymp_B \frac{t^2 h(\mathcal{J})}{\varphi(m)}. \tag{43.15}$$

For the full coprime family this is $\asymp \eta_1(m) t^3/\varphi(m)$. This theorem inherits the provisional status of Theorem 9.4.

Proof ledger; provisional status retained. The incidence second moment is unchanged after restricting the outer k-set to $(k,m) = 1$: it sees only $k \mid u + cv$, and its lcm boundary is at most K^2. Markov pruning therefore removes $o(t^2 h(\mathcal{J}))$. Identity (43.6) leaves at least $1/\varphi(m)$ of the retained harmonic mass. The retained multiplicity is $2^{\omega(uv)} t^4$, while BV runs over $muv \le t^B x^{1/3}$; increasing its fixed logarithmic saving by B makes the error $o(x \log x\, h(\mathcal{J})/\varphi(m))$. Deduplication is (43.12), and the upper bound is Lemma 11.5. These are all entries at which m changes the proof of Theorem 9.4. □

For completeness, the prime-power moment cancellation also survives. Condition on $(n, P_y) = 1$ and add a compatible atom with multiplier k to a set whose multiplier lcm is R. With $g = (k, R)$, the exact density ratio is $q_y(k) b_y(g)/(k\ell)$, while compatibility fixes one unit residue modulo g. Lemma 11.5 supplies $1/\varphi(g)$, leaving

$$\frac{b_y(g)}{\varphi(g)} = \prod_{p \mid g,\ p > y} \frac{p}{p-1}. \tag{43.16}$$

Primes dividing m do not divide k. At an unmodified such prime, deleting the multiplier Euler factor gives $p^2/(p^2 + p - 1)$; if $p \le y$, it instead gives the smaller $p/(p+1)$. Thus the total deletion factor is at most $\eta_2(m)$, and

$$\sum_{\substack{k \le K \\ (k,m)=1}} \frac{\varphi(k)^2}{k^3} q_y(k) \prod_{p \mid (k,R),\ p > y} \frac{p}{p-1} \ll \eta_2(m) \log K. \tag{43.17}$$

The ordered induction, including unequal prime-power exponents, gives

$$\mathbb{E}(H_{m,X})_j \le \left(C \frac{\eta_2(m)}{\varphi(m)} t^3 \right)^j \le (Ct^3)^j. \tag{43.18}$$

Theorem 11.7 (source Theorem 43.8; CLAIMED/PROVISIONAL)**.** *For every fixed $m \ge 3$, and uniformly for each fixed $\varepsilon > 0$ in $3 \le m \le (\log N)^{3/4 - \varepsilon}$,*

$$E_m(N) \ll N \exp\left\{ -c_0 \frac{\eta_1(m)}{\varphi(m)} (\log N)^{3/4} \right\}. \tag{43.28}$$

This holds for primes and all denominators and inherits the provisional status of Theorems 9.4 and 10.7.

It is the direct §39 replay with $y, r \asymp t^3$; it remains valid, but is superseded by the thinned-window theorem below.

11.3. **Pomerance–Weingartner comparison.** Pomerance–Weingartner's Theorem 1.3 [20] states that there is an absolute $C > 0$ such that, for $4 \le m \le (\log N)^2$,

$$E_m(N) \le N \exp\left\{-C\left(\frac{(\log N)^2}{\varphi(m)}\right)^{1/3}\right\}. \tag{43.32}$$

Their prime mass is $(\log X)^2/\varphi(m)$, so no $(\log\log N)^{1/3}$ is hidden in (43.32). For the optimized provisional layer, write $L = \log N$ and compare the exponent scales

$$R = \left(\frac{\eta_2(m)}{\varphi(m)}\right)^{1/4} L^{3/4}, \qquad P = \frac{L^{2/3}}{\varphi(m)^{1/3}}. \tag{43.33}$$

Then

$$\frac{R}{P} = \{\eta_2(m)^3\varphi(m)L\}^{1/12}, \qquad R \ge P \iff L \ge \frac{1}{\eta_2(m)^3\varphi(m)}. \tag{43.34}$$

Unknown absolute constants prevent reading (43.34) as a literal finite threshold. In the common stated domain $4 \le m \le L^2$, the formal new scale is larger once (43.34) holds, and it eventually does for each fixed m. Theorem 11.8 continues through every fixed-gap range $m \le L^{3-\varepsilon}$, where the quoted published theorem is not stated once $m > L^2$; at $m \asymp L^3$ the new saving stops growing, so no claim is made there or beyond. This scale comparison does not erase the provisional-versus-published evidentiary distinction. The first layer is uniform for primes and all denominators through $m \le L^{2-\varepsilon}$ and has exponent ratio $(\eta_1(m)\log L)^{1/3}$ relative to (43.32).

11.4. **The attested-blind thinned window.** The protocol and adjudication are recorded in Subsection 10.8. In particular, the blind construction's finite relative-mass inequality was adjudicated line by line, but neither that internal verdict nor the attestation upgrades the inherited status.

Theorem 11.8 (source Theorem 43.12; CLAIMED/PROVISIONAL)**.** *Put $L = \log N$. For every fixed $\varepsilon > 0$, uniformly for $3 \le m \le L^{3-\varepsilon}$,*

$$E_m(N) \ll_\varepsilon N \exp\left\{-c_\varepsilon\left(\frac{\eta_2(m)L^3}{\varphi(m)}\right)^{1/4}\right\}. \tag{43.35}$$

The same formula holds for exceptional primes. It inherits the provisional Theorem 9.4 through Theorem 11.6, together with every maximum-severity qualification in Subsection 10.7 (source §39.7); the internal wave-16 adjudication changes the exponent and range, not the external status.

Internal argument; provisional status retained. Put

$$\theta_m = \frac{\eta_2(m)}{\varphi(m)}, \qquad \Theta_m = \theta_m^{-1} \asymp m, \qquad M = \theta_m t^3.$$

The shrinking-window input is a finite relative-mass inequality. If $H_m(K) = \sum_{k\le K,(k,m)=1}\varphi(k)/k^2$, then for every prime $p \nmid m$,

$$\sum_{\substack{k\le K,(k,m)=1\\ p\mid k}} \frac{\varphi(k)}{k^2} = \sum_{e\ge 1}\frac{1-p^{-1}}{p^e}\sum_{\substack{a\le K/p^e\\ (a,pm)=1}}\frac{\varphi(a)}{a^2} \le \frac{H_m(K)}{p}. \tag{43.36}$$

For $p \mid m$ the left side is zero. Reveal the multiplier coordinates, put $\mathcal{J}_c = \{k \le K : (k,m) = (k,c) = 1\}$, and set $Z(c) = \sum_{y<p\le K,\ p\mid c,\ p\mid L_{m,K}} 1/p$. The union bound and (43.36) give $h(\mathcal{J}_c) \ge H_m(K)(1 - Z(c))$. Independence above y gives

$$\mathbb{E}\mathrm{e}^{yZ} \le \exp\left\{Cy\sum_{p>y} p^{-2}\right\} = \mathrm{e}^{O(1)}, \qquad \Pr\{Z > \eta_0\} \le \mathrm{e}^{-\eta_0 y + O(1)} \tag{43.37}$$

for a fixed $0 < \eta_0 < 1$.

Take an integer $y \asymp B_0 M$, choosing the absolute B_0 after the good-fibre constant. On $Z \le \eta_0$, Theorem 11.6 gives active prime-coordinate mass

$$\frac{t^2 h(\mathcal{J}_c)}{\varphi(m)} \gg \frac{\eta_1(m)}{\varphi(m)} t^3 \asymp M,$$

so the conditioned void is e^{-cM}; (43.37) makes the bad-fibre cost the same size. The estimate is uniform because $y\sum_{p>y} p^{-2} = O(1)$ and $\eta_1 \asymp \eta_2$. Equation (43.18) supplies every factorial moment with base CM. Taking r to be the least even integer at least $D_0 M$, with fixed sufficiently large D_0, Stirling and the exact even-Bonferroni identity give

$$\mathbb{E}_{\mathrm{CRT}}\{S_y Q_r(H_{m,X})\} \le \mathrm{e}^{-cM} + \frac{(CM)^{r+1}}{(r+1)!} \le \mathrm{e}^{-c'M}. \tag{43.38}$$

The resulting expansion has the genuinely thinned ledger

$$\begin{aligned} \deg &= O(M), \\ \log d_{\mathrm{term}} &\le O(y) + r(t + \log K) = O(Mt), \\ \log \sum |c_{\mathrm{term}}| &= O(rt) = O(Mt). \end{aligned} \tag{43.39}$$

Here $\log|\mathcal{A}_{m,X}| = O(t)$, and the atom modulus is $k\ell$, not $mk\ell$: m selects atoms but contributes no ledger factor. Thus finite-interval rounding is absorbed when

$$L \ge C_0 M t = C_0 \theta_m t^4. \tag{43.40}$$

Choose $t = \alpha(L/\theta_m)^{1/4}$ with fixed sufficiently small α. Then

$$M \asymp \theta_m^{1/4} L^{3/4}, \tag{43.41}$$

which proves (43.35) for primes.

The fixed-gap range supplies every analytic constraint. Since $\Theta_m \asymp m \le L^{3-\varepsilon}$, $t/L \asymp (\Theta_m/L^3)^{1/4} \ll L^{-\varepsilon/4}$ and $M \gg L^{\varepsilon/4}$. Also $m \le t^4$ after a harmless fixed enlargement, $K = \mathrm{e}^{\kappa t} \gg m$, muv remains exponentially below the Bombieri–Vinogradov level, $2 \le y < X^{1/2}$, and $\max(m, K, y) = \mathrm{e}^{o(L)}$. Shrinking y exposes more primes, but (43.16)–(43.18) already sum their local factors uniformly.

For the all-denominator transfer fix a final margin $0 < \varepsilon_0 < 3$, choose $0 < \varepsilon' < \varepsilon_0/(4 - \varepsilon_0)$, and start the prime theorem at $u_0 = \log x_0 \asymp \Theta_m^{1/(3-\varepsilon')}$. With $g_m(u) = \Theta_m^{-1/4} u^{3/4}$ and $\delta \asymp \Theta_m^{-1/4} L^{-1/4}$,

$$\delta u_0 \ll L^{-\gamma}, \qquad \gamma = \frac{\varepsilon_0(1+\varepsilon') - 4\varepsilon'}{4(3-\varepsilon')} > 0. \tag{43.42}$$

Thus declaring all smaller primes exceptional costs only $\exp\{O(\log \Theta_m)\} = \mathrm{e}^{o(g_m(L))}$. Above x_0, $g_m(u)/u$ decreases and $g_m(u_0) = \Theta_m^{\varepsilon'/(4(3-\varepsilon'))}$ beats the partial-summation Jacobian uniformly. The semigroup argument of Theorem 10.7 preserves the exponent and the full $L^{3-\varepsilon_0}$ fixed-gap range. □

No pointwise equidistribution among the classes $k \pmod m$ is used, and no growing-saving claim is made at $m \asymp L^3$ or beyond.

12. Pointwise Type-I slices

12.1. **Exact vanishing and finite conspiracies.** For an odd prime p, put

$$\mathcal{B}_p = \left\{ (c,k) : 1 \le k \le \lfloor 2p/3 \rfloor,\ 1 \le c \le \left\lfloor \frac{2p+k}{4k} \right\rfloor,\ (p, ck) = 1 \right\}.$$

For $(c,k) \in \mathcal{B}_p$, let $h = 4ck$, $N_{c,k} = p^2 + 4ck^2$, and define the raw ordered Type-I slice

$$M_{c,k}(p) = \#\{D : D \mid N_{c,k},\ D \equiv -p \pmod h\}. \tag{44.1}$$

Each such divisor gives the ordered row $a = (D+p)/h$, $b = (N_{c,k}/D + p)/h$ in $p(a+b) = k(4abc-1)$, and every row arises this way.

If $N_{c,k} = \prod_j \ell_j^{e_j}$, character orthogonality gives the exact vanishing law

$$\begin{aligned} M_{c,k}(p) &= \frac{1}{\varphi(h)} \sum_{\chi \pmod h} \overline{\chi(-p)} \prod_j (1 + \chi(\ell_j) + \cdots + \chi(\ell_j)^{e_j}), \\ M_{c,k}(p) = 0 &\iff \nexists (0 \le u_j \le e_j) : \prod_j \ell_j^{u_j} \equiv -p \pmod{4ck}. \end{aligned} \tag{44.2}$$

The second line is an exponent-box condition, not merely subgroup membership; vanishing need not be caused by one character factor. This is the raw count. The primitive count additionally requires $(a,b) = 1$ and can vanish when the raw count does not: for $(p,c,k) = (241,7,3)$ the two target divisors give $(a,b) = (3,66), (66,3)$, so $M_{7,3}(241) = 2$ but the primitive mass is zero.

The deterministic obstruction is a moving genus character. Write $c = st^2$ with s squarefree, let $\Delta_s < 0$ be the fundamental discriminant of $\mathbb{Q}(\sqrt{-s})$, and put

$$\chi_s(n) = \left(\frac{\Delta_s}{n} \right). \tag{48.1}$$

Its conductor divides $4s$.

Theorem 12.1 (moving genus obstruction; source Theorem 48.1)**.** *Let $p \equiv 1 \pmod{24}$ and $(c,k) \in \mathcal{B}_p$. Then*

$$\chi_s(p) = 1 \quad \Longrightarrow \quad M_{c,k}(p) = 0. \tag{48.2}$$

The restriction of χ_s to the hard progression is identically one if and only if

$$s \in \{1,2,3,6\}. \tag{48.3}$$

For every other squarefree s, the hard primes with $\chi_s(p) = 1$ have relative Dirichlet density $1/2$, apart from the finitely many ramified primes; admissibility for fixed (c,k) removes only the additional finite set dividing ck.

Proof. Every prime $\ell \mid p^2 + 4s(tk)^2$ is odd and prime to stk, and $(p(2tk)^{-1})^2 \equiv -s \pmod \ell$. Hence $\chi_s(\ell) = 1$, and every divisor D of the norm has $\chi_s(D) = 1$. Since the conductor divides $4s \mid 4ck$, a target divisor would instead satisfy $\chi_s(D) = \chi_s(-p) = -\chi_s(p)$, proving (48.2). The primitive quadratic character is constant on the kernel of reduction modulo 24 exactly when its conductor divides 24. The negative fundamental discriminants with that property are $-4, -8, -3, -24$, which give precisely (48.3). Otherwise the restriction is a nonprincipal quadratic character,

so half of the compatible reduced classes have value one; Dirichlet's theorem gives the density assertion. □

Thus all eight slices $ck \le 4$ and the 80 slices $ck \le 30$ with $s \in \{1,2,3,6\}$ vanish on every hard prime. For each of the other 31 slices, Theorem 12.1 still forces a positive-density vanishing subfamily, but it is only a sufficient source of zeros.

The remaining slices can conspire beyond fixed congruence data: $193 \equiv 1033 \pmod{840}$, both the $(5,1)$ and $(7,1)$ slices vanish at 193, but the $(5,1)$ slice is positive at 1033. Conversely, every prime $p \equiv 97 \pmod{120}$ has a positive $(5,1)$ slice via $D = 3$; this is the already known multiplier progression, not a new supply.

An exact finite census over the 385 primes $p < 30000$, $p \equiv 1 \pmod{24}$, gives

C	4	5	10	15	20	25	30
$\#\{(c,k) : ck \le C\}$	8	10	27	45	66	87	111
$\#\{p : \text{all slices vanish}\}$	385	220	161	63	41	30	15.

This is a finite exact factorization census, not an asymptotic theorem. The 15 primes for which all 111 slices vanish include 2521; its known twelve ordered Type-I rows all lie outside $ck \le 30$.

12.2. **Fixed divisors, residue-one escape, and depth.** The positive progression just displayed is one instance of an exact law.

Theorem 12.2 (fixed-divisor progression law; source Theorem 48.4)**.** *Fix (c,k), put $h = 4ck$, and fix a positive integer d prime to h. Let $L_d = \operatorname{lcm}(24, h, d)$. A reduced class $r \pmod{L_d}$ is a hard-prime class on which this fixed d guarantees $M_{c,k}(p) > 0$ if and only if*

$$r \equiv 1 \pmod{24}, \qquad r \equiv -d \pmod{h}, \qquad r^2 \equiv -4ck^2 \pmod{d}. \tag{48.16}$$

Every prime in such a class for which $(c,k) \in \mathcal{B}_p$ has

$$e = \frac{p^2 + 4ck^2}{d}, \qquad a = \frac{p+d}{h}, \qquad b = \frac{p+e}{h}, \qquad p(a+b) = k(4abc - 1). \tag{48.17}$$

Within the proof shape "one specified divisor has the target grade," these conditions are also necessary; classes modulo multiples of L_d are exactly refinements of them.

Proof. The last congruence in (48.16) is $d \mid p^2 + 4ck^2$, and the middle one is $d \equiv -p \pmod{h}$. Conversely those two facts give a target divisor in (44.1). Since the norm is $p^2 \pmod{h}$, its cofactor e has the same target grade, so (48.17) is integral; direct expansion proves its final identity. The first congruence is exactly the hard-prime restriction. □

Theorem 12.3 (residue-one escape; source Theorem 48.5)**.** *No class in (48.16) contains $1 \pmod{L_d}$. Consequently, for every Q, the union of all fixed-divisor slice guarantees with $L_d \le Q$ misses the infinite prime class*

$$p \equiv 1 \pmod{\Lambda_Q}, \qquad \Lambda_Q = \operatorname{lcm}\{24, L_d : L_d \le Q \text{ occurs above}\}. \tag{48.20}$$

Proof. If residue one were guaranteed, then $d \equiv -1 \pmod{h}$ and $d \mid 1 + 4ck^2 = 1 + hk$. The cofactor $e = (1+hk)/d$ would also be $-1 \pmod{h}$, so $d, e \ge h - 1$, whereas

$$(h-1)^2 - (1+hk) = h(h-k-2) > 0, \qquad h - k - 2 = k(4c-1) - 2 \ge 1,$$

a contradiction. There are finitely many triples with $L_d \le Q$, and Dirichlet's theorem supplies infinitely many primes in (48.20). □

This is the slice analogue of the residue-one compactness wall for the paper's fixed identity classes. It does not control a divisor chosen from the moving factorization of each prime, and "not guaranteed" does not mean "vanishing." The later Theorem 17.3 instead quantifies the related residue-one *genus* construction of Theorem 13.5; it proves actual slice vanishing through a logarithmic box, not a consequence of this fixed-divisor non-guarantee.

For scale only, let $ck_{\min}(p)$ be the first positive admissible slice whose squarefree c-core is not in $\{1,2,3,6\}$ and put $D(p) = ck_{\min}(p) - 1$. The exact streamed census finds $ck_{\min}(p) \le 77$ for all 385 hard primes below 30000, with maximum $D = 76$ at $p = 12289$; the 15 full-$ck \le 30$ conspiracies have first positive product between 31 and 77. The optional census finds $ck_{\min}(p) \le 103$ for all 1181 hard primes below 10^5, with maximum $D = 102$ at $p = 92401$. These are finite computations, not a growth law or pointwise progress. Proving $ck_{\min}(p) < \infty$ for every hard prime would already prove the Type-I form of Erdős–Straus and hence the conjecture; it remains open.

13. Conditional pointwise routes through the slice frame

Scope and verdict. This section isolates a one-prime-factor event sufficient for a Type-I solution and states the corresponding conditional theorem. Its hypothesis is deliberately labelled BESPOKE AND UNPROVED: none of Bateman–Horn, Linnik, GRH for Dirichlet, Hecke, or ray-class L-functions, Elliott–Halberstam, or Duke-type equidistribution has this pointwise factorization conclusion in its standard form. The audit makes the quantifier failure exact. Two proved walls sharpen the earlier slice analysis: prime values of the norm have the wrong grade, and any bounded collection of squarefree cores is simultaneously genus-forced on infinitely many hard primes. This section's deliverable is a rigorous conditional reduction and a map of its missing pointwise input. The later stacked sieve, Theorem 16.4, proves an almost-all form but still does not settle that pointwise quantifier or claim that a standard conjecture secretly settles Erdős–Straus.

13.1. **The good prime-factor class and exact reduction.** Fix a hard prime p, an admissible slice $(c,k) \in \mathcal{B}_p$, and write

$$c = st^2, \quad s = \mathrm{sf}(c), \qquad h = 4ck, \qquad N = N_{c,k} = p^2 + 4ck^2. \tag{50.1}$$

Inflate the primitive genus character χ_s to $(\mathbb{Z}/h\mathbb{Z})^\times$; its conductor divides $4s \mid h$. Put

$$K_s(h) = \ker \chi_s, \qquad \mathcal{G}_p(c,k) = \{-p \pmod h\} \cap K_s(h). \tag{50.2}$$

Every prime $q \mid N$ lies in $K_s(h)$, while

$$\mathcal{G}_p(c,k) = \begin{cases} \{-p \pmod h\}, & \chi_s(p) = -1, \\ \varnothing, & \chi_s(p) = 1, \end{cases} \tag{50.3}$$

because $\chi_s(-1) = -1$. Thus the moving-genus theorem is exactly the empty-good-set case. On an unforced slice there is still only one good ray grade, not half the factor grades; a split prime is the wrong kind of divisor unless it is also $-p \pmod h$.

Theorem 13.1 (one-prime-factor slice criterion; source Theorem 50.1)**.** *Let p be an odd prime and $(c,k) \in \mathcal{B}_p$. If a prime q satisfies*

$$q \mid p^2 + 4ck^2, \qquad q \pmod{4ck} \in \mathcal{G}_p(c,k), \tag{50.4}$$

then the (c,k)-slice is positive and gives a Type-I Erdős–Straus representation. Explicitly, with

$$e = \frac{p^2+4ck^2}{q}, \qquad a = \frac{p+q}{4ck}, \qquad b = \frac{p+e}{4ck}, \tag{50.5}$$

one has $a,b \in \mathbb{Z}_{>0}$ and

$$p(a+b) = k(4abc-1), \qquad \frac{4}{p} = \frac{1}{ack} + \frac{1}{bck} + \frac{1}{pabc}. \tag{50.6}$$

Primality of q does not imply $(a,b)=1$, and no primitivity is needed. For example, $q=11$ at $(p,c,k)=(241,7,3)$ gives $(a,b)=(3,66)$.

Proof. Admissibility gives $(p,ck)=1$, hence $(N,h)=1$ and $(q,h)=1$. The target grade is $q \equiv -p \pmod h$, while $N \equiv p^2 \pmod h$; therefore $e = Nq^{-1} \equiv -p \pmod h$. This proves the integrality and positivity in (50.5). Substitution of $q = ha-p$, $e = hb-p$, and $qe = N$, followed by cancellation of p^2, gives $4ck^2 = h^2ab - hp(a+b)$. Division by $h = 4ck$ gives the first identity in (50.6), and the unit-fraction identity follows over the common denominator $pabck$. Reduction of the first identity modulo p gives $4abc \equiv 1 \pmod p$, so exactly the denominator $pabc$ is divisible by p. This is Type I. □

The raw/primitive boundary is intentional: the selected prime is a raw target divisor, and a nonprimitive row still proves the displayed identity. The canonical decomposition can recover a primitive Type-I tuple if wanted.

Hypothesis 13.2 (slice prime factor $H_{\mathrm{SPF}}(A)$; bespoke and unproved)**.** *Fix $A>0$. There is a constant $p_0 = p_0(A)$ such that every prime $p > p_0$, $p \equiv 1 \pmod{24}$, admits positive integers c,k and a prime q with*

$$\boxed{ck \le (\log p)^A, \qquad q \mid p^2+4ck^2, \qquad q \equiv -p \pmod{4ck}.} \tag{50.7}$$

This is a quantitative sufficient hypothesis, not the logically weakest one-prime-factor statement: deleting the log-power bound is weaker. It puts no bound on q, prescribes no slice, imposes no primitivity, and needs no extra genus condition, which is automatic from (50.3). It is stronger than slice positivity, since a product of wrong-grade prime factors can hit the target even when no one prime does. Theorem 17.3 proves that $H_{\mathrm{SPF}}(A)$ is false for every $0 < A < 1$; exactly the exponents $A \ge 1$ remain open.

Theorem 13.3 (conditional on $H_{\mathrm{SPF}}(A)$; $H_{\mathrm{SPF}}(A)$ unproved; source Theorem 50.2)**.** *If $H_{\mathrm{SPF}}(A)$ holds for some A, then every sufficiently large hard prime has a Type-I Erdős–Straus representation.*

Proof. For sufficiently large p, (50.7) gives $ck < p$, $3k \le 2p$, and $4ck \le 2p+k$. Hence $(p,ck)=1$ and $(c,k) \in \mathcal{B}_p$. The congruence in (50.7) makes q the unique good class in (50.2), so Theorem 13.1 applies. □

If q is fixed first, (50.4) is exactly the pair of root progressions

$$p \equiv -q \pmod{4ck}, \qquad p \equiv \pm 2k\sqrt{-c} \pmod q, \qquad p \equiv 1 \pmod{24}. \tag{50.8}$$

The fixed-divisor law preassigns q and obtains a fixed CRT class; (50.7) asks the actual factorization attached to each p to supply some q after p is known. The residue-one escape walls bounded collections of the former, not the latter.

13.2. **Prime values, bounded cores, and least nonresidues.**

Lemma 13.4 (prime norms are wrong-grade; source Lemma 50.3)**.** *For a hard prime p and any admissible (c,k), if $N_{c,k}$ is prime then the slice vanishes. More generally, a good prime factor in (50.4) is* $3 \pmod 4$*, its complementary divisor is also* $3 \pmod 4$*, and hence $N_{c,k}$ must be composite.*

Proof. The target grade is $-p \equiv 3 \pmod 4$, whereas $N_{c,k} \equiv 1 \pmod 4$. If N is prime, both divisors $1, N$ have grade $1 \pmod 4$. Under (50.4), both q and N/q have the target grade by Theorem 13.1. □

Assessment 50.4 (Bateman–Horn). Standard Bateman–Horn gives an asymptotic in the variable for a fixed finite list of irreducible integer polynomials. It neither controls every external coefficient p uniformly down to a $(\log p)^A$ interval nor prescribes a prime factor of a composite value. Applied literally to $p^2 + 4ck^2$ as a prime-value polynomial, success triggers Lemma 13.4, not (50.7). A uniform assertion that every p has a logarithmic-box value with a factor in the moving grade $-p \pmod{4ck}$ would imply Theorem 13.3, but merely restates H_{SPF}; it is not standard Bateman–Horn. The same quantifier defect applies to Hardy–Littlewood prime tuples. Standard smooth-value conjectures control factor sizes, not the bounded exponent box of grades in $(\mathbb{Z}/h\mathbb{Z})^\times$.

Theorem 13.5 (bounded-core genus escape; source Theorem 50.5)**.** *For every fixed B, infinitely many hard primes satisfy*

$$\chi_s(p) = 1 \quad \text{for every squarefree } s \le B. \tag{50.9}$$

For each such prime, every admissible slice whose squarefree c-core is at most B vanishes, for all k. Consequently no absolute bound for a least active core, and no proof using only boundedly many cores with arbitrary k, can hold.

Proof. Let $R_B = 24 \prod_{\substack{\ell \le B \\ \ell \text{ prime}}} \ell$. Dirichlet's theorem gives infinitely many primes $p \equiv 1 \pmod{R_B}$. For every odd prime $\ell \le B$, quadratic reciprocity and $p \equiv 1 \pmod 4$ give $(\ell/p) = (p/\ell) = 1$; the factor 24 gives $(2/p) = (-1/p) = 1$. Multiplicativity proves (50.9), and Theorem 12.1 makes every asserted slice zero. □

Theorem 13.6 (GRH gives an active slice only; source Theorem 50.6)**.** *Conditional on the unproved Generalized Riemann Hypothesis, every sufficiently large hard prime p has a squarefree $s \ll (\log p)^2$ with $\chi_s(p) = -1$; hence $(c,k) = (s,1)$ is an admissible unforced slice in a logarithmic box. This does* not *assert that the slice is positive.*

Proof. The standard Ankeny consequence of GRH gives an integer $n \ll (\log p)^2$ with $(n/p) = -1$. Its squarefree kernel s has the same Legendre symbol and is no larger. Since $p \equiv 1 \pmod 8$, $\chi_s(p) = (-s/p) = (s/p) = -1$. Admissibility is automatic for large p; the residual exponent-box condition in (44.2) remains untouched. □

Thus least-nonresidue technology supplies permission for a slice, not the divisor in it. Linnik similarly finds a small prime in $q \equiv -p \pmod h$ but gives no reason for that prime to divide the fixed integer $N_{c,k}$. Asking for both properties is a bespoke form of (50.7), not an existing Linnik hypothesis.

13.3. **The GRH/Chebotarev compositum wall.** With the notation (50.1), put

$$K = \mathbb{Q}(\sqrt{-s}), \qquad \alpha = p + 2tk\sqrt{-s}, \qquad N_{K/\mathbb{Q}}(\alpha) = N_{c,k}. \tag{50.10}$$

For a prime $q \nmid 2ck$,

$$q \mid N_{c,k} \iff \text{some } \mathfrak{q} \mid q \text{ in } K \text{ divides } (\alpha) \iff p \equiv \pm 2k\sqrt{-c} \pmod{q}. \tag{50.11}$$

In particular q splits in K, but the converse is much weaker: (50.11) selects a prime-ideal divisor of the one specified principal ideal (α). Ring-class Chebotarev controls primes represented by form classes, not the factorization of a fixed represented integer. Combining (50.11) with the ray condition gives (50.8), a condition on p modulo q when q is fixed, not a Frobenius condition on p in one fixed field as q varies. Effective Chebotarev under GRH does not turn the infinite disjunction over actual divisors of (α) into a pointwise factorization theorem.

A quantitative check exposes the scale. Restrict $ck \le L$ and fixed candidate primes $q \le Z$. Before deduplication, (50.8) supplies at most two classes modulo $4ckq$, so

$$\begin{aligned} \mu_{\mathrm{raw}}(L, Z) &\le \frac{1}{2} \sum_{ck \le L} \frac{1}{ck} \sum_{\substack{q \le Z \\ q \equiv 3 \ (4) \\ q \text{ prime}}} \frac{1}{q} \\ &= \left(\frac{1}{8} + o(1) \right) (\log L)^2 \log\log Z. \end{aligned} \tag{50.12}$$

This is a class-mass envelope, not a coverage theorem. Although there are $L \log L + O(L)$ slices in the box, all k over one core share one genus bit.

Assessment 50.7 (the compositum wall). A literal joint-Chebotarev implementation introduces cyclotomic conductors containing $\mathrm{lcm}(1, \dots, 4L)$ and candidate primes through Z; its naive compositum degree is exponential on the $L + Z$ scale. The same GRH-effective bookkeeping as the earlier Chebotarev audit limits a direct compositum to $L + Z = O(\log N)$ when $p \le N$ is counted. At $L, Z \asymp \log N$, even an ideal independent-events model applied to (50.12) predicts only

$$\exp\{-O((\log\log N)^2 \log\log\log N)\} \tag{50.13}$$

for the survivor proportion. Allowing an average sieve to take $Z = N^\theta$ changes the model exponent only to $O((\log\log N)^3)$. Both leave $N^{1-o(1)}$ possible survivors and are much weaker in shape than the CLAIMED/PROVISIONAL bound $N \exp\{-c(\log N)^{3/4}\}$. GRH supplies no independence theorem for the overlapping root classes, so (50.13) is an audit ceiling, not a conditional exceptional-set theorem. There is no GRH exceptional-set improvement to record here. Elliott–Halberstam and Kloosterman-refined levels of distribution change average ranges but retain an exceptional set and do not imply (50.7) in standard form.

13.4. **The exact genus pairing and the Duke wall.** For a product cutoff L, the exact active-box mass is

$$T_L^-(p) = \sum_{\substack{ck \le L, \ (c,k) \in \mathcal{B}_p \\ \chi_{\mathrm{sf}(c)}(p) = -1}} \frac{1}{\varphi(4ck)} \sum_{\chi \pmod{4ck}} \overline{\chi(-p)} \prod_{\ell^e \| p^2 + 4ck^2} (1 + \chi(\ell) + \cdots + \chi(\ell)^e). \tag{50.14}$$

Inflate χ_s to modulus h. Every factor $\ell \mid N$ has $\chi_s(\ell) = 1$, while an active slice has $\chi_s(-p) = 1$. Therefore

$$\text{the terms indexed by } \chi \text{ and } \chi\chi_s \text{ are equal.} \tag{50.15}$$

For a genus-forced slice they are opposites and cancel. Restricting to active slices removes that one cancellation but leaves every character of the residual quotient of $K_s(h)$; it does not make the principal character dominant. The principal/genus pair contributes

$$P_L^-(p) = \sum_{\substack{ck \le L,\ (c,k)\in\mathcal{B}_p \\ \chi_{\mathrm{sf}(c)}(p)=-1}} \frac{2\tau(p^2+4ck^2)}{\varphi(4ck)} \le p^{o(1)}(\log L)^2 \log\log(3L) \tag{50.16}$$

when $L = p^{o(1)}$. It is positive when the active box is nonempty, but the remaining ray characters can cancel it exactly. At $p = 2521$, for example, active cores $11, 17, 19, 22, 23, 29$ occur in $ck \le 30$, yet every unforced slice in that box vanishes.

Assessment 50.8 (Duke/class-group GRH). Summing (50.14) does not create a fixed theta series: the discriminant, ray modulus, norm, and active-core selection all move. The Hurwitz–Kronecker identity restores class numbers only after removing the ray projector and changing to a weighted all-divisor mass. Equation (50.15) removes exactly the genus character and no other part of that projector. GRH for class-group or ray-class L-functions controls prime or ideal sums, not the target coefficient of the finite Euler product of one norm. A Duke-type theorem would therefore need a new uniform statement for the whole moving family (50.14), with error $o(P_L^-)$; this is a genuinely new hypothesis, not a recognized consequence of GRH for the relevant L-functions.

13.5. **Conditional frontier and census.** The audit is summarized by

input	standard output	missing condition
Bateman–Horn	prime polynomial values on average in the variable	a composite value with a good factor, uniformly per p
Linnik	a prime $q \equiv -p \pmod h$	$q \mid N_{c,k}$
GRH/Chebotarev least nonresidue	split/ray primes in fixed fields	$\mathfrak{q} \mid (\alpha)$ for each fixed α
under GRH	$\chi_s(p) = -1$	the residual exponent-box hit
EH/GEH/Kloosterman	average distribution in p unprojected or averaged	an empty exceptional set
Duke/class-group GRH	form mass	the moving ray projector

(50.17)

The proved obstructions are Lemma 13.4, Theorem 13.5, the fixed-divisor escape Theorem 12.3, and the exact pairing (50.15). The claims about named hypotheses are assessments of their standard conclusions, not logical independence theorems. Within the pointwise frontier (50.7), Theorem 17.3 refutes every $0 < A < 1$ and leaves exactly $A \ge 1$ open, while Corollary 16.5 proves that the failure set is triple-logarithmically small. Replacing "prime factor" by "divisor" returns the open unrestricted depth criterion.

For finite comparison define

$$ck_{\mathrm{pr}}(p) = \min\{ck : (c,k) \in \mathcal{B}_p,\ \exists \text{ prime } q \mid p^2+4ck^2,\ q \equiv -p \pmod{4ck}\}. \tag{50.18}$$

Computational 50.9 (exact stated range; informational). A streamed exact computation finds $ck_{\mathrm{pr}}(p)$ for all 385 hard primes below 30000, checks the complete

exponent box at every tested norm, and reconstructs (50.5)–(50.6) for each good-prime event. Its distribution is

ck_{pr}	5	7	10	11	13	14	17	19	21	22	23
#	156	30	36	35	13	15	16	4	15	9	6
ck_{pr}	26	28	29	31	33	34	35	37	38	39	42
#	10	4	3	1	2	3	1	1	4	2	5
ck_{pr}	43	46	55	62	66	67	69	70	77	78	
#	1	1	2	1	4	1	1	1	1	1	

(50.19)

and the strict records are

p	73	193	241	1201	2521	4729	7489	9601
ck_{pr}	7	10	21	34	38	66	70	78.

(50.20)

The prime-factor minimum equals the unrestricted $ck_{\min}$ for 311 primes and is larger for 74. The largest gap is 55 at $p = 23689$: $ck_{\min} = 11$ but $ck_{\mathrm{pr}} = 66$, witnessed by $(c, k, q) = (33, 2, 77951)$. The largest ck_{pr} is 78 at $p = 9601$. The optional memory-bounded scan over the 1181 hard primes below 10^5 has maximum 282 at $p = 83689$. These observations support no pointwise growth law; the later almost-all theorem does not use them. The verification factors one norm at a time and stores at most one residue set of size $4ck$.

14. Witness-modulus tails and the effectivity repair

For a prime, or any positive integer, p, define

$$W(p) = W_{\mathrm{II}}(p) = \min k\ell, \tag{51.1}$$

where the minimum is over positive k, ℓ, u, v, w satisfying

$$k\ell \equiv 3 \pmod 4, \qquad \frac{k\ell + 1}{4} = uvw, \qquad p \equiv -uv^{-1} \pmod{k\ell}. \tag{51.2}$$

Put $W(p) = +\infty$ if there is no such tuple; no general finiteness assertion is made. Lemma 3.4 says exactly that $W(p) \le T$ supplies an explicit representation with multiplier modulus at most T. The definition allows composite ℓ, although the proof uses a prime subfamily.

14.1. **Parameterized truncation and the master tail.** Write $t = \log X$, let $K \ge K_0$, and suppose that the canonical dyadic boxes (39.2)–(39.3) furnish, uniformly in every $\mathcal{J} \subseteq \{k \le K : k \equiv 1 \pmod 4\}$ containing 1 and every reduced compatible fibre c,

$$\sum_{X^{1/2} < \ell \le X} \frac{f_{c;\mathcal{J}}(\ell)}{\ell} \asymp t^2 h(\mathcal{J}), \qquad h(\mathcal{J}) = \sum_{k \in \mathcal{J}} \frac{\varphi(k)}{k^2}. \tag{51.3}$$

Put

$$\mu = t^2 \log K. \tag{51.4}$$

Lemma 14.1 (truncated c-free assembly; proved internally, with the provisional review status of the fixed-c-free chain)**.** *Assume (51.3), $KX \le T$, $\log K \ll t$, and*

$$\log N \ge C\mu t. \tag{51.5}$$

Then, with effective constants whenever the supply constants in (51.3) are effective,

$$\#\{p \le N : p \text{ prime},\ W(p) > T\} \ll N \exp\{-c\mu\}. \tag{51.6}$$

Proof. Use the atoms (39.2)–(39.4), all of modulus at most $KX \le T$. Lemma 10.1's geometry needs only $u, v \le X^{1/6}$ and $4uv > 4H^2 > K$. The profile and exact CRT/Euler induction replay as

$$W_{k,a}(g) \ll \frac{t^2}{\varphi(g)} \frac{\varphi(k)^2}{k^3}, \qquad \mathbb{E}(H_X)_m \le (C\mu)^m. \tag{51.7–51.8}$$

for every needed order. With $y = B\mu$, the bad fibres cost $\exp\{-\eta B\mu + O(1)\}$, while the others have $h(\mathcal{J}_c) \gg \log K$; (51.3) therefore gives $\Pr(H_X = 0 \mid S_y = 1) \le e^{-c\mu}$. The fixed-polylog lower proof of Lemma 5.2 and the cubic proof of Theorem 9.4 both supply exactly these canonical dyadic boxes.

Take the least even $r \ge D_B\mu$. The same Bonferroni factorial tail has CRT mean $O(e^{-c\mu})$ and ledger

$$\deg = O(\mu), \qquad \log d_{\mathrm{term}} = O(\mu t), \qquad \log \sum |c_{\mathrm{term}}| = O(\mu t). \tag{51.11}$$

Since $y < X^{1/2}$ and the atom count has logarithm $O(t)$, (51.5) absorbs rounding and the primes at most $\max(K, y)$. □

Theorem 14.2 (master witness-modulus tail; proved internally with the exact status below)**.** *There are effectively computable positive constants such that the following hold. Every set counted with variable p in this theorem and Corollary 14.3 is restricted to primes.*

1. ***Fixed-polylogarithmic supply, without Theorem 9.4.*** *Uniformly when*

$$3 \le T, \qquad \log N \ge C\{1 + (\log T)^3 \log(2 + \log T)\}, \tag{51.12}$$

one has

$$\#\{p \le N : W(p) > T\} \ll N \exp\{-c(\log T)^2 \log(2 + \log T)\}. \tag{51.13}$$

This is unconditional in the no-unproved-hypothesis sense and does not inherit Theorem 9.4; it does inherit the campaign's provisional-review qualification on the Section 10 moment and Bonferroni machinery.

2. ***Cubic supply.*** *Fix $0 < \kappa < 1/240$. Uniformly when*

$$3 \le T, \qquad \log N \ge C_\kappa\{1 + (\log T)^4\}, \tag{51.14}$$

one has

$$\#\{p \le N : W(p) > T\} \ll_\kappa N \exp\{-c_\kappa(\log T)^3\}. \tag{51.15}$$

This statement inherits Theorem 9.4 and is CLAIMED/PROVISIONAL.

Both estimates are uniform throughout the simpler common range

$$3 \le T \le \exp\{c(\log N)^{1/4}\}. \tag{51.16}$$

Proof. For (1), set $X = T^{1/2}$ and $K = \lfloor (\log X)^5 \rfloor$, with fixed adjustments below the asymptotic threshold. Then $KX \le T$, $\log K \asymp \log t$, and the canonical-box proof of (21) supplies (51.3). Thus $\mu \asymp t^2 \log t$ and (51.5) is (51.12). For (2), use the same X and $K = \lfloor X^\kappa \rfloor$. Theorem 9.4 supplies (51.3), $\mu \asymp t^3$, and (51.5) becomes (51.14). Bounded T is absorbed by enlarging the implied constants. □

At $T = \exp\{\alpha(\log N)^{1/4}\}$, (51.15) is $N \exp\{-c(\log N)^{3/4}\}$, exactly the prime part of (39.35) up to constants. This is a consistency check, not a new record.

Corollary 14.3 (polylogarithmic tails; same status as Theorem 14.2)**.** *For each fixed $A > 0$,*

$$\#\{p \le N : W(p) > (\log N)^A\} \ll_A N \exp\{-c_A(\log\log N)^2 \log\log\log N\}, \quad (51.17)$$

unconditionally and without Theorem 9.4. The cubic supply gives the additional CLAIMED/PROVISIONAL *estimate*

$$\#\{p \le N : W(p) > (\log N)^A\} \ll_A N \exp\{-cA^3(\log\log N)^3\}. \quad (51.18)$$

The $(\log\log N)^3$ exponent shape at polylogarithmic depth has the near-simultaneous independent antecedent Dahan [4, Theorem 4.17] for a different two-parameter (u, a) search-depth statistic without a genus gate; his printed $o(1)$ is explicitly ineffective. Equations (51.17)–(51.18) instead bound the minimal congruence modulus W, and Lemmas 14.4– 14.5 give effective constants subject to the labels displayed here. This priority distinction is part of the statement, not an assertion that the statistics are interchangeable.

The formulas are interpreted beyond a fixed large threshold. The same bounds hold with cutoff $(\log p)^A$: split at $N^{1/2}$, use comparable logarithms on the upper half, and absorb the lower half. Define

$$H_{\mathrm{MOD}}(A): \qquad W(p) \le (\log p)^A \quad \text{for every sufficiently large prime } p. \quad (51.19)$$

Theorem 17.1 proves that this hypothesis is false for every $0 < A < 1$; exactly the exponents $A \ge 1$ remain open. For any A for which it holds, this unproved pointwise hypothesis implies eventual prime solvability by Lemma 3.4; the upper tail does not make its failure set empty. It is not equivalent to $H_{\mathrm{SPF}}(A)$: the former is the Type-II multiplier statistic (51.1), while the latter is the Type-I factor-in-one-specified-grade assertion (50.7). Neither hypothesis implies the other as stated, although each implies an eventual representation. Thus (39.35) and H_{MOD} are respectively a tail and a pointwise question for the same W; H_{SPF} is a different frontier.

14.2. **Effective class supply away from one conductor.** The lower half of the fixed-polylogarithmic supply uses Bombieri–Vinogradov, not merely the elementary box calculation. Large composite progression moduli do not remove a possible Siegel zero, because their characters may be induced from a small primitive conductor. The needed effective form is the following.

Lemma 14.4 (effective Bombieri–Vinogradov away from one conductor; standard cited input with dependency proof)**.** *For every $A > 0$ there are effectively computable B, C, x_0 such that, for $x \ge x_0$ and $Q \le x^{1/2}/(\log x)^B$, there is either no exceptional conductor or one real primitive conductor $r_x \ge 3$ for which*

$$\sum_{\substack{q \le Q \\ r_x \nmid q}} \max_{(a,q)=1} \left| \psi(x; q, a) - \frac{x}{\varphi(q)} \right| \le \frac{Cx}{(\log x)^A}. \quad (51.20)$$

The same statement holds for dyadic π-differences by effective partial summation.

Proof. Lenstra–Pomerance [17, Lemma 11.2, pp. 1258–1260] give computable absolute constants, one excluded $s(x) \in [(\log x)^{1/2}, e^{\sqrt{\log x}}]$, maxima over $2 \le y \le x$ and reduced residues, and error $Cx^{1/2}Q(\log x)^5 + Cxe^{-c\sqrt{\log x}}$ over $s(x) \nmid q$. Restricting to $Q \le x^{1/2}(\log x)^{-B}$, and enlarging smaller Q to $x^{1/3}\log x$, proves (51.20); the maximum in y fixes one conductor through a dyadic interval.

For completeness, set $R = (\log x)^{B_0}$. Away from Page, the effective explicit formula gives $|\psi(y,\chi^*)| \ll y e^{-c\sqrt{\log x}}$ uniformly on $[x,2x]$, while $\sum_{q\le Q,\ r\mid q} 1/\varphi(q) \ll \log(2Q)/\varphi(r)$. At most $\varphi(r)$ primitive characters have conductor r, so all $r \le R$ cost $O(R\log(2Q))$; the exponential saving absorbs this. Vaughan's identity and the large sieve handle larger conductors. Fixed conductors, including 4, are checked effectively, and selecting Page at $2x$ prevents two endpoint deletions. □

Lemma 14.5 (exceptional-conductor deletion tolerance; proved)**.** *In the boxes used for (21), let $r > 1$ be a primitive real character conductor. Apart from the explicitly harmless conductor 4, there is a prime $p \mid r$ (or $p = 2$ when the required extra factor is the third power of 2) such that*

$$p \nmid uv \quad\Longrightarrow\quad r \nmid 4uv. \tag{51.21}$$

Uniformly in c, k, the pairs satisfying (51.21) retain a fixed positive proportion of the harmonic box mass. The low-ω cutoff may be chosen so that it and (51.21) retain a fixed positive proportion simultaneously. The same remains true after the low-congestion pruning of Theorem 9.4.

Proof. A primitive real conductor is the absolute value of a fundamental discriminant. Choose an odd prime factor, or $p = 2$ for conductor 8; conductor 4 is handled effectively directly. Fix k. If $p \mid k$, the condition $(uv,k) = 1$ already excludes p. If $p \nmid k$ and $p \le Ck$, the box count with the extra reduced conditions modulo p retains the local proportion

$$\frac{(1-1/p)^2}{1-1/p^2} = \frac{p-1}{p+1} \ge \frac{1}{3}, \tag{51.22}$$

and $H = K^{10}$ absorbs the boundary for combined modulus at most Ck^2. If $p > Ck$, direct counting of the discarded pairs with $p \mid u$ or $p \mid v$ costs $O(1/(pk))$ against main term $\asymp \varphi(k)/k^2$, a ratio $O(k/(p\varphi(k))) = O(1/C)$; endpoint errors are absorbed as before. Choose C large. Increasing the fixed Rankin cutoff makes the high-ω tail smaller than any prescribed fixed fraction, and the high-incidence triples have $o(1)$ of the original mass by Lemma 9.3. Both deletions therefore leave the same order of mass. □

Theorem 14.6 (effectivity of the repaired chains; proved internally, record-adjacent and therefore CLAIMED/PROVISIONAL)**.** *The constants in the source class-mass conclusions (16.8) and (34.18)–(34.19), in source Theorems 16.4 and 39.7 (paper Theorems 9.1 and 10.7), and in Theorem 14.2 may all be taken effectively computable. The record statement retains every correctness and priority qualification already attached to Theorem 10.7.*

Proof. In each dyadic prime block apply Lemma 14.4; if its exceptional conductor exists, retain only triples with $r_x \nmid 4uv$. Lemma 14.5 preserves the full order of box mass, uniformly in every subfamily and fibre. This retained subfamily proves the original lower class-mass conclusion. All downstream choices in the larger sieve, exact CRT moments, Chernoff estimate, Bonferroni polynomial, integer class counting, and Rankin semigroup transfer are finite and effective. □

The perimeter is deliberately narrow. The first sentence of Theorem 9.4 also contains the source Theorem 34.8 ancillary $o(1)$ bound for the absolute progression-error sum over *all* low-congestion triples. Excluded-conductor deletion effectivizes

the class supply (34.18)–(34.19), not that all-triples $o(1)$ as printed. The exceptional-set chain uses the class supply only. Thus Theorem 14.6 is not an effectivity upgrade of every clause of Theorem 9.4.

14.3. **Pomerance–Weingartner, priority, and blind adjudication.** Pomerance–Weingartner [20, §4] prove for their $k = 1$ forced classes

$$\sum_{\ell \le X} \frac{f(\ell)}{\ell} \asymp (\log X)^2. \tag{51.24}$$

Stopping their larger-sieve argument at prime modulus T gives

$$\#\{p \le N : W_{\mathrm{PW}}(p) > T\} \ll N \exp\{-c(\log T)^2\}, \qquad \log N \ge C(\log T)^3, \tag{51.25}$$

where W_{PW} measures that subfamily. It is also a weaker tail for W. Their lower class mass explicitly uses Bombieri–Vinogradov after a polylogarithmic multiplicity bound, their upper mass uses Brun–Titchmarsh, and their N-side is the elementary larger sieve and Rankin truncation. Thus PW already contains the square-log truncated tail; the campaign-derived one-conductor deletion makes the fixed-$m = 4$ supply effective, but PW do not print that repair, and no uniform varying-m effectivity is claimed when the exceptional conductor divides m. The resulting tail exponents are $(\log T)^2$ for the proved PW $k = 1$ truncation, $(\log T)^2 \log\log T$ for the unconditional fixed-polylog multipliers with the Section 10 review qualification, and $(\log T)^3$ for the CLAIMED/PROVISIONAL power multipliers. Vaughan's 1970 paper remains inaccessible to this campaign; this is a comparison with PW's reconstruction, not a direct audit of Vaughan. Searches found no paper explicitly claiming an effective Vaughan-strength constant or a Siegel-free proof, but that is a search result, not a literature guarantee. PW Theorem 1.3 says only "There is an absolute positive constant C," not that it or the threshold is computable; their Section 4 says "We now use the Bombieri–Vinogradov theorem." They do not perform the effectivity repair. Dahan's statistic and cubic-shape priority are distinguished immediately after (51.18).

The attested-blind protocol used branch `blind51`; base `4dcf3b7e1e10e31f529f5e2862a2aa23c0820bb6`; freeze `7ecd7d514ae379e9c90d6a7b292d11e80840b360`; and artifact `blind51.md`, SHA-256 `b0a4a27dba52fb563b4667427f2200cc98e670c27e8fff390aa846b1039c8b7c`. Read isolation is self-attested. Adjudication was CONVERGED-WITH-DIVERGENCES: both tails, windows, the Section 10 replay, conductor repair, (39.35), and PW truncation converged, but the blind side overclaimed effectivity of the ancillary all-triples $o(1)$. The paper's narrower perimeter is correct; no label changed.

15. Per-slice vanishing frequency and the empty third layer

The moving genus obstruction is the last fixed-congruence obstruction for one fixed slice. On every reduced hard class where the genus permits positivity, an upper-bound sieve makes residual vanishing relatively sparse. This is a proved, nonprovisional per-slice almost-all result, not a pointwise bound or a stack over growing slices. Here "hard" still means prime and $1 \pmod{24}$.

15.1. **The fixed-slice sieve.** Fix positive c, k and put

$$c = st^2, \qquad s = \mathrm{sf}(c), \qquad h = 4ck, \qquad Q = \mathrm{lcm}(24, h). \tag{52.1}$$

Theorem 15.1 (per-slice sieve upper bound; proved)**.** *Suppose $s \notin \{1,2,3,6\}$. Fix a reduced class $a \pmod Q$ such that*

$$a \equiv 1 \pmod{24}, \qquad (a,h)=1, \qquad \chi_s(a) = -1. \tag{52.2}$$

Then

$$\#\{p \le X : p \text{ prime},\ p \equiv a \pmod Q,\ M_{c,k}(p) = 0\} \ll_{c,k} \frac{X}{(\log X)^{1+2/\varphi(h)}}. \tag{52.3}$$

The finitely many p for which $(c,k) \notin \mathcal{B}_p$ are included in the implied constant, which is uniform over the finitely many classes $a \pmod Q$ satisfying (52.2).

Proof. The contrapositive of Theorem 13.1, followed by the genus calculation for every good $q \nmid Q$, is

$$M_{c,k}(p) = 0 \Longrightarrow p^2 + 4ck^2 \text{ has no prime divisor } q \equiv -a \pmod h,$$
$$\chi_s(q) = \chi_s(-a) = 1, \qquad \left(\frac{-4ck^2}{q}\right) = \left(\frac{\Delta_s}{q}\right) = 1, \tag{52.4–52.7}$$
$$n^2 \equiv -4ck^2 \pmod q.$$

The last congruence has exactly two distinct nonzero roots because $4, k^2, t^2$ are nonzero squares modulo q.

Sieve $\mathcal{A} = \{n \le X : n \equiv a \pmod Q\}$ by removing zero at every $\ell \nmid Q$ and the two roots at every good q. For squarefree d prime to Q, CRT and $\rho(\ell) \le 3$ give

$$\#\mathcal{A}_d = \frac{X\rho(d)}{Qd} + O(\rho(d)), \qquad \sum_{d \le D} \mu^2(d)\rho(d) \ll D(\log D)^2. \tag{52.8–52.10}$$

The exact dimension is $\kappa = 1 + 2/\varphi(h) < 2$. With fixed sufficiently large $u = u(\kappa)$, $D = X^{1/2}$, and $z = D^{1/u} = X^{\eta}$, the upper fundamental lemma gives

$$S(\mathcal{A}, z) \ll_\kappa \frac{X}{Q} V(z) + \sum_{d \le D} \mu^2(d)|r_d|. \tag{52.12}$$

This avoids asserting that the endpoint sieve parameter 2 works for every fractional dimension. The local product is

$$V(z) = \prod_{\substack{\ell \le z \\ \ell \nmid Q}} (1 - 1/\ell) \prod_{\substack{q \le z \\ q \equiv -a \ (h) \\ q \nmid Q}} \frac{1 - 3/q}{1 - 1/q}. \tag{52.13}$$

Mertens in the fixed ray says $\sum_{q \le z,\ q \equiv -a\ (h)} q^{-1} = \varphi(h)^{-1} \log\log z + C(h,-a) + o(1)$, whence $V(z) \asymp_{c,k,a} (\log z)^{-1-2/\varphi(h)}$. The elementary remainder is $O(X^{1/2}(\log X)^2)$ and the $O(z)$ small primes are negligible, proving (52.3). □

The only analytic inputs are the upper-bound fundamental lemma, ordinary Mertens, and Mertens in one fixed reduced progression. No Bombieri–Vinogradov, Siegel–Walfisz, or prime level of distribution is hidden: primality is majorized by the zero-class sieve. For explicitly fixed (c,k,a) these constants are effective in principle.

Corollary 15.2 (the congruence-level third layer is empty; proved)**.** *Under Theorem 15.1, let M' be any fixed multiple of Q and let $a' \pmod{M'}$ be a reduced refinement of $a \pmod Q$. The set*

$$\{p : p \equiv a' \pmod{M'},\ M_{c,k}(p) = 0\} \tag{52.16}$$

does not contain all sufficiently large primes in that progression. Consequently no prime-bearing arithmetic progression inside an unforced slice-class is an identically vanishing progression.

Proof. The fixed-progression prime number theorem gives

$$\#\{p \le X : p \equiv a' \pmod{M'}\} \sim \frac{X}{\varphi(M')\log X}. \tag{52.17}$$

The vanishing set in the refinement is bounded by (52.3), and the extra factor $(\log X)^{-2/\varphi(h)}$ tends to zero. □

Thus the universal cores $1, 2, 3, 6$ and the moving genus half in Theorem 12.1 are the complete congruence-level vanishing laws for a fixed slice. The complementary half has no identically vanishing reduced subprogression. This does not exclude infinitely many residual vanishers or a positive slice guaranteed by a fixed divisor.

Proposition 15.3 (polylogarithmic uniformity with an epsilon loss; proved, ineffective)**.** *Fix $\theta > 0$ and $0 < \epsilon < 1$. Uniformly for slices and reduced classes satisfying (52.1)–(52.2) and*

$$h \le (\log X)^{\theta}, \tag{52.18}$$

one has

$$\#\{p \le X : p \equiv a \pmod{Q},\ M_{c,k}(p) = 0\} \ll_{\theta,\epsilon} \frac{X}{\varphi(Q)(\log X)^{1+2(1-\epsilon)/\varphi(h)}}. \tag{52.19}$$

The implied constant and threshold are ineffective. No uniformity for arbitrary $h = X^{o(1)}$ is asserted.

Proof. Retain the two extra root classes only for good primes in

$$y = \exp\{(\log X)^{\epsilon}\} < q \le z = X^{\eta}. \tag{52.20}$$

For $q \ge y$, the hypothesis gives $h \le (\log q)^{\theta/\epsilon}$, so Siegel–Walfisz with this fixed parameter and partial summation yield uniformly

$$\sum_{\substack{y<q\le z\\ q\equiv -a\ (h)}} \frac{1}{q} = \frac{(1-\epsilon)\log\log X + O(1)}{\varphi(h)} + o(1). \tag{52.21}$$

Ordinary Mertens with primes dividing Q omitted contributes $\ll Q/(\varphi(Q)\log z)$. The coarse uniform dimension-three upper sieve and the elementary remainder $O(X^{1/2}(\log X)^2)$ now give (52.19). The usual all-moduli Siegel–Walfisz constants are ineffective, which is exactly the label in the statement. □

15.2. **Finite residual structure and its limits.** The memory-bounded verifier block `(ay)` factors one norm at a time and checks all 31 unforced slices with $ck \le 30$ over the 385 hard primes below 30000; the optional full scan uses all 1181 below 10^5. The residual frequencies broadly track the fixed-slice log-power scale, but this is finite calibration only.

The 74 strict gaps $ck_{\mathrm{pr}} > ck_{\min}$ in the first range contain 98 canonical target divisors at the minimal positive product; the full range has 221 events and 308 divisors. Individual divisor shapes are $68, 24, 6$ below 30000 and $199, 93, 16$ below 10^5 for respectively two distinct primes, at least three prime factors, and a prime power. Thus the majority mechanism is a product of two individually wrong-grade

primes, but products of three or more factors and exponent caps are genuine. For example $D = 155 = 5 \cdot 31$ occurs at $(p, c, k) = (241, 11, 1)$, $D = 75 = 3 \cdot 5^2$ at $p = 409$ on that slice, and $D = 27 = 3^3$ at $(p, c, k) = (5953, 5, 1)$.

After conditioning on both genus signs being negative, define

$$R(A, B) = \frac{\Pr(M_A = 0, M_B = 0)}{\Pr(M_A = 0)\Pr(M_B = 0)}. \tag{52.26}$$

Most tested pairs are near one, but stable anticorrelations remain: $(5, 1)$ with $(5, 2)$ gives $.529, .577$ and $(5, 1)$ with $(10, 1)$ gives $.714, .699$ in the two ranges. Hence approximate independence is a useful first calibration for many pairs, not a blanket theorem. No asymptotic constant, simultaneous-factorization law, lower bound for residual vanishing, or pointwise conclusion follows. In particular, the sieve is upper-bound only and does not cross the prime-pair parity barrier.

16. The stacked slice sieve and almost-all H_{SPF}

Distinct slice labels can define the same norm: $(5, 2)$ and $(20, 1)$ both have $ck^2 = 20$. Their root classes then coincide whenever both ray conditions accept the prime. Quotienting duplicate radicands gives the correct fixed-box exponent. A canonical squarefree-core subfamily has no duplicates and proves, for every fixed $A < 1$, the effective headline

$$\#\mathcal{F}_A(X) \ll_A \frac{X}{\log X} \exp\left\{-c_A \frac{(\log\log X)^{3/2}}{\sqrt{\log\log\log X}}\right\}, \tag{53.1}$$

where $\mathcal{F}_A(X)$ is the set of hard primes at which $H_{\mathrm{SPF}}(A)$ fails. It is an almost-all theorem only. The square-root loss relative to the provisional Type-II cubic tail is caused by the Type-I genus-permission gate.

16.1. **Duplicate-radicand geometry and the fixed stack.** For fixed $T \geq 1$, put

$$Q_T = \operatorname{lcm}(24, \{4ck : ck \leq T\}). \tag{53.2}$$

For integral $T \geq 6$, $Q_T = 4\operatorname{lcm}(1, \dots, T)$ and Chebyshev's bounds give $\log Q_T = \psi(T) + O(1) = O(T)$. Let $C \pmod{Q_T}$ be reduced and $C \equiv 1 \pmod{24}$. Call (c, k) with $ck \leq T$ active at C when $\chi_{\mathrm{sf}(c)}(C) = -1$. The raw sum

$$m_{\mathrm{raw}}(C) = \sum_{\substack{ck \leq T \\ (c,k) \text{ active at } C}} \frac{2}{\varphi(4ck)} \tag{53.3}$$

is not the local-density mass. Put $d = ck^2$ and define

$$\mathcal{H}_d(C) = \{4ck : ck \leq T,\ ck^2 = d, (c, k) \text{ active at } C\}. \tag{53.4}$$

For reduced $b \pmod{Q_T}$ set

$$\begin{aligned} J_C(b) &= \#\{d : \exists h \in \mathcal{H}_d(C),\ b \equiv -C \pmod{h}\}, \\ m_{\#}(C) &= \frac{2}{\varphi(Q_T)} \sum_{b \in (\mathbb{Z}/Q_T\mathbb{Z})^\times} J_C(b). \end{aligned} \tag{53.5}$$

A radicand is thereby counted once whenever at least one of its rays accepts the prime.

Lemma 16.1 (root overlap; proved)**.** *Put $d_i = c_i k_i^2$. If a root class of*

$$x^2 \equiv -4d_1 \pmod{q} \tag{53.6}$$

coincides with a root class for d_2, then

$$q \mid 4(d_1 - d_2). \tag{53.7}$$

Consequently, if $d_1 \neq d_2$, all root pairs belonging to slices $c_i k_i \leq T$ are disjoint at every odd prime $q > T^2$, and no root is zero there. If $d_1 = d_2$, the root sets coincide identically.

Proof. Square a common root and subtract. At odd q this gives $q \mid d_1 - d_2$. Since $d_i = (c_i k_i)k_i \leq T^2$, a nonzero difference has absolute value less than T^2. A zero root similarly forces $q \mid d_i$; equality of radicands gives identical congruences. □

For $(5, 2)$ and $(20, 1)$ the moduli are 40 and 80. On an active class the modulus-80 ray lies inside the modulus-40 ray, so the corrected contribution is $2/\varphi(40) = 1/8$, not the raw $3/16$.

Let $E_C(X, z)$ count primes $p \leq X$, $p \equiv C \pmod{Q_T}$, for which there is no prime $q \leq z$ and active (c, k), $ck \leq T$, with

$$q \mid p^2 + 4ck^2, \qquad q \equiv -C \pmod{4ck}. \tag{53.8}$$

Theorem 16.2 (fixed-box stacked sieve; proved, effective)**.** *For fixed T there is an effective $\eta_T > 0$ such that, uniformly over the finitely many reduced hard classes $C \pmod{Q_T}$,*

$$E_C(X, X^{\eta_T}) \ll_T \frac{X}{(\log X)^{1+m_\#(C)}}. \tag{53.9}$$

The same estimate holds for primes having no good prime factor of any size in the box.

Proof. Take $T_0 > \max(Q_T, T^2)$ and sieve $\mathcal{A}_C = \{n \leq X : n \equiv C \pmod{Q_T}\}$. Remove zero at ordinary primes and accepted root pairs at $q > T_0$. The genus calculation supplies two roots, and Lemma 16.1 makes distinct radicands disjoint, so

$$\rho_C(q) = 1 + 2J_C(q \bmod Q_T), \qquad \#(\mathcal{A}_C)_d = \frac{X\rho_C(d)}{Q_T d} + O_T(\rho_C(d)) \tag{53.11–53.12}$$

for squarefree d prime to Q_T, with multiplicative ρ_C. Effective fixed-class Mertens gives

$$\sum_{T_0 < q \leq z} \frac{2J_C(q \bmod Q_T)}{q} = m_\#(C) \log\log z + O_T(1), \qquad V_C(z) \asymp_T (\log z)^{-1-m_\#(C)}. \tag{53.13–53.14}$$

Apply Friedlander–Iwaniec [12, Theorem 6.9] with $D = X^{1/2}$ and $z = D^{1/u_T} = X^{\eta_T}$ for sufficiently large fixed u_T:

$$S(\mathcal{A}_C, z) \ll_T \frac{X}{Q_T} V_C(z) + \sum_{d \leq D} \mu^2(d) O_T(\rho_C(d)). \tag{53.15}$$

Since $\rho_C(\ell) = O_T(1)$, the remainder and small primes are negligible. □

For one active slice, $m_\# = 2/\varphi(4ck)$, so Theorem 15.1 is exactly the one-slice specialization. If a class has no active slice, $m_\# = 0$ and only the ordinary one-dimensional bound remains. Such escape classes exist for every fixed T: $C = 1 \pmod{Q_T}$ is one, and Theorem 13.5 supplies infinitely many corresponding primes.

16.2. **Exact genus bits and deterministic permission mass.** The full raw box has the record-only Selberg–Delange asymptotic

$$\mathfrak{M}(T) = \sum_{ck \le T} \frac{2}{\varphi(4ck)} = \sum_{n \le T} \frac{2\tau(n)}{\varphi(4n)} \sim C_{\mathrm{sl}} (\log T)^2,$$
$$C_{\mathrm{sl}} = \frac{1}{2} \prod_{p \text{ odd}} (1-p^{-1})^2 \left(1 + \frac{p(2p-1)}{(p-1)^3}\right) > 0. \tag{53.16–53.17}$$

Only the effective bound $\mathfrak{M}(T) \ll (1+\log T)^2$ is used below. Retain

$$\mathcal{S}_T(C) = \{(s,k) : s \text{ squarefree},\ sk \le T,\ \chi_s(C) = -1\}, \qquad m_0(C) = \sum_{(s,k)\in\mathcal{S}_T(C)} \frac{2}{\varphi(4sk)}. \tag{53.19–53.20}$$

Their radicands sk^2 are distinct. Quadratic reciprocity gives exactly

$$\chi_s(C) = \prod_{\substack{q \mid s \\ q \ge 5}} \left(\frac{C}{q}\right). \tag{53.21}$$

As C ranges uniformly over reduced hard classes modulo Q_T, the bits (C/q) for distinct primes $5 \le q \le T$ are independent uniform signs. This is a finite Chinese-remainder theorem identity, not a heuristic: the $-1, 2, 3$ factors are pinned by $C \equiv 1 \pmod{24}$, and each odd prime-power coordinate is independent.

Lemma 16.3 (permission-pair mass; proved)**.** *Suppose $5 \le q \le T/2$ is prime and $\chi_q(C) = -1$. Then*

$$m_0(C) \gg \frac{\log^2(T/q)}{q}. \tag{53.22}$$

The constant is absolute. The same estimate, with another absolute constant, survives deleting every slice whose modulus $4sk$ is divisible by one prescribed primitive real-character conductor r, provided $q \nmid r$.

Proof. Pair each squarefree $u \le T/q$, $(u,q) = 1$, with qu. Equation (53.21) makes exactly one core active. For $k \le T/(qu)$ its weight is at least $1/(2quk)$, and Möbius summation gives

$$m_0(C) \gg \frac{1}{q} \sum_{\substack{u \le T/q \\ u \text{ squarefree} \\ (u,q)=1}} \frac{1}{u} \log \frac{T/q}{u} \gg \frac{\log^2(T/q)}{q}. \tag{53.24}$$

If an exceptional conductor has odd prime factor ℓ, restrict to $\ell \nmid uk$; if it is 8, restrict the relevant cores and k to odd integers. These restrictions retain fixed Euler factors and ensure $r \nmid 4sk$. Conductor 4 is effectively controlled directly. □

Using only core q would provide one logarithm. Pairing u with qu provides the second regardless of every later genus bit.

16.3. **The effective growing stack.** Fix $0 < A < 1$ and put

$$T = (\log X)^A, \qquad L = \log\log X, \qquad K_T = C_0(1+\log^2 T),$$
$$D = X^{1/3}, \qquad z = \exp\left\{\frac{\log D}{BK_T}\right\}. \tag{53.25}$$

where C_0, B are sufficiently large absolute constants. Then

$$\log z \asymp \frac{\log X}{L^2}, \qquad \log\log z = L + O(\log L), \qquad \log\log Q_T = AL + O(1), \tag{53.26}$$

and $Q_T < z$ for every fixed $A < 1$ once X is large.

Theorem 16.4 (growing stacked slice sieve; proved, effective)**.** *For every fixed $0 < A < 1$ there are effectively computable $c_A > 0$ and X_A such that the following holds for $X \ge X_A$. Let $\mathcal{E}_A(X)$ be the hard primes $p \le X$ for which there do not exist an active (c,k) with $ck \le (\log X)^A$ and a prime $q \le z$ satisfying $q \mid p^2 + 4ck^2$ and $q \equiv -p \pmod{4ck}$. Then*

$$\#\mathcal{E}_A(X) \ll_A \frac{X}{\log X} \exp\left\{-c_A \frac{L^{3/2}}{\sqrt{\log L}}\right\}. \tag{53.27}$$

The proof uses only the squarefree-c subfamily, so the assertion for all slices follows a fortiori.

Proof. Partition into the $\varphi(Q_T)/\varphi(24)$ reduced hard classes modulo $Q_T = \exp\{O(T)\} = X^{o(1)}$. For ray moduli $4sk \le 4T$, effective Landau–Page-repaired Siegel–Walfisz leaves at most one primitive real conductor $r = r_X$, $|r| \le 4T$. If it exists retain only $r \nmid 4sk$; otherwise delete nothing. Uniformly for retained moduli and $x \ge Q_T$, the effective explicit formula is $\psi(x;h,a) = x/\varphi(h) + O(xe^{-c\sqrt{\log x}})$.

Write $m_r(C)$ for the retained mass and start root primes above Q_T. Distinct canonical radicands and partial summation give, uniformly in C,

$$\sum_{\substack{Q_T < q \le z \\ q \equiv -C \ (4sk)}} q^{-1} = \frac{\log\log z - \log\log Q_T}{\varphi(4sk)} + o(1/\varphi(4sk)).$$

The errors over all $O(T\log T)$ labels and the quadratic local terms are $o(1)$. Hence ordinary roughness and the added roots give

$$V_C(z) \ll \frac{Q_T}{\varphi(Q_T)\log z} e^{-m_r(C)(\log\log z - \log\log Q_T) + o(1)} \ll_A \frac{L^{O(1)}}{\log X} e^{-c_A m_r(C) L}. \tag{53.30}$$

The load-bearing uniform-in-dimension input is Friedlander–Iwaniec [12, Theorem 6.9]. Under

$$\prod_{w \le \ell < y} (1 - g_C(\ell))^{-1} \le K_0 \left(\frac{\log y}{\log w}\right)^{\kappa_C}, \tag{53.31}$$

if $s = \log D/\log z > 9\kappa_C + 1$, its main multiplier is $1 + O(e^{9\kappa_C - s} K_0^{10})$ with an absolute constant. Here the interval prime estimates give absolute K_0, $\kappa_C \ll 1 + m_r(C) \le K_T$, and $s = BK_T$; choose C_0 and then $B > 9$ with room. The class remainders, crudely dominated with $R_T = O(T\log T)$, and the resulting class average are

$$\begin{aligned} Q_T \sum_{d \le D} \mu^2(d) O(\rho_C(d)) &\le X^{1/3 + o(1)}, \\ \#\mathcal{E}(X) &\ll_A \frac{XL^{O(1)}}{\log X} \frac{1}{\#\mathcal{C}_T} \sum_{C \in \mathcal{C}_T} e^{-c_A m_r(C) L}. \end{aligned} \tag{53.32–53.33}$$

This sieves integers in each class; no prime number theorem modulo growing Q_T is used.

Set $y = L^{3/2}\sqrt{\log L}$ and ignore the $O(L)$ primes dividing r. Exact genus-bit independence handles delayed permission, while Lemma 16.3 handles every other class:

$$\begin{aligned} &2^{-\pi(y)+O(L)} \le e^{-cy/\log y} \le e^{-cL^{3/2}/\sqrt{\log L}}, \\ &m_r(C) \gg_A L^2/y, \qquad e^{-c_A m_r(C) L} \le e^{-c_A L^3/y} = e^{-c_A L^{3/2}/\sqrt{\log L}}. \end{aligned} \tag{53.35–53.36}$$

Absorb $L^{O(1)}$ in the exponent. Every threshold after the one-conductor deletion is effective. □

The balance is exact: delaying the first negative bit beyond y costs $\exp\{-cy/\log y\}$, while permission by y costs $\exp\{-c_A L^3/y\}$. The endpoint $A = 1$ is excluded both because $Q_T = X^{o(1)}$ is then lost and because the remaining Mertens length $(1-A)L + O(\log L)$ vanishes.

Corollary 16.5 (almost-all H_{SPF}; proved, effective)**.** *For each fixed $0 < A < 1$, if $\mathcal{F}_A(X)$ is the set of hard primes $p \le X$ at which the event (50.7) fails, then*

$$\#\mathcal{F}_A(X) \ll_A \frac{X}{\log X} \exp\left\{-c_A \frac{(\log\log X)^{3/2}}{\sqrt{\log\log\log X}}\right\}. \tag{53.37}$$

Proof. A failure of (50.7) has no good prime of any size. Fix $A' < A$. On $[X/2, X]$, eventually $(\log X)^{A'} \le (\log p)^A$, so each failure is counted by Theorem 16.4 with A'. Summation over dyadic intervals gives (53.37), with constants renamed in terms of A. □

16.4. **The genus-permission asymmetry.** The honest comparison is: Type I has failure exponent $(\log\log X)^{3/2}/\sqrt{\log\log\log X}$ and requires independent genus bits; fixed-polylog Type II has exponent $(\log\log X)^2 \log\log\log X$ and no permission gate; cubic Type II has exponent $(\log\log X)^3$, no gate, and remains CLAIMED/PROVISIONAL. Theorem 13.5's escape classes are exactly the fixed-box extreme of the Type-I row. The Type-II stack reaches cubic scale because it has no permission gate; Type I pays the square root while waiting for the first negative genus bit. Dahan [4, Theorem 4.17] obtains the cubic shape for his different stacked-shift two-parameter (u, a) depth, again without a genus gate, and labels its error ineffective. His statistic and the Type-I statistic here are not interchangeable; the Landau–Page deletion makes (53.27) effective.

Corollary 16.5 does not prove the pointwise hypothesis, bound $D(p)$, remove fixed-box escape progressions, or cross the parity barrier.

17. The lower tail and the two-sided frontier

The upper tails above leave exceptional sets. Residue-one progressions, combined with effective Linnik, prove that those sets are genuinely nonempty infinitely often below logarithmic exponent one. These are escape theorems, not nonsolvability theorems.

17.1. **Type II: the minimal multiplier modulus.** For real $T \ge 3$, put

$$M(T) = \mathrm{lcm}(1, 2, \dots, \lfloor T \rfloor), \qquad \log M(T) = \psi(\lfloor T \rfloor). \tag{54.1}$$

The full least common multiple is needed because (51.1) ranges over every product $k\ell$, with no primality condition on ℓ.

Theorem 17.1 (Type-II logarithmic lower tail; proved, effective; source Theorem 54.1)**.** *Every prime $p \equiv 1 \pmod{M(T)}$ satisfies*

$$W(p) > T. \tag{54.2}$$

There is an effectively computable absolute constant C_0 such that, for all sufficiently large T, one can choose such a prime with

$$p \le C_0 M(T)^{5.2} = \exp\{(5.2 + o(1))T\}. \tag{54.3}$$

Consequently these primes are unbounded and

$$\limsup_{\substack{p\to\infty \\ p \text{ prime}}} \frac{W(p)}{\log p} \ge \frac{1}{5.2} = 0.192307\ldots > 0.19. \tag{54.4}$$

Proof. Suppose data in (51.2) have $m = k\ell \le T$. Then $m \equiv 3 \pmod 4$, $uvw = (m+1)/4$, and $p \equiv 1 \pmod m$. If they witnessed p, multiplying $1 \equiv -uv^{-1} \pmod m$ by v would give $m \mid u + v$. But

$$2 \le u + v \le uv + 1 \le \frac{m+1}{4} + 1 < m;$$

this includes $m = 3$, where the upper bound is $2 < 3$. Thus (54.2) holds. The class $1 \pmod{M(T)}$ is reduced. Xylouris's effective Linnik theorem [26] gives its least prime at most $C_0 M(T)^{5.2}$, and the effective prime number theorem gives $\psi(T) = T + o(T)$. Any prime in the class is at least $M(T) + 1$, so primes selected along unbounded T are unbounded. Equations (54.2)–(54.3) then give (54.4). □

Corollary 17.2 (pointwise Type-II frontier; proved; source Corollary 54.2)**.** *For every $0 < A < 1$, $H_{\mathrm{MOD}}(A)$ is false. Thus, in its defined positive-exponent range, exactly $A \ge 1$ remains open.*

Proof. Along Theorem 17.1's primes, $W(p) > (1/(5.2 + o(1))) \log p > (\log p)^A$. □

If

$$E_A(N) = \{p \le N : p \text{ prime},\ W(p) > (\log N)^A\},$$

then Corollary 14.3 and Theorem 17.1 give the two-sided summary

$$\begin{array}{ll} \#E_A(N) = o(\pi(N)) & (A > 0), \\ \#E_A(N) \ge 1 \text{ for infinitely many } N & (0 < A < 1). \end{array} \tag{54.5}$$

The first line retains the internal-review qualification of (51.17), and the additional cubic upper bound (51.18) remains CLAIMED/PROVISIONAL. For fixed T, the exhibited progression has relative Dirichlet density $1/\varphi(M(T)) = \exp\{-(1+o(1))T\}$ among the primes. Thus the lower side asserts only nonemptiness infinitely often; the true size of $E_A(N)$ is open.

17.2. **Type I: simultaneous genus forcing.** Put

$$R(T) = \operatorname{lcm}\left(24,\ \prod_{\substack{\ell \le T \\ \ell \text{ prime}}} \ell\right), \qquad \log R(T) = \vartheta(T) + \log 4. \tag{54.6}$$

The identity holds for $T \ge 3$.

Theorem 17.3 (Type-I logarithmic lower tail; proved, effective; source Theorem 54.3)**.** *For every $T \geq 3$ and every prime $p \equiv 1 \pmod{R(T)}$,*

$$\chi_s(p) = 1 \quad (1 \leq s \leq T,\ s \text{ squarefree}), \qquad ck_{\mathrm{pr}}(p) > T, \qquad ck_{\min}(p) > T. \tag{54.7}$$

In fact $M_{c,k}(p) = 0$ for every admissible slice with $ck \leq T$. There are unboundedly many such primes satisfying

$$p \leq C_0 R(T)^{5.2} = \exp\{(5.2 + o(1))T\}, \qquad \limsup_{\substack{p \to \infty \\ p \equiv 1 \ (24)}} \frac{ck_{\min}(p)}{\log p} \geq \frac{1}{5.2} > 0.19. \tag{54.8}$$

The convention $ck_{\min}(p) = +\infty$ is allowed.

Proof. For every odd prime $\ell \leq T$, quadratic reciprocity and $p \equiv 1 \pmod 4$ give

$$\left(\frac{\ell}{p}\right) = \left(\frac{p}{\ell}\right) = 1. \tag{54.9}$$

The congruence modulo 24 also gives $(2/p) = (-1/p) = (3/p) = 1$. The sign and factor 4 in the fundamental discriminant Δ_s therefore have symbol one at p; multiplicativity proves $\chi_s(p) = 1$ for every squarefree $s \leq T$. If $ck \leq T$, then $s = \mathrm{sf}(c) \leq T$, so (50.3) is empty and Theorem 12.1 gives $M_{c,k}(p) = 0$.

All positive pairs $ck \leq T$ are admissible. Bertrand's postulate, with $3 \leq T < 7$ checked directly, gives $R(T) > 4T$, hence $p > 4T$; this supplies the coprimality and size conditions in $\mathcal{B}_p$. Apply effective Linnik to $1 \pmod{R(T)}$ and use $\vartheta(T) = T + o(T)$. Since every prime in the class is at least $R(T) + 1$, the argument for (54.4) proves (54.8). □

Corollary 17.4 (pointwise Type-I frontier; proved; source Corollary 54.4)**.** *For every $0 < A < 1$, $H_{\mathrm{SPF}}(A)$ is false; exactly $A \geq 1$ remains open. More strongly, infinitely many hard primes have no positive slice at all with*

$$ck \leq (1/(5.2 + o(1))) \log p. \tag{54.10}$$

For each $0 < A < 1$, these primes make the failure set in Corollary 16.5 nonempty at infinitely many heights, while (53.37) gives its effective almost-all upper bound. This quantifies Theorem 13.5's genus-forced zeros; it is strictly stronger than the fixed-divisor non-guarantee in Theorem 12.3.

The classical effectivity register is exact. Xylouris [26] proves the usable exponent 5.2, refining Heath–Brown [14]; the implied constant and threshold are effectively computable, with no ineffective Siegel lower bound. Rosser–Schoenfeld [21] give the convenient explicit bounds $\psi(x) < 1.03883x$ and $\vartheta(x) < 1.01624x$. The sharper $(1 + o(1))x$ used above comes from the classical effective prime number theorem. No numerical value of C_0 is extracted here.

17.3. **The two-sided frontier.** The census puts much of the observed mass at small constant depths, while (51.17) says that for every fixed $\varepsilon > 0$, all but $o(\pi(N))$ primes have $W(p) \leq (\log p)^{\varepsilon}$. Theorem 17.1 supplies the effective opposite extreme. The resulting proved sandwich is

$$\begin{aligned} &\text{typical:} && W(p) \leq (\log p)^{\varepsilon} \quad \text{for almost all primes, every fixed } \varepsilon > 0, \\ &\text{extremal:} && W(p) \geq c \log p \quad \text{for infinitely many primes, effectively.} \end{aligned} \tag{54.11}$$

The finite W census, Dahan's different two-parameter depth statistic [4], and the different statistic measured in the auro-zera file suggest logarithmic, logarithm–logarithm, and logarithm-squared normalizations worth plotting, but the statistics

are not interchangeable and no finite envelope supplies a growth law. The exact pointwise frontier is only

$$\boxed{\underbrace{A<1}_{\text{proved impossible pointwise}} \quad \Big| \quad \underbrace{A\geq 1}_{\text{still open pointwise}} \quad .} \tag{54.12}$$

The same proved lower edge holds for H_{SPF}. The extended census and its deliberately separate finite-search panel appear in (56.16); in particular, its symbol $\lesssim 2$ is not a proved upper bound or a conjectured limiting exponent. Covering every $m \leq T$ by residue one forces divisibility by $M(T)$, whose logarithm is $T + o(T)$, so the full-lcm escape route is intrinsically logarithmic. Theorems 18.1–18.2 sharpen this observation: logarithmic order is optimal inside the complete-class-plus-generic-least-prime architecture, with a factor-two gap in the proved Type-II modulus constant and exact saturation in Type I. They do not close any part of the open pointwise range in (54.12). None of these lower-tail primes is asserted to be an Erdős–Straus exception.

18. The congruence-certificate ceiling and extremal census

This section closes the elementary two-sided certificate question left by Sections 14 and 17. A complete congruence class certifying escape through depth T must itself have exponentially large modulus. The Type-II statement is the prime-progression extension of the all-integer cylinder wall in Theorem 8.2, not a new independent supply theorem. The Type-I statement uses Corollary 15.2 to turn every unforced prime core into a required conductor coordinate. The theorems are asymptotic and effective. The censuses below are exact only in their stated ranges and imply no growth law.

18.1. **Type II: every certificate carries the** 3 (mod 4) **primes.** Call $a \pmod Q$ a *prime congruence certificate for* $W > T$ if $(a, Q) = 1$ and every sufficiently large prime $p \equiv a \pmod Q$ satisfies $W(p) > T$. An all-integer certificate has the analogous meaning with every sufficiently large positive integer in the class; it need not be reduced. Put

$$P_3(T) = \prod_{\substack{\ell \leq T \\ \ell \text{ prime},\ \ell \equiv 3 \ (4)}} \ell. \tag{56.1}$$

Theorem 18.1 (Type-II congruence-certificate ceiling; proved, effective; source Theorem 56.1)**.** *If $a \pmod Q$ is either a prime or an all-integer congruence certificate for $W > T$, then*

$$P_3(T) \mid Q, \qquad \log Q \geq \vartheta(T; 4, 3) = \left(\frac{1}{2} + o(1)\right) T. \tag{56.2}$$

Consequently, for every fixed $\epsilon > 0$, there is an effectively computable T_ϵ such that $Q \geq \exp\{(1/2 - \epsilon)T\}$ for $T \geq T_\epsilon$.

Proof. Fix a prime $\ell \leq T$, $\ell \equiv 3 \pmod 4$, and put $A = (\ell+1)/4$. The divisor $D = 1$ in Lemma 4.1 is not merely a formal intrinsic class: its Lemma 3.4 datum is

$$k = 1, \quad \ell = \ell, \quad (u, v, w) = (1, A, 1), \qquad -uv^{-1} \equiv -A^{-1} \equiv -4 \pmod \ell. \tag{56.3}$$

Thus every positive integer $n \equiv -4 \pmod \ell$ has $W(n) \leq \ell \leq T$.

Suppose $\ell \nmid Q$. The Chinese remainder theorem gives infinitely many positive integers satisfying both $n \equiv a \pmod{Q}$ and $n \equiv -4 \pmod{\ell}$, contradicting an all-integer certificate. In the prime version their joint class modulo $Q\ell$ is reduced: it is reduced modulo Q by hypothesis and $-4 \not\equiv 0 \pmod{\ell}$. Dirichlet's theorem therefore gives infinitely many primes in it, again contradicting the certificate. Hence every factor in (56.1) divides Q. The final estimate is the effective prime number theorem in the fixed progression $3 \pmod 4$. This is exactly the mechanism of Theorem 8.2, with the prime-bearing conclusion added. □

The theorem uses only one harvested class at each prime modulus. It does *not* prove that the full $\operatorname{lcm}(1,\dots,T)$ of Section 17 is necessary for an arbitrary certificate. Composite moduli can meet a progression in only one coset modulo (Q, M), and the full harvested set need not cover all such cosets. Thus replacing the constant $1/2$ in (56.2) by 1 would require an additional argument.

18.2. **Type I: every prime core is forced into the conductor.** A *hard Type-I congruence certificate through* T is a reduced class $a \pmod{Q}$ with

$$24 \mid Q, \qquad a \equiv 1 \pmod{24}, \tag{56.4}$$

such that every sufficiently large prime $p \equiv a \pmod{Q}$ has $ck_{\min}(p) > T$. Equivalently, every admissible slice with $ck \le T$ vanishes on every sufficiently large prime in the class.

Theorem 18.2 (Type-I congruence-certificate ceiling; proved, effective; source Theorem 56.2)**.** *Every hard Type-I congruence certificate through T satisfies*

$$\prod_{\substack{5 \le \ell \le T \\ \ell \text{ prime}}} \ell \mid Q, \qquad \log Q \ge \vartheta(T) - \log 6 = T + o(T). \tag{56.5}$$

In particular, for every $\epsilon > 0$, effectively for all sufficiently large T, one has $Q \ge \exp\{(1-\epsilon)T\}$.

Proof. Let $5 \le \ell \le T$ be prime and suppose $\ell \nmid Q$. The primitive quadratic character attached to the core $s = \ell$ is nonconstant modulo its prime coordinate. Concretely, choose a nonzero quadratic nonresidue $b \pmod{\ell}$, and use the Chinese remainder theorem to refine the certificate to

$$p \equiv a \pmod{Q}, \qquad p \equiv b \pmod{\ell}. \tag{56.6}$$

This is a reduced class modulo $Q\ell$, hence contains infinitely many primes. For every such hard prime, quadratic reciprocity and $p \equiv 1 \pmod 4$ give

$$\chi_\ell(p) = \left(\frac{\Delta_\ell}{p}\right) = \left(\frac{-\ell}{p}\right) = \left(\frac{p}{\ell}\right) = -1. \tag{56.7}$$

The slice $(c, k) = (\ell, 1)$ is therefore unforced and is admissible for all sufficiently large primes in (56.6). Since $Q\ell$ is a multiple of $\operatorname{lcm}(24, 4\ell)$, Corollary 15.2 says that this reduced refinement cannot be identically vanishing: arbitrarily far out it contains a prime with $M_{\ell,1}(p) > 0$. That prime has $ck_{\min}(p) \le \ell \le T$, contradicting the certificate. Therefore $\ell \mid Q$ for every prime $5 \le \ell \le T$. Taking logarithms and using the effective prime number theorem proves (56.5). □

The primitivity point in this proof is additive, not multiplicative: the progression modulo a conductor fills residue classes modulo every conductor coordinate omitted from Q. Equation (56.6) exhibits both character signs directly. No claim that a multiplicative subgroup generated by Q controls the progression is used.

Corollary 18.3 (congruence/Linnik recipe; proved within the stated recipe; source (56.8))**.** *Suppose a certificate construction at depth T uses $Q = \exp\{(\alpha + o(1))T\}$, and a least-prime theorem with exponent L selects a certified prime $p \le Q^{L+o(1)}$. Here "certified" includes that p lies beyond any finite exceptional prefix allowed by the certificate definition; a least-prime bound alone does not ensure this for an arbitrary eventual certificate. Then*

$$W(p) > T \quad \text{or} \quad ck_{\min}(p) > T, \qquad T \ge \frac{1+o(1)}{\alpha L}\log p. \tag{56.8}$$

Theorems 18.1 and 18.2 force respectively $\alpha \ge 1/2$ and $\alpha \ge 1$. Thus the coefficient available from this generic recipe is at most $2/L$ in Type II and at most $1/L$ in Type I. The residue-one choices of Theorems 17.1 and 17.3 have $\alpha = 1$ in both frames. They therefore saturate the Type-I certificate rate and are within a factor two of the proved Type-II ceiling; in both frames they establish the optimal logarithmic order for a certificate-plus-least-prime argument, up to the least-prime exponent. With the conservative effective Linnik value $L = 5.2$, their coefficient is $1/5.2$.

Remark 18.4 (GRH least-prime consequence; conditional; source (56.9))**.** *Conditional on the unproved Generalized Riemann Hypothesis, the standard least-prime bound permits $L = 2 + \epsilon$. The same residue-one constructions therefore give, respectively along unbounded prime sequences,*

$$W(p) > (1/2 - \epsilon)\log p, \qquad ck_{\min}(p) > (1/2 - \epsilon)\log p, \tag{56.9}$$

after harmlessly changing ϵ. This statement is conditional; (56.2) and (56.5) are unconditional and effective.

Assessment 56.2a (the congruence wall). The proved conclusion is that a *uniform complete-class* certificate through T has exponential modulus. Consequently, inside the Section 17 architecture—complete congruence certificate, modulus $\exp\{\Theta(T)\}$, then a generic least-prime theorem—a superlogarithmic lower bound would have to abandon at least one ingredient. The natural alternative is a noncongruence certificate on a sifted set of primes, in the critical-window territory of Section 8. The ceiling does not logically prohibit every congruence-informed construction: it says nothing about an accidentally tiny first prime in a chosen class, or a modulus much larger than $\exp\{\Theta(T)\}$. Accordingly this is an architecture wall, not a theorem that every possible extremal proof beyond $C\log p$ must be noncongruential.

18.3. **The exact W census. Computational 56.3 (exact stated ranges; informational).** Verifier block `(bc)` harvests the complete intrinsic class set of Lemmas 3.4 and 4.1, $\{-4D \pmod M : D \mid ((M+1)/4)^2\}$, for every $M \le 3000$, $M \equiv 3 \pmod 4$. It then scans every hard prime in increasing order and stops at the first harvested modulus. All 9,732 hard primes below 10^6 resolve within the cap. Their strict records are

p	73	193	1201	2521	3361	33289	90841
$W(p)$	7	15	31	47	99	155	167

p	144169	167521	225289	361321	915961	954409
$W(p)$	191	259	279	287	303	335

(56.10)

With `ES_FULL_SCAN=1`, the streamed, chunked scan covers all 82,887 hard primes below 10^7. The only later records are

$$(p, W(p)) = (1853329, 383), \qquad (2031121, 2495), \tag{56.11}$$

and the running maximum remains 2495 through the endpoint. These values are exact minima in the harvested family because every smaller eligible modulus was tested; the cap is not hit.

The requested normalized maxima, using natural logarithms, are INFO-ONLY:

p-range	$\max W/\log p$	$\max W/(\log p \log\log p)$	$\max W/(\log p)^2$
$p < 10^6$	24.330286 (954409, 335)	9.277839 (954409, 335)	1.836625 (225289, 279)
$p < 10^7$	171.783468 (2031121, 2495)	64.198698 (2031121, 2495)	11.827479 (2031121, 2495)

(56.12)

Each parenthesis is $(p, W(p))$. At $p = 10^7$, the unconditional Section 17 coefficient gives $(1/5.2)\log p = 3.0996\ldots$. This comparison must not mix registers: Section 17 guarantees rare residue-one escapes at a height bounded by $\exp\{(5.2 + o(1))T\}$, whereas (56.10)–(56.12) record whatever finite primes happen to be extremal below a moderate endpoint. For example, the generic height attached to $T = 2495$ is astronomically beyond this scan. The census neither samples that constructed sequence nor estimates a tail density, and the large isolated jump in (56.11) supports no growth law.

18.4. **The Type-I extremal extension. Computational 56.4 (exact stated ranges; informational).** The same block factors one norm at a time, skips the exact genus-forced signs before factorization, and streams unresolved primes in product order. For the 3,202 hard primes below $3 \cdot 10^5$, the strict ck_{pr} records are

p	73	193	241	1201	2521	4729	7489
ck_{pr}	7	10	21	34	38	66	70

p	9601	31081	51769	83689	113161	171481
ck_{pr}	78	110	249	282	378	461

(56.13)

The unrestricted conspiracy-depth records in the same range are

p	73	193	241	769	1321	2281
$D(p)$	6	9	10	12	20	25

p	2521	9601	12289	55441	92401
$D(p)$	37	66	76	82	102

(56.14)

With `ES_FULL_SCAN=1`, all 9,732 hard primes below 10^6 are scanned. The later ck_{pr} records are $(319489, 878), (538561, 898)$; the sole later depth record is $(414241, 217)$. Every prime resolves by these maxima. The normalized maxima are again INFO-ONLY:

statistic and range	$\max X/\log p$	$\max X/(\log p \log\log p)$	$\max X/(\log p)^2$
$ck_{\mathrm{pr}},\ p < 3 \cdot 10^5$	38.250190 (171481, 461)	15.366153 (171481, 461)	3.173703 (171481, 461)
$ck_{\mathrm{pr}},\ p < 10^6$	69.273069 (319489, 878)	27.277261 (319489, 878)	5.465556 (319489, 878)
$D,\ p < 3 \cdot 10^5$	8.920846 (92401, 102)	3.661213 (92401, 102)	0.857113 (12289, 76)
$D,\ p < 10^6$	16.777222 (414241, 217)	6.553922 (414241, 217)	1.297121 (414241, 217)

(56.15)

Here X in the header denotes the statistic in the first column, and each parenthesis is $(p, X(p))$. These are census maxima, not estimates of H_{SPF}, $ck_{\min}$, or a population exponent. The block's persistent state is linear in the hard-prime count: it keeps the hard-prime tuple, unresolved sets, per-prime minimum maps, and a small core cache. It keeps only one bounded residue set for the current norm. Prime generation is chunked; no prime-by-slice factor table or interval-sized residue array is retained.

18.5. **The remaining window.** The exact logical frontier and the finite plotting window can be displayed together only if their labels remain separate:

$$\underbrace{A<1}_{\substack{\text{impossible pointwise}\\ \text{Theorems 17.1, 17.3}}} \;\Big|\; \underbrace{1\le A \lesssim 2}_{\substack{\textbf{Assessment: finite-search window only}\\ \log p,\ \log p\log\log p,\ (\log p)^2 \text{ are the census anchors}}} \;\Big|\; \underbrace{A\ge 1}_{\substack{H_{\mathrm{MOD}}(A),H_{\mathrm{SPF}}(A)\\ \text{still open pointwise.}}} \tag{56.16}$$

The symbol $\lesssim 2$ in the middle is deliberately not a proved upper bound, a conjectured limiting exponent, or even a monotonic trend. It only names the normalization panel in (56.12) and (56.15). The proved lower wall is $A = 1$; no finite census supplies the other side. The pointwise-frontier statements in Sections 14 and 17 remain unchanged: Theorems 18.1 and 18.2 constrain only the complete-class certificate architecture.

19. The supercritical window and count-below-one reduction

Status. The count-below-one implication below is **proved conditional** on explicit, falsifiable window hypotheses, which are UNPROVED. Its fixed-polylogarithmic use inherits the provisional external-review qualification on the Section 10 moment/Bonferroni machinery; its cubic use additionally inherits the CLAIMED/PROVISIONAL Theorem 9.4. The only currently realized common power window is $\theta = 1/4$ in Theorem 14.2, with exactly those qualifications, and it is below both thresholds. Proposition 19.5 is proved only inside its explicitly defined absolute-remainder architecture. Statements that every sieve or linear combination belongs to that architecture are Assessments, not theorems. The crossing inputs are open hypotheses. Nothing here gives unconditional pointwise progress on the Erdős–Straus conjecture.

Put $L = \log N$, $t = \log T$, and

$$\mathcal{E} = \{p : p \text{ is prime and } 4/p \text{ has no three-unit-fraction representation}\}.$$

For every finite T there is the load-bearing containment

$$\mathcal{E} \subseteq \{p : W(p) = +\infty\} \subseteq \{p : W(p) > T\}. \tag{57.1}$$

Indeed, finite $W(p)$ supplies minimizing data (51.2), and Lemma 3.4 gives the explicit representation (4). That lemma has no primality, parity, or hard-class hypothesis. Thus every prime carrying a datum is covered, including one also soluble by an easy identity; no classification is assumed. The empty-datum convention $W = +\infty$ in (51.1) is essential.

19.1. **Count below one.** The endpoint constant cannot be suppressed in the literal hypotheses.

Hypothesis 19.1 ($H_{\mathrm{WIN}}(\theta; b)$; cubic window; open)**.** *For some fixed $b, c, C > 0$ and N_0, uniformly for $N \ge N_0$ and $3 \le T \le \exp\{bL^\theta\}$,*

$$\#\{p \le N : p \text{ prime},\ W(p) > T\} \le CN\exp\{-c(\log T)^3\}. \tag{57.2}$$

The effective version requires all four constants to be effective.

Hypothesis 19.2 ($H'_{\mathrm{WIN}}(\theta; b)$; square-log window; open)**.** *With the same quantifier pattern and some fixed positive constants,*

$$\#\{p \le N : p \text{ prime},\ W(p) > T\} \le CN\exp\{-c(\log T)^2\log(2+\log T)\}. \tag{57.3}$$

Using $\log\log T$ *is asymptotically equivalent; the displayed form is positive throughout the stated range and matches (51.13). Write* $H_{\mathrm{WIN}}(\theta)$ *or* $H'_{\mathrm{WIN}}(\theta)$ *when the corresponding assertion holds for some* $b > 0$.

Theorem 19.3 (count-below-one; proved conditional; source 57.1)**.** *1. If* $H_{\mathrm{WIN}}(\theta)$ *holds with* $\theta > 1/3$*, every sufficiently large prime satisfies the Erdős–Straus conjecture.*

2. If $H'_{\mathrm{WIN}}(\theta)$ *holds with* $\theta \geq 1/2$*, the same conclusion follows. In particular,* $\theta = 1/2$ *itself suffices; no* $+\epsilon$ *is needed.*

The threshold is effective if the constants in the relevant hypothesis are effective. At the cubic endpoint $\theta = 1/3$*,* $cb^3 > 1$ *is sufficient. If* $cb^3 = 1$*, the right side is the constant* C*, so it forces zero exactly when* $C < 1$*; if* $cb^3 < 1$*, it grows. Without a favorable endpoint-constant condition, one honestly needs* $\theta > 1/3$.

Proof. Assume first $H_{\mathrm{WIN}}(\theta; b)$. Fixed constants $b, c, C > 0$ and N_0 work for every $N \geq N_0$ and every $3 \leq T \leq \exp\{b(\log N)^\theta\}$. For such N, choose the single admissible endpoint $T_N = \exp\{bL^\theta\}$ once it is at least 3. The logarithm of the right side of (57.2) is

$$\log C + L - cb^3 L^{3\theta}. \tag{57.4}$$

For $\theta > 1/3$ this is negative for every sufficiently large L, after accounting for both L and $\log C$. Hence the nonnegative integer $A_N = \#\{p \leq N : p \text{ prime},\ W(p) > T_N\}$ is less than one and is zero. At $\theta = 1/3$, (57.4) is $\log C + (1 - cb^3)L$, giving exactly the three endpoint cases in the statement.

For $H'_{\mathrm{WIN}}(\theta; b)$ the same fixed-constant and endpoint-choice quantifiers give logarithm

$$\log C + L - cb^2 L^{2\theta} \log(2 + bL^\theta). \tag{57.5}$$

It tends to $-\infty$ relative to L for $\theta > 1/2$. At $\theta = 1/2$, its negative term is $(cb^2/2 + o(1))L \log L$, which dominates $L + \log C$ for every fixed $b, c > 0$. For $\theta < 1/2$ the saving is $o(L)$, so this inequality alone cannot force the count below one.

Finally, let p be a counterexample prime in a sufficiently large dyadic range and choose dyadic N with $p \leq N < 2p$. The definition (51.1) and Lemma 3.4 give

$$p \in \mathcal{E} \Longrightarrow W(p) = +\infty \Longrightarrow W(p) > T_N \Longrightarrow p \text{ is counted by } A_N.$$

The first implication holds for every counterexample prime and the second for every finite T_N, not only for this endpoint. But $A_N = 0$. Successive dyadic N cover every sufficiently large prime. Every threshold is computable when b, c, C, N_0 are. □

Calibration against the present internally proved/provisional window. Theorem 14.2 supplies both tails only in the common power window

$$t \leq bL^{1/4} \tag{57.6}$$

for sufficiently small effective b; the cubic row is CLAIMED/PROVISIONAL, while the square-log row has only the Section 10 machinery-review qualification. This is a window exponent, not $H_{\mathrm{WIN}}(1/4; 1)$: the endpoint constant matters. The square-log row is stronger than (57.6), since (51.12) permits

$$t^3 \log(2 + t) \ll L, \qquad t \ll (L/\log L)^{1/3}. \tag{57.7}$$

Thus it already has power-window exponent $1/3$, up to the displayed $\log L = \log\log N$ factor, but its count crosses one only when $t^2 \log t \gg L$, at scale $(L/\log L)^{1/2}$.

The cubic row is valid at $t^4 \ll L$ and crosses at $t^3 \gg L$. Both realized rows pay exactly one extra factor t beyond supply mass. This calibrates a conditional reduction; it is not an unconditional Erdős–Straus result.

19.2. **The mass-invariant gap.** The antecedents are not new here. The proof of (21) records the prime-slice supply wall and its summed Bombieri–Vinogradov multiplicity (source §18.2, Assessment 18.3(b)). Hypothesis 6.1 and Theorem 6.2 are the paper's earlier mass/product-tail budget and cubic optimization; source §18.3 sharpens that budget to $\min(K, J\log K) + Jt \lesssim L$ in (18.15), and Assessment 18.4 performs both its square-log and cubic optimizations. Lemma 8.4 and Corollary 8.5 are the paper's corrected form of source Lemma 33.3 and its conditional finite-transfer ledger (33.6). Most directly, the actual Section 14 assembly has

$$\deg = O(\mu), \qquad \log d_{\mathrm{term}} = O(\mu t), \qquad \log \sum |c_{\mathrm{term}}| = O(\mu t) \tag{57.8}$$

by (51.11), source (51.11). The next definition isolates only the arithmetic common to those antecedents.

Definition 19.4 (absolute-remainder mass-level architecture)**.** At depth $T = \mathrm{e}^{O(t)}$, an $\mathrm{AR}(\mu, t)$ argument has a nonnegative majorant $\nu = \sum_S c_S 1_{a_S \ (d_S)}$ for the relevant void or conspiracy set, uses truncation degree $J \asymp \mu$, and has CRT mean at most $\mathrm{e}^{-c_0\mu}$. It transfers to $[1, N]$ using each class evaluation only through

$$\#\{n \le N : n \equiv a_S \pmod{d_S}\} = \frac{N}{d_S} + O(1) \tag{57.9}$$

and the triangle inequality over the coefficient ledger. The architecture books level and absolute-remainder cost $\mathrm{e}^{O(Jt)}$ and declares the clean main-term estimate valid only in the standard window

$$L \ge C_0 \mu t. \tag{57.10}$$

This formalizes one proof architecture; it does not assert that every linear combination has this ledger or must be evaluated absolutely.

Proposition 19.5 (mass-invariant gap; proved arithmetic in the AR architecture; source Proposition 57.2)**.** *Suppose the supply mass is $\mu = \mu(t)$ and an $\mathrm{AR}(\mu, t)$ argument gives a tail $\ll N\mathrm{e}^{-c_0\mu}$ in (57.10).*

1. *If the implicit tail constant is $A > 0$, the exact requirement from this upper bound is $c_0\mu > L + \log A$. A convenient fixed-margin* sufficient *condition is*

$$c_0\mu \ge (1+\epsilon)L \tag{57.11}$$

for some fixed $\epsilon > 0$. Either form implies $\mu \gg L$ and is incompatible with (57.10) as $t \to \infty$.

2. *If $\mu(t) \asymp t^a$ for fixed $a > 0$, the largest saving exponent certified in the validity window is*

$$\mu \ll L^{a/(a+1)}, \qquad \theta_{\mathrm{AR}} = \frac{a}{a+1} < 1. \tag{57.12}$$

Thus increasing the polynomial supply mass alone never reaches a pointwise conclusion in this architecture.

Proof. The exact count-below-one inequality implies $\mu \gg L$ for large L; with (57.10), this gives $L \gg \mu t \gg Lt$, impossible for unbounded t. If $\mu \asymp t^a$, then (57.10) says $t^{a+1} \ll L$; substitution proves (57.12). □

For orientation, restoring the slowly varying $\log t$, $a = 2$ is the square-log $2/3$ scale of source (18.16); $a = 3$ is the cubic $3/4$ scale of Assessment 18.4 and Theorem 14.2. The exponent changes with mass, but the one-factor gap does not.

Proved device check (Bonferroni–CRT). The Sections 10 and 14 construction fits the formal architecture as an implemented certificate: it uses $J \asymp \mu$, and (51.11) gives the booked costs $\mathrm{e}^{O(\mu t)}$. This is only a ledger statement; the tail retains the labels in this section's status paragraph.

Assessment 57.3 (two further standard devices, and the general moral). In the standard growing-dimension fundamental-lemma/beta-sieve model, a void saving $\mathrm{e}^{-c\mu}$ requires truncation order proportional to dimension or mass, while its sieve level is a product of that many modulus scales, $D = \mathrm{e}^{\Theta(\mu t)}$. Its usual parameter condition $s = \log D/t \gtrsim \mu$ reproduces (57.10). This assesses that named uniform-in-dimension implementation; it is not a no-go theorem for all hypergraph or arithmetic sieves.

The product-modulus Montgomery large sieve has the same arithmetic. A product of J modulus scales of entropy t has $Q = \mathrm{e}^{\Theta(Jt)}$; the useful range $Q^2 \le N$ gives

$$J \le \frac{L}{2t}. \tag{57.13}$$

In the usual truncated-exponential model, the available partial saving is

$$F(J) = J \log \frac{\mathrm{e}\mu}{J}, \qquad J \le \min(\mu, L/(2t)). \tag{57.14}$$

For $\mu = t^a$, put $t_0 = (L/2)^{1/(a+1)}$. If $t \le t_0$, the maximum is $F(\mu) = \mu = O(L^{a/(a+1)})$. If $t = t_0 u$, $u > 1$, the cap gives

$$F \le \frac{t_0^a}{u} \log(\mathrm{e}u^{a+1}) = O_a(t_0^a), \tag{57.15}$$

since the last factor is maximized at $u = \mathrm{e}^{a/(a+1)}$. Thus this modeled saving is $O_a(L^{a/(a+1)}) = o(L)$ and cannot make the count less than one. The calculus is proved; the Assessment is that this product-modulus large-sieve implementation models a proposed assembly. None of these three checks excludes a sparse formula, signed cancellation, or input seeing the actual remainder locations. In particular, the broader claim "no absolute-remainder assembly can ever work" is an **Assessment**, not Proposition 19.5.

19.3. **Where the crossing lives.** For the c-free atoms, every compatible set of r distinct atoms has distinct prime coordinates by Lemma 10.1, each in $(X^{1/2}, X]$. Hence

$$d_S \ge \prod_{A \in S} \ell_A \ge X^{r/2}, \qquad \log d_S \ge \frac{rt}{2}. \tag{57.16}$$

In the witness-tail polynomial, the matching upper bound follows from $k\ell = \mathrm{e}^{O(t)}$. At void-detecting degree $r \asymp \mu$, near crossing $\mu \asymp L$, top-degree moduli satisfy $d_S = \mathrm{e}^{\Theta(\mu t)} \gg N$. A nonempty class modulo such d_S contains zero or one integer of $[1, N]$, and its $O(1)$ remainder is no longer a small perturbation of N/d_S.

For a finite congruence expansion define

$$\mathrm{Rem}_\nu(N) = \sum_S c_S \left(\#\{n \le N : n \equiv a_S \pmod{d_S}\} - \frac{N}{d_S} \right). \tag{57.17}$$

Then exactly

$$\sum_{n\le N}\nu(n)=N\mathbb{E}_{\mathrm{CRT}}\nu+\mathrm{Rem}_{\nu}(N). \tag{57.18}$$

Below crossing, Section 14 bounds the remainder by $\sum|c_S|$. **Assessment (crossing model).** Above crossing, the missing information is cancellation among signed zero-or-one remainders across an exponentially large subledger, modeled at void scale as $\mathrm{e}^{\Theta(\mu)}$ selected classes. This is an upper-transfer mirror of source Lemma 33.3, which transfers a positive CRT lower bound only while its absolute ledger is affordable.

Fix in advance a family $\mathscr{F}$ of pairs (N,T) with $N,T\to\infty$ and a specified mass $\mu(T)\to\infty$.

Hypothesis 19.6 ($H_{\mathrm{XW}}(\mathscr{F},\mu)$; supercritical remainder cancellation; open)**.** *There are fixed $c_1,\epsilon,A>0$ and specified functions $\eta(T)\ge 0$, $\eta(T)\to 0$, and $h(T)\ge 0$, $h(T)=o(\mu(T))$, such that for every sufficiently large $(N,T)\in\mathscr{F}$, writing $L=\log N$, if*

$$\mu(T)\le AL,\qquad c_1\mu(T)\ge(1+\epsilon)L,$$

there is a finite nonnegative congruence expansion $\nu_{N,T}$, at least one on every prime $\mathrm{e}^{h(T)}<p\le N$ with $W(p)>T$, for which

$$\mathbb{E}_{\mathrm{CRT}}\nu_{N,T}\le\mathrm{e}^{-c_1\mu(T)},\qquad \mathrm{Rem}_{\nu_{N,T}}(N)\le N\mathrm{e}^{-c_1\mu(T)+\eta(T)\mu(T)}. \tag{57.19}$$

All constants, the family, and both error functions are quantified before (N,T). The remainder bound is deliberately one-sided; absolute cancellation would be stronger than needed.

Proposition 19.7 (crossing implication; proved conditional; source Proposition 57.4)**.** *Under $H_{\mathrm{XW}}(\mathscr{F},\mu)$, every sufficiently large covered pair meeting the two crossing conditions has no prime $\mathrm{e}^{h(T)}<p\le N$ with $W(p)>T$. If $\mathscr{F}$ contains such a pair for every sufficiently large dyadic N, the Erdős–Straus conjecture holds for all sufficiently large primes.*

Proof. Equations (57.18)–(57.19) give the honest two-term bound

$$\sum_{n\le N}\nu_{N,T}(n)\le 2N\mathrm{e}^{-c_1\mu(T)+\eta(T)\mu(T)}.$$

Since $\mu(T)\le AL$, its logarithm is at most $\log 2-\epsilon L+\eta(T)AL<0$ eventually. The majorant excludes every stated tail prime. Under dyadic coverage, the two crossing inequalities give $\mu\asymp L$, hence $h=o(L)$ and $\mathrm{e}^h<N/2$ eventually. Every large prime lies in some $(N/2,N]$, and (57.1) excludes a counterexample. □

Alternatively, let $\mathcal{C}_T$ be the union of exact CRT classes avoiding *all* Lemma 3.4 witness classes of modulus at most T, and let δ_T be its density in the finite CRT product. Thus $W(n)>T$ implies $n\in\mathcal{C}_T$.

Hypothesis 19.8 ($H_{\mathrm{EQ}}(\mathscr{F},\mu)$; fair-share conspiracy classes; open)**.** *For a fixed family and mass, there are fixed $c_2,\epsilon,A>0$ and a specified $\eta(T)\ge 0$, $\eta(T)\to 0$, such that for every sufficiently large $(N,T)\in\mathscr{F}$ satisfying $\mu(T)\le A\log N$, $c_2\mu(T)\ge(1+\epsilon)\log N$, and $\delta_T\le\mathrm{e}^{-c_2\mu(T)}$,*

$$\#\{p\le N: p\text{ prime},\ p\in\mathcal{C}_T\}\le\mathrm{e}^{\eta(T)\mu(T)}\delta_T\pi(N). \tag{57.20}$$

This is falsifiable once $\mathscr{F}, \mu, \eta$ are specified. Its right side is at most $\exp\{-\epsilon L + \eta(T)AL\} < 1$ eventually, so it gives no tail prime at each covered pair and, with dyadic coverage, the same large-prime conclusion as Proposition 19.7. Neither crossing hypothesis is asserted.

Assessment (localization of the missing input). The residue-dispersion target (37.27), reduced to the short-cofactor endpoint (46.13), is an earlier pair-level request for cancellation across arithmetic residue fibres. Both remain OPEN; even proving them would not itself give the all-degree cancellation in (57.19). They nevertheless locate the same missing information: mass and absolute ledgers do not control where compatible residues land. The all-avoider cylinder wall, Theorem 8.2 (source Theorem 33.1), is consistent only at this diagnostic level. It concerns the complete intrinsic avoider, not every c-free family or every possible supercritical assembly.

The lower-tail primes of Section 17 do not contradict the proposed crossing. For the least prime $p_T \equiv 1 \pmod{M(T)}$, Theorem 17.1 gives

$$\begin{gathered} T + o(T) \le \log p_T \le (5.2 + o(1))T, \qquad \log T = \log\log p_T + O(1), \\ \mu(T) \asymp (\log\log p_T)^3 = o(\log p_T). \end{gathered} \tag{57.21}$$

The last mass is the cubic calibration with its inherited status; the square-log mass is smaller, $\asymp (\log\log p_T)^2 \log\log\log p_T$. These residue-one escapes are subcritical, not counterexamples to either crossing hypothesis.

19.4. **Finite arithmetic replay. Computational 57.6 (exact stated range; informational).** Verifier block `(bd)` computes with exact rational arithmetic

$$m(T) = \sum_{\substack{M \le T \\ M \equiv 3 \ (4)}} \frac{\omega(M)}{M}, \qquad \omega(M) = \#\{-4D \pmod M : D \mid ((M+1)/4)^2\}, \tag{57.22}$$

using the cached complete-harvest helper from block `(bc)`, and performs a chunked scan through $3 \cdot 10^5$. The displayed mass decimals come from exact fractions. The tails include all positive integers or all primes:

T	$m(T)$	$\#\{n \le 3\cdot 10^5 : W(n) > T\}$	$\#\{p \le 3\cdot 10^5 : W(p) > T\}$	$\rho_{\mathrm{int}}/\mathrm{e}^{-m}$	$\rho_{\mathrm{prime}}/\mathrm{e}^{-m}$
100	4.248031290751	5516	76	1.28647	0.20454
300	7.349236274282	1064	5	5.51510	0.29907
1000	12.171587320339	569	0	366.47489	0
3000	18.126592298585	550	0	136621.93932	0

(57.23)

The density denominators are 300000 and the exact 25997 primes. These ratios are INFO, not estimates or growth laws. The next section explains almost all high-T integer excess: all 547 positive squares at most 300000 have $W = +\infty$, accounting for 547 of 569 survivors at $T = 1000$ and 547 of 550 at $T = 3000$. Conversely, the isolated $(p, W) = (2031121, 2495)$ in (56.11) warns that a larger prime box may contain an escape far above a naive small-box fair-share picture. Neither finite observation decides (57.19) or (57.20).

Block `(bd)` also replays (57.4)–(57.5) at $L = 10^3, 10^4, 10^5, 10^6$ with constants normalized to one, including all three C-cases at the cubic endpoint. Its additional asymptotic, fixed-margin surrogates are $t^3/(2L)$ and $t^2 \log t/(2L)$ at $t = L^\theta$. In that normalization, cubic $\theta = 0.40$ crosses at $L = 32$, while $\theta = 0.25$ decreases, and square-log $\theta = 1/2$ crosses at $L = \mathrm{e}^4$. These are surrogate crossings only; $L = \mathrm{e}^4$ is not an exact crossing for $\log(2+t)$ in (57.5). This arithmetic verifies neither window hypothesis.

Computational 57.7 (exact toy only). For every distinct harvested atom through $X = 15, 23, 31$, block `(bd)` enumerates compatible sets and counts exact lcms above $N_{\text{toy}} = X^2$:

X	r	# atoms	# compatible	$\#\{d_S > X^2\}$
15	2	11	41	0
15	3	11	57	45
15	4	11	18	18
23	2	23	200	0
23	3	23	846	765
23	4	23	1809	1809
31	2	31	393	0
31	3	31	2653	2418
31	4	31	10389	10371

(57.24)

This is a finite structural illustration of (57.16), not asymptotic evidence.

A sequence violating (57.2) or (57.3) in its moving window falsifies that hypothesis. For fixed $(\mathscr{F}, \mu)$, universal failure of all permitted candidates in (57.19), or of every allowed fixed-constant/error-function choice in (57.20), would falsify the corresponding crossing hypothesis; one bad expansion does not falsify existential H_{XW}. Proposition 19.5 excludes neither a sparse identity, nonabsolute transfer, nor a new distribution theorem, and the finite tables support neither hypothesis. The proved content is the conditional count-below-one implication, the one-factor arithmetic gap in the stated architecture, and the exact remainder identity (57.18). There is no universal localization theorem and no unconditional pointwise Erdős–Straus claim.

20. The inverse census and the integer/prime dichotomy

Status and headline. The definitions, inverse equivalences, Jacobi argument, and bounds below are proved. The censuses are **Computational** and exact only on their stated grids. The Iwaniec statement was checked in full secondary sources archived under `sources/jacobsthal-literature/`; the publisher did not deliver the 1978 primary PDF, so no primary-source verification is claimed. The main structural fact is stronger than a generic Jacobsthal estimate: the full harvested system misses every square, so $W(m^2) = +\infty$ for every positive integer m. The useful all-integer inverse is therefore near-linear, while finding a prime survivor remains open below the exponential congruence construction. No finite table below suggests a growth law.

20.1. **Moving inverses and an exact finite census.** The literal least-positive-integer definition is corrupted at once:

$$W(1) = W(4) = +\infty, \qquad W(2) = 3, \qquad W(3) = 7. \tag{58.1}$$

Thus $\min\{n \geq 1 : W(n) > T\} = 1$ for every T. A fixed prime with infinite W, if one existed, would similarly corrupt a raw prime inverse without bearing on eventual pointwise behavior. Throughout this section use the **moving convention**

$$\begin{aligned} L_{\text{int}}(T) &= \min\{n > T : W(n) > T\}, \\ L_p(T) &= \min\{p > T : p \text{ prime}, \ W(p) > T\}, \\ L_h(T) &= \min\{p > T : p \equiv 1 \pmod{24}, \ p \text{ prime}, \ W(p) > T\}. \end{aligned} \tag{58.2}$$

The minima are finite by Theorem 20.1. The hard-prime inverse L_h is included because (56.10)–(56.11) use that class; it must not be substituted for the all-prime inverse L_p. For example,

$$L_p(335) = 22621, \qquad L_h(335) = 1853329.$$

Lemma 4.1 makes the inverse condition exact:

$$W(n) > T \quad \Longleftrightarrow \quad n \not\equiv -4D \pmod{M} \quad \left(\begin{array}{l} M \le T,\ M \equiv 3 \pmod 4, \\ D \mid ((M+1)/4)^2 \end{array} \right). \tag{58.3}$$

This ranges over the original data, not a selected subfamily. Data (51.2) with $M = k\ell$ and $uvw = (M+1)/4$ give $D = u^2 w$ and $-uv^{-1} \equiv -4D \pmod{M}$; conversely, the prime-by-prime construction in Lemma 4.1 recovers positive u, v, w for every such D, while $k = 1, \ell = M$ is allowed. Thus no factorization, composite modulus, or duplicated class is lost. To certify an inverse row it is enough to harvest all moduli through that row's threshold; no claim about a later or nonexistent witness is used.

Computational 58.1 (exact grid; informational). Verifier block `(be)` harvests (58.3), streams integers and primes upward, and retains every unresolved threshold. The default grid reaches $T = 2000$; every entry and bracketed witness value is exact, with the bracket giving W at the prime:

T	$L_{\mathrm{int}}(T)$	$L_p(T)$ $[W]$	$L_h(T)$ $[W]$
3	4	7 [11]	73 [7]
7	9	37 [15]	193 [15]
15	16	79 [23]	1201 [31]
31	36	211 [43]	2521 [47]
63	64	1381 [83]	3361 [99]
100	121	10399 [103]	33289 [155]
200	225	22621 [419]	167521 [259]
335	336	22621 [419]	1853329 [383]
382	400	22621 [419]	1853329 [383]
500	529	206299 [695]	2031121 [2495]
750	784	2031121 [2495]	2031121 [2495]
1000	1024	2031121 [2495]	2031121 [2495]
1250	1296	2031121 [2495]	2031121 [2495]
1500	1521	2031121 [2495]	2031121 [2495]
1750	1764	2031121 [2495]	2031121 [2495]
2000	2025	2031121 [2495]	2031121 [2495]

(58.4)

The independent hard-prime replay gives exactly

$$L_h(T) = 1853329 \qquad (335 \le T \le 382), \tag{58.5}$$

as required by the two records (56.10)–(56.11). With `ES_FULL_SCAN=1`, the grid adds

T	2200	2400	2494
$L_{\mathrm{int}}(T)$	2209	2401	2500
$L_p(T) = L_h(T)$	2031121	2031121	2031121.

(58.6)

For reference, each normalization cell below is $(\log L/\log T, \log L/T)$, with natural logarithms:

T	L_{int}	L_p	L_h
31	(1.044, .11560)	(1.558, .17264)	(2.281, .25266)
100	(1.041, .04796)	(2.008, .09249)	(2.261, .10413)
335	(1.001, .01736)	(1.725, .02993)	(2.482, .04308)
382	(1.008, .01568)	(1.686, .02625)	(2.427, .03778)
500	(1.009, .01254)	(1.969, .02447)	(2.337, .02905)
1000	(1.003, .00693)	(2.103, .01452)	(2.103, .01452)
2000	(1.002, .00381)	(1.911, .00726)	(1.911, .00726)

(58.7)

The integer first coordinate near one has a theorem behind it. The prime columns are step functions: decreases across plateaux and jumps at a record are finite-table shapes only. In particular, the isolated $W(2031121) = 2495$ jump permits no asymptotic extrapolation.

20.2. **Square escape and shifted constraints.**

Theorem 20.1 (square escape and two inverse scales; proved; source Theorem 58.1)**.** *For every positive integer m, $W(m^2) = +\infty$. Consequently, for every real $T \geq 3$,*

$$T < L_{\mathrm{int}}(T) \leq (\lfloor \sqrt{T} \rfloor + 1)^2 \leq T + 2\sqrt{T} + 1, \tag{58.8}$$

and also

$$L_{\mathrm{int}}(T) \leq M(T) + 1 = \exp\{(1 + o(1))T\}. \tag{58.9}$$

Moreover,

$$L_{\mathrm{int}}(T) \leq L_p(T) \leq \exp\{(5.2 + o(1))T\}. \tag{58.10}$$

The last upper bound also holds for $L_h(T)$. All asymptotic upper bounds are effective in the same sense as Theorem 17.1.

Proof. We replay the full square quantifier chain. Fix arbitrary $M = 4A - 1$ and $D \mid A^2$. Since $(A, M) = 1$, also $(D, M) = 1$; thus the apparently degenerate case in which a prime dividing M also divides D is impossible. A factor 2 in D causes no common factor because M is odd. Hence every class $-4D \pmod M$ is a unit.

For every odd prime $r \mid A$, quadratic reciprocity, $M \equiv -1 \pmod r$, and $M \equiv 3 \pmod 4$ give

$$\left(\frac{r}{M}\right) = \left(\frac{M}{r}\right)(-1)^{(r-1)(M-1)/4} = \left(\frac{-1}{r}\right)(-1)^{(r-1)/2} = 1.$$

If $2 \mid A$, then $M \equiv 7 \pmod 8$, so $(2/M) = 1$. Multiplicativity therefore gives $(D/M) = 1$ for every divisor $D \mid A^2$, whether or not D is a square and with every possible 2-adic exponent. Since $(-1/M) = -1$ and $(4/M) = 1$, one has $(-4D/M) = -1$.

For a local reading, write $M = \prod q^{e_q}$. The last Jacobi sign says that for at least one q with odd e_q, $(-4D/q) = -1$; hence this class is not a square modulo q, and therefore not modulo q^{e_q} or M. One must not insist that this obstructing q is itself $3 \pmod 4$: for $(M, D) = (15, 2)$, the class is a square modulo 3 and a nonsquare modulo 5. The condition $M \equiv 3 \pmod 4$ forces the *global* Jacobi sign. Finally, a nonunit square cannot be congruent to the unit $-4D$. Thus every square avoids every class for every eligible M, D; (58.3) gives $W(m^2) = +\infty$.

The strict lower bound in (58.8) is definitional, not a claim that all integers at most T were witnessed. Taking $s = \lfloor \sqrt{T} \rfloor + 1$ gives the upper bounds.

For (58.9), $M(T) + 1 \equiv 1 \pmod m$ for every eligible $m \leq T$. If its datum had $uvw = (m + 1)/4$, the witness congruence would give $m \mid u + v$, whereas

$$2 \leq u + v \leq uv + 1 \leq (m + 1)/4 + 1 < m.$$

This is the arithmetic part of Theorem 17.1 and uses no primality. Finally, primes are a subset of the integers, and Theorem 17.1 supplies a prime in $1 \pmod{M(T)}$ below $\exp\{(5.2 + o(1))T\}$. For $T \geq 8$ that class is also $1 \pmod{24}$, proving the hard-prime assertion. □

No lower bound $L_{\mathrm{int}}(T) \gg T \log T$, much less a superpolynomial one, is possible under (58.2). The following elementary shifts explain why a naive lower-bound search can still look tempting.

Lemma 20.2 (two necessary shifted-sieve conditions; proved; source Lemma 58.2)**.** *If $W(n) > T$, then*

$$\begin{aligned} \ell \nmid n+4 \quad (\ell \le T,\ \ell \equiv 3 \pmod 4,\ \ell \text{ prime}),\\ \ell \nmid n+8 \quad (\ell \le T,\ \ell \equiv 7 \pmod 8,\ \ell \text{ prime}). \end{aligned} \tag{58.11}$$

Proof. For the first line choose $D = 1$ in (58.3). For a prime $\ell \equiv 3 \pmod 4$, put $A = (\ell+1)/4$. The divisor $D = 2$ is eligible exactly when $2 \mid A^2$, equivalently A is even, equivalently $\ell \equiv 7 \pmod 8$. Its class is $-8 \pmod \ell$. □

The first line says that every $3 \pmod 4$ prime factor of $n+4$ exceeds T. If $T \ge x+4$, the $n \le x$ passing this one test are shifts of numbers supported on 2 and primes $1 \pmod 4$. The contextual Selberg–Delange calculation has order $x/\sqrt{\log x}$, from the Dirichlet series

$$\prod_{q=2 \text{ or } q\equiv 1 \ (4)} (1-q^{-s})^{-1} = \zeta(s)^{1/2} H(s), \tag{58.12}$$

where H is holomorphic and positive at 1. This classical calculation is not used as a new theorem. It already shows why one shift cannot force a superpolynomial first survivor. The full conjunction is stronger, but Theorem 20.1 gives an exact common escape.

20.3. **What Jacobsthal does, and what it does not do.** For $q > 0$, let $g(q)$ be the least G such that every G consecutive integers contain one coprime to q; up to the conventional endpoint difference of one, this is the maximal gap between consecutive integers coprime to q.

Theorem 20.3 (Iwaniec's Jacobsthal bound; secondary-source-verified; source Theorem 58.3)**.** *If $k = \omega(q) \ge 2$, uniformly in q,*

$$g(q) \ll (k \log k)^2. \tag{58.13}$$

Equivalently, $g(q) \ll (k \log(2k))^2$ for all $q > 1$, including the one-prime-factor case. The constant is absolute but was not explicit in the checked formulation. Page 1 of the archived full Costello–Watts paper [3] calls this the best asymptotic *upper bound, displays $g(n) \le X(k \log k)^2$ with unknown X, and attributes it to Iwaniec [15]. As an independent scope check, pages 2–3 of the archived Ford–Green–Konyagin–Maynard–Tao paper [9] define the endpoint-shifted maximal-gap function j, identify $Y(x) = j(\prod_{p \le x} p) - 1$, and record Iwaniec's consequence $Y(x) \ll x^2$. The primary DOI is 10.1515/dema-1978-0121; automated retrieval reached the publisher's human-verification page rather than a PDF. This is secondary-source verification, not inspection of the 1978 primary paper.*

There is a precise one-way map to (58.3). For $P_3(T)$ from (56.1), avoiding only $D = 1$ is exactly

$$(n+4, P_3(T)) = 1. \tag{58.14}$$

Since $\omega(P_3(T)) = \pi(T; 4, 3)$, (58.13) says every interval of length $O(T^2)$ contains a survivor of this subsystem. This does *not* upper-bound $L_{\mathrm{int}}(T)$: avoiding the full system is harder, so the full survivor set is a subset of the set in (58.14). Reversing this inclusion is the Jacobsthal direction error.

The full system removes, for every composite as well as prime $M \equiv 3 \pmod 4$, the structured family

$$\{-4D \pmod M : D \mid ((M+1)/4)^2\}. \tag{58.15}$$

This is not deletion of one class modulo each prime. A composite class in (58.15) is an intersection of coordinate conditions, not a union of fixed forbidden classes at its prime factors, so Iwaniec's theorem does not apply verbatim. Ford–Konyagin–Maynard–Pomerance–Tao [10] allow bounded sets I_p at each prime under a one-dimensional average hypothesis, but their theorem constructs *large gaps* in a prime-indexed sifted set; it neither bounds the first survivor nor covers (58.15). The same warning applies to the lower bounds in [9]: they concern a maximal, worst-position gap, not this specified least element.

For the full all-integer system the Jacobsthal comparison is only historical context, and potentially context for prime- or nonsquare-restricted variants. Theorem 20.1 instead proves

$$L_{\mathrm{int}}(T) = T + O(\sqrt{T}). \tag{58.16}$$

This means exactly the two-sided interval (58.8), not an asymptotic constant for the difference.

Corollary 20.4 (integer/prime count-below-one dichotomy; proved; source Corollary 58.4)**.** *For all $N > T$,*

$$\begin{aligned} \#\{T < n \le N : W(n) > T\} &\ge \max\{0, \lfloor\sqrt{N}\rfloor - \lfloor\sqrt{T}\rfloor\}, \\ N_0 := L_{\mathrm{int}}(T) \le S &:= (\lfloor\sqrt{T}\rfloor + 1)^2, \\ S \le T + 2\sqrt{T} + 1, &\qquad W(S) = +\infty. \end{aligned} \tag{58.17}$$

The moving count excludes $n \le T$, so its nonemptiness is not the vacuous result of $W(1) = W(4) = +\infty$. At $N = N_0$ the left side is at least one by definition, and at $N = S$ the square is a structural survivor. For fixed T, or $T = o(N)$, the all-integer tail contains $(1 + o(1))\sqrt{N}$ squares. Thus an all-integer moving-tail bound below one cannot hold at $N = N_0$ or $N = S$. Any count-below-one route to a pointwise prime theorem must use primality essentially: S is composite for every $T \ge 3$.

This dichotomy retroactively explains the prime restriction in Theorem 14.2; an all-integer analogue could not support the count-below-one step in Section 19.

20.4. **Prime survivors: congruence wall and technology audit.** Squares settle the integer inverse but provide no prime. Theorem 18.1 says a complete congruence certificate for $W > T$ has modulus divisible by $P_3(T)$ and therefore at least $\exp\{(1/2 + o(1))T\}$. Coupling such a certificate to a generic least-prime theorem remains an $\exp\{\Theta(T)\}$ architecture. Residue one and Linnik give (58.10).

Assessment 58.1 (the missing prime technology). Density alone gives no least-prime theorem for an arbitrary sifted set, and ordinary upper-bound sieves have the parity obstruction: they bound survivors without producing a prime. Bombieri–Vinogradov-type distribution handles many small progression conditions on average, but no theorem found in this audit produces a prime avoiding the complete T-dependent family (58.15) below $\exp\{o(T)\}$. Work on divisors or smoothness of one shifted prime concerns sizes and locations of factors of $p + a$, not a least-prime theorem for all $p + 4D$ here. In particular, the first condition in (58.11) is not a substitute for the full harvest.

The Generalized Riemann Hypothesis improves least primes in a specified progression but does not remove parity for this union of sifted classes. A Cramér-type independent-prime model would put a first prime near the reciprocal of modeled survivor density, up to logarithmic factors. That is a **Heuristic**, not a conditional theorem here. The only proved prime bounds are

$$T < L_{\mathrm{int}}(T) \le L_p(T) \le \exp\{(5.2 + o(1))T\}. \tag{58.18}$$

Beating the exponential upper end requires aggregate information about primes in this sifted set, not merely one complete residue class.

20.5. **Exact inverse dualities and the A-window.** The slogan that superlogarithmic records are equivalent to $L_p(T) = \exp\{o(T)\}$ needs a quantifier correction.

Lemma 20.5 (record and pointwise inverse dualities; proved; source Lemma 58.5)**.** *Under (58.2):*

1. *There are primes $p_j \to \infty$ with $W(p_j)/\log p_j \to \infty$ if and only if*

$$\liminf_{T\to\infty} \frac{\log L_p(T)}{T} = 0. \tag{58.19}$$

   The global statement $L_p(T) = \exp\{o(T)\}$, with a limit rather than a subsequence, is sufficient but is not equivalent without regularity controlling record gaps.
2. *For every fixed $A > 0$,*

$$H_{\mathrm{MOD}}(A) \quad\Longleftrightarrow\quad L_p((\log x)^A) > x \quad \text{for all sufficiently large } x. \tag{58.20}$$

   Equivalently, $H_{\mathrm{MOD}}(A)$ fails exactly when $L_p(T) \le \exp(T^{1/A})$ for arbitrarily large T.

Proof. Given a superlogarithmic sequence, choose $T_j < \min(W(p_j), p_j)$ with $T_j/\log p_j \to \infty$; this is possible because both $W(p_j)/\log p_j$ and $p_j/\log p_j$ diverge. Then $L_p(T_j) \le p_j$, so $0 \le \log L_p(T_j)/T_j \le \log p_j/T_j \to 0$. Conversely, choose $T_j \to \infty$ along the liminf and put $p_j = L_p(T_j)$. Then $p_j > T_j \to \infty$, $W(p_j) > T_j$, and $W(p_j)/\log p_j > T_j/\log L_p(T_j) \to \infty$. This includes $W = +\infty$.

For (58.20), negate both sides. If $H_{\mathrm{MOD}}(A)$ fails at arbitrarily large primes p, put $x = p$. Since $(\log p)^A < p$ eventually, that prime is admissible and $L_p((\log x)^A) \le x$. Conversely, if this inverse inequality holds at unbounded x, put $q = L_p((\log x)^A) \le x$. Then $q \to \infty$ and $W(q) > (\log x)^A \ge (\log q)^A$, so $H_{\mathrm{MOD}}(A)$ fails. Substituting $T = (\log x)^A$ gives failures $L_p(T) \le \exp(T^{1/A})$ at arbitrarily large T; conversely, any unbounded such T gives $x = \exp(T^{1/A})$ and a failure of the right side of (58.20). □

Formula (58.20) is the exact inverse form of the remaining window. For $A < 1$, Theorem 17.1 gives $L_p(T) \le \exp\{(5.2 + o(1))T\} \ll \exp(T^{1/A})$, reproving failure of $H_{\mathrm{MOD}}(A)$. At $A = 1$, the hypothesis asks for the eventual lower bound $L_p(T) > \mathrm{e}^T$; for $A > 1$, it asks for $L_p(T) > \exp(T^{1/A})$. Neither (58.10) nor the census decides any $A \ge 1$. The $1 \le A \lesssim 2$ display (56.16) remains a finite normalization window, not a claimed exponent.

21. Polynomial escape: complete through degree two, cyclotomic in general

Status and stakes. The local reduction, the qualitative Chebotarev obstruction, the complete degree-at-most-two classification, and the all-integer tail obstruction

below are **proved**. The classification in degree greater than two remains **Conjectural**; in particular, this section does not claim a full polynomial classification. The finite censuses are **Computational** and exact only in their displayed ranges. The algebraic results use Theorem 20.1, qualitative Chebotarev, Schur's prime-divisor theorem, and the standard conductor description of quadratic fields. They do not use Theorem 9.4 or any of the Section 10 moment/Bonferroni machinery, and inherit none of their provisional status. The Chebotarev obstruction is qualitative and **ineffective**: no effective least prime is extracted.

The stakes are asymmetric. The square family makes an all-integer cubic tail impossible as soon as its saving passes the one-half-density crossing: there are about $N^{1/2}$ structural survivors. Thus the prime restriction in Theorems 14.2 and 19.3 is necessary, not cosmetic. In the other direction, a uniform escape polynomial taking prime values infinitely often would give infinitely many primes with $W = +\infty$, permanently outside the Type-II frame and violating every pointwise $H_{\mathrm{MOD}}(A)$. A proof that every such polynomial is square-valued would close this polynomial-family counterexample flank, because a positive square greater than one is composite. Theorem 21.4 closes it only through degree two; the higher-degree flank and nonpolynomial thin families remain open.

21.1. **Uniform escape and its exact local form.** For $D = \prod r^{e_r}$, put

$$R(D) = \prod_r r^{\lceil e_r/2 \rceil}. \tag{59.1}$$

Thus $D \mid A^2$ if and only if $R(D) \mid A$; this is the same radical with half-rounded exponents denoted $R_0(D)$ earlier.

Definition 21.1 (uniform polynomial escape; source Definition 59.1)**.** Let $f \in \mathbb{Z}[X]$ have degree at least one and be eventually positive on the positive integers. It is a *uniform polynomial escape* if there is an m_0 such that

$$W(f(m)) = +\infty \qquad (m \geq m_0). \tag{59.2}$$

The threshold permits the finitely many negative, zero, or exceptional initial values and nothing else. A polynomial identity $f = g^2$, with $g \in \mathbb{Z}[X]$ nonconstant, is a uniform escape by Theorem 20.1; take m_0 beyond the finitely many integral zeros of g. Constants are excluded. A positive constant c would qualify exactly when $W(c) = +\infty$, so including degree zero would build the open constant classification into the definition. Known examples include all positive square constants, in particular the anomalies 1 and 4 in (58.1).

Lemma 21.2 (exact local reduction; proved; source Lemma 59.1)**.** *An eventually positive $f \in \mathbb{Z}[X]$ of positive degree is a uniform polynomial escape if and only if, for every $M \equiv 3 \pmod 4$ and every $D \mid ((M+1)/4)^2$,*

$$f(x) \equiv -4D \pmod M \tag{59.3}$$

has no solution $x \pmod M$.

Proof. If (59.3) has a solution, its residue class contains infinitely many positive integers m, including arbitrarily large ones beyond both the positivity and escape thresholds. Equation (58.3) then gives $W(f(m)) \leq M$, a contradiction. Conversely, any finite witness for a tail value $f(m)$ supplies exactly one congruence (59.3) by the complete original-data/divisor-form equivalence in (58.3). Thus absence of every local solution is equivalent to (59.2). □

There is a distinct progression variant. Fix $a \pmod q$ and require (59.2) only for sufficiently large $m \equiv a \pmod q$. The corresponding criterion is that (59.3) have no integral solution satisfying

$$x \equiv a \pmod{(M, q)}. \tag{59.4}$$

Indeed, (59.4) is exactly the Chinese-remainder compatibility condition between a solution modulo M and the chosen class modulo q. This variant can admit more escape mechanisms and is not classified below. It must also be separated from *finite-depth* escape: for fixed T, the choice $q = \operatorname{lcm}(1, \dots, T)$, $a = 1$, and $f(X) = X$ gives the Section 17 class with $W(m) > T$. It is not an infinite-depth escape. For any fixed q, a prime $\ell \equiv 3 \pmod 4$, $\ell \nmid q$, lets the Chinese remainder theorem combine $m \equiv a \pmod q$ with $m \equiv -4 \pmod \ell$, and the $D = 1$ class then hits that progression.

21.2. **The cyclotomic obstruction in every degree.** The $D = 1$ subsystem already has a precise Galois meaning. The other shifts give a useful strengthened form of the same statement.

Theorem 21.3 (root-field/cyclotomic obstruction; proved using qualitative Chebotarev; ineffective; source Theorem 59.2)**.** *Fix $D \geq 1$, put $Q_D = 4R(D)$, and let $C_D = \mathbb{Q}(\zeta_{Q_D})$. Let τ_D be complex conjugation on C_D. For a root α of*

$$h_D(X) = f(X) + 4D, \tag{59.5}$$

put $F_\alpha = \mathbb{Q}(\alpha)$. Apart from the finite ramified and leading-coefficient primes, the polynomial h_D has no root modulo any prime

$$p \equiv -1 \pmod{Q_D} \tag{59.6}$$

if and only if

$$\tau_D|_{F_\alpha \cap C_D} \neq 1 \quad \text{for every root } \alpha. \tag{59.7}$$

Consequently every uniform polynomial escape satisfies (59.7) for every D. At $D = 1$, this specializes to

$$\mathbb{Q}(i) \subseteq \mathbb{Q}(\alpha) \quad \text{for every root of } f(X) + 4. \tag{59.8}$$

Proof. The eligibility condition for a prime modulus is

$$D \mid ((p+1)/4)^2 \iff R(D) \mid (p+1)/4 \iff p \equiv -1 \pmod{Q_D}. \tag{59.9}$$

Suppose first that τ_D is trivial on $F_\alpha \cap C_D$. The identity on F_α and τ_D on C_D then agree on the intersection, so they glue on $F_\alpha C_D$. Extend the resulting automorphism to a Galois closure of h_D composited with C_D. It fixes α and restricts to τ_D on the cyclotomic field. Qualitative Chebotarev supplies infinitely many unramified primes in its conjugacy class. Their cyclotomic Frobenius is -1, hence they satisfy (59.6), while their polynomial Frobenius fixes a root, so h_D has a root modulo each of them.

Conversely, a good prime in (59.6) at which h_D has a root has a Frobenius element restricting to τ_D and fixing some root. Its restriction forces τ_D to be trivial on that root field's intersection with C_D. This proves the all-but-finitely-many equivalence. Lemma 21.2 and the prime data (59.9) give the necessary condition for a uniform escape. When $D = 1$, the cyclotomic field is $\mathbb{Q}(i)$; its intersection with a root field is either $\mathbb{Q}$ or $\mathbb{Q}(i)$, and complex conjugation is nontrivial only in the second case. This is (59.8). □

No effective least prime is extracted here. In permutation language, if K is the splitting field of $f+4$, the union of conjugacy classes in $\mathrm{Gal}(K/\mathbb{Q})$ which fix a root lies inside $\mathrm{Gal}(K/\mathbb{Q}(i))$. The root-field formulation (59.8), rather than the looser statement merely that K contains i, is the load-bearing conclusion. For example, the nonsquare polynomial $f = X^4 - 3$ passes the $D = 1$ test because $f + 4 = X^4 + 1$ has root field $\mathbb{Q}(\zeta_8)$. It is not a uniform escape: $(M, D, x) = (7, 2, 2)$ solves (59.3). Thus the other D's really are needed.

21.3. **Complete classification through degree two.**

Theorem 21.4 (linear and quadratic uniform escapes; proved; source Theorem 59.3)**.** *Let $f \in \mathbb{Z}[X]$ be eventually positive and $1 \le \deg f \le 2$. Then f is a uniform polynomial escape if and only if*

$$f(X) = g(X)^2 \tag{59.10}$$

for some $g \in \mathbb{Z}[X]$. In particular there is no linear uniform escape. The identity (59.10) is an identity of polynomials, not merely an agreement after the threshold m_0.

Proof. If f is linear, $f + 4$ has a root modulo every prime not dividing its leading coefficient. Infinitely many primes are $3 \pmod 4$, and only finitely many divide that coefficient, so one of them contradicts the eligible $D = 1$ subsystem. Thus no coprimality assumption on the leading coefficient is hidden here.

Now write

$$f(X) = aX^2 + bX + c. \tag{59.11}$$

Eventual positivity gives $a > 0$; thus the $a < 0$ sign is excluded before any character argument. Theorem 21.3 at $D = 1$ says that the root field of $f + 4$ contains $\mathbb{Q}(i)$. A reducible quadratic, including the degenerate discriminant-zero case, has a rational root field and is therefore impossible. An irreducible quadratic has only its own quadratic root field, so that field must equal $\mathbb{Q}(i)$. This rules out every positive discriminant and every negative square class other than that of -1. Equivalently its integral discriminant is

$$\Delta_1 = b^2 - 4a(c + 4) = -r^2 \tag{59.12}$$

for a nonzero integer r.

For a general fixed D, the discriminant of $f + 4D$ is

$$\Delta_D = -S_D, \qquad S_D = r^2 + 16a(D - 1) = 16aD + B, \qquad B = r^2 - 16a. \tag{59.13}$$

Suppose $B \neq 0$. Choose an odd prime $q \nmid 2aB$. The linear congruence $16aD + B \equiv 0 \pmod q$ has q lifts modulo q^2, exactly one of which also vanishes modulo q^2. Choose a positive D in one of the other lifts. Then

$$q \parallel S_D, \qquad q \nmid D, \qquad q \nmid Q_D = 4R(D). \tag{59.14}$$

The quadratic field $\mathbb{Q}(\sqrt{-S_D})$ consequently has fundamental conductor divisible by q, so it is not contained in $\mathbb{Q}(\zeta_{Q_D})$. Its intersection with that cyclotomic field is therefore $\mathbb{Q}$. Complex conjugation acts trivially on the intersection, contradicting Theorem 21.3. Hence $B = 0$, so $r^2 = 16a$. Write $r = 4d$; then $a = d^2$. Equation (59.12) now reduces to $b^2 = 4d^2c$. Thus $b = 2de$ and $c = e^2$ for an integer e, and

$$f(X) = (dX + e)^2. \tag{59.15}$$

The divisibilities follow prime by prime from $4d^2 \mid b^2$. Conversely every polynomial (59.15) escapes by Theorem 20.1. □

The proof shows why using only a short fixed list of shifts would be misleading: the decisive D is selected from the coefficients through a new prime q. Concrete nonsquare replays which pass the *whole* $D = 1$ prime layer are

f	$\mathrm{disc}(f+4)$	(M, D, x) hitting $f(x) + 4D$
$2X^2 + 2X - 3$	-4	$(11, 3, 3)$
$X^2 - 4X + 1$	-4	$(7, 2, 5)$
$X^2 - 6X + 30$	-100	$(11, 3, 3)$.

Here no prime $3 \pmod 4$ divides the displayed square root of the negative discriminant. Verifier block `(bf)` checks the $D = 1$ assertion over a finite prime range and each exact later-shift hit. The theorem, not that finite replay, proves that every nonsquare quadratic is eventually hit.

Conjecture C_{POLY} (full polynomial escape classification). Every uniform polynomial escape in Definition 21.1 is a square in $\mathbb{Z}[X]$.

This is a falsifiable higher-degree extension of Theorem 21.4, not a theorem. Theorem 21.3 is only a necessary local structure theorem there. Root fields such as $\mathbb{Q}(\zeta_8)$ show that its $D = 1$ condition alone does not imply a square polynomial; pinning all shifts (59.5) in arbitrary degree is the remaining algebraic problem.

Corollary 21.5 (prime values in the classified range; proved; source Corollary 59.4)**.** *No positive-degree uniform escape of degree at most two takes a positive prime value. Hence no linear or quadratic polynomial family can produce an unbounded sequence of primes with $W = +\infty$. Conditional on C_{POLY}, the same statement holds for every polynomial degree.*

Proof. By Theorem 21.4 every positive value is an integer square. It is then either 1 or composite. The conditional extension is identical. □

This is the Type-II polynomial-level mirror of Corollary 15.2 only in a qualified sense. Type I has no third fixed congruence-level vanishing law; Type II does have the square polynomial law, now proved unique through degree two. Corollary 21.5 closes the polynomial-family route to permanent Type-II-less primes only in that range. Exponential sequences, recurrence orbits, arbitrary thin sets, the progression variant (59.4), and all higher-degree polynomials are outside its unconditional scope.

21.4. **The all-integer cubic tail fails at the square crossing.**

Proposition 21.6 (integer-tail obstruction; proved; source Proposition 59.5)**.** *For every $N > T \geq 1$,*

$$\#\{n : T < n \leq N,\ W(n) > T\} \geq \lfloor \sqrt{N} \rfloor - \lfloor \sqrt{T} \rfloor. \tag{59.16}$$

Consequently an all-integer analogue of the cubic hypothesis (57.2),

$$\#\{n : T < n \leq N,\ W(n) > T\} \leq CN \exp\{-c(\log T)^3\}, \tag{59.17}$$

cannot hold uniformly in any window containing

$$\log T = \lambda(\log N)^{1/3} \quad \textit{with} \quad \lambda > (1/(2c))^{1/3}. \tag{59.18}$$

Any unshifted all-integer analogue also bounds the moving set in (59.17), so the same contradiction applies. In particular it fails in every power window $\log T \leq b(\log N)^\theta$ with $b, c > 0$ and $\theta > 1/3$. At $\theta = 1/3$, the square obstruction contradicts (59.17) when $cb^3 > 1/2$; equality leaves the prefactor and endpoint floors to decide. The constant $(1/(2c))^{1/3}$ is the exact crossing for this square-lower-bound comparison, not a claim that no other integer obstruction acts earlier.

Proof. Every square in $(T, N]$ has $W = +\infty$ by Theorem 20.1, which proves (59.16). Put $L = \log N$ and $t = \log T$. At $t = \lambda L^{1/3}$, one has $T = o(N)$, and the logarithms of the square lower bound and the proposed upper bound are respectively

$$\frac{1}{2}L + o(L), \qquad \log C + L - c\lambda^3 L. \tag{59.19}$$

The upper bound is smaller when $c\lambda^3 > 1/2$, a contradiction. If $\theta > 1/3$, choose any fixed $\lambda > (1/(2c))^{1/3}$. Then $t = \lambda L^{1/3}$ lies inside the window $t \le bL^\theta$ for all large L, and still has $T < N$; applying the bound at that interior point gives the contradiction. □

Thus Section 19's count-below-one implication is prime-essential before one even asks whether its open window hypothesis can be proved. This sharpens the integer/prime dichotomy (58.17): for the supercritical extension needed there, the prime restriction in Theorem 14.2 is logically necessary, not cosmetic. The presently realized $\theta = 1/4$ window and all its inherited status labels are unchanged.

21.5. **The refuted complete-square proposal and finite census. Former Conjecture C_{SQ} (complete square escape; refuted in Section 22).**

$$\{n \in \mathbb{Z}_{>0} : W(n) = +\infty\} = \{m^2 : m \in \mathbb{Z}_{>0}\}. \tag{59.20}$$

Source Section 59 proposed (59.20), whose right side already includes 1 and 4. The computer-assisted Theorem 22.3 now refutes it by proving three nonsquare escapes. Its implication structure remains useful: had (59.20) held, it would have been *strictly stronger* than Erdős–Straus, because every prime is a nonsquare and a finite Type-II witness yields an explicit three-unit-fraction representation by Lemma 3.4. The reverse implication was never claimed: Erdős–Straus alone neither forces this witness type nor says anything about composite nonsquares. The former conjecture would also have implied C_{POLY} by the following standard lemma. The repaired open statement C'_{SQ} appears in (60.22); C_{POLY} remains open.

Lemma 21.7 (eventual square values force a square polynomial; proved; source Lemma 59.6)**.** *If $f \in \mathbb{Z}[X]$ is eventually positive and $f(n)$ is a square for all sufficiently large integers n, then $f = g^2$ for some $g \in \mathbb{Z}[X]$.*

Proof. Remove the square factors of f in $\mathbb{Q}[X]$. If the remaining squarefree polynomial H is nonconstant, Schur's theorem gives infinitely many prime divisors of its values. Choose one avoiding the leading coefficient, discriminant, and the fixed rational content. It has a simple root $a \pmod p$. Of the p lifts of a modulo p^2, exactly one makes $p^2 \mid H(n)$; any other lift contains arbitrarily large n and makes the valuation of $f(n)$ odd, a contradiction. Thus the squarefree part is constant. That constant must be a rational square, because one nonzero square value has its square class. Hence f is a square in $\mathbb{Q}[X]$. Gauss's lemma then makes its square root integral: writing it as $(u/v)G$ with $(u, v) = 1$ and primitive $G \in \mathbb{Z}[X]$, integrality of $(u^2/v^2)G^2$ forces $v = 1$. □

Lemma 21.8 (twisted-square survivor reduction; proved; source Lemma 59.7)**.** *Put $n = sm^2$, where s is squarefree. If eligible (M, D), with $M = 4A - 1$ and $D \mid A^2$, hits n, then*

$$(M, sm) = 1, \qquad \left(\frac{s}{M}\right) = -1, \qquad s \nmid D. \tag{59.21}$$

Moreover, if $t \mid m$ and $t^2 \mid D$, then the hit is equivalent to the eligible auxiliary hit

$$M \mid s(m/t)^2 + 4D/t^2. \tag{59.22}$$

Conversely an auxiliary hit with divisor E lifts to sm^2 through $D = t^2E$ exactly when $t^2E \mid A^2$.

Proof. The proof of Theorem 20.1 gives $\left(\frac{D}{M}\right) = 1$ for every eligible D, while $\left(\frac{-1}{M}\right) = -1$. A hit makes $-4D$, hence n, a unit modulo M. Taking Jacobi symbols in $sm^2 \equiv -4D \pmod{M}$ gives the middle assertion of (59.21). If $D = sE$, then $M \mid s(m^2 + 4E)$; cancelling the unit s would make the square m^2 hit the eligible divisor E, contrary to Theorem 20.1. Finally

$$sm^2 + 4D = t^2\{s(m/t)^2 + 4D/t^2\}.$$

Here t is a unit modulo M, because $t^2 \mid D$ and $(D, M) = 1$. This proves both directions and the stated lift condition. □

For the three finite-census survivors, Lemma 21.8 makes the historical searches exact and substantially thinner.

- For $288 = 2 \cdot 12^2$, one must have $M \equiv 3 \pmod 8$, $3 \nmid M$, and D odd, and the residual congruence is $D \equiv -72 \pmod M$. Directly, if $D = 2E$, then $288 + 4D = 2(12^2 + 4E)$, an impossible square hit; the redundant deeper layer $D = 8E$ gives $8(6^2 + 4E)$. Thus no even-D layer was left in the scan.
- For $336 = 21 \cdot 4^2$, one must have $(M, 21) = 1$, $\left(\frac{M}{3}\right)\left(\frac{M}{7}\right) = -1$, $21 \nmid D$, and $D \equiv -84 \pmod M$. Here quadratic reciprocity gives $\left(\frac{21}{M}\right) = \left(\frac{M}{3}\right)\left(\frac{M}{7}\right)$. If $D = 21E$, then $336 + 4D = 21(4^2 + 4E)$, which is impossible. The exact scaling chain is $4 \mid D \Rightarrow 336 \to 84$ and $16 \mid D \Rightarrow 336 \to 21$. Although $W(21) = 11$ via $(M, E) = (11, 3)$, lifting that datum would require $16E = 48 \mid A^2 = 9$, which fails.
- For $4545 = 505 \cdot 3^2$, one must have $(M, 3 \cdot 5 \cdot 101) = 1$, $\left(\frac{M}{5}\right)\left(\frac{M}{101}\right) = -1$, and $505 \nmid D$, with $D \equiv -4545A \pmod M$. The forbidden layer is $4545 + 4(505E) = 505(3^2 + 4E)$. If $9 \mid D$, the datum scales down to one for 505. The least such datum, again $(11, 3)$, does not lift: $9 \cdot 3 \nmid 3^2$.

These are necessary filters plus an exact replay of the residual divisor congruence, not by themselves an infinitude proof. In particular the character-permitted moduli are nonempty, and the argument does not force a nonsquare at one prime factor of every such M. At the source-Section-59 stage these reductions therefore made no $W = +\infty$ claim; the additional finite normal form in Theorem 22.1 is what closes the search in Theorem 22.3.

Computational 59.8 (exact finite scan; historical C_{SQ} stress test). Verifier block `(bf)` replays the complete divisor harvest and, below $3 \cdot 10^5$, removes the known squares from the tail. The exact nonsquare counts are

T	100	300	1000	3000
$\#\{n \le 3 \cdot 10^5 : n \ne \square,\ W(n) > T\}$	4969	517	22	3.

(59.23)

The three nonsquare survivors in the last column, now proved by Theorem 22.3 to have $W = +\infty$, are

$$288, \qquad 336, \qquad 4545. \tag{59.24}$$

The default independent smallest-prime-factor scan uses Python integers for every divisor product and checks every eligible modulus through $3 \cdot 10^6$. At the Section-59

stage it proved only the finite statement

$$W(288), W(336), W(4545) > 3 \cdot 10^6. \tag{59.25}$$

The wave-23 deep run checked all moduli through $1.5 \cdot 10^8$, after the character filters above, and enumerated 2,878,826,874 divisor values without a hit. Thus the stronger exact computational frontier at that stage was

$$W(288), W(336), W(4545) > 1.5 \cdot 10^8. \tag{59.26}$$

The global ceilings in (60.19) now show retrospectively that this finite run had crossed the whole witness space, while Theorem 22.3 supplies the proof through the finite normal form and an independent exhaustion. The historical run used Python integers throughout; it peaked below 170 MB. The modulus/divisor phase streams one value of A at a time and retains no old divisor list. Previous external quick-probe values $506063, 754379, 652271$, respectively, came from fixed-width integer overflow, including impossible negative divisors. They are false and are explicitly rejected: they are not W-values. Setting `ES_FULL_SCAN=1` reproduces (59.26). It also raises the population endpoint to 10^6, where the nonsquare counts at the same threshold grid are $(17007, 1841, 77, 4)$. The fourth depth-3000 survivor is 643245 and has the exact witness $W(643245) = 3119$.

As a family control, block `(bf)` computes every least finite W for $1 \le m \le 60$ in the three natural twisted-square families. The table records histograms $\{W : \#m\}$; the single lower-bound entry in each row is the former finite-census survivor in (59.24).

n	$\{W(n) : \#m\}, \quad 1 \le m \le 60$	deep survivor
$2m^2$	$\{3 : 40, 11 : 11, 19 : 4, 35 : 1, 59 : 2, 131 : 1\}$	$m = 12$
$21m^2$	$\{11 : 33, 19 : 9, 23 : 15, 31 : 1, 71 : 1\}$	$m = 4$
$505m^2$	$\{11 : 33, 23 : 23, 47 : 1, 87 : 1, 107 : 1\}$	$m = 3.$

(59.27)

Thus the factorization $n = sm^2$ is not by itself an escape mechanism: 177 of the 180 controls are hit by $M \le 131$. The three exceptional multipliers are highly atypical in this finite sample. This observation is not the proof of their escape; Theorem 22.3 uses the separate exhaustive normal-form certificate. The same block checks Lemma 21.8 exactly over a bounded complete harvest, pins all 180 table entries, tests three nonsquare quadratics passing the $D = 1$ layer, replays the discriminant-character dichotomy, and verifies the coefficient-dependent construction (59.14). None of these finite checks substitutes for Chebotarev or for an unbounded classification.

Walls and relation to the frontier. The three values in (59.24) were the live finite stress test for C_{SQ}, not positive evidence for it. Section 22 now proves that all three have $W = +\infty$, refutes C_{SQ}, and replaces it by open C'_{SQ}. This does not alter the degree-two classification or prove that there are no further nonsquare escapes. It also yields no unconditional pointwise prime conclusion. In higher polynomial degree, Theorem 21.3 remains only a necessary, qualitatively ineffective obstruction, and C_{POLY} remains open.

22. Witness duality, finite decidability, and the survivor frontier

Status and correction to the proposed frontier. The dual criterion, its finite normal form, the bound on every possible datum, and the partial swap symmetry

below are proved. The resolution of the three Section 21 survivors is a **computer-assisted theorem**: the reduction is proved, and verifier block `(bg)` exhausts its small exact integer certificate by two parameterizations. The displayed replay and near-miss tables are **Computational**, exact only in their stated ranges. No prime-tail or Erdős–Straus theorem follows.

The suggestion that fixed small a can correspond to arbitrarily large M is false for fixed n. Eligibility on the M-side forces a second divisor condition on the a-side. Together the two conditions put every datum for fixed n in a finite box. This correction resolves, rather than merely extends, the survivor frontier.

22.1. **Exact duality, cancellation, and the finite normal form.** Call (M, D) a witness datum for n when

$$M \equiv 3 \pmod 4, \qquad A = \frac{M+1}{4}, \qquad D \mid A^2, \qquad M \mid n + 4D. \tag{60.1}$$

This is the complete harvested criterion (58.3), not a selected Type-II subfamily. It reparameterizes Lemma 3.4's intrinsic divisor system; it is not a new Egyptian-fraction identity family. The new content is the dual cancellation ledger and finite normal form.

Theorem 22.1 (witness duality and finite normal form; proved; source Theorem 60.1)**.** *Let $n \geq 1$.*

(1) A datum (60.1) exists if and only if there are positive integers a, D such that, on putting

$$a \equiv -n \pmod 4, \qquad h = \frac{n+a}{4}, \qquad A = \frac{D+h}{a}, \tag{60.2}$$

one has

$$D \mid h^2, \qquad a \mid D + h, \qquad D \mid A^2. \tag{60.3}$$

The two maps are

$$a = \frac{n+4D}{M}, \qquad M = 4A - 1 = \frac{n+4D}{a}, \tag{60.4}$$

and are inverse. Thus all quantities in (60.2)–(60.4) are positive integers.

(2) Once $D \mid h^2$ and $a \mid D + h$ hold, the exact cancellation ledger is

$$D \mid h^2 \iff \frac{D}{(D, a^2)} \mid A^2, \qquad h \equiv aA \pmod D. \tag{60.5}$$

Hence the last check $D \mid A^2$ in (60.3) is redundant when $(a, D) = 1$, but not in general. If $q \mid (a, D)$, then $q \mid n$. More precisely, for every prime q, with $d = v_q(D)$, $\alpha = v_q(a)$, and $\beta = v_q(A)$, the dual condition certifies

$$\max(d - 2\alpha, 0) \leq 2\beta, \tag{60.6}$$

whereas eligibility requires $d \leq 2\beta$. This includes $q = 2$: if D is even, no cancellation of the powers of 2 in a^2 is free.

(3) There is a canonical normal form involving no divisors of large numbers. Witness data are in bijection with positive integers (g, u, v, d) satisfying

$$(u, v) = 1, \qquad d \mid g, \qquad a := 4gv - n \geq 1, \qquad au = d + v, \tag{60.7}$$

through

$$A = gu, \qquad h = gv, \qquad D = gd, \qquad M = 4gu - 1. \tag{60.8}$$

If $B = \lfloor (n+1)/3 \rfloor$, *every datum satisfies*

$$g, v \le B, \qquad a \le 2B, \qquad u \le 2B, \qquad M \le 8B^2 - 1. \tag{60.9}$$

Proof. From (60.1), define a by (60.4). Reduction modulo 4 gives $a \equiv -n \pmod 4$. Direct expansion, stronger than merely squaring a congruence, gives

$$aA = \frac{a(M+1)}{4} = D + \frac{n+a}{4} = D + h. \tag{60.10}$$

Thus $h \equiv aA \pmod D$, and $D \mid A^2$ implies $D \mid h^2$. Conversely, (60.2)–(60.3) make A integral and positive. With $M = 4A - 1$, equation (60.10) and $4h = n + a$ give $aM = n + 4D$; the last condition in (60.3) gives eligibility. This proves the first equivalence.

Prime by prime, $D \mid a^2A^2$ says $d \le 2\alpha + 2\beta$. Removing from D the part already supplied by a^2 gives exactly (60.5)–(60.6). Also $(a, D) \mid aM - 4D = n$. This proves the cancellation claims, including the even branch.

For the normal form, start with a datum and put

$$g = (A, h), \qquad u = A/g, \qquad v = h/g. \tag{60.11}$$

The divisor D divides both A^2 and h^2, hence $D \mid g^2$. Equation $D = aA - h$ also gives $g \mid D$. Therefore $D = gd$ with $d \mid g$, and division of (60.10) by g gives $au = d + v$. Conversely, (60.7)–(60.8) give $D \mid g^2 \mid A^2$ and (60.10), hence a witness. The coprimality in (60.7) makes the construction canonical.

Finally $au = d + v \le g + v$, so $a \le g + v$ and $u \le g + v$. Since $a = 4gv - n$,

$$4gv \le n + g + v. \tag{60.12}$$

Using first $v \ge 1$ and then $g \ge 1$ in (60.12) gives $3g \le n + 1$ and $3v \le n + 1$. Thus $g, v \le B$, and the remaining bounds in (60.9) follow from $u, a \le g + v$ and $M = 4gu - 1$. No comparison or swap between a and M is used. □

The factor 4 in the seed congruence can therefore be removed without any oddness assumption: (60.10) always gives the necessary condition $D \mid ((n+a)/4)^2$. Oddness matters only when one tries to cancel a and infer eligibility from that condition.

For $n = 288$, an eligible D must be odd. Indeed, if $D = 2E$, then $E \mid A^2$ and

$$M \mid 288 + 4D = 2(144 + 4E),$$

so the odd M would make the square $144 = 12^2$ hit the eligible divisor E, contradicting Theorem 20.1. Lemma 21.8 gives $M \equiv 3 \pmod 8$, and the defining equation then gives $a \equiv 4 \pmod 8$. Write $a = 4r$, with r odd. Theorem 22.1 becomes the exact criterion

$$\begin{gathered} D \mid (72 + r)^2, \qquad r \mid D + 72, \qquad M = \frac{D + 72}{r} \equiv 3 \pmod 8, \\ D \mid \left(\frac{M+1}{4} \right)^2. \end{gathered} \tag{60.13}$$

The last line is redundant if $(r, D) = 1$. In the noncoprime branch, $(r, D) \mid 72$, and oddness reduces this to $(r, D) \mid 9$; a 3-adic eligibility check remains essential. Bound (60.9) gives $a \le 192$, hence only odd $r \le 47$. The apparently unbounded quadratic sequence in (60.13) is finite once both sides of the duality are enforced.

For 336, write $a = 4r$ and $h = 84 + r$. The exact quotient conditions are

$$D \mid (84 + r)^2, \quad r \mid D + 84, \quad M = (D + 84)/r \equiv 3 \pmod 4, \quad D \mid ((M+1)/4)^2, \tag{60.13a}$$

in addition to the proved necessary conditions $(M, 21) = 1$, $(M/3)(M/7) = -1$, and $21 \nmid D$. Here $(a, D) \mid 336$, so powers of 2, 3, and 7 are precisely the possible cancellation branches; (60.9) gives $a \leq 224$. For 4545, one has $a \equiv 3 \pmod 4$, $h = (4545 + a)/4$, and the unshortened criterion (60.2)–(60.3), with $(a, D) \mid 4545 = 3^2 \cdot 5 \cdot 101$. Its additional necessary filters are $(M, 4545) = 1$, $(M/5)(M/101) = -1$, and $505 \nmid D$, and $a \leq 3030$. Since this last a is odd, its 2-adic dual condition already equals 2-adic eligibility; only primes 3, 5, and 101 can be lost to cancellation.

22.2. **What swaps, and what does not.** The equation in (60.4) is symmetric in a, M, but its eligibility data are not automatically symmetric.

Lemma 22.2 (partial witness involution; proved; source Lemma 60.2)**.** *Let (M, D, a) be a witness triple for n. Swapping M and a gives another witness triple for the same n if and only if*

$$a \equiv 3 \pmod 4, \qquad D \mid \left(\frac{a+1}{4}\right)^2. \tag{60.14}$$

On this bi-eligible subset the map

$$(M, D, a) \longmapsto (a, D, M) \tag{60.15}$$

is a genuine involution, and $W(n) \leq \min(a, M)$. Such a swap can occur only when $n \equiv 1 \pmod 4$.

Proof. The swapped equation is the same equation. Conditions (60.14) are exactly the modulus class and harvested-divisor eligibility conditions for modulus a. Applying the swap twice is the identity. Finally $a \equiv -n \pmod 4$, so $a \equiv 3 \pmod 4$ forces $n \equiv 1 \pmod 4$. □

For example, $(M, D, a) = (7, 1, 3)$ and $(3, 1, 7)$ are the paired data for $n = 17$. The triple $(7, 1, 7)$ for $n = 45$ is a fixed point. There is no universal involution even when $n \equiv 1 \pmod 4$: $(23, 36, 7)$ is a datum for 17, but $36 \nmid ((7 + 1)/4)^2 = 4$.

Lemma 3.4 is the honest source of eligibility on each side: the bi-eligible set is an intersection of two intrinsic harvested-divisor systems. It is not that lemma's divisor involution $D \mapsto A^2/D$, nor the four-parameter Case-B dictionary: equation (60.4) merely records the positive quotient of one intrinsic residue congruence. Thus the raw symmetry was latent in that congruence, while closure under the swap is the additional condition (60.14), not a universal symmetry.

22.3. **The survivor frontier closes.** Before using the finite normal form, the suggested two-front scan has the wrong geometry. From $D \mid h^2$,

$$M = \frac{n + 4D}{a} \leq \frac{a}{4} + \frac{n}{2} + \frac{n^2/4 + n}{a}. \tag{60.16}$$

From the original eligibility $D \leq ((M + 1)/4)^2$,

$$a \leq \frac{M}{4} + \frac{1}{2} + \frac{n + 1/4}{M}. \tag{60.17}$$

Consequently fixed small a never sees arbitrarily large M for fixed n. An M-scan through M_0 and an a-scan through A_0 cover exactly

$$\{(M, D, a) : M \leq M_0 \text{ or } a \leq A_0\}; \tag{60.18}$$

before using further structure; the uncovered region is exactly $M > M_0, a > A_0$. It is not a mysterious middle-divisor regime. Theorem 22.1 puts the whole witness space in (60.9), so $A_0 = 2B$, or $M_0 = 8B^2 - 1$, leaves no uncovered region.

For the former finite-census survivors, the rigorous all-data bounds are

n	B	$a_{\max} = 2B$	$M_{\max} = 8B^2 - 1$
288	96	192	73,727
336	112	224	100,351
4545	1515	3030	18,361,799

(60.19)

Thus even the historical exact $M \le 1.5 \cdot 10^8$ scan in (59.26) crossed each now-proved global ceiling. Block `(bg)` supplies a much smaller independent certificate in the new coordinates.

Theorem 22.3 (three nonsquare escapes; computer-assisted, proved; source Theorem 60.3)**.**

$$W(288) = W(336) = W(4545) = +\infty. \tag{60.20}$$

Proof. Theorem 22.1 proves that every possible datum occurs with $1 \le a \le 2B$. Block `(bg)` enumerates every admissible a in those three finite intervals, every positive divisor $D \mid h^2$, and checks both $a \mid D + h$ and the uncancelled eligibility $D \mid A^2$, using Python integers. It finds no eligible row. Independently, the same block enumerates every (g, v) in (60.9), every $d \mid g$, and equation (60.7), again finding no row. These are exhaustive finite certificates, not extrapolations from a cutoff. Equivalence in Theorem 22.1 proves (60.20). □

Here "escape" means escape from the harvested Lemma 3.4 witness statistic W, not escape from Egyptian-fraction solvability. None of these composite integers is an Erdős–Straus counterexample; explicitly,

$$\frac{4}{288} = \frac{1}{144}+\frac{1}{216}+\frac{1}{432}, \qquad \frac{4}{336} = \frac{1}{168}+\frac{1}{252}+\frac{1}{504}, \qquad \frac{4}{4545} = \frac{1}{1515}+\frac{1}{6060}+\frac{1}{18180}. \tag{60.20a}$$

Computational 60.1 (exact survivor certificate and near misses). Verifier block `(bg)` gives

n	$D \mid h^2$	$a \mid D + h$	§59 necessary filters	$D \mid A^2$	(g, v) pairs	(g, v, d) tests
288	756	43	4	0	66	209
336	912	65	37	0	78	262
4545	22,995	188	62	0	1,152	5,405

(60.21)

The Section 21 filter column applies, respectively, the odd-D, $M \equiv 3 \pmod 8$, and unit conditions; the 21-Jacobi and $21 \nmid D$ conditions; and the 505-Jacobi and $505 \nmid D$ conditions. Every row surviving that column has $(a, D) > 1$, and every one fails exactly the uncancelled prime-power eligibility check. For 288, all four have $(a, D) = 9$ and fail at 3. This is the gcd caveat in (60.5), not an overflow artifact.

Assessment 60.1 (the failed character route). There is no global mod-8 obstruction at the proposed first layer: for example, $D = 1$ and $r \equiv 3 \pmod 8$ already satisfy $D \mid (72 + r)^2$ and $D \equiv 3r \pmod 8$. The complete conditions add the quotient and uncancelled eligibility constraints. Theorem 22.1 makes their range finite, and (60.21) shows that eligibility, not the Jacobi filters, is the final killer. An asymptotic expected-count argument over unbounded r therefore addresses a nonexistent frontier.

Heuristic 60.1 (uniform-residue comparison only). If the divisors in the first numerical column of (60.21) were uniform modulo each a, the naive expected counts for $a \mid D+h$, namely $\sum_a \tau(h^2)/a$, would be $13.266827, 15.617351, 49.909726$. The observed exact counts are $43, 65, 188$. This mismatch is unsurprising for divisors of a square; neither list predicts an eligible witness. The comparison is a finite diagnostic, not evidence for a tail law.

Conjecture C_{SQ} in (59.20) is therefore refuted. The minimal repair supported by current proofs is:

Conjecture C'_{SQ} (repaired complete escape set; open).

$$\{n \geq 1 : W(n) = +\infty\} = \{m^2 : m \geq 1\} \cup \{288, 336, 4545\}. \tag{60.22}$$

Only the inclusion from right to left is proved. No claim is made that the three sporadics form a family or that there are no further nonsquare escapes. Since all three are composite, (60.22), if proved, would still imply Erdős–Straus. It would also imply C_{POLY}: an unbounded polynomial cannot remain in the three-value sporadic set, and Lemma 21.7 handles eventual square values.

22.4. **The statistic and the prime side.** Theorem 22.1 gives an exact terminating algorithm for $W(n)$. Set $B = \lfloor (n+1)/3 \rfloor$ and enumerate

$$1 \leq a \leq 2B, \qquad a \equiv -n \pmod 4, \qquad h = (n+a)/4, \qquad D \mid h^2. \tag{60.23}$$

Retain $a \mid D+h$ and $D \mid ((D+h)/a)^2$, and minimize $M = 4(D+h)/a - 1$. An empty list proves $W(n) = +\infty$. Its divisor work is exactly

$$\sum_{\substack{a \leq 2B \\ a \equiv -n \ (4)}} \tau\big(((n+a)/4)^2\big), \tag{60.24}$$

which has the usual average order $O(n(\log n)^2)$, apart from factoring cost; in particular the corresponding per-prime decision at $n = p$ has $O(p(\log p)^2)$ average divisor work. A smallest-prime-factor sieve suffices at moderate scale, while a segmented sieve keeps large censuses memory-bounded. By contrast, deciding from the original side using only (60.9) scans $O(n^2)$ moduli and has divisor work of average order $O(n^2(\log n)^2)$. The a-algorithm is a genuine dimension reduction, not an extension to large M at fixed small a.

For an odd prime p, the gcd caveat disappears completely. Indeed, $(a, D) \mid p$, while (60.9) gives $a \leq 2\lfloor (p+1)/3 \rfloor < p$. Hence $(a, D) = 1$, and (60.5) makes the last eligibility check redundant. We obtain the proved equivalence

$$\begin{aligned} W(p) < +\infty \quad \Longleftrightarrow \quad & \exists a, D : \ 1 \leq a \leq 2\lfloor (p+1)/3 \rfloor, \quad a \equiv -p \pmod 4, \\ & h = (p+a)/4, \quad D \mid h^2, \quad a \mid D+h, \\ & M = 4(D+h)/a - 1, \end{aligned} \tag{60.25}$$

and the least such M is $W(p)$. The prime $p = 2$ has the known separate datum $(M, D, a) = (3, 1, 2)$. Formula (60.25) is a cheaper exact prime test and a structural simplification of the cancellation branches. It does not prove that the finite list is nonempty for every prime, does not control pointwise tails, and does not prove Erdős–Straus. In particular, the gcd-structural mechanism behind the three sporadics is unavailable to odd primes.

22.5. **Verification and finite coverage ledger. Computational 60.2 (exact bounded replay).** Verifier block `(bg)` replays all 33,882 incidences (n, M, D) with $n \le 2000$, $M \le 3000$, and $D \mid ((M+1)/4)^2$ from the complete `(bc)` harvest. Every incidence maps through (60.2)–(60.8) and back exactly. Of these, 11,451 have $(a, D) > 1$, so the cancellation branch is materially represented rather than spot-checked.

The same block brute-enumerates 1,070,466 divisor values for $M \le 10^5$ and $n \in \{2, 17, 45, 288, 336, 4545\}$. There are nine witness data in that toy box. The union of the $M \le 100$ scan and the $a \le 30$ scan contains exactly the rows satisfying (60.18): eight arrive from the M-side, nine from the a-side, and eight are in the overlap. This is a finite double-count check of the region identity, not a proof of its unbounded form; (60.18) itself is set-theoretic.

Block `(bg)` also pins the swap examples, tests the swap criterion on the complete 33,882-incidence replay, pins (60.21), the two independent survivor exhaustions, and the heuristic sums. Of the replayed incidences, 1,196 are bi-eligible under (60.14), including 54 fixed points; all have $n \equiv 1 \pmod 4$. Block `(bf)` retains the independent Python-integer M-scan through $3 \cdot 10^6$ by default and through $1.5 \cdot 10^8$ under `ES_FULL_SCAN=1`. The latter is now a redundant historical cross-check beyond all three ceilings in (60.19), not the logical frontier.

Internal review attestation (wave 24). The maximum-severity review returned SOUND-AFTER-REPAIRS. Its repairs were expository/register repairs plus a malformed TeX spacing command; no mathematical conclusion or ceiling changed. It rederived $D \mid (A, h)^2$, $(A, h) \mid D$, and hence the canonical $A = gu, h = gv, D = gd$, $d \mid g$, $au = d + v$ form without assuming or swapping $a \le M$. Thus $au = d + v \le g + v$ gives $4gv \le n + g + v$, then $g, v \le \lfloor (n+1)/3 \rfloor$, and all four bounds in (60.9) are unconditional. As a separate stress test, a direct M-harvest for every represented $n \le 2000$ through $M = 7{,}118{,}223 > 2(8\lfloor 2001/3 \rfloor^2 - 1)$ checked 44,202 incidences, found no bound violation, and found no incidence past the global ceiling.

A fresh memory-bounded implementation factored each $A = (M+1)/4$, generated every positive divisor of A^2 as a Python integer, and tested $M \mid n + 4D$, using no character or dual filter. At the exact ceilings 73,727, 100,351, and 18,361,799 for 288, 336, and 4545, it tested respectively 748,810, 1,074,878, and 409,000,770 divisor incidences and found no witness. It streamed one A at a time, with at most 6,561 divisors for one A. This independently confirms Theorem 22.3.

Prime-adic rederivation confirmed (60.5)–(60.6), including at 2, and proved $(a, D) \mid n$. A separate dual scan reproduced the stage counts $(756, 43, 4, 0)$, $(912, 65, 37, 0)$, and $(22{,}995, 188, 62, 0)$. All filtered rows fail eligibility; the four $n = 288$ rows have $(a, D) = 9$ and fail at 3. An independent test of all 33,882 replay incidences found exactly 1,196 bi-eligible rows and 54 fixed points, with (60.14) necessary and sufficient and every such $n \equiv 1 \pmod 4$.

The full prescribed verifier completed under its 460-second timeout. Isolated block `(bg)` runs in under one second on the review host and pins the 33,882/11,451 replay, survivor counts, swap sample, and 1,070,466-divisor toy union. The optional historical full scan remains gated. These are internal exact checks, not external refereeing. As throughout, identifying witness data with W uses the already-proved Lemma 3.4/(58.3) harvested-family equivalence; nothing here upgrades that family to all Egyptian-fraction decompositions.

23. The complete decidable census, twisted-square laws, and the prime criterion

Status and headline. The forward normal-form sieve, the exact specializations, and the hard-prime criterion below are **proved**. The bounded censuses and tables are **Computational**, exact only in their displayed ranges. The default complete census reaches $n \le 200000$; the optional full replay reaches $n \le 10^6$. In both ranges the set with $W = +\infty$ is exactly the squares together with

$$288, \qquad 336, \qquad 4545. \tag{61.1}$$

Thus Conjecture C'_{SQ}, (60.22), survives this stress test; it is not proved. Every integer has a unique form sm^2 with s squarefree, so the coordinate description "twisted square" is not by itself a mechanism. The three squarefree kernels and multipliers are $(s, m) = (2, 12), (21, 4), (505, 3)$.

23.1. **A forward finite decision and the complete census.** The normal form can be run simultaneously in the opposite direction, without a separate quadratic modulus scan for each n.

Lemma 23.1 (forward normal-form sieve; proved; source Lemma 61.1)**.** *Fix $X \ge 1$. Every witness datum for some $n \le X$, and no false datum, occurs among*

$$\begin{gathered} 1 \le g \le \lfloor (X+1)/3 \rfloor, \qquad 1 \le v \le \left\lfloor \frac{X+g}{4g-1} \right\rfloor, \\ d \mid g, \qquad u \mid d+v, \qquad (u, v) = 1, \\ a = \frac{d+v}{u}, \qquad n = 4gv - a, \qquad M = 4gu - 1. \end{gathered} \tag{61.2}$$

Consequently, marking every resulting $n \le X$, while retaining its least M, decides $W(n)$ simultaneously for all $n \le X$.

Proof. Equations (60.7)–(60.8) give $a = (d+v)/u$, $n = 4gv - a$, and $M = 4gu - 1$, so every tuple in (61.2) is a canonical datum. It has $a \le d + v \le g + v < 4gv$, hence $n > 0$. Conversely every datum has $g \le \lfloor (n+1)/3 \rfloor$ by (60.9), and

$$n = 4gv - \frac{d+v}{u} \ge 4gv - g - v.$$

Thus $g \le \lfloor (X+1)/3 \rfloor$ and $(4g-1)v \le X + g$, exactly the bounds in (61.2). Also $gv \ge g + v - 1$, so the latter inequality gives

$$g + v \le \lfloor (X+4)/3 \rfloor = \lfloor (X+1)/3 \rfloor + 1;$$

this is the divisor-table bound used in verifier block `(bh)`. The remaining conditions are precisely the canonical normal form. □

The number of (g, v) pairs is hyperbolic, of order $X \log X$, rather than X^2. This preserves an exhaustive certificate while making a simultaneous census practical.

Computational 61.1 (complete bounded $W = +\infty$ census). Block `(bh)` applies Lemma 23.1 with Python integers throughout. Its exact results are

X	$\#\{n \le X : W(n) = +\infty\}$	# squares	nonsquares	(g, v) pairs	(g, v, d, u) tests
200000	450	447	288, 336, 4545	599581	24246108
1000000	1003	1000	288, 336, 4545	3400244	173713414

(61.3)

The first row is the default committed range. The second is reproduced only with `ES_FULL_SCAN=1`; it streams tuples and retains only the divisor sieve and least-W

array, so it remains memory-bounded. This path is the simultaneous canonical (g, v, d, u) sieve (61.2), not a per-n dual a-scan or an original-side M-sieve. Thus the exact headline is three nonsquare escapes through 200000 by default and through 10^6 in the optional full scan. This is finite evidence for C'_{SQ}, not evidence that its exceptional list is complete without a bound on n.

The committed two-method overlap harvests the original (M, D) side through $M = 100351$, tests 1074878 Python-integer divisor values and 73105 incidences, and compares every $n \le 3000$. It finds exactly the 54 squares and 288,336 unresolved. Every other nonsquare has an explicit least datum, while the two sporadics are scanned through their exact ceilings in (60.19); Theorem 20.1 supplies the square side. The forward and original methods agree on every finite least W, not merely on the survivor count. As a wider independent-method cross-check, the default block runs the forward form through $n \le 300000$ and compares its exact minimum with the original-side harvest for all 3202 primes $p \equiv 1 \pmod{24}$ there. Every value agrees. The census and tuple counts in (61.3) remain restricted to their displayed endpoints.

23.2. **The exact twisted-square criterion and first layers.** The character filter of Lemma 21.8 and the finite duality combine without discarding the cancellation branch.

Theorem 23.2 (finite twisted-square vanishing criterion; proved; source Theorem 61.2)**.** *Let $s > 1$ be squarefree, $m \ge 1$, $n = sm^2$, and $B = \lfloor (n+1)/3 \rfloor$. Then $W(sm^2) = +\infty$ if and only if the following finite set is empty:*

$$\mathcal{F}_{s,m} = \{(a, D) : 1 \le a \le 2B,\ a \equiv -sm^2 \pmod 4,\ h = (sm^2 + a)/4, \\ D \mid h^2, \quad a \mid D + h, \quad D \mid ((D+h)/a)^2\}. \tag{61.4}$$

For every row, with $A = (D+h)/a$ and $M = 4A - 1$, one automatically has

$$(M, sm) = 1, \qquad \left(\frac{s}{M}\right) = -1, \qquad s \nmid D, \qquad (a, D) \mid sm^2. \tag{61.5}$$

Thus the first three conditions in (61.5) are sound early filters, but the last test in (61.4) may be deleted only when $(a, D) = 1$.

Proof. Apply Theorem 22.1 with $n = sm^2$ to obtain the equivalence and finite bound. Lemma 21.8 gives the first three necessary conditions in (61.5), and (60.5) gives the gcd assertion and the exact uncancelled eligibility test. No parity case is omitted: (61.4) retains all powers of 2 in D. □

The three sporadics do not have one coarse local signature:

(s, m)	s mod 24	$\omega(s)$	$\gcd(s, m)$	$\#\{a \text{ classes}\}$	gcds of filtered near misses
$(2, 12)$	2	1	2	48	9 (4 times)
$(21, 4)$	21	2	1	56	3, 4, 6, 7, 8, 14, 16, 24, 28, 48, 112
$(505, 3)$	1	2	1	757	5, 9, 15, 45, 101, 303, 909

(61.6)

Here ω counts distinct prime factors. Block `(bh)` pins the complete gcd multiplicity distributions; only their supports are displayed. Neither s mod 24, factor count, (s, m), nor one fixed gcd explains all three. What they share is narrower: quotient-congruence near misses exist, every character-filtered near miss has $(a, D) > 1$, and every one fails the uncancelled eligibility test. This is a finite diagnosis, not a family theorem. In particular $m = 24, 36, 48$ in the $s = 2$ family are witnessed, so $m = 12$ is not one instance of a "multiple of 12" escape law.

There is nevertheless an exact infinite low-modulus law in that family.

Theorem 23.3 (the first layers for $2m^2$; proved; source Theorem 61.3)**.** *For every $m \geq 1$,*

$$\begin{aligned} W(2m^2) = 3 &\iff 3 \nmid m, \\ W(2m^2) = 11 &\iff 3 \mid m \quad \textit{and} \quad m \bmod 11 \in \{\pm 2, \pm 3, \pm 4\}. \end{aligned} \tag{61.7}$$

Moreover, if $3 \mid m$, the second residue condition fails, and

$$m \bmod 19 \in \{\pm 3, \pm 6, \pm 8\}, \tag{61.8}$$

then $W(2m^2) = 19$.

Proof. At $M = 3$, one has $A = D = 1$, and the hit condition is $3 \mid 2m^2 + 4$, equivalently $m^2 \equiv 1 \pmod 3$. This is exactly $3 \nmid m$. If it fails, the only additional modulus below 11 is 7; Lemma 21.8 excludes it because $(2/7) = 1$.

At $M = 11$, one has $A = 3$ and $D \in \{1, 3, 9\}$. The hit condition is $m^2 \equiv -2D \pmod{11}$. For $D = 1, 3, 9$ its roots are respectively $\pm 3, \pm 4, \pm 2$. Their union is exactly the six classes in (61.7). The already imposed $3 \mid m$ excludes $M = 3$, and the character calculation excludes $M = 7$, so these hits have least modulus 11; conversely a least-11 hit must lie in one of the three displayed root sets.

If all six mod-11 classes fail, the only additional modulus below 19 is 15, and it is excluded by $(2/15) = 1$. At $M = 19$, one has $A = 5$ and $D \in \{1, 5, 25\}$. The congruences $m^2 \equiv -2D \pmod{19}$ have roots respectively $\pm 6, \pm 3, \pm 8$. Hence (61.8) supplies a hit at 19, while the preceding arithmetic excludes every smaller eligible modulus. □

The theorem constructs many witnesses but does not classify vanishing in the remaining residue classes. In particular, it does not prove that $m = 12$ is the only escape in the $s = 2$ family.

Computational 61.2 (complete twisted-square box). Block `(bh)` computes the exact least $W(sm^2)$ for every squarefree $2 \leq s \leq 200$ and $1 \leq m \leq 30$, a total of 3630 entries. Only

$$(s, m) = (2, 12), (21, 4) \tag{61.9}$$

vanish in this box. Across the 3628 finite entries, the most frequent least values are

W	3	7	11	15	19	23	31	35	39	43	47	55
#	940	1064	449	154	177	334	86	31	51	30	104	24

(61.10)

The remaining 184 entries have W between 59 and 727. In the canonical encoding `s:W(s·1²),...,W(s·30²);...`, ordered by increasing squarefree s and with 0 for $+\infty$, the full 3630-entry codebook has SHA-256 digest (with the displayed line break ignored)

`3c6c31f96adce8ea9180cc86e93da841`
`e49d71f7a25a8d094e58c05fd5047007`.

The verifier pins this digest, the displayed histogram, and the per-s vanishing sets.

The third sporadic kernel, $s = 505$, is explicitly outside the box $s \leq 200$. Its first thirty entries are

$$\begin{aligned} (11, 11, +\infty, 23, 11, 11, 23, 47, 11, 11, 23, 11, 11, 23, 107, \\ 11, 11, 23, 23, 11, 11, 23, 11, 11, 23, 87, 11, 11, 23, 23), \end{aligned} \tag{61.11}$$

so only $m = 3$ vanishes there. Equations (61.9)–(61.11) show finite isolation, not a shared congruence family. The repaired global conjecture remains exactly C'_{SQ},

(60.22). Only the inclusion from right to left is proved:

$$\{m^2 : m \geq 1\} \cup \{288, 336, 4545\} \subseteq \{n \geq 1 : W(n) = +\infty\}$$

is proved. The census is stress evidence for the reverse inclusion, not a proof, and no C''_{SQ} is introduced.

23.3. **The hard-prime criterion and the missing third mechanism.** For a hard prime the cancellation check disappears identically, not merely for most rows.

Theorem 23.4 (standalone hard-prime divisor criterion; proved; source Theorem 61.4)**.** *Let $p \equiv 1 \pmod{24}$ be prime and put $B = \lfloor (p+1)/3 \rfloor$. Then a Type-II witness exists if and only if there are*

$$1 \leq a \leq 2B, \qquad a \equiv 3 \pmod 4, \qquad h = \frac{p+a}{4}, \tag{61.12}$$

with

$$D \mid h^2, \qquad D \equiv -h \pmod a. \tag{61.13}$$

Every such pair gives

$$A = \frac{D+h}{a}, \qquad M = 4A - 1 \equiv 3 \pmod 4, \qquad aM = p + 4D, \tag{61.14}$$

and $D \mid A^2$ automatically. Moreover

$$W(p) = \min_{(a,D) \text{ satisfying } (61.12)\text{–}(61.13)} \left(4\frac{D+h}{a} - 1 \right), \tag{61.15}$$

with the minimum of the empty set interpreted as $+\infty$.

Proof. Theorem 22.1 gives necessity, the bound, and all the formulas. Conversely, suppose (61.12)–(61.13) hold. Any common divisor of a and D also divides h, hence divides $4h - a = p$. But $a \leq 2\lfloor (p+1)/3 \rfloor < p$, so $(a, D) = 1$. Since $h \equiv aA \pmod D$ and $D \mid h^2$, multiplication by the inverse of a modulo every prime power dividing D gives $D \mid A^2$. Equivalently, this is the no-cancellation case of (60.5), and it makes the pair sufficient.

Nothing in this argument assumes that D is odd. When D is even, the odd integer $a \equiv 3 \pmod 4$ is a unit modulo every power of 2 in D; therefore the same inference retains the entire even-D branch. Formula (61.14) gives the witness modulus, and minimizing it gives (61.15). □

The same proof works for every odd prime, with $a \equiv -p \pmod 4$. The hard class is stated because it is the unresolved pointwise frontier.

Corollary 23.5 (the two known escape mechanisms are unavailable for primes; proved; source Corollary 61.5)**.** *An odd prime is not in the square family of Theorem 20.1, and no prime candidate passing the quotient congruence can be killed by the cancellation/gcd failure in (60.5). Therefore, if an odd prime had $W(p) = +\infty$, every divisor of every h^2 in its finite range would have to avoid the single class $-h \pmod a$.*

This does *not* prove that a prime cannot fail. **Assessment 61.1 (mechanism trichotomy).** Squares fail by a global quadratic-character obstruction. The three known nonsquare failures reach the quotient stage only in noncoprime branches and then fail uncancelled eligibility. Corollary 23.5 removes both diagnosed mechanisms

for primes, so a prime failure would require a third, currently unstructured mechanism: complete quotient-class avoidance across all the varying (a, h). The word "third" classifies only the mechanisms currently exhibited; it is not a theorem that every possible obstruction has been classified.

This is the pointwise form of the localization problem behind H_{EQ} in Section 19. A prime with $W = +\infty$ would lie in every finite complete avoider $\mathcal{C}_T$, while (61.12)–(61.13) says pointwise that its entire finite divisor system has zero hits. The fair-share hypothesis would control such residue concentration statistically. The criterion supplies no such theorem, because $h = (p + a)/4$ and its divisors vary with p.

Heuristic 61.1 (divisor-chance mass, not independence). Define

$$R(p) = \sum_a \tau(h^2), \qquad E(p) = \sum_a \frac{\tau(h^2)}{a}, \tag{61.16}$$

where a runs through (61.12). There are about $p/6$ admissible a's. Treating each divisor residue as uniform would make $E(p)$ the expected quotient-row count. Divisors of squares are neither uniform nor independent, so this is only a mass diagnostic. For comparison, the three composite sporadics have heuristic masses $13.266827, 15.617351, 49.909726$, but have $43, 65, 188$ exact quotient rows; gcd eligibility kills all of them. For a prime, every exact quotient row is already a witness.

Computational 61.3 (hard-prime replays). Block `(bh)` compares the exact forward-duality minimum with the independent original-side `(bc)` harvest for all 3202 hard primes through $3 \cdot 10^5$, maps every minimum back to a row of (61.12)–(61.14), and finds complete agreement. The largest minimum is 279, at $p = 225289$; the sum of the minima is 44426. Among the selected minimum rows, 1941 have even D, so the even branch is materially exercised. The block also exhausts the divisor criterion itself, rather than merely a modulus cutoff, for three later record primes. In the table, $E(p)$ is rounded to six decimals and every integer column is exact:

p	$R(p)$	$E(p)$	# quotient rows	$W(p)$	a	D	h
954409	11465930	192.502348	62	335	2855	504	239316
1853329	24242528	247.180398	112	383	4839	2	464542
2031121	26874324	222.590291	36	2495	815	576	507984

(61.17)

Each last quadruple satisfies (61.12)–(61.14), and exhaustive minimization in (61.15) gives the displayed W. The final prime is literally the Section 17 residue-one instance at depth 10:

$$2031121 = 806 \cdot 2520 + 1, \qquad M(10) = 2520.$$

Theorem 17.1 therefore proves only $W(2031121) > 10$, not the observed value 2495. Its late resolution uses $a = 815$, $D = 576$, and h just above $p/4$; only 36 quotient rows survive more than 26 million divisor trials. The other record rows have different a, D. This small sample shows how residue-one lower-tail forcing and eventual dual resolution can coexist; it supports no asymptotic law. Section 24 recasts this exact finite criterion in ratio coordinates and refines the exhibited-mechanism register without changing its no-claim scope.

23.4. **Verification pedigree and exact scope.** Block `(bh)` checks the complete forward census, the independent original-side overlap, the 3202-prime original/duality comparison, all 3630 twisted-square values and the $s = 505$ control, both residue laws, and the three exhaustive prime-criterion replays. The default block is memory-bounded and uses no fixed-width products. The million-point census is gated by `ES_FULL_SCAN=1`.

The wave-25 maximum-severity review returned SOUND-AFTER-REPAIRS. Its fresh Python-integer implementation used only the original $D \mid ((M+1)/4)^2$ harvest and classified every $n \le 5000$: 4927 values were finite, with largest least $W = 727$, and exactly 73 were unresolved, namely the 70 squares and $288, 336, 4545$. The square cases were settled by an independent recheck of the Jacobi proof of Theorem 20.1. Streaming through the three exact normal-form ceilings tested respectively 748810, 1074878, and 409000770 divisor values and found no datum.

With Python's seeded sampler and seed 61250025, 2000 distinct integers in $[5000, 200000]$ gave 1996 direct harvested witnesses and four squares, with no other unresolved value and largest least $W = 983$. With seed 61250026, 500 distinct integers in $[200000, 10^6]$ all resolved, with largest least $W = 239$. Neither sample disagreed with block `(bh)`. The optional complete forward run independently reproduced the million row: $1003 = 1000 + 3$ infinities, namely 1000 squares and the same three nonsquares, from 3400244 (g, v) pairs and 173713414 (g, v, d, u) tests.

An independent original harvest reproduced all 3630 twisted-square box entries: 3628 finite, maximum least value 727, and only $(2, 12), (21, 4)$ vanishing. It separately checked that $s = 505$ is outside the box and that only $m = 3$ vanishes among its first 30 multipliers. The review also rederived every root set and every lower-modulus exclusion in (61.7)–(61.8).

For primes, the review rederived $(a, D) = 1$ from $p = 4h - a$ and $a < p$, then proved $D \mid A^2$ prime-power by prime-power without a parity assumption. The repaired verifier compares all 3202 hard primes through 300000 against the independent original harvest, obtaining maximum 279, sum 44426, and 1941 even-D selected minima. It also checks $2031121 = 806 \cdot 2520 + 1$ against the independently minimized $W = 2495$ row. The full prescribed verifier completed under its 460-second review timeout, while isolated block `(bh)` took about 13 seconds on the review host. The largest independent scan streamed one A at a time with at most 6561 live divisors; it used neither fixed-width arithmetic nor a Cartesian-product allocation. These checks and the review verdict are internal validation, not external refereeing.

The census found no new nonsquare sporadic, so C'_{SQ} survives and no C''_{SQ} is introduced. Theorem 23.3 is only a constructive classification of the first layers of one family; no unbounded classification of twisted-square escapes has been proved. Most importantly, Theorem 23.4 is a terminating prime test and structural simplification, not a proof that its finite set is always nonempty. No pointwise prime bound, fair-share estimate, or new Erdős–Straus theorem is claimed.

24. The divisor-ratio spectrum and the finite conspiracy

Status and headline. The ratio-spectrum law and the six small-a laws below are **proved**. They are elementary consequences of Theorems 22.1 and 23.4 and inherit none of Theorem 9.4 or the provisional machinery behind it. The bounded replays and staged hunt are **Computational**, exact only in their stated ranges. The ratio spectrum is an exact change of coordinates, not a new supply theorem; the Erdős–Straus conjecture is untouched.

24.1. **The exact ratio spectrum.** For a positive integer h coprime to an odd integer a, define

$$\mathrm{Rat}_a(h) = \left\{ \prod_{q \mid h} q^{f_q} \pmod a : -v_q(h) \le f_q \le v_q(h) \right\}. \tag{62.1}$$

Negative powers in (62.1) are taken in $(\mathbb{Z}/a\mathbb{Z})^*$. This is the *divisor-ratio spectrum* of h modulo a.

Theorem 24.1 (ratio-spectrum law; proved; source Theorem 62.1)**.** *Let $p \equiv 1 \pmod{24}$ be prime, put $B = \lfloor (p+1)/3 \rfloor$, and let*

$$1 \le a \le 2B, \qquad a \equiv 3 \pmod 4, \qquad h = \frac{p+a}{4}. \tag{62.2}$$

Then

$$\begin{aligned} &\text{there is a witness at this } a \\ \iff\quad & \exists D \mid h^2 :\ D \equiv -h \pmod a \\ \iff\quad & -1 \pmod a \in \mathrm{Rat}_a(h). \end{aligned} \tag{62.3}$$

Moreover the divisors in the middle line pair under

$$D \longleftrightarrow D' = \frac{h^2}{D}, \tag{62.4}$$

and the target class $-h \pmod a$ is preserved by this pairing. Each hit supplies

$$A = \frac{D+h}{a}, \qquad M = 4A - 1, \qquad aM = p + 4D, \tag{62.5}$$

and the minimum of these M's over all admissible a is $W(p)$. Consequently

$$W(p) < +\infty \quad \iff \quad \exists a \text{ as in (62.2) with } -1 \in \mathrm{Rat}_a((p+a)/4). \tag{62.6}$$

Proof. Since $p \equiv 1 \pmod 4$, integrality of h forces $a \equiv -p \equiv 3 \pmod 4$; conversely this congruence makes h an integer. The bound on a is the finite normal-form bound (60.9). In particular $a < p$. Since a is odd and $4h = p + a$,

$$(a, h) = (a, 4h) = (a, p) = 1.$$

Thus every divisor of h^2 is a unit modulo a.

If $DD' = h^2$ and $D \equiv -h \pmod a$, inversion gives

$$D' \equiv h^2 D^{-1} \equiv -h \pmod a.$$

The converse follows by applying the same involution, proving (62.4). Write $h = \prod q^{v_q}$ and $D = \prod q^{e_q}$, where $0 \le e_q \le 2v_q$. Then

$$Dh^{-1} \equiv \prod_{q \mid h} q^{e_q - v_q} \pmod a, \qquad -v_q \le e_q - v_q \le v_q. \tag{62.7}$$

Every bounded exponent in (62.1) arises uniquely by taking $e_q = v_q + f_q$. Hence

$$\{Dh^{-1} \pmod a : D \mid h^2\} = \mathrm{Rat}_a(h), \tag{62.8}$$

and $D \equiv -h$ is equivalent to $Dh^{-1} \equiv -1$. Theorem 23.4 identifies the middle line of (62.3) with a witness and proves (62.5), including automatic eligibility $D \mid A^2$. Its minimization formula (61.15) proves (62.6). □

24.2. **Six exact small-a laws.** Here and below "failure at a" means that (62.3) fails for that fixed admissible a; it says nothing about the other values of a. The following notation makes the laws finite and explicit. For a prime $\ell \in \{3, 7, 11, 19, 23\}$, use the primitive root g_ℓ in the second column below and put

$$V_{\ell,r}(h) = \sum_{\substack{q^e \parallel h \\ q \bmod \ell \in \{g_\ell^r, g_\ell^{-r}\}}} e, \qquad 1 \le r \le \frac{\ell - 1}{2}. \tag{62.9}$$

The inverse-paired classes, in increasing r, are

ℓ	g_ℓ	$\{g_\ell^r, g_\ell^{-r}\}$ $(1 \le r \le (\ell-1)/2)$
3	2	2
7	3	3/5, 2/4, 6
11	2	2/6, 3/4, 7/8, 5/9, 10
19	2	2/10, 4/5, 8/12, 6/16, 3/13, 7/11, 14/15, 9/17, 18
23	5	5/14, 2/12, 7/10, 4/6, 15/20, 3/8, 17/19, 13/16, 11/21, 9/18, 22

(62.10)

A slash denotes a two-element pair, not division.

Theorem 24.2 (small-a failure laws; proved; source Theorem 62.2)**.** *Assume* $(a, h) = 1$.

1. *At* $a = 3$*, success occurs if and only if some prime* $q \mid h$ *has* $q \equiv 2 \pmod 3$*. Thus failure occurs exactly when every prime divisor of* h *is* $1 \pmod 3$*; the assertion is vacuous for* $h = 1$*.*
2. *At* $a = 7$*, writing* $V_r = V_{7,r}(h)$*, success occurs exactly when*
$$V_3 \ge 1, \quad \text{or} \quad V_1 \ge 1 \text{ and } V_2 \ge 1, \quad \text{or} \quad V_1 \ge 3. \tag{62.11}$$
Failure is the negation of these three conditions.
3. *At* $a = 11$*, success occurs if* $V_{11,5} \ge 1$*. Otherwise it occurs exactly when* $(V_{11,1}, \dots, V_{11,4})$ *componentwise dominates at least one row of*
$$\begin{array}{lllll} (0,0,1,2) & (0,0,3,1) & (0,0,5,0) & (0,1,1,0) & (1,0,0,1) \\ (1,0,2,0) & (1,2,0,0) & (2,0,1,0) & (3,1,0,0) & (5,0,0,0). \end{array} \tag{62.12}$$
Again failure is the exact negation.
4. *For* $a = 19$ *or* 23*, put* $m = a - 1$*. The completely explicit law is*
$$\boxed{\text{failure at } a \iff \frac{m}{2} \notin \left\{ \sum_{r=1}^{m/2} r j_r \pmod m : |j_r| \le V_{a,r}(h) \right\}.} \tag{62.13}$$
Together with the residue rows in (62.10), this is a finite calculation using only the residue classes and multiplicities of primes dividing h*; no discrete logarithm remains to be found.*
5. *At* $a = 15$*, define*
$$\begin{aligned} U &= \sum_{q^e \| h,\ q \equiv 2,8 \ (15)} e, & V &= \sum_{q^e \| h,\ q \equiv 7,13 \ (15)} e, \\ I_c &= \mathbf{1}_{\exists q \mid h:\ q \equiv c \ (15)} & & (c = 4, 11, 14). \end{aligned} \tag{62.14}$$
Success occurs exactly when $I_{14} = 1$*, or* (U, V, I_4, I_{11}) *dominates one of*
$$(0,0,1,1), \quad (0,2,0,1), \quad (1,1,0,0), \quad (2,0,0,1). \tag{62.15}$$
Failure is the negation of this list.

Proof. In the cyclic prime cases, take discrete logarithms to base g_ℓ. The primes in the two classes indexed by r contribute $\pm r f_q$. Since each f_q independently runs through a complete integer interval, their sum runs through every integer in $[-V_{\ell,r}, V_{\ell,r}]$. Since $-1 = g_\ell^{(\ell-1)/2}$, (62.13) is necessary and sufficient. For $\ell = 3, 7$, reducing this one-line congruence modulo 2 and 3 gives (1)–(2). For $\ell = 11$, replace each j_r by a representative modulo 10 with absolute value at most 5. The finite 6^4 budget check gives exactly the ten componentwise-minimal rows in (62.12); each displayed row visibly has a signed weighted sum congruent to 5, and testing the six possible capped budgets in each coordinate leaves no other minimal row.

For 15, identify

$$(\mathbb{Z}/15\mathbb{Z})^* = \langle 2\rangle \times \langle 11\rangle \cong C_4 \times C_2.$$

The classes $2/8, 4, 11, 7/13, 14$ have coordinates, respectively, $(1,0), (2,0), (0,1), (1,1), (2,1)$, up to sign. The target $-1 = 14$ is $(2,1)$. If class 14 is absent, solving the two coordinate congruences produces exactly the four minimal rows in (62.15). This also proves that all valuation budgets, including repeated use of one prime, have been retained. □

The budget cannot in general be replaced by the subgroup generated by the prime classes. For example, $a = 7, h = 17$ has the single class $17 \equiv 3 \pmod 7$, which generates the whole unit group, but its valuation is one and

$$\mathrm{Rat}_7(17) = \{3^{-1}, 1, 3\} = \{5, 1, 3\}; \tag{62.16}$$

this misses $-1 = 6$. Equivalently, the required exponent 3 is outside $[-1, 1]$. Thus bounded multiplicity is already essential at $a = 7$.

Assessment 62.1 (the first-layer density scale). The law at 3 is the classical multiplicative-semigroup condition with allowed primes $1 \pmod 3$. Landau–Selberg–Delange theory gives

$$\#\{h \le H : (h,3) = 1,\ h \text{ fails at } 3\} \sim C_3 \frac{H}{\sqrt{\log H}} \tag{62.17}$$

for a positive constant C_3. This contextual asymptotic is not used in any proof here. Block `(bi)` exhaustively checks every unit $h \le 10^5$ against all six laws. The numbers checked, followed by the exact failures, are

a	3	7	11	15	19	23
# checked	66667	85715	90910	53333	94737	95653
# failures	8814	25359	34046	26423	46679	54165.

(62.18)

This finite table is **Computational**; it is not a density estimate.

24.3. **The finite conspiracy and late resolution.** For a hard prime define

$$\mathcal{A}(p) = \{a : 1 \le a \le 2\lfloor (p+1)/3\rfloor,\ a \equiv 3 \pmod 4\}, \qquad F_a(p) = \{-1 \notin \mathrm{Rat}_a((p+a)/4)\}.$$

Theorem 24.1 gives the exact finite-conspiracy form

$$\boxed{W(p) = +\infty \iff F_a(p) \text{ holds for every } a \in \mathcal{A}(p).} \tag{62.19}$$

Thus the Type-II pointwise problem is: some shifted value $h = (p + a)/4$, with $a \le 2B$, must represent $-1 \pmod a$ by bounded products of its own prime factors.

Assessment 62.2 (mechanism register). Squares fail by the global quadratic-character obstruction of Theorem 20.1. The three known nonsquare failures have quotient rows, but only in noncoprime branches, and uncancelled eligibility kills them (Section 22). For primes that gcd branch is unavailable. Accordingly a prime failure would require complete ratio-class avoidance in (62.19), the third mechanism described in Assessment 61.1. This is a classification of the mechanisms presently exhibited, not a theorem that every conceivable obstruction has been classified.

Heuristic 62.1 (quantified independence model). Let $\rho_a(H)$ denote a model probability that a unit h of size H fails its a-law, and let $\mathcal{A}_Y = \{a \le Y : a \equiv 3$

(mod 4)}. If the shifted events were independent, then

$$\Pr\left(\bigcap_{a\in\mathcal{A}_Y} F_a\right) \approx \prod_{a\in\mathcal{A}_Y} \rho_a(H) = \exp\{-S(Y,H)\}, \quad S(Y,H) = \sum_{a\in\mathcal{A}_Y} \log\frac{1}{\rho_a(H)}. \tag{62.20}$$

The $a = 3$ law alone contributes $\frac{1}{2}\log\log H + O(1)$ to S, by (62.17). More aggressively, if a fixed fraction δ of the roughly $Y/4$ admissible values had $\rho_a(H) \le \rho < 1$, then this model would give

$$S(Y,H) \ge \frac{\delta Y}{4}\log(1/\rho) + O(1), \qquad \Pr(\text{all fail}) \lesssim e^{-cY}. \tag{62.21}$$

This is only a **Heuristic**. The shifts $(p+a)/4$ share the same p, the moduli share factors, and the events are far from demonstrably independent. Taking $Y \asymp p$ in (62.21) therefore does not constitute a tail bound, much less a pointwise proof.

Computational 62.1 (anatomy of late resolution; exact displayed rows). Block `(bi)` gives both the first witnessing a and the generally different a attached to the minimum W:

p	W	a_W	D_W	h_W factorization	a_{first}	M_{first}	D_{first}	h_{first} factorization
225289	279	811	245	$5^2\cdot 7\cdot 17\cdot 19$	31	7335	524	$2\cdot 5\cdot 43\cdot 131$
954409	335	2855	504	$2^2\cdot 3\cdot 7^2\cdot 11\cdot 37$	3	318495	269	$269\cdot 887$
1853329	383	4839	2	$2\cdot 13\cdot 17\cdot 1051$	3	617815	29	$13\cdot 29\cdot 1229$
2031121	2495	815	576	$2^4\cdot 3\cdot 19\cdot 557$	11	185279	1737	$3\cdot 193\cdot 877$

(62.22)

For $a = 3, 7, 11, 15, 19, 23$, respectively, the exact outcomes are

p	3	7	11	15	19	23
225289	F	F	F	F	F	F
954409	P_{318495}	F	F	F	F	F
1853329	P_{617815}	P_{264783}	P_{168503}	F	F	F
2031121	F	F	P_{185279}	F	P_{107255}	F

(62.23)

Here F means failure of the exact law and P_M means success whose least modulus at that fixed a is M. To make every F auditable, the corresponding budget vectors from (62.9), or from (62.14) at 15, are

```
225289: 3:(0); 7:(0,3,0); 11:(0,2,0,2,0); 15:(2,0,0,0,0);
        19:(0,1,1,0,1,0,0,0,0); 23:(0,3,0,0,0,1,0,0,0,0,0)
954409: 7:(0,3,0); 11:(0,1,0,1,0); 15:(2,0,0,0,0);
        19:(0,1,2,0,0,0,0,0,0); 23:(0,4,0,0,0,2,0,0,0,0,0)
1853329: 15:(4,0,0,0,0); 19:(0,0,0,0,0,1,1,0,0);
         23:(0,1,0,1,0,2,0,0,0,0,0)
2031121: 3:(0); 7:(0,3,0); 15:(4,0,0,0,0);
         23:(1,1,0,0,0,1,0,0,0,0,0)
```

The rows show an important distinction: a late minimum W need not mean that all small a's fail. A small a may hit the spectrum only through a very large divisor and hence a very large M; minimization can instead choose a much larger a and a small M. These four records support no asymptotic law.

24.4. **Where the ratio frame degrades for the sporadics.** For composites, $(a,h) = 1$ is no longer automatic, so $h^{-1} \pmod a$ and (62.1) may not exist. The raw congruence $D \equiv -h \pmod a$ remains valid, but it no longer makes $(a,D) = 1$, and the final condition $D \mid ((D+h)/a)^2$ must be restored. Block `(bi)` gives the

complete ledger

n	$\#a$	$(a,n)>1$	$(a,h)>1$	# quotient rows	# eligible	# unit quotient rows
288	48	48	32	43	0	0
336	56	56	40	65	0	0
4545	757	357	357	188	0	0

(62.24)

For 288 and 336 every admissible a is a multiple of 4; for 4545 the nonunit branches are exactly those in which a shares one of $3, 5, 101$ with n, with powers retained. Every quotient row lies in a nonunit branch and every one fails uncancelled eligibility. After the additional character filters, the surviving gcd supports and counts are exactly those in (60.21) and (61.6), including the four gcd-9 failures for 288. Thus the ratio frame does not explain away the sporadics: it stops precisely at their diagnosed gcd/eligibility branch.

24.5. **A staged fourth-sporadic stress. Computational 62.2 (twisted-square staged hunt; exact box).** A streamed Python-integer scan covered every unique twisted square

$$n = sm^2 \in (10^6, 10^8], \qquad 2 \le s \le 1000 \text{ squarefree}, \qquad 1 \le m \le 300. \tag{62.25}$$

This is exactly the intersection with the stated (s,m)-box, not the full interval: values whose unique coordinates have $s > 1000$ or $m > 300$ are outside the scan. There are 607 squarefree kernels and 146016 pairs, hence 146016 distinct integers, in the box. Stage 1 harvested every modulus $M \le 10^4$, $M \equiv 3 \pmod 4$, and found a witness for every pair. Hence the survivor lists passed to the $M \le 10^6$ stage and to the complete finite dual a-decision were both empty. The largest exact least value in the box was

$$W(9{,}028{,}800) = 3359, \quad 9{,}028{,}800 = 627 \cdot 120^2, \quad (M, D, a) = (3359, 48, 2688). \tag{62.26}$$

All smaller eligible moduli were harvested, so the minimum in (62.26) is exact. No fourth sporadic was found. Thus C'_{SQ} survives this finite stress and is not proved; no C''_{SQ} is introduced.

Block `(bi)` keeps the slice $s \le 200, m \le 150$ by default. It contains 5105 pairs in (62.25), has no stage-1 survivor, and has maximum $W(1{,}666{,}170) = 659$ at $(s,m) = (170, 99)$. With `ES_FULL_SCAN=1` it replays all of (62.25) and pins the counts and maximum above. The research run used the same staged code; because stage 1 had no survivor, the deeper two branches were vacuous rather than cutoff claims.

24.6. **Verification, review pedigree, and honest walls. Computational 62.3 (ratio replay; exact stated ranges).** Block `(bi)` checks (62.8) and the pairing on 500 seeded random unit pairs (h,a), and checks each law in Theorem 24.2 by direct divisor enumeration for every unit $h \le 10^5$. In prime chunks it then recomputes the exact minimum via the ratio form for all 1181 hard primes through 10^5 and compares every value with the independent `(bc)`/`(bh)` harvest. It tests 27107184 possible (a, M, D) rows, obtains maximum 167 and sum 15779, and finds no mismatch. The block also pins (62.22)–(62.24) and the staged hunt. Divisor products are Python integers; only one factorization/divisor list is live at a time.

Review attestation (wave 26). The maximum-severity review returned SOUND-AFTER-REPAIRS; no mathematical counterexample or changed conclusion was found.

It independently rederived the ratio law and pairing, all six local laws, the record-prime anatomy, and the staged-hunt maximum; fresh enumeration for every unit $h \le 10^5$ reproduced (62.18), and a separate full dual scan reproduced (62.22)–(62.23), with respectively 24, 62, 112, and 36 quotient rows. An independent original-coordinate harvest rechecked all 146016 distinct members of the stated box and a seeded 2000-member witness sample, again finding the unique maximum $W(9028800) = 3359$ with $(M, D, a) = (3359, 48, 2688)$. The repaired text and executable pins make clear that the hunt covers the stated box subset rather than the whole interval, that C'_{SQ} remains open with only its right-to-left inclusion proved, and that the mechanism register classifies only exhibited mechanisms. The full command completed green in 131 seconds under its 480-second timeout; the default block took about 7.3 seconds on the review host. Block `(bi)` uses Python integers and bounded factor/divisor state, and the control-byte, malformed-TeX, and divisibility-command sweeps were clean. This attestation and its finite checks are internal validation, not external refereeing.

Honest walls. The spectrum does not improve the witness-modulus tail, show pointwise nonemptiness, or control correlations among the shifts. The small-a laws are exact local tests, but failure at finitely many small values is compatible with later success, and success at small a can produce a huge M. The finite-conspiracy display is the terminal distillation of the Type-II pointwise problem and clarifies the three known mechanisms; it does not solve that problem, prove C'_{SQ}, or prove Erdős–Straus. Sections 25–27 only reexpress this finite system, diagnose its gcd failures, and extend bounded censuses; they do not strengthen the finite-conspiracy equivalence.

25. Algebraic witness taxonomy, square-root coverage, and the first-witness index

Status and scope. The DIV and D1 families, fixed-D laws, universal a-law, square-root bound, and effective exceptional-set estimate below are **proved**. The two first-witness censuses are **Computational**, exact only in their displayed ranges. This taxonomy consolidates Theorems 22.1, 23.4, and 24.1 in two coordinate systems. It creates no new identity family or pointwise result and does not improve the witness-modulus tails of Section 14.

25.1. **Elementary families and the fixed-D coordinate.** Unless stated otherwise in this section, let $p \equiv 1 \pmod{24}$ be prime and put

$$B = \left\lfloor \frac{p+1}{3} \right\rfloor, \qquad 1 \le a \le 2B, \qquad a \equiv 3 \pmod 4, \qquad h = \frac{p+a}{4}. \tag{63.1}$$

Theorem 23.4 says that this fixed a succeeds exactly when some $D \mid h^2$ satisfies $D \equiv -h \pmod a$.

Lemma 25.1 (DIV family; proved; source Lemma 63.1)**.** *If $e \mid h$ and $e \equiv -1 \pmod a$, then*

$$D = \frac{h}{e} \tag{63.2}$$

is a witness at a.

Proof. The positive integer D divides h^2, and $h = De \equiv -D \pmod a$. Theorem 23.4 supplies the witness, including $D \mid A^2$. □

This family is sufficient, not necessary: a general $D \mid h^2$ can have prime exponents larger than those in h.

Lemma 25.2 (D1 family; proved; source Lemma 63.2)**.** *For (63.1),*

$$h \equiv -1 \pmod{a} \quad \Longleftrightarrow \quad a \mid p+4. \tag{63.3}$$

When these conditions hold, both $D = 1$ and $D = h^2$ are witnesses. The first has

$$M = \frac{p+4}{a}. \tag{63.4}$$

Proof. Since $4(h+1) = p + a + 4$ and a is odd, multiplication by 4 is invertible modulo a, proving (63.3). Then $1 \equiv h^2 \equiv -h \pmod{a}$, and both divisors divide h^2. The criterion proves both claims, while $aM = p + 4D$ gives (63.4) for $D = 1$. □

Corollary 25.3 (explicit square-root law; proved; source Corollary 63.3)**.** *If $p + 4$ has a prime factor congruent to 3 modulo 4, then*

$$W(p) \le \sqrt{p+4}. \tag{63.5}$$

Proof. Put $N = p + 4 \equiv 1 \pmod 4$. If $q \equiv 3 \pmod 4$ divides N, then

$$q \equiv N/q \equiv 3 \pmod 4. \tag{63.6}$$

The second factor is not 1. Let a be the larger factor and M the smaller. Then $a, M \equiv 3 \pmod 4$, $aM = N$, and $M \le \sqrt{N}$. Also $M \ge 3$, so $a \le N/3$. Writing $p = 3r + 1$ gives $B = r$, and the hard-prime range has $r \ge 2$; hence

$$a \le \frac{p+4}{3} = r + \frac{5}{3} \le 2r = 2B. \tag{63.7}$$

Thus (63.1) holds, and Lemma 25.2 gives exactly $M = N/a$. □

The same datum reads the chosen factor of $p + 4$ as a and its complement as M; choosing the larger factor as a gives the square-root bound.

Theorem 25.4 (fixed-D divisor laws; proved; source Theorem 63.4)**.** *Fix a positive integer D. A witness for p with this D and modulus M exists if and only if*

$$M \mid p + 4D, \qquad M \equiv 3 \pmod 4, \qquad D \mid \left(\frac{M+1}{4}\right)^2. \tag{63.8}$$

Its dual coefficient and auxiliary integer are

$$a = \frac{p+4D}{M}, \qquad A = \frac{M+1}{4}, \qquad aM = p + 4D. \tag{63.9}$$

If D is squarefree, the eligibility condition is exactly

$$M \equiv -1 \pmod{4D}. \tag{63.10}$$

In particular,

D	*shifted value*	*exact class of M*
1	$M \mid p + 4$	$M \equiv 3 \pmod 4$
2	$M \mid p + 8$	$M \equiv 7 \pmod 8$
3	$M \mid p + 12$	$M \equiv 11 \pmod{12}$
4	$M \mid p + 16$	$M \equiv 7 \pmod 8$.

(63.11)

Proof. The witness definition is precisely (63.8), and its first condition defines the positive a in (63.9); Theorem 22.1 supplies all finite range conditions. For squarefree D, primewise valuations give $D \mid A^2$ if and only if $D \mid A$, equivalent to $4D \mid M+1$. This gives the first three rows. For $D = 4$, however, $4 \mid A^2$ if and only if $2 \mid A$, hence only $8 \mid M+1$; this gives the final row and retains the full 2-adic branch. □

Thus the fixed-D coordinate asks for an eligible divisor of $p + 4D$, whereas the fixed-a coordinate asks whether -1 lies in the bounded ratio spectrum of $(p+a)/4$. Equations (60.4) and (60.10) identify the rows bijectively. They are the same finite system in dual coordinates, not additional witness supply.

25.2. **The universal bounded-coefficient law.**

Theorem 25.5 (universal a-law; proved; source Theorem 63.5)**.** *Let $a > 1$ be odd, let $(a, h) = 1$, and write*

$$a = \prod_{i=1}^{r} \ell_i^{k_i}, \qquad m_i = \varphi(\ell_i^{k_i}). \tag{63.13}$$

Choose a primitive root g_i modulo each odd prime power. For every prime $q \mid h$, define

$$\boldsymbol{\lambda}(q) = \left(\log_{g_i}(q \bmod \ell_i^{k_i}) \bmod m_i\right)_{i=1}^{r}, \qquad \boldsymbol{\epsilon} = (m_i/2)_{i=1}^{r}. \tag{63.14}$$

Then $-1 \in \mathrm{Rat}_a(h)$, equivalently success at a when h comes from (63.1), is equivalent to

$$\boxed{\boldsymbol{\epsilon} \in \left\{ \sum_{q \mid h} f_q \boldsymbol{\lambda}(q) : -v_q(h) \le f_q \le v_q(h) \right\} \subseteq \prod_{i=1}^{r} \mathbb{Z}/m_i\mathbb{Z}.} \tag{63.15}$$

The same integer f_q is used in every CRT component of $\boldsymbol{\lambda}(q)$.

Proof. The unit group decomposes as

$$(\mathbb{Z}/a\mathbb{Z})^* \cong \prod_i (\mathbb{Z}/\ell_i^{k_i}\mathbb{Z})^*, \tag{63.16}$$

and each factor is cyclic. The chosen logarithms send -1 to $\boldsymbol{\epsilon}$. Formula (62.1) says that a ratio-spectrum element is $\prod_{q \mid h} q^{f_q}$ with exactly the bounds in (63.15); taking all component logarithms proves necessity, and exponentiating followed by the Chinese remainder theorem proves sufficiency. Each q divides the single integer h, so one integer f_q, not independent component exponents, is chosen. Theorem 24.1 identifies this with success at a. □

For $a = 3, 7, 11, 19, 23$ this is the cyclic budget law of (62.9)–(62.13); for $a = 15$ it is the coupled $C_2 \times C_4$ calculation (62.14)–(62.15). For fixed small a, one may tabulate unit residues and update a reachable set; this finite calculation is not an asymptotic algorithmic claim.

25.3. **Explicit coverage and the half-dimensional shifted layer.**

Corollary 25.6 (D1 congruence layer; proved; source Corollary 63.6)**.** *Let $\ell \equiv 3 \pmod 4$ be prime. Every hard prime satisfying*

$$p \equiv -4 \pmod{\ell} \tag{63.17}$$

has a witness at $a = \ell$ with $D = 1$. In particular, $\ell = 7$ covers $p \equiv 3 \pmod 7$. The value $\ell = 3$ never applies, because a hard prime is 1 modulo 3 whereas -4 is 2 modulo 3.

Proof. The congruence says $\ell \mid p + 4$. Its complementary factor is also $3 \pmod 4$ and at least 3, so $\ell \le (p+4)/3 \le 2B$ by (63.7). Lemma 25.2 applies. □

This is exactly the classical $D = 1$ atom supply in dual coordinates; no novelty is claimed for the progression coverage.

Define the checkable exceptional set

$$\begin{aligned}\mathcal{E}(N) = \{p \le N : p \equiv 1 \pmod{24},\ p \text{ prime},\\ p+4 \text{ has no prime factor } q \equiv 3 \pmod 4\}.\end{aligned} \tag{63.18}$$

Theorem 25.7 (effective half-dimensional shifted sieve; proved; source Theorem 63.7)**.** *With an effectively computable absolute implied constant,*

$$\#\mathcal{E}(N) \ll \frac{N}{(\log N)^{3/2}}. \tag{63.19}$$

Proof. Write $p = 24m + 1$, $m \le N/24$. Apart from $2, 3$, exclude modulo each prime ℓ the class $24m + 1 \equiv 0 \pmod \ell$, and for $\ell \equiv 3 \pmod 4$ also exclude $24m + 5 \equiv 0 \pmod \ell$. Their difference is 4, so for $\ell \ge 5$ the local count is

$$\rho(\ell) = \begin{cases} 1, & \ell \equiv 1 \pmod 4, \\ 2, & \ell \equiv 3 \pmod 4. \end{cases} \tag{63.20}$$

Every member of $\mathcal{E}(N)$ with $p > z$ avoids these classes for $\ell < z$.

Put $X = \lfloor (N-1)/24 \rfloor$. For squarefree d supported on these primes, the Chinese remainder theorem gives $\rho(d) = \prod_{\ell \mid d} \rho(\ell)$ removed classes and the exact interval count

$$\frac{X\rho(d)}{d} + O(\rho(d)).$$

Since $\rho(\ell) \le 2$, $\sum_{d \le D} \mu^2(d)\rho(d) \ll D \log D$. Apply the fixed-dimensional upper-bound fundamental lemma in the form used in (52.8)–(52.12), with $D = X^{1/2}$ and $z = D^{1/u} \asymp N^{\eta}$, for fixed sufficiently large u and fixed $\eta > 0$. It gives

$$\#\mathcal{E}(N) \ll X \prod_{5 \le \ell < z} \left(1 - \frac{\rho(\ell)}{\ell}\right) + O(D \log D) + O(z). \tag{63.21}$$

The final term covers $p \le z$, and both errors are $o(N/(\log N)^{3/2})$. This is an integer-progression upper-bound sieve; no level of distribution for primes is assumed.

Effective Mertens estimates in the two fixed classes modulo 4 give

$$\sum_{5 \le \ell < z} \frac{\rho(\ell)}{\ell} = \sum_{5 \le \ell < z} \frac{1}{\ell} + \sum_{\substack{5 \le \ell < z \\ \ell \equiv 3 (4)}} \frac{1}{\ell} = \frac{3}{2} \log\log z + O(1). \tag{63.22}$$

Thus the product in (63.21) is $O((\log z)^{-3/2}) = O((\log N)^{-3/2})$. All constants and the threshold are effective. □

Corollary 25.8 (effective square-root coverage; proved; source Corollary 63.8)**.** *Every hard prime $p \le N$ outside $\mathcal{E}(N)$ satisfies (63.5). Hence the number not certified by the explicit square-root law is $O(N/(\log N)^{3/2})$, effectively.*

Proof. Outside $\mathcal{E}(N)$, Corollary 25.3 applies; the count is Theorem 25.7. □

The factorization condition (63.18) is also a direct decision test for this elementary certificate.

Assessment 63.1 (honest comparison). The exact comparison is

result	threshold controlled	exceptional bound	status
Corollary 25.8	$\sqrt{p+4}$	$N/(\log N)^{3/2}$	classical sieve; effective; explicit exception
Theorem 51.2(1)	$T \le \exp\{c(\log N)^{1/4}\}$	$N\exp\{-c(\log T)^2\log(2+\log T)\}$	proved internally; Section 39 review qualification
Theorem 51.2(2)	$T \le \exp\{c(\log N)^{1/4}\}$	$N\exp\{-c(\log T)^3\}$	CLAIMED/PROVISIONAL cubic supply
Corollary 51.3	$T = (\log N)^A$	$N\exp\{-c_A(\log\log N)^2 \cdot \log\log\log N\}$	fixed-polylogarithmic variant.

(63.23)

The Section 14 events are stronger on both axes: they control much smaller moduli and have much smaller exceptional sets. The advantages here are only relative elementarity, effectivity, and the explicit factorization test. This is not a tail record or pointwise progress. Moreover $\sqrt{p}$ lies far beyond the top proven Section 51 window $\exp\{\alpha(\log N)^{1/4}\}$, so the thresholds are not directly comparable. Within this campaign it is only a weak almost-all datum in a new regime.

The $D = 1$ gap at the late primes is exact:

p	$p+4$
225289	$225293 = 37 \cdot 6089$
954409	$954413 = 181 \cdot 5273$
1853329	1853333 is prime
2031121	$2031125 = 5^3 \cdot 16249$.

(63.24)

Every displayed prime factor is 1 (mod 4), so D1 misses all four globally, not merely below a cutoff. This is the full-range version of the necessary condition in Lemma 20.2; it does not suggest that these are the only elements of (63.18).

25.4. **The first-witness index.** Define, on first use,

$$a_1(p) = \min\left\{a \in \mathcal{A}(p) : -1 \in \mathrm{Rat}_a\left(\frac{p+a}{4}\right)\right\}, \tag{63.25}$$

with $+\infty$ when the set is empty. Thus a_1 minimizes the dual coefficient and ignores the resulting modulus. By contrast, let $a_W(p)$ be the least a among data attaining $W(p)$, and then choose the least D if that pair also ties.

Computational 63.1 (exact a_1 censuses). Block `(bj)` factors $h = (p+a)/4$ for $a = 3, 7, 11, \ldots$, applies Theorem 25.5, and stops at the first hit. Independently it obtains W, a_W, D_W by an M-ascending complete class scan through $M = 3000$. The default range is all 9732 hard primes through 10^6; the `ES_FULL_SCAN=1` range is all 82887 hard primes through 10^7. The exact histograms are

a_1	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63	71	107
$p \le 10^6$	5192	3551	584	131	113	96	8	33	4	9	1	3	2	2	2	1	0	0
$p \le 10^7$	47137	28606	4419	961	766	637	63	183	27	44	7	23	2	4	4	1	2	1.

(63.26)

The maxima and raw Pearson correlations are

range	$\max a_1$	p	$\max W$	p	$\mathrm{corr}(a_1, W)$
10^6	63	87481	335	954409	0.354836
10^7	107	8803369	2495	2031121	0.271029.

(63.27)

The correlations use untransformed integers and are descriptive only. The four late rows are

p	a_1	W	a_W	D_W
225289	31	279	811	245
954409	3	335	2855	504
1853329	3	383	4839	2
2031121	11	2495	815	576.

(63.28)

This includes $a_1(225289) = 31 > 23$ and the three requested small values.

Computational 63.2 (dual-minimum separation; exact displayed ranges). Every row has $a_W > a_1$: 9732/9732 by default and 82887/82887 in the research run. The largest ratios are

$$\max_{p\le 10^6} \frac{a_W}{a_1} = 47597 \quad (p = 999529;\ a_1 = 3, W = 7, a_W = 142791, D_W = 2), \tag{63.29}$$

$$\max_{p\le 10^7} \frac{a_W}{a_1} = \frac{1428563}{3} \quad (p = 9999937;\ a_1 = 3, W = 7, a_W = 1428563, D_W = 1). \tag{63.30}$$

The largest differences are 142788 and 1428560. At the opposite extreme, $a_1(8803369) = 107$ has $(W, a_W, D_W) = (139, 63335, 49)$. These are coordinate-separation anomalies, not large-W anomalies: a small a can first hit through huge D, M, while a tiny modulus can have a of order p. Hence a_1 does not bound W, and these finite ratios support no growth law.

25.5. **Honest walls and verification. Assessment 63.2 (stacking wall).** For prime $a \equiv 3 \pmod 4$, $a > 3$, the D1 progression has relative prime density about $1/(a-1)$; $a = 3$ is incompatible with the hard class. Thus

$$\sum_{\substack{3<a\le Y\\ a\equiv 3(4)\\ a \text{ prime}}} \frac{1}{a-1} = \frac{1}{2}\log\log Y + O(1). \tag{63.31}$$

Without correlation control, stacking recovers only log-log-strength exponents. The richer ratio laws share the moving p and are not independent. Nothing here improves Theorem 51.2 or Corollary 51.3. Nor does finite feasibility imply a hit for every prime; the count-below-one and absolute-remainder mass gap of Proposition 19.5 remain. The index a_1 is infinite exactly when the finite conspiracy survives and, when finite, says nothing quantitative about the first modulus.

For the direct positivity question put

$$N_{\text{hit}}(p) = \sum_{a\in\mathcal{A}(p)} \#\left\{ D \mid \left(\frac{p+a}{4}\right)^2 : D \equiv -\frac{p+a}{4} \pmod a \right\}. \tag{63.32}$$

Then Theorem 23.4 proves

$$N_{\text{hit}}(p) > 0 \iff W(p) < +\infty. \tag{63.33}$$

Assessment 63.3 (no hidden positivity identity). The summands are the raw Type-II witness count in dual coordinates. The Case-B dictionary and the Elsholtz–Tao reconciliation show that, after canonical ordering and primitivity conventions, this is the same solution structure measured by $f_{II}(p)$. An exact rearrangement may exist, but an automatically positive lower bound for every hard prime would

prove Type-II positivity and settle the remaining Erdős–Straus problem. None is known. The square vanishing of f_I, f_{II} warns that a formally positive mass cannot ignore the moving congruence projector. This is not a no-go theorem for a new signed, spectral, or geometric identity.

Computational 63.3 and review pedigree (block `bj`; exact stated ranges). The Python-integer block checks D1, the square-root construction, and the full original datum on 512 seeded hard primes; 256 proper-divisor DIV rows; all 7146 direct fixed-D rows; and the CRT-coupled law against direct spectra on 360 seeded composite-modulus pairs, including an uncoupled-exponent false positive. It recomputes all 3202 hard-prime minima through 300000: 1517 shifts have a 3 (mod 4) factor and obey (63.5); the only values with $W(p) > \sqrt{p+4}$ are $(p, W, p+4) = (193, 15, 197)$ and $(3361, 99, 5 \cdot 673)$. It pins (63.24) and every census, correlation, separation, and late-row datum. The 10^7 run is gated; generation is chunked, with one factorization or divisor list live at a time.

The wave-27 maximum-severity review returned SOUND-AFTER-REPAIRS. It rederived the elementary range endpoints, fixed-D valuations including the $D = 2, 4$ 2-adic branches, and the single-exponent CRT coupling; rebuilt the dimension-3/2 sieve from exact $O(\rho(d))$ interval remainders without a prime level of distribution; and independently reproduced the default 9732-prime census, its histogram, maxima, correlation, strict separation, square-root census, and all late rows. A seeded 200-prime sample from $(10^6, 10^7]$ cross-checked direct and spectrum paths, while the gated block replayed all 82887 research rows. The repaired isolated block took about 2.1 seconds by default and 8.7 seconds gated; the prescribed full suite passed in 201 seconds. This is internal validation only. The residual uncertainty is exactly the stated open mathematics: these local laws and censuses do not force a witness for every hard prime, and the half-dimensional result is only an almost-all bound for the explicit $D = 1$ layer.

26. The gcd mechanism: cancellation covers and shifted purity

Status and headline. The cancellation-cover criterion, both shifted support laws, the dyadic and twisted-square exclusions, and the composite unit-branch ratio law are **proved**. The sporadic ledgers and hunts are **Computational**, exact only in their stated finite ranges. Failure after the quotient stage has an exact necessary-and-sufficient description: every quotient row must lose eligibility at a prime dividing the candidate. This identifies the gcd mechanism, but does not prove that every nonsquare failure has a quotient row, that only three sporadics exist, or the Erdős–Straus conjecture. The cancellation-immune $D = 1, 2$ laws below are necessary conditions, not a characterization; all three known sporadics pass them.

26.1. **The exact cancellation cover.** Put $B = \lfloor (n+1)/3 \rfloor$ and define the finite quotient-row set

$$\begin{aligned} \mathcal{Q}(n) = \{(a, D) : 1 \le a \le 2B, \quad a \equiv -n \pmod 4, \quad h = (n+a)/4, \\ D \mid h^2, \qquad a \mid D + h\}. \end{aligned} \tag{64.1}$$

For a row put

$$A = \frac{D+h}{a}, \qquad M = 4A - 1. \tag{64.2}$$

Only $D \mid A^2$ remains to make such a quotient-stage row a witness.

Theorem 26.1 (gcd-supported cancellation cover; proved; source Theorem 64.1)**.** *For every positive integer n,*

$$W(n) = +\infty \quad \Longleftrightarrow \quad \text{every } (a,D) \in \mathcal{Q}(n) \text{ has a prime } q \mid n \text{ for which}$$

$$2v_q(A) < v_q(D) \le 2v_q(a) + 2v_q(A), \qquad q \mid (a,D). \tag{64.3}$$

The universal statement includes $\mathcal{Q}(n) = \varnothing$. Consequently one unit quotient row, $(a,D) = 1$, proves $W(n) < +\infty$. More generally, every failure of the last eligibility check is supported on primes dividing n; no outside prime can kill a quotient row.

Proof. For a row, $h = aA - D$, so $h \equiv aA \pmod{D}$. Since $D \mid h^2$, one has $D \mid a^2A^2$. At a prime q, writing $d = v_q(D)$, $\alpha = v_q(a)$, and $\beta = v_q(A)$, this is $d \le 2\alpha + 2\beta$. Eligibility fails exactly when some $d > 2\beta$; then $\alpha > 0$ and $q \mid (a,D)$. Also $aM = n + 4D$, so $(a,D) \mid n$. Thus every failure has exactly the window in (64.3). If no prime fails, $D \mid A^2$ and Theorem 22.1 gives a witness; conversely every witness is a row with no failed prime. □

The mechanism can kill only an existing row, through prime powers already in n. It does not make a row exist; $\mathcal{Q}(n) = \varnothing$ is the quotient-avoidance alternative in the finite conspiracy.

26.2. **Two cancellation-immune shifted laws.** For an integer, "odd-prime support" means its set of odd prime divisors, without multiplicity.

Theorem 26.2 (exact $D = 1$ and $D = 2$ support laws; proved)**.** *(1) The $D = 1$ layer gives a witness for n if and only if $n + 4$ has a divisor $M \equiv 3 \pmod 4$. Equivalently it does so if and only if some prime $q \equiv 3 \pmod 4$ divides $n + 4$, and its least modulus is the least such prime. In particular,*

$$W(n) = +\infty \quad \Longrightarrow \quad \text{every odd prime dividing } n+4 \text{ is } 1 \pmod 4. \tag{64.4}$$

(2) The $D = 2$ layer gives a witness if and only if $n + 8$ has a divisor $M \equiv 7 \pmod 8$. Equivalently, its odd-prime support contains a prime $7 \pmod 8$, or contains both a prime $3 \pmod 8$ and a prime $5 \pmod 8$. Therefore

$$W(n) = +\infty \quad \Longrightarrow \quad \begin{cases} q \not\equiv 7 \pmod 8 & \text{for every odd } q \mid n+8, \\ \text{not both residues } 3,5 \pmod 8 & \text{occur in the support.} \end{cases} \tag{64.5}$$

Equivalently, that support is contained in $\{1,3\} \pmod 8$ or in $\{1,5\} \pmod 8$. Every witness in part 1 has $M \le n+4$, and every witness in part 2 has $M \le n+8$.

Proof. For $D = 1$, eligibility is automatic and the datum equation is $M \mid n + 4$. Conversely any such $M \equiv 3 \pmod 4$, with $A = (M+1)/4$, is a datum; $a = (n+4)/M$ satisfies $a \equiv 3n \equiv -n \pmod 4$. A divisor $3 \pmod 4$ contains a prime in that class, which is itself a smaller allowed divisor. This proves part 1.

For $D = 2$, eligibility is $2 \mid A^2$, equivalent to A even and hence to $M = 4A - 1 \equiv 7 \pmod 8$; the datum equation is $M \mid n + 8$. In the Klein four-group of odd residues modulo 8, a divisor is 7 exactly when one can select either a prime $7 \pmod 8$ or one prime from each of the classes $3, 5 \pmod 8$. This proves part 2, with multiplicities unrestricted.

For either converse, $a = (n + 4D)/M$ has the required congruence and $a \le 2\lfloor (n+1)/3 \rfloor$ by Theorem 22.1. For $D = 1$, $(a,D) = 1$; for $D = 2$, any common factor 2 divides n, and the full 2-adic condition is exactly that A be even. Thus no composite normal-form condition was omitted. □

For the known sporadics,

n	$n+4$	$n+8$
288	$2^2 \cdot 73$	$2^3 \cdot 37$
336	$2^2 \cdot 5 \cdot 17$	$2^3 \cdot 43$
4545	4549	$29 \cdot 157$.

(64.6)

Every middle-column odd factor is $1 \pmod 4$; the last-column supports use only 5, only 3, and only $5 \pmod 8$, respectively. Thus all three pass both necessary laws.

Corollary 26.3 (explicit forbidden families; proved; source Corollary 64.3)**.** *For every $e, k \geq 0$, whenever the displayed n is positive,*

$$\begin{aligned} n &\equiv 3 \cdot 2^e - 4 \pmod{2^{e+2}} \implies W(n) < +\infty, \\ n &\equiv 7 \cdot 2^e - 8 \pmod{2^{e+3}} \implies W(n) < +\infty. \end{aligned} \tag{64.7}$$

The first line uses $(n+4)/2^e \equiv 3 \pmod 4$, and the second uses $(n+8)/2^e \equiv 7 \pmod 8$. In particular no failure lies in $n \equiv 3 \pmod 4$, $2 \pmod 8$, $8 \pmod{16}$, $6 \pmod{16}$, $20 \pmod{32}$, or $48 \pmod{64}$, with overlaps harmless.

Corollary 26.4 (twisted-square $D = 1$ exclusion; proved; source Corollary 64.4)**.** *Let $s > 1$ be squarefree, let $q \equiv 3 \pmod 4$ be prime with $q \nmid s$, and suppose*

$$\left(\frac{s}{q}\right) = -1.$$

Then $sm^2 \equiv -4 \pmod q$ has exactly two classes of solutions for $m \pmod q$, and every integer m in either class satisfies

$$W(sm^2) \leq q. \tag{64.8}$$

Consequently a twisted-square failure must avoid those two classes for every such q. At $q = 3$ this recovers $s \equiv 2 \pmod 3$, $3 \nmid m \Rightarrow W(sm^2) = 3$.

Proof. Solubility is equivalent to

$$\left(\frac{-4s^{-1}}{q}\right) = \left(\frac{-1}{q}\right)\left(\frac{s}{q}\right) = -\left(\frac{s}{q}\right) = 1.$$

A nonzero quadratic residue has two roots. For each, $q \mid sm^2 + 4$, so Theorem 26.2 gives the $D = 1, M = q$ datum. At 3 there is no smaller allowed modulus. □

These infinite-family exclusions leave many classes and do not classify the $s = 2, 21, 505$ families.

26.3. **Unit branches and complete sporadic ledgers.**

Theorem 26.5 (composite unit-branch ratio law; proved; source Theorem 64.5)**.** *Let n be odd, let*

$$1 < a \leq 2\lfloor (n+1)/3 \rfloor, \qquad a \equiv -n \pmod 4, \qquad h = (n+a)/4, \qquad (a, n) = 1.$$

Then $(a, h) = 1$, and a witness exists at this a if and only if

$$-1 \in \mathrm{Rat}_a(h). \tag{64.9}$$

Every ratio hit is automatically eligible, even for composite n. In particular, if $n \geq 5$, $n \equiv 1 \pmod 4$, $3 \nmid n$, and $h = (n+3)/4$ has a prime divisor $2 \pmod 3$, then

$$W(n) < +\infty. \tag{64.10}$$

Proof. Both a, n are odd and $(a, 4h) = (a, n + a) = (a, n) = 1$, so $(a, h) = 1$. The ratio bijection (62.7)–(62.8) identifies a hit with a quotient row. Its congruence and $(a, h) = 1$ force $(a, D) = 1$, so Theorem 26.1 makes eligibility automatic. The converse uses the same bijection. For $a = 3$, the first small-a law gives (64.10). □

Even n has only even admissible a, so this shortcut is unavailable; that parity distinction does not explain odd 4545.

For a quotient row set

$$F(a, D) = \{q : v_q(D) > 2v_q(A)\}. \tag{64.11}$$

Computational 64.1 (complete sporadic cancellation ledgers). Block `(bk)` exhausts every row of $\mathcal{Q}(n)$. Writing $g : c$ when exactly c rows have $(a, D) = g$, the full gcd distributions are

```
288 : 3:5, 4:2, 6:4, 9:6, 12:3, 16:3, 18:3, 24:2, 36:3, 96:12
336 : 2:2, 3:6, 4:2, 6:8, 7:4, 8:3, 14:2, 16:1, 21:6,
      24:2, 28:1, 42:4, 48:14, 112:10
4545: 3:15, 5:13, 9:18, 15:73, 45:35, 101:3, 303:20,
      909:3, 1515:8
```

The failed-prime-set counts are

$$\begin{array}{r|l} 288 & \{2\} : 8,\ \{3\} : 17,\ \{2, 3\} : 18 \\ 336 & \{2\} : 12,\ \{3\} : 11,\ \{7\} : 10,\ \{2, 3\} : 15,\ \{2, 7\} : 10,\ \{3, 7\} : 5,\ \{2, 3, 7\} : 2 \\ 4545 & \{3\} : 46,\ \{5\} : 35,\ \{101\} : 8,\ \{3, 5\} : 73,\ \{3, 101\} : 18,\ \{5, 101\} : 2,\ \{3, 5, 101\} : 6. \end{array} \tag{64.12}$$

The corresponding $(\#a, \#\mathcal{Q}, \#\text{eligible})$ triples are $(48, 43, 0)$, $(56, 65, 0)$, and $(757, 188, 0)$. Thus the row totals are $43, 65, 188$, all failed sets are nonempty, and their unions are exactly the prime supports $\{2, 3\}$, $\{2, 3, 7\}$, and $\{3, 5, 101\}$. The character-filtered subledgers remain (60.21) and (61.6). For $288, 336$ every admissible a shares a factor with n. For 4545, exactly 400 of 757 admissible a's are unit branches; none has a quotient row, while all 188 rows occur among the other 357 values and are killed by cancellation. Thus each sporadic passes both shifted tests, has quotient rows, and has every row covered by a failed prime dividing n. This is a necessary-and-sufficient diagnosis of these inputs, not a congruence classification of all failures.

26.4. **Finiteness wall and the wider hunt. Assessment 64.1 (finiteness remains open).** No finiteness theorem is obtained for nonsquares with $W(n) = +\infty$. The finite box in Theorem 22.1 grows with n; the shifted laws are only necessary and their survivor set contains every square; Theorem 26.1 neither forces a quotient row nor an uncancelled or unit row, and even inputs have no unit a. Chebotarev, the generalized Riemann hypothesis (GRH), and standard shifted-prime conjectures do not by themselves control all pointwise divisor congruences in $\mathcal{Q}(n)$. Nor is C_{POLY} sufficient: an infinite sparse set need contain no full polynomial or progression. C'_{SQ} would imply finiteness, but is precisely the open classification. No unconditional or standard-conjectural finiteness claim is made.

Heuristic 64.1 (uniform-independent class model). For $S_M = \{-4D \pmod M : D \mid ((M + 1)/4)^2\}$, uniformity of a nonsquare modulo each M, independence across M, and no additional squarefree-kernel character correlation would give

$$\Pr(W(n) > 10^4) \approx \prod_{\substack{M \le 10^4 \\ M \equiv 3 (4)}} \left(1 - \frac{|S_M|}{M}\right) = 1.033524640428 \cdot 10^{-12}. \tag{64.13}$$

There are 2500 moduli and $\sum |S_M| = 64978$; the toy prediction is about $2.51 \cdot 10^{-7}$ stage-1 survivors among the 242837 new points below, compatible with observing zero. This is only a **Heuristic**: squares violate it globally, twisted squares have real character correlations, every modulus reuses the same integer, and no summable uniform error is known. It cannot imply finiteness.

Computational 64.2 (expanded twisted-square hunt; exact box). With `ES_FULL_SCAN=1`, block `(bk)` scans every unique

$$n = sm^2 \in (10^6, 10^8], \quad 2 \le s \le 2000 \text{ squarefree}, \quad 1 \le m \le 600, \tag{64.14}$$

except the already-scanned subbox $s \le 1000, m \le 300$. Its exact ledger is

squarefree kernels through 2000	1214
new pairs	242837
pass the $n+4$ support law	50633
pass both shifted support laws	18943
survive all $M \le 10^4$	0
survive all $M \le 10^6$	0
complete-normal-form infinities	0.

(64.15)

Together with the 146016 pairs of (62.25), this covers all 388853 pairs in the enlarged box. This is the coordinate-box subset (64.14), not a scan of the full interval $(10^6, 10^8]$. The unique largest exact least value among new pairs is

$$\begin{aligned} W(3{,}201{,}660) &= 5303, & 3{,}201{,}660 &= 15 \cdot 462^2, \\ (M, D, a) &= (5303, 338, 604). \end{aligned} \tag{64.16}$$

Every smaller eligible modulus was checked, and the row satisfies $D \mid ((M+1)/4)^2$ and $n + 4D = aM$. Stage 1 had no survivor, so the $M \le 10^6$ and complete-decision stages received empty lists; their emptiness is exact, not a cutoff inference. The default new subbox $s \le 400, m \le 350$ has 11833 pairs, with 2406 passing (64.4), 872 passing both laws, and none surviving $M \le 10^4$. Its unique maximum is $W(1{,}359{,}015) = 2147$ at $(s, m) = (15, 301)$ and $(D, a) = (9, 633)$.

Computational 64.3 (thin slice beyond 10^8). The optional run also scans

$$2 \le s \le 50 \text{ squarefree}, \quad 1 \le m \le 2000, \quad 10^8 < sm^2 \le 10^9. \tag{64.17}$$

There are 30 kernels and 4992 pairs; 904 pass the first shifted law, 314 pass both, and all resolve by $M \le 10^4$. The unique maximum is

$$\begin{aligned} W(108{,}868{,}200) &= 599, & 108{,}868{,}200 &= 42 \cdot 1610^2, \\ (M, D, a) &= (599, 7500, 181800). \end{aligned} \tag{64.18}$$

This is a thin coordinate slice, not the full interval through 10^9. Thus C'_{SQ} survives finite stress and remains open; only its right-to-left inclusion is proved, and no C''_{SQ} is introduced.

26.5. **Verification and review pedigree.** Block `(bk)` uses Python integers throughout. It pins all 64978 classes through 10^4, compares the two finite normal forms for $n \le 80$, checks 3523 quotient rows including all 2347 cancellation failures, checks both shifted laws for every $n \le 5000$, replays 460 dyadic rows, 359 twisted-square rows, and 300 seeded unit branches, and recomputes all three ledgers. The default hunt is streamed; the 242837-point expansion and thin slice are gated.

The wave-27 maximum-severity review returned SOUND-AFTER-REPAIRS. It rederived the exact eligibility-failure window, including even D, every modulo-4 and modulo-8 support law, the Legendre-symbol roots, and composite unit-branch admissibility. A fresh Python-integer implementation rebuilt all 64978 classes, sampled 3000 new-box pairs, directly cross-checked 64 minima, and independently resolved all 242837 new pairs with the unique maximum (64.16). It reproduced the old/new/enlarged counts 146016/242837/388853, proved all enlarged values distinct by recovering their squarefree kernels, matched every gcd and failed-prime multiplicity, and matched all prefilter and thin-slice counts. The repaired isolated block took about 1.03 seconds by default and 9.01 seconds gated; the prescribed suite passed in 185.25 seconds. This is internal validation only. The hunts prove no growth law or unbounded classification; $W = +\infty$ is failure of the harvested Type-II mechanism, not failure of an Egyptian-fraction representation.

27. The two-decade hard-prime witness census and record anatomy

Computational 65.1 (exact range, informational). This census extends

$$W(p) = \min\{M \equiv 3 \pmod 4 : p \equiv -4D \pmod M \text{ for some } D \mid ((M+1)/4)^2\} \tag{65.1}$$

through every hard prime $p < 10^8$. The original transient runner reported all 7500 eligible rows through $M = 30000$, containing 244216 residue classes, a segmented NumPy prime bitmap, an increasing-modulus walk, exactly 719781 hard primes, no unresolved prime, and Python-integer divisor arithmetic. It was not committed, so its NumPy dtype, first-hit logic, and emergency-extension check cannot be source-audited and its timing below is historical.

The committed independent replay `scripts/review65_independent.py` supplies the durable full check. It imports neither `verify.py` nor SymPy. Its private segmented Eratosthenes sieve uses explicit `int64` on the prime side; trial division and divisor expansion use Python integers; and an independent shrinking-survivor walk visits M strictly upward and assigns each prime once. An explicit leftover assertion fails if a prime requires $M > 30000$. It independently obtains all 244216 classes, all 719781 primes and minima, and largest needed modulus 2495.

The transient staged $p < 10^7$ run reported all 82887 primes and the old record tables; its claimed second implementation on 1000 seeded primes was also not preserved. Block `(bl)` durably rebuilds both class systems and, between default and optional modes, replays the displayed records, four late rows, and 1000 seeded hard primes below 10^5. The independent review script instead recomputes the complete 10^8 census and every table below from scratch.

27.1. **The complete strict-record ledger.** The complete strict-record table is

p	$W(p)$	a_W	D_W	$p+4$	$W(p) > \sqrt{p+4}$
73	7	11	1	$7 \cdot 11$	N
193	15	15	8	197	Y
1201	31	39	2	$5 \cdot 241$	N
2521	47	55	16	$5^2 \cdot 101$	N
3361	99	39	125	$5 \cdot 673$	Y
33289	155	215	9	$13^2 \cdot 197$	N
90841	167	551	294	$5 \cdot 18169$	N
144169	191	755	9	144173	N
167521	259	647	13	$5^2 \cdot 6701$	N
225289	279	811	245	$37 \cdot 6089$	N
361321	287	1259	3	$5^2 \cdot 97 \cdot 149$	N
915961	303	3023	2	$5 \cdot 29 \cdot 6317$	N
954409	335	2855	504	$181 \cdot 5273$	N
1853329	383	4839	2	1853333	N
2031121	2495	815	576	$5^3 \cdot 16249$	Y.

(65.2)

There is no new strict record in $10^7 \le p < 10^8$; the running maximum remains 2495. At a minimum modulus with several firing divisors, the selected witness is the least positive firing D, equivalently the least (a, D) used by the Section 23 replay. On all 82887 rows below 10^7 this happens also to be first in the existing factor-expansion order, but that coincidence is not the definition. Every record satisfies $a_W W = p + 4D_W$ and is a valid original-coordinate witness. The displayed a_W's have minimum 11, nearest-rank median 647, and maximum 4839: five are below 100, six lie in $[100, 1000)$, and four are at least 1000. The D_W's split eight even and seven odd. These are finite record-row descriptions, not distribution laws.

Exactly three records, $193, 3361, 2031121$, satisfy $W(p) > \sqrt{p+4}$. Their complete displayed factorizations contain only prime factors $1 \pmod 4$, so the $p+4$ prediction has zero violations in all 15 rows. This is a numerical check of the proved implication, not a proof of a parallel argument.

27.2. **Minimum-row parity and distribution below 10^7. Computational 65.2 (exact $p < 10^7$).** Under the selected-divisor convention, 49975 of 82887 minimum rows have even D and 32912 have odd D: 60.293% versus 39.707%. The $p < 3 \cdot 10^5$ split was 1941/3202, or 60.618% even. There are 458 rows where both parities fire at the same minimum M; the least firing D is odd in all 458. Intrinsically, 49975 minima have only even firing divisors, 32454 only odd, and 458 both; none has more than two firing divisors.

The exact histogram and nearest-rank percentiles are

W interval	$\#p$	percentile	W
$[1, 7]$	41577	50%	7
$[8, 15]$	26959	75%	15
$[16, 31]$	8632	90%	23
$[32, 63]$	3989	95%	39
$[64, 127]$	1463	99%	87
$[128, 255]$	223	99.9%	203
$[256, 511]$	40	99.99%	351
$[512, 1023]$	3	100%	2495
$[1024, \infty)$	1		

(65.3)

The 82887 values occupy 67 eligible moduli. The counts sum to the exact population; prose percentages and quantile labels are descriptive only.

27.3. **Blockwise normalized trajectories. INFO 65.1 (finite block maxima only).** Each parenthesis below is $(p, W(p))$, logarithms are natural, and the final block is clipped at the exact census endpoint.

p range	$\#p$	$\sup W/\log p$	$\sup W/(\log p \log\log p)$
$[64, 128)$	2	1.631527 (73, 7)	1.120251 (73, 7)
$[128, 256)$	2	2.850253 (193, 15)	1.716356 (193, 15)
$[256, 512)$	5	2.449106 (457, 15)	1.351360 (457, 15)
$[512, 1024)$	6	2.303530 (673, 15)	1.229462 (673, 15)
$[1024, 2048)$	16	4.371794 (1201, 31)	2.231858 (1201, 31)
$[2048, 4096)$	31	12.192127 (3361, 99)	5.821495 (3361, 99)
$[4096, 8192)$	58	4.521754 (5569, 39)	2.098591 (5569, 39)
$[8192, 16384)$	99	4.973014 (12721, 47)	2.214045 (12721, 47)
$[16384, 32768)$	200	8.451130 (29569, 87)	3.624593 (29569, 87)
$[32768, 65536)$	375	14.885265 (33289, 155)	6.352935 (33289, 155)
$[65536, 131072)$	704	14.627482 (90841, 167)	6.006953 (90841, 167)
$[131072, 262144)$	1331	22.636661 (225289, 279)	9.012698 (225289, 279)
$[262144, 524288)$	2548	22.426217 (361321, 287)	8.797177 (361321, 287)
$[524288, 1048576)$	4815	24.330286 (954409, 335)	9.277839 (954409, 335)
$[1048576, 2097152)$	9147	171.783468 (2031121, 2495)	64.198698 (2031121, 2495)
$[2097152, 4194304)$	17541	25.905959 (3587809, 391)	9.544481 (3587809, 391)
$[4194304, 8388608)$	33433	48.296787 (5214049, 747)	17.634931 (5214049, 747)
$[8388608, 16777216)$	64072	39.184586 (13383241, 643)	14.005192 (13383241, 643)
$[16777216, 33554432)$	123271	53.961431 (21475609, 911)	19.092786 (21475609, 911)
$[33554432, 67108864)$	236663	52.790268 (39203761, 923)	18.449734 (39203761, 923)
$[67108864, 10^8)$	225462	55.021338 (88808281, 1007)	18.927125 (88808281, 1007).

(65.4)

Over the complete range, both normalized suprema remain at $(p, W) = (2031121, 2495)$, respectively 171.783468 and 64.198698. Later block maxima are local maxima, not strict records of W.

27.4. **Replay scope, review pedigree, and honest walls. Computational 65.3 (bounded replay).** By default block `(bl)` freshly builds both Python-integer class systems through $M = 3000$, asserts rowwise equality, replays every record through 10^6 and all four late rows including exact fraction sums, compares 1000 seeded hard-prime minima below 10^5, pins the small-prefix parity count $(706, 475)$, and checks the $p + 4$ prediction on records through 10^6. With `ES_FULL_SCAN=1`, it uses $M \le 30000$ and replays all 15 record primes below 10^8, certifying absence of smaller moduli for those primes only. It does *not* rerun the 719781-prime census; that is the separate committed independent script.

The original author reported 1.37 seconds for the 10^7 process and 2.65 seconds with 120960 KiB maximum resident memory for 10^8; because the runner was not preserved, these are historical, not reproducible from those commits. Block `(bl)` was reported at 0.029 seconds by default and 0.663 seconds in isolation with the full path. Scan speed reflects tiny typical minima and replaces none of the exact checks.

The wave-27 maximum-severity review returned SOUND-AFTER-REPAIRS. Its durable from-scratch script enumerated exactly 719781 hard primes in the exclusive range $73 \le p < 100000000$, ending at 99999721, produced all 7500 rows and 244216 classes, resolved every prime with last necessary row $M = 2495$, and reproduced all 15 records. It matched every one of the 21 block counts and both normalized columns to six decimals, the global suprema 171.783468/64.198698, selected parity 49975/82887, intrinsic split $(49975, 32454, 458)$, and zero $p + 4$-prediction violations. The review also repaired the selected-D convention and narrowed claims about the missing transient source without changing a table entry. The independent census passed in 1.39 seconds with 108424 KiB peak resident memory; bounded block `(bl)` remained deterministic and did not claim to rerun it. These are internal computational checks, not external validation.

Honest walls. This census sharpens only the historical finite-normalization panel denoted $1 \le A \lesssim 2$ in (56.16). Section 29 records the exact interpretation: every $A \ge 1$ remains open, while 1–2 is not an upper frontier. No new record after 10^7,

parity stability, and normalized block values prove nothing about $H_{\mathrm{MOD}}(A)$: they give no upper bound for any $A \geq 1$ and do not alter the proved refutation for $A < 1$. No growth law, asymptotic fit, independence assertion, or tail-density claim follows from (65.2)–(65.4). Every conclusion is Computational/INFO finite-range evidence; H_{MOD} and the Erdős–Straus conjecture are untouched.

28. Twisted-family escape cylinders and the identity wall

Status and headline. The finite-depth class law, the asymptotic size of its period, the resulting unboundedness in all three proved sporadic families, the family certificate ceiling, and the anchored polynomial-identity obstruction below are **proved**. The coefficient hunt, projected residue panel, and three bounded censuses are **Computational**, exact only in their stated ranges. In particular,

$$\limsup_{m\to\infty} W(2m^2) = \limsup_{m\to\infty} W(21m^2) = \limsup_{m\to\infty} W(505m^2) = +\infty. \tag{66.1}$$

This produces no fourth value with $W = +\infty$: unbounded finite depth is not pointwise infinite escape.

28.1. **Finite-depth class decidability.** For real $T \geq 0$, with the least common multiple of an empty set defined to be 1, put

$$L(T) = \operatorname*{lcm}_{\substack{M\leq T\\ M\equiv 3\ (4)}} M, \qquad S_M = \{-4D \pmod M : D \mid ((M+1)/4)^2\}. \tag{66.2}$$

All divisors in this section are positive, and no primality condition is imposed on M.

Theorem 28.1 (twisted-family class decidability; proved; source Theorem 66.1)**.** *Let $s \geq 2$ be squarefree and let $T \geq 0$. Membership in*

$$\{m \geq 1 : W(sm^2) \leq T\} \tag{66.3}$$

depends only on $m \pmod{L(T)}$. More explicitly, (66.3) is the finite union, over $M \leq T$, $M \equiv 3 \pmod 4$, and $D \mid ((M+1)/4)^2$, of the solution classes of

$$sm^2 \equiv -4D \pmod M. \tag{66.4}$$

Proof. The inequality holds exactly when one eligible modulus $M \leq T$ fires, which is exactly (66.4) for one eligible D. There are finitely many such pairs, and truth for a fixed pair depends only on $m \pmod M$. Every such M divides $L(T)$. Conversely, every class in the union is a witness datum, so none has been added. □

Lemma 28.2 (exact exponential scale of the class period; proved; source Lemma 66.2)**.** *As $T \to \infty$,*

$$\begin{aligned} \log L(T) &= \vartheta(T;4,3) + \vartheta(T/3;4,1) + O(\sqrt{T}\log T) \\ &= \left(\frac{2}{3} + o(1)\right) T. \end{aligned} \tag{66.5}$$

In particular $P_3(T) \mid L(T)$, with $P_3(T)$ as in (56.1), and $\log L(T) \geq (1/2 + o(1))T$. More strongly, for every $\epsilon > 0$, effectively for all sufficiently large T, $L(T) \geq \exp((2/3 - \epsilon)T)$.

Proof. No factor 2 occurs in $L(T)$. For an odd prime power p^e, the least positive multiplier r for which $rp^e \equiv 3 \pmod 4$ is 1 when $p^e \equiv 3 \pmod 4$, and 3 when $p^e \equiv 1 \pmod 4$. Hence

$$p^e \mid L(T) \quad \Longleftrightarrow \quad \begin{cases} p^e \le T, & p^e \equiv 3 \pmod 4, \\ 3p^e \le T, & p^e \equiv 1 \pmod 4. \end{cases} \tag{66.6}$$

First powers contribute every prime $3 \pmod 4$ through T and every prime $1 \pmod 4$ through $T/3$; higher powers contribute $O(\sqrt{T}\log T)$. The prime number theorem in the two fixed progressions gives $T/2 + o(T)$ and $T/6 + o(T)$. Formula (66.6) also gives the divisibility by $P_3(T)$, and effective fixed-progression estimates give the final claim. □

Thus the leading constant is $2/3$, not merely the valid lower bound $1/2$ used in Theorem 18.1; the extra $1/6$ comes from multiplying primes $1 \pmod 4$ through $T/3$ by 3.

28.2. **Unbounded depth in the three sporadic families.**

Theorem 28.3 (twisted-family unboundedness; proved; source Theorem 66.3)**.** *For every $T \ge 0$,*

s	m_0	*an infinite depth-T class*
2	12	$m \equiv 12 \pmod{L(T)}$
21	4	$m \equiv 4 \pmod{L(T)}$
505	3	$m \equiv 3 \pmod{L(T)}$.

(66.7)

Every positive m in the displayed class satisfies $W(sm^2) > T$. Consequently all three limsups in (66.1) are infinite. Quantitatively, for each row and every $X \ge 1$,

$$\#\{m \le X : W(sm^2) > T\} \ge \frac{X}{L(T)} - 1. \tag{66.8}$$

Thus the supplied lower-density bound is $1/L(T)$, where $\log L(T) = (2/3 + o(1))T$.

Proof. Theorem 22.3 gives $W(2 \cdot 12^2) = W(21 \cdot 4^2) = W(505 \cdot 3^2) = +\infty$. Each anchor is therefore outside (66.3), and Theorem 28.1 puts its whole class modulo $L(T)$ outside. The progressions contain arbitrarily large positive integers, proving (66.1), and counting one class gives (66.8). □

This differs from Theorem 17.1: there the input itself is put in the residue-one class modulo $\mathrm{lcm}(1, \dots, \lfloor T \rfloor)$ and Linnik selects a prime. Here no prime selection occurs; the cylinders lie inside the thin families sm^2 in their parameter m, anchored at proved $W = +\infty$ points.

Corollary 28.4 (family-level finite-certificate ceiling; proved; source Corollary 66.4)**.** *No finite collection of fixed harvested congruence laws proves*

$$W(2m^2) < +\infty \qquad (m \ne 12). \tag{66.9}$$

Equivalently, at every finite harvested depth T, infinitely many parameters $m \ne 12$ remain uncovered, including every $m = 12 + kL(T)$, $k \ge 1$. Any proof of $W(2m^2) = +\infty \Longleftrightarrow m = 12$ must use structure other than a finite cover by fixed (M, D) congruences.

Proof. A finite collection has a largest modulus T, and Theorem 28.3 puts every $12 + kL(T)$, $k \ge 1$, beyond all its firing conditions. □

This is the family-parameter analogue of Section 18, not a restatement of it. It uses an actual cylinder based at 288 and only rules out finite collections of fixed harvested (M, D) laws. The older prime-identity and quadratic-form finite-cover obstructions motivate the same warning but do not prove this corollary.

28.3. **Polynomial identities in the residual classes.** For this hunt, an *admissible polynomial datum* on $m(t) = 12 + Kt$ consists of $M, Q \in \mathbb{Z}[t]$ and $A, D, E \in \mathbb{Q}[t]$, with the latter three integer-valued on integers, such that

$$M = 4A - 1, \qquad A^2 = DE, \qquad 2m^2 + 4D = MQ \tag{66.10}$$

as polynomial identities, with $2m^2 + 4D \in \mathbb{Z}[t]$, and such that for every integer $t \geq 0$,

$$M(t) > 0, \qquad M(t) \equiv 3 \pmod 4, \qquad D(t) > 0. \tag{66.11}$$

Thus $D(t) \mid A(t)^2$ with integer cofactor $E(t)$ at every argument. Polynomial divisibility, positivity, and integrality at every nonnegative integer are part of the definition.

Proposition 28.5 (anchored polynomial-identity obstruction; proved; source Proposition 66.5)**.** *For every $K \geq 1$, no admissible polynomial datum (66.10)–(66.11) exists on $m(t) = 12 + Kt$. The analogous statements hold on $m(t) = 4 + Kt$ for $s = 21$ and on $m(t) = 3 + Kt$ for $s = 505$, after replacing the leading 2 in (66.10) by the relevant kernel.*

Proof. Specialization at $t = 0$ would give a positive eligible datum for 288, 336, or 4545, contradicting Theorem 22.3. No degree or coefficient bound is used. □

This all-degree obstruction applies only when eligibility includes the anchor. By itself it does *not* extend to a forward ray: polynomial equalities holding at every sufficiently large integer extend identically back to the anchor, but positivity and positive-divisor eligibility need not; without them, specialization there is not a datum for W. Section 31 closes precisely this gap for polynomial data by first collapsing integer-wise divisibility to polynomial divisibility with integer-valued cofactors and then propagating positivity backward. Its scope does not include arbitrary non-polynomial witness functions.

Computational 66.1 (coefficient/resultant identity hunt; exact stated ansatz). Block `(bm)` used

$$K \in \{12, 24, 44, 84, 132, 231\}, \qquad m(t) = 12 + K(t + e), \quad e \in \{0, 1\}, \tag{66.12}$$

where $231 = 3 \cdot 7 \cdot 11$. It exhausted $\deg M \in \{1, 2\}$, $1 \leq d \leq 500$, and

$$D = d, \qquad D = dA, \qquad D = A^2/d. \tag{66.13}$$

For each shape it imposed pointwise integer eligibility exactly. If $d = \prod p^v$, then $d \mid A(t)^2$ for every integer t is equivalent to $\prod p^{\lceil v/2 \rceil} \mid A(t)$ for every t; the first three values decide this for integer-valued quadratic A. The identities reduce to

$$M \mid 2m^2 + 4d, \qquad M \mid 2m^2 + d, \qquad M \mid 8dm^2 + 1. \tag{66.14}$$

Each right side is a positive quadratic and has no real linear factor. In degree 2, coefficient comparison fixes M up to a positive scalar dividing the content of all three coefficients; every such scalar was enumerated, so there was no hidden cutoff on M or the quotient.

There were exactly 54,990 candidates $(K, e, d, \text{shape}, \text{scalar})$; 6,810 made $M(t) \equiv 3 \pmod 4$ identically, and none passed the remaining identity and eligibility conditions. The anchored half had 27,495 and 3,405 candidates respectively. The forward-ray half also had no hit, but this is only a negative result for (66.12)–(66.13), not an all-shape theorem. No sampled-value test substitutes for the exact coefficient checks.

28.4. **The depth-1000 projection and bounded censuses.** Let $\mathcal{R}_{2,T} \subseteq \mathbb{Z}/L(T)\mathbb{Z}$ be the exact residual set defined by $W(2m^2) > T$. The full $L(1000)$ has 287 decimal digits and

$$\log L(1000) = 659.7431249956927\ldots. \tag{66.15}$$

Computational 66.2 (exact depth-1000 projection). Put $K = 3 \cdot 11 \cdot 19 = 627$, which divides $L(1000)$. Block `(bm)` computes

$$\begin{aligned} \pi_K(\mathcal{R}_{2,1000}) = \{r \pmod{627} : {} & r \equiv 0 \pmod 3, \\ & r \pmod{11} \in \{0, \pm 1, \pm 5\}, \\ & r \pmod{19} \in \{0, \pm 1, \pm 2, \pm 4, \pm 5, \pm 7, \pm 9\}\}. \end{aligned} \tag{66.16}$$

These are exactly 65 classes; the other 562 are covered by the $M = 3, 11, 19$ rows. The only additional eligible modulus through 1000 dividing 627 is 627, and it removes none. For every displayed class the block finds an explicit positive lift with no firing $M \le 1000$, which certifies reverse inclusion in the full projection. The least searched lifts have maximum 175,539, in class 606 with lift index 279; their ordered SHA-256 digest is `7f385ae9b7ddb042e873f52883d4f90f43ddb551b34f89b1a377c2a57a0c4389`. This is a projection statement, not a claim that the full residual set has period 627; its full period is $L(1000)$.

Computational 66.3 (three multiplier censuses; exact stated box and cap). Block `(bm)` walks every $M \equiv 3 \pmod 4$ upward through 10^4 with Python integers. For each $s = 2, 21, 505$ and every $1 \le m \le 2000$, it finds the exact minimum whenever a row fires:

s	no hit through 10^4	$\max\{W(sm^2) : m \le 2000,\ W < \infty\}$	where
2	12	4019	1530
21	4	695	851
505	3	479	1239.

(66.17)

The cap survivors are exactly the proved infinite anchors; no second survivor occurs in any box, and infinity is not inferred from the cap. Every other entry fires by 10^4, and its minimum is exact because every smaller eligible modulus was checked. At depth 1000, the only additional $s = 2$ parameters in the box are

$$W(2 \cdot 264^2) = 1499, \qquad W(2 \cdot 1050^2) = 1259, \qquad W(2 \cdot 1530^2) = 4019. \tag{66.18}$$

For $s = 21, 505$, only the relevant anchor survives depth 1000. This extends earlier, differently sliced finite codebooks and remains a finite census. On the research host block `(bm)` took 0.151 seconds; the complete prescribed suite remained green in 210.08 seconds with 342,224 KiB peak resident memory. Timings are informational.

28.5. **Honest walls and review attestation.** The quantifiers proved are

$$\forall T\ \exists \text{ infinitely many } m : \quad W(2m^2) > T, \tag{66.19}$$

not $\exists m \ne 12 : W(2m^2) = +\infty$. A fixed $m = 12 + kL(T)$ may acquire a deeper witness, and the class changes with T. Neither (66.16) nor (66.17) changes this.

Accordingly,

$$W(2m^2) = +\infty \quad \Longleftrightarrow \quad m = 12 \tag{66.20}$$

remains open. It requires either, for every fixed $m \neq 12$, one eligible pair $M \equiv 3 \pmod 4$, $D \mid ((M+1)/4)^2$ with $M \mid 2m^2+4D$, or an explicit second parameter whose complete finite normal form is empty. Finite covers and bounded searches do not exchange the pointwise $\forall m\, \exists (M, D)$ quantifiers. Thus the restricted classification and C'_{SQ} remain open.

Review attestation (wave 28). The maximum-severity verdict was SOUND-AFTER-MINOR-REPAIRS. The review independently rederived the class quantifiers and the $1/2 + 1/6 = 2/3$ period constant, reproduced the 287-digit $L(1000)$ and $\log L(100)/100 = 0.5808244316\ldots$, $\log L(1000)/1000 = 0.6597431250\ldots$, and confirmed that the limsups and (66.8) create no new $W = +\infty$ member. It restricted the certificate ceiling to finite fixed harvested laws and identified the exact forward-ray leak: equalities extend backward but positivity and eligibility need not. Section 31 subsequently closes that leak for a single polynomial datum, without reaching non-polynomial supply. The shifted $e = 1$ result remains only the bounded three-shape computation.

A fresh ascending-M Python-integer replay reproduced sole survivors $(2, 12), (21, 4), (505, 3)$, maxima $(4019, 1530), (695, 851), (479, 1239)$, every depth-1000 exception, all 65 projected classes, maximum least lift $(606, 175539, 279)$, and the digest. A separate coefficient implementation replayed all 54,990 candidates in the twelve (K, e) families and all three shapes; 6,810 had the mod-4 law and none was eligible. Block `(bm)` is deterministic, has no random seed, uses Python integers, and pins complete boxes rather than samples. It took 0.30 seconds on the review host, and the full prescribed verification command reran green. This is internal validation, not external refereeing.

29. Record-stall anatomy and the honest exponent window

Status. The extremal anatomy, residue-one examples, and exponent table below are **Computational**, exact in their stated finite sets. The record-spacing calculation is explicitly **Heuristic**. The final window statement is an **Assessment** of existing quantifiers, not a new bound. No growth law, tail theorem, or pointwise conclusion is claimed.

29.1. **The top-ten obstruction anatomy.** Let

$$\mathcal{A}_{200} = \{3, 7, 11, \ldots, 199\}, \qquad S_{200}(p) = \{a \in \mathcal{A}_{200} : -1 \in \mathrm{Rat}_a((p+a)/4)\}\,. \tag{67.1}$$

Thus $\mathcal{A}_{200} \setminus S_{200}(p)$ is the complete 50-law failure profile. For a second, fixed-D diagnostic define the exhaustive purity depth

$$\tau(p) = \sup\left\{T \in \mathbb{Z}_{\geq 0} : \begin{array}{l}\text{for every } 1 \leq D \leq T,\ p+4D \text{ has no divisor } M \text{ with} \\ M \equiv 3 \pmod 4,\ M > 4D, \quad D \mid ((M+1)/4)^2\end{array}\right\} \in \mathbb{Z}_{\geq 0} \cup \{+\infty\}. \tag{67.2}$$

The inequality $M > 4D$ is part of the diagnostic, so τ is not W and ignores eligible divisors on the other side. When $\tau < +\infty$, write (D_*, M_*) for the first failing pair after the pure prefix, with $D_* = \tau + 1$ and the least eligible M_* in a tie. The decision is finite: the cofactor $(p + 4D)/M$ is 3 (mod 4) and at least 3, so $M > 4D$ forces $D < p/8$. Infinity genuinely occurs: exhaustive use of this bound gives $\tau(193) = +\infty$, although $W(193) = 15$ via $(D, M) = (8, 15)$, excluded by $M > 4D$.

Computational 67.1 (complete $p < 10^8$ top ten; exact). The Section 27 replay has 98 hard primes with $W \geq 400$. Its ten largest values, ordered by decreasing W then increasing p, have the two-frame anatomy

p	W	a_1	$S_{200}(p)$	τ	(D_*, M_*)
2031121	2495	11	11, 19, 39, 55, 59, 95, 111, 139, 167, 179	28	(29, 52083)
88808281	1007	3	3, 15, 19, 23, 35, 43, 59, 127, 131, 135, 159	7	(8, 2783)
39203761	923	7	7, 11, 27, 47, 71, 107, 127, 143, 159, 167	10	(11, 94467)
43371241	923	7	7, 35, 43, 71, 103, 179, 187, 199	1	(2, 1223)
21475609	911	7	7, 27, 35, 51, 71, 103, 107, 111, 119, 151, 183, 191	9	(10, 15439)
23836201	911	11	11, 19, 27, 31, 35, 47, 55, 59, 63, 71, 87, 111, 131, 135, 183	1	(2, 1759)
62850769	911	3	3, 23, 43, 47, 51, 55, 67, 71, 75, 83, 95, 99, 103, 111, 119, 131, 151	6	(7, 1987)
22706161	783	11	11, 19, 47, 55, 59, 79, 131, 143	13	(14, 783)
5214049	747	3	3, 7, 11, 15, 39, 47, 71, 83, 95, 131, 191	14	(15, 27299)
5309329	719	7	7, 11, 23, 31, 39, 63, 107, 111, 131, 171	1	(2, 1047)

(67.3)

The rows fail respectively $40, 39, 40, 42, 38, 35, 33, 42, 39, 40$ of the 50 laws. A large W need not require all small a to fail: an early spectrum hit can use a large divisor and produce a much larger modulus.

The exact four-shift ledger used by the fixed-D side is

p	$p+4$	$p+8$	$p+12$	$p+16$
2031121	$5^3 \cdot 16249$	$3^3 \cdot 75227$	$13 \cdot 156241$	2031137
88808281	$5 \cdot 349 \cdot 50893$	$3 \cdot 17 \cdot 1741339$	$7 \cdot 331 \cdot 38329$	88808297
39203761	$5 \cdot 401 \cdot 19553$	$3 \cdot 11 \cdot 1187993$	$7^2 \cdot 800077$	$67 \cdot 585131$
43371241	$5 \cdot 8674249$	$3 \cdot 1223 \cdot 11821$	43371253	43371257
21475609	21475613	$3 \cdot 7158539$	21475621	$5^4 \cdot 34361$
23836201	$5 \cdot 4767241$	$3 \cdot 1759 \cdot 4517$	23836213	$433 \cdot 55049$
62850769	62850773	$3 \cdot 11 \cdot 601 \cdot 3169$	$7^2 \cdot 211 \cdot 6079$	$5 \cdot 17 \cdot 101 \cdot 7321$
22706161	$5 \cdot 4541233$	$3 \cdot 17 \cdot 41 \cdot 10859$	$7 \cdot 3243739$	$13 \cdot 1746629$
5214049	$13 \cdot 17 \cdot 23593$	$3 \cdot 1738019$	5214061	$5 \cdot 89 \cdot 11717$
5309329	5309333	$3 \cdot 11 \cdot 349 \cdot 461$	$19 \cdot 103 \cdot 2713$	$5 \cdot 1061869$

(67.4)

Every $p+4$ here is free of prime factors 3 (mod 4), so the global $D = 1$ law misses all ten. This is a common mechanistic first filter. The exact $D = 2$ class $M \equiv 7 \pmod 8$ fires next only for the three rows with $\tau = 1$. Thereafter depth does not order W: the two $W = 923$ rows have depths $10, 1$, the three $W = 911$ rows have depths $9, 1, 6$, and the smaller $W = 747$ row has depth 14. The isolated record has the largest depth 28, but the other nine give no monotone relation. These finite facts partly explain the gate, not the duration of the stall.

29.2. **The residue-one construction at its first primes.** Section 17 uses exactly

$$M(T) = \operatorname{lcm}(1, 2, \ldots, \lfloor T \rfloor), \tag{67.5}$$

not merely eligible 3 (mod 4) moduli. For the four values below, $24 \mid M(T)$, so combining $p \equiv 1 \pmod{M(T)}$ with the hard-prime condition changes no modulus. Exhausting $p = 1 + kM(T)$ upward and then scanning eligible witness moduli gives

T	$M(T)$	k	p	$W(p)$	$W - T$	a_1	$\frac{\log p}{5.2 \log M(T)}$
15	360360	12	4324321	23	8	3	0.229656
19	232792560	1	232792561	183	164	27	0.192308
23	5354228880	2	10708457761	47	24	7	0.198258
27	80313433200	5	401567166001	71	44	7	0.204634

(67.6)

The last column compares only logarithmic sizes with the bare $M(T)^{5.2}$ scale and omits the unknown effective constant C_0; it is not a sharper Linnik estimate. The guaranteed $W > T$ is strict in every row. The excesses are $8, 164, 24, 44$, with the $T = 19$ row a conspicuous finite outlier. Small k puts these examples far below the generic ceiling by arithmetic luck, not an improvement to the theorem.

29.3. **A deliberately fragile record-spacing model. Heuristic 67.1 (calibrated independent-hit Poisson model).** For eligible M, define all model inputs by

$$\begin{aligned} \mathcal{C}_M &= \{-4D \pmod M : D \mid ((M+1)/4)^2\}, \\ \mathcal{R}_M &= \{r \pmod M : (r,M)=1,\ r \equiv 1 \pmod{(M,24)}\}, \\ u_M &= |\mathcal{C}_M \cap \mathcal{R}_M|/|\mathcal{R}_M|. \end{aligned} \tag{67.7}$$

The raw model falsely treats all modulus hits as independent, suppressing the shared-p, shared-factor, and cross-a dependencies, and uses

$$P_{\mathrm{raw}}(W > T) = \prod_{\substack{M \le T \\ M \equiv 3 \ (4)}} (1 - u_M). \tag{67.8}$$

At $T = 2495$, direct class enumeration gives

$$-\log P_{\mathrm{raw}} = 20.522039361\ldots, \qquad P_{\mathrm{raw}} = 1.222901698 \cdot 10^{-9}. \tag{67.9}$$

At the five lower, nonzero-count calibration thresholds—not at 2495—the raw values are below observed finite frequencies. Replacing the raw model by $P_\beta = P_{\mathrm{raw}}^{\beta}$ gives

T	$\#\{p < 10^8 : W(p) > T\}$	$P_{\mathrm{raw}}(W > T)$	$\beta = \frac{\log(\#/719781)}{\log P_{\mathrm{raw}}(W>T)}$
255	421	$1.955681544 \cdot 10^{-4}$	0.871712
383	109	$3.757844433 \cdot 10^{-5}$	0.863214
511	45	$1.049874640 \cdot 10^{-5}$	0.844367
639	16	$3.501544731 \cdot 10^{-6}$	0.852878
767	8	$1.397322616 \cdot 10^{-6}$	0.846176

(67.10)

Thus $\beta = 0.85$ is a finite calibration, not a constant of nature. As a separate warning, multiplying the six uniform-h failure frequencies from the local-law census gives 0.002025, whereas $361/82887 = 0.004355$ hard primes have $a_1 > 23$; the populations differ, so even this factor-two discrepancy is only diagnostic.

At 2495 the calibrated per-prime probability is exactly the toy output $P_{0.85} = 2.656334824 \cdot 10^{-8}$. Holding it fixed and using

$$\frac{\pi(10^9) - \pi(10^8)}{8} = \frac{50{,}847{,}534 - 5{,}761{,}455}{8} = 5{,}635{,}759.875$$

as the future hard-prime proxy gives

p interval	# hard primes used	$E\#\{W > 2495\}$	$\beta \in [0.80, 0.90]$	observed
$(10^7, 10^8)$	636894	0.0169180	$[0.0061, 0.0472]$	0
$(10^8, 10^9)$	5,635,759.875 (proxy)	0.1497047	$[0.0537, 0.4177]$	$--$

(67.11)

The formal first-row Poisson zero mass is $e^{-0.0169} = 0.9832$. It is not a p-value and supplies no evidence for or against the stall. Independence is false, calibration stops at 767, extrapolation crosses an unobserved tail, one eighth is only a proxy despite exact ordinary prime counts, and selection of the old record is ignored. Nothing about a tail density, next-record location, growth law, or support for the model is inferred.

29.4. **The empirical exponent and eventual window.** All logarithms here are natural. For a finite row put

$$A_{\mathrm{emp}}(p) = \frac{\log W(p)}{\log \log p}. \tag{67.12}$$

It is the exponent making $W(p) = (\log p)^{A_{\mathrm{emp}}(p)}$ at that one prime, not a fitted population exponent.

Computational 67.3 (all 15 strict records; exact inputs).

p	W	A_{emp}	p	W	A_{emp}
73	7	1.336115	167521	259	2.234072
193	15	1.630725	225289	279	2.242045
1201	31	1.753095	361321	287	2.220056
2521	47	1.870574	915961	303	2.181299
3361	99	2.194077	954409	335	2.217096
33289	155	2.152501	1853329	383	2.228161
90841	167	2.101766	2031121	2495	2.923244
144169	191	2.122345			

(67.13)

Assessment 67.1 (the honest A-window). The hypothesis $H_{\mathrm{MOD}}(A)$ asks for $W(p) \le (\log p)^A$ for every sufficiently large prime. Theorem 17.1 refutes every $A < 1$, while every $A \ge 1$ remains open. Thus the $1 \le A \lesssim 2$ display (56.16) has two registers: $A \ge 1$ is the exact eventual frontier, while 1–2 is only the historical finite normalization panel, not an upper frontier. The finite value $A_{\mathrm{emp}}(2031121) = 2.923244$ is ignored by every sufficiently-large quantifier. The complete census has no later W-record through 10^8; since later p are larger, it also has no later A_{emp} record. These finite facts do not choose among eventual $\limsup A_{\mathrm{emp}}$ equal to 1, any larger finite value, or infinity. No eventual upper exponent is known.

29.5. **Verification, review, and scope.** Block `(bn)` rebuilds every displayed top-ten minimum, small-a spectrum, purity depth, and four-shift factorization with Python integers. It separately recomputes W, a_1, S_{200} and exhaustive τ for all 15 strict records, including $\tau(193) = +\infty$; exhausts all four least-prime searches; replays all residue-one and exponent rows; performs seeded dual/class checks; and rebuilds (67.7)–(67.11). The full $p < 10^8$ ranking and calibration counts are gated by `ES_FULL_SCAN=1` and use the memory-bounded shrinking-survivor census in `scripts/review65_independent.py`. Default mode checks displayed primes but does not claim a ranking rescan. One shifted factorization and divisor list is live at a time. Post-review timings were 0.34 seconds by default, 1.39 seconds gated, and 215.85 seconds for the complete default suite under the 700-second timeout.

Review attestation (wave 28). The maximum-severity verdict was SOUND-AFTER-REPAIRS. A fresh standard-library implementation with independent trial factorization, divisor expansion, primality testing, and ascending-modulus scan reproduced all displayed finite arithmetic. The substantive repair made τ an integer supremum with an infinity convention and proved the exhaustive $D < p/8$ bound. The heuristic repair confined underestimation to the five nonzero calibration thresholds, exposed every input, called the one-eighth quantity a proxy, and removed all evidential language. The eventual window was corrected to every $A \ge 1$ open, with 1–2 only finite normalization; direct recomputation gives $\log 2495/\log\log 2031121 = 2.923244274\ldots$.

For all 15 records the independent check found

$$a_1 = (7, 7, 23, 23, 3, 3, 7, 15, 15, 31, 19, 11, 3, 3, 11)$$

and

$$\tau = (0, \infty, 1, 7, 28, 7, 7, 4, 12, 31, 2, 1, 7, 1, 28).$$

The complete replay retained 719,781 hard primes, 98 with $W \geq 400$, exactly the displayed top ten, and no record in $(10^7, 10^8)$. It reproduced all four tuples $(T, M, k, p, W, W-T)$ in (67.6), $P_{\text{raw}} = 1.222901698 \cdot 10^{-9}$, $P_{0.85} = 2.656334824 \cdot 10^{-8}$, and the toy expectations $0.0169180, 0.1497047$. The full suite passed with "all checks passed" in 215.85 seconds at 340,808 KiB peak resident memory; AST, control-byte, conflict-marker, and diff checks were clean. These are reproducible toy outputs and internal finite validation, not a tail claim or external refereeing. No eventual upper exponent, tail density, next record, or pointwise Erdős–Straus conclusion follows.

30. The 2025 Pomerance–Weingartner exceptions paper

Source, version, and scope. The current arXiv:2511.16817v2 is archived verbatim in `sources/pomerance-weingartner-2511.16817/`; the full reading audit is `reviews/external-pw-2511.16817-wave28.md`. The older `sources/pomerance-weingartner-2025.pdf` is the November 2025 version of the same paper, not a second paper. This section refreshes the source already underlying Sections 11 and 14; Theorem 1.3 is unchanged in substance. All verdicts below are bounded to this PDF and its bibliography, not complete literature certification.

30.1. **Exact statements and proof routes. Upper bound.** Pomerance–Weingartner Theorem 1.3 says (quote): "There is an absolute positive constant C such that for each pair m, N with $4 \leq m \leq \log^2 N$ the number of $n \leq N$ with m/n not the sum of 3 unit fractions is at most $N/\exp(C \log^{2/3}(N)/\varphi(m)^{1/3})$" (p. 2). In the notation of Section 11, this is exactly

$$E_m(N) \ll N \exp\{-C(\log N)^{2/3}/\varphi(m)^{1/3}\}, \qquad 4 \leq m \leq (\log N)^2. \tag{68.1}$$

It counts all integer denominators. The paper says only "absolute positive constant," not effective or computable. Its lower class-mass proof invokes Bombieri–Vinogradov (Lemma 4.1, p. 9), and the denominator side invokes the large sieve (p. 10); it supplies no exceptional-conductor repair.

Lower exceptions. Theorem 1.1 says (quote): "For each $\epsilon > 0$ there is a bound $m(\epsilon)$ such that for each $m \geq m(\epsilon)$ there is some $n > \exp(m^{1/3-\epsilon})$ with m/n not the sum of 3 unit fractions" (p. 2). The headline does not say prime. Theorem 3.1 gives the stronger count

$$> \exp\{c\varphi(m)^{1/3}/(\log m)^{2/3}\} \tag{68.2}$$

of *prime* exceptions for every $m \geq 8$, most in $(N_0/2, N_0]$, where

$$\log N_0 = \{\varphi(m)/(C \log^2 m)\}^{1/3}. \tag{68.3}$$

The complete prime-denominator split is used. Type I forces $p \equiv -f \pmod{mad}$ with $f \mid ma^2d + 1$; Type II forces $p \equiv -e \pmod{mab}$ with $e \mid a + b$ (Corollaries 2.2 and 2.4 as used in Theorem 3.1, pp. 6–7). Brun–Titchmarsh shows that the union of these solution-admitting families misses most primes in the selected interval. This avoids every admitting class; it is not one fixed congruence obstruction.

Theorem 1.2 says (quote): "For each integer $m \geq 6.52 \times 10^9$ there is a prime p in $(m^2, 2m^2)$ for which m/p is not the sum of 3 unit fractions" (p. 2). Proposition 7.5 bounds the Type-I primes (p. 20), Corollary 9.1 bounds the Type-II primes (p. 23), and their sum is compared with an explicit prime-count lower bound (p. 23). Section 6 only computationally supports the conjectural extension $m \geq 20$; it is not a theorem.

At $m = 4$, (68.1) has exactly Vaughan's 2/3 exponent. The paper states no stronger $m = 4$ all-denominator upper bound; its $m = 4, 5$ claims beyond this are finite verifications. Theorem 1.4 says (quote): "For each pair of positive integers j, k, there is a number $m(j, k)$ such that for each $m \geq m(j, k)$, we have $m/(km + 1)$ not the sum of j unit fractions" (p. 3). Its proof is a discreteness argument for fixed-length sums (Lemmas 5.1–5.2, pp. 11–12), not a modification of the three-summand Type-I/Type-II split.

30.2. **Range consistency and crossover.** Fix the fixed-gap parameter $0 < \sigma < 3$ in Theorem 11.8. Its range is

$$m \leq (\log N)^{3-\sigma} \quad \Longleftrightarrow \quad \log N \geq m^{1/(3-\sigma)}, \qquad \frac{1}{3-\sigma} - \frac{1}{3} = \frac{\sigma}{3(3-\sigma)} > 0. \tag{68.4}$$

The exceptions at (68.3) have $\log p = m^{1/3+o(1)}$, hence

$$\frac{m}{(\log p)^{3-\sigma}} = m^{\sigma/3+o(1)} \longrightarrow \infty. \tag{68.5}$$

They lie outside every fixed-gap range of Theorem 11.8 and the smaller Layer-1 range $m \leq L^{2-\epsilon}$. The explicit $p \in (m^2, 2m^2)$ has $\log p = 2 \log m + O(1)$ and lies farther outside.

Section 35 requires a separate reverse check. At $N = 2m^2$, condition (55.11) forces $\log T = O(\log m)$, so $m \leq (\log T)^B$ fails for every fixed B; no Section 55 tail applies. At the larger scale (68.3), a tail can formally enter its fixed-B parameter range when $B > 3$, but is vacuous at the exponent scale. With $t = \log T \leq O(\log N_0)$,

$$\lambda_m t^2 \log(2 + t) = m^{-1/3+o(1)} \longrightarrow 0, \qquad \theta_m t^3 \leq \frac{\eta_2(m)}{C \log^2 m} \longrightarrow 0. \tag{68.5a}$$

Theorem 35.2 only upper-bounds the number of exceptions; it never asserts zero. Thus neither Section 43 nor Section 55 implies representability in a Pomerance–Weingartner construction regime. There is no contradiction. Saying the asymptotic regime was outside Section 55's range would be false; its saving masses instead vanish.

Since (68.1) is the existing benchmark, the crossover is unchanged. With $L = \log N$, published scale P, and the CLAIMED/PROVISIONAL Theorem 11.8 scale R,

$$P = \frac{L^{2/3}}{\varphi(m)^{1/3}}, \qquad R = \left(\frac{\eta_2(m) L^3}{\varphi(m)} \right)^{1/4}, \qquad \frac{R}{P} = \{\eta_2(m)^3 \varphi(m) L\}^{1/12}. \tag{68.6}$$

Formally $R \geq P$ exactly when $L \geq 1/(\eta_2(m)^3 \varphi(m))$; below that scale P is larger. Unknown constants preclude a finite crossover. The published theorem is proved on $m \leq L^2$. Theorem 11.8 reaches every fixed-gap $m \leq L^{3-\epsilon}$ but inherits Theorems 9.4 and 11.6 and Section 10.7; it remains CLAIMED/PROVISIONAL. No comparison row changes.

30.3. **Priority impact and Vaughan secondary evidence.**

- **Theorem 39.7.** No priority threat was found in this PDF. It quotes Vaughan's $N/\exp(c \log^{2/3} N)$ bound, calls the improvement "strongly improved, though not recently," and retains 2/3 at $m = 4$. It states no larger $m = 4$ all-denominator upper exponent. This strengthens a secondary-source search record, not a literature guarantee. Theorem 10.7 remains CLAIMED/PROVISIONAL.

- **Theorem 43.12.** Equation (68.1) is the already-audited benchmark and (68.2)–(68.5a) are range-consistent lower results. Status and crossover are unchanged; Theorem 11.8 remains CLAIMED/PROVISIONAL.
- **Theorems 51.2/51.6.** Section 4's $k = 1$ good-class family is the already recorded square-log truncation. The paper does not define W, state a tail in T, claim polylogarithmic witness depth, address computability, or cite Dahan. The cubic part of Theorem 14.2 and record-adjacent Theorem 14.6 retain their CLAIMED/PROVISIONAL qualifications.

All three verdicts are source-bounded, not a complete priority search; a later audit (November 2026) found the earlier preprint [16] with the exponent 3/4 for $m = 4$ and for each fixed $m \geq 4$.

For the inaccessible Vaughan original, the five substantive secondary descriptions are: "A result of Vaughan is that for each m, most n's have m/n representable; we make the dependence on m explicit"; "In 1970, Vaughan [13] gave the upper bound $N/\exp(c\log^{2/3} N)$"; "Exploiting the large sieve, the proof is largely derivative of Vaughan's theorem"; "Our proof largely follows the argument in Vaughan"; and "As shown in [13] for each p there are at least $f(p)$ residue classes mod p" forcing representations. Section 4 proves $\sum f(p)/p \asymp (\log x)^2/\varphi(m)$ and applies the large sieve. This updates the access record without claiming the 1970 primary source was read.

30.4. **Review attestation and source register. Review attestation (wave 28).** The hostile maximum-severity verdict was CONFIRMED AFTER REPAIRS, SOURCE-BOUNDED ONLY. The reviewer read all 25 pages of archived v2, searched its full extracted text, compared v1 and v2, audited every campaign citation site, and replayed the range algebra. No theorem, conditional result, heuristic, or cited third-party result in this source threatens the three provisional priority positions. This does not certify the full literature; Theorems 10.7, 11.8, 14.2, and 14.6 remain CLAIMED/PROVISIONAL where so labelled.

The PDF audit confirmed the exact quantifiers above: Theorem 1.1 has every $\epsilon > 0$, all sufficiently large m, and one $n > \exp(m^{1/3-\epsilon})$; Theorem 3.1 has more than (68.2) prime exceptions for every $m \geq 8$; Theorem 1.2 has exact threshold $6.52 \cdot 10^9$ and interval $(m^2, 2m^2)$; Theorem 1.3 has range $4 \leq m \leq \log^2 N$, all integer $n \leq N$, and exponent $C\log^{2/3} N/\varphi(m)^{1/3}$; and Theorem 1.4 has fixed positive j, k and denominator $km+1$. It also confirmed the Type-I/Type-II congruences and that Section 6 proves only finite checks while conjecturing $m \geq 20$. The $\exp(m^{1/2+o(1)})$ discussion is explicitly a Poisson heuristic.

The crossover redivision gives the η_2, φ, L powers $1/4, 1/12, 1/12$. Equations (68.4)–(68.5) are exact. The reverse audit found and repaired the Section 55 scope error: the asymptotic construction can enter fixed-B range, and (68.5a), not range disjointness, explains why it is harmless. At the explicit polynomial scale, both Sections 43 and 55 are out of range. Verifier block `(bo)` locks these distinctions as symbolic exponent checks.

All 13 bibliography entries on pp. 24–25 were classified. Elsholtz–Tao, Obláth, and Vaughan touch asymptotic exceptional sets or admitting classes; Mihnea–Dumitru and Salez are finite-verification references. Aigner and Nakayama [19] concern parametrizations, while Gottschlich [11] supplies only a generic Hardy–Ramanujan-style transfer lemma. Dusart, Koukoulopoulos, Nicolas–Robin, Erdős–Graham, and Guy are tools or surveys. No item concerns the minimal witness modulus W,

and Dahan is absent. The newly registered adjacent references are Gottschlich and Nakayama; neither is a post-Vaughan exceptional-set upper bound. Mihnea–Dumitru's former placeholder is resolved as arXiv:2509.00128 and a 10^{18} finite bound. No new priority reference was found.

The older PDF has the same title, authors, 25-page length, abstract theorem capsule, numbering, Theorem 1.3, and Section 4 argument. A fresh arXiv-v1 fetch has SHA-256 `cb4654e1b329ed5ff8c246ec766901d5d2aae0f720bfc105077ab6951f4c398d`; it is not byte-identical to the author PDF because arXiv regenerated it, but the source creation instant and full mathematical text match. Thus "v1" is a content identity, not a binary-file claim. V2 repairs the abstract's numerical-support threshold from $m \geq 19$ to $m \geq 20$, adds an introductory pre-Vaughan sieve explanation and Gottschlich citation, and otherwise changes notation/typesetting in no cited proof.

Every Pomerance citation site survived the version audit. The HathiTrust and zbMATH access-log entries record only that the parent session found no viewable Vaughan copy or link; they claim neither a saved artifact nor a direct reading. Core quotations precede campaign paraphrase, record language remains would-be/CLAIMED/PROVISIONAL, and every verdict in this section remains expressly source-bounded.

31. Forward-ray identities: pointwise divisibility collapses, and positivity propagates backward

Status and headline. The integer-wise polynomial-divisibility lemma, the shadow classification, and the forward-ray obstruction below are **proved**. They close the one-sided gap left in Proposition 28.5: no polynomial witness datum, even one required to be eligible only on a forward tail, exists on any of the three anchored rays. The finite coefficient ledgers and the projected class calculation are **Computational**, exact only in their stated boxes. No new witness, fourth value with $W = +\infty$, or classification of $W(2m^2)$ follows.

31.1. **The pointwise definition and its algebraic closure.** Write $\mathrm{Int}(\mathbb{Z})$ for the rational polynomials which take integer values at every integer. A *forward-ray polynomial datum* for

$$n(t) = sm(t)^2, \qquad m(t) = m_0 + Kt \tag{69.1}$$

consists initially of $M, D \in \mathbb{Q}[t]$, integer-valued for every integer $t \geq 1$, such that at every such integer

$$\begin{gathered} M(t) \equiv 3 \pmod 4, \quad M(t) \geq 3, \quad D(t) \geq 1, \\ A(t) = \frac{M(t)+1}{4} \in \mathbb{Z}, \quad D(t) \mid A(t)^2, \quad M(t) \mid sm(t)^2 + 4D(t). \end{gathered} \tag{69.2}$$

This pointwise definition is the right generality. Requiring coefficients in $\mathbb{Z}$ would unnecessarily omit such standard integer-valued polynomials as $t(t-1)/2$; requiring polynomial cofactors turns out not to be an extra assumption.

Lemma 31.1 (integer-wise polynomial divisibility collapses; proved; source Lemma 69.1)**.** *Let $F, G \in \mathbb{Q}[t]$, with $G \neq 0$, be integer-valued at every sufficiently large integer, and suppose $G(k) \neq 0$ and $G(k) \mid F(k)$ there. Then*

$$F = GH \quad \text{in } \mathbb{Q}[t] \tag{69.3}$$

for a polynomial $H \in \mathrm{Int}(\mathbb{Z})$. In particular the quotient is an integer at every integer, including all arguments before the range on which divisibility was assumed.

Equivalently, if $g = (F, G)$ in $\mathbb{Q}[t]$, $G = ge$, and $F = gb$, then the coprime residual factor e is constant. Before this conclusion, clearing coefficients to coprime integer polynomials e_1, b_1 gives the useful resultant bound

$$e_1(k) \mid \mathrm{Res}(e_1, b_1) \tag{69.4}$$

at every point of integer-wise divisibility. Thus the proposed "polynomial-divisibility core times a bounded factor" has no nonconstant bounded factor: a polynomial bounded on a forward sequence is constant.

Proof. First, a rational polynomial integral at all sufficiently large integers is integral at every integer. Indeed its finite-difference Newton expansion based at any sufficiently large integer has integral coefficients in the binomial basis, and the binomial polynomials are integral on all integers.

Divide in $\mathbb{Q}[t]$: $F = GH + R$, where $\deg R < \deg G$. Choose a positive integer c clearing the coefficients of H. At a sufficiently large integer k,

$$\frac{R(k)}{G(k)} = \frac{F(k)}{G(k)} - H(k) \in \frac{1}{c}\mathbb{Z}.$$

If G is nonconstant, the left side tends to zero. It is therefore zero for all sufficiently large k, so R has infinitely many zeros and is the zero polynomial. If G is constant, (69.3) is immediate. In either case $H(k) = F(k)/G(k)$ is integral on the assumed tail, and the first paragraph gives $H \in \mathrm{Int}(\mathbb{Z})$.

For the equivalent ledger, clear a common coefficient denominator from the coprime e, b, without changing b/e, to obtain $e_1, b_1 \in \mathbb{Z}[t]$. Whenever $b(k)/e(k)$ is integral, $e_1(k) \mid b_1(k)$. The integral Bezout identity for the nonzero resultant then gives (69.4). Hence $|e_1(k)|$ is bounded on the tail; so e_1, and therefore e, is constant. This is also a second proof of the divisibility assertion. □

The conclusion is deliberately in $\mathbb{Q}[t]$ with an integer-*valued* quotient, not coefficient-wise divisibility in $\mathbb{Z}[t]$. Thus $G = 2$, $F = t(t+1)$ gives the legitimate quotient $H = t(t+1)/2$. As an adversarial nonconstant example, for $e = t^2 + t + 2$ and $b = (t+1)^2$,

$$4 = (t+3)e - (t+2)b,$$

so every integer point with $e(k) \mid b(k)$ has $e(k) \mid 4$; this is (69.4) with $|\mathrm{Res}(e, b)| = 4$. In particular, because $t^2 + t + 2$ is irreducible over $\mathbb{Q}$, applying Lemma 31.1 to $D = t^2 + t + 2$ and $F = A^2$ forces $D \mid A$ in $\mathbb{Q}[t]$. There is no residue-class exception hidden by coefficient content.

Apply Lemma 31.1 first to $D(t) \mid A(t)^2$, and then to $M(t) \mid sm(t)^2 + 4D(t)$. The congruence in (69.2) also puts $(M-3)/4$ in $\mathrm{Int}(\mathbb{Z})$. Consequently every pointwise ray datum automatically supplies $A, D, E, M, Q \in \mathrm{Int}(\mathbb{Z})$ with

$$M = 4A - 1, \qquad A^2 = DE, \qquad sm^2 + 4D = MQ \tag{69.5}$$

as polynomial identities. In particular, specializing at zero gives

$$M(0) \mid sm_0^2 + 4D(0), \qquad M(0) \equiv 3 \pmod 4. \tag{69.6}$$

Lemma 31.2 (exact shadow blockage classification; proved; source Lemma 69.2)**.** *For a datum (69.5) which is eligible at every positive integer, the specialization at*

zero satisfies the congruence and divisibility parts of a witness. If it is not a positive witness datum, then exactly the range/positivity obstruction

$$M(0) \le -1 \quad \text{or} \quad D(0) < 0 \tag{69.7}$$

occurs. Divisor eligibility cannot fail by itself. More explicitly, if $D(0) \neq 0$*, then* $D(0) \mid A(0)^2$ *because* $A(0)^2 = D(0)E(0)$*; if* $D(0) = 0$*, then* $A(0) = 0$ *and* $M(0) = -1$*. Also* $M(0) \equiv 3 \pmod 4$*, so a nonpositive value of* $M(0)$ *is at most* -1*, while a positive value is at least* 3*.*

Proof. All five values in (69.5) are integers at zero by Lemma 31.1 and the finite-difference observation. Equations (69.5) give both claims in (69.6) and the asserted divisor relation when $D(0) \neq 0$. The zero case and the mod-4 alternatives are as stated. Thus $M(0) \ge 3, D(0) \ge 1$ would be a witness, and every failure has exactly one of the forms in (69.7). □

This corrects the apparent split between two divisibility regimes. For polynomial D, integer-wise divisibility is no broader than polynomial divisibility with an integer-valued cofactor. Computational 66.1 searched only the three special shapes $d, dA, A^2/d$, not every polynomial divisor of A^2; Proposition 28.5 covered every shape only when eligibility was assumed at $t = 0$. Neither result by itself disposed of (69.7).

31.2. **One-step positivity and the all-ray obstruction.** The missing ingredient is a real-sign consequence of the two identities, not a larger coefficient search.

Lemma 31.3 (one-step positivity propagation; proved; source Lemma 69.3)**.** *Let* $s > 0$*, and let real polynomials satisfy (69.5). Suppose* $M(1) > 0$ *and* $D(1) > 0$*. If* $M(0) \neq 0$*, then*

$$M(0) > 0, \qquad D(0) > 0. \tag{69.8}$$

Proof. At every real zero r of M, one has $A(r) = 1/4$ and hence $D(r)E(r) = 1/16$. On the other hand (69.5) gives

$$0 = sm(r)^2 + 4D(r). \tag{69.9}$$

Thus $D(r) \neq 0$, $m(r) \neq 0$, and necessarily $D(r) < 0$.

Suppose first that $M(0) < 0$. There is a zero of M in $(0, 1)$; let r be the largest one. Then $M > 0$ throughout $(r, 1]$. But $D(r) < 0 < D(1)$, so D has a zero $u \in (r, 1)$. The identity $A^2 = DE$ gives $A(u) = 0$, whence $M(u) = -1$, a contradiction.

Now suppose $M(0) > 0$ but $D(0) < 0$. A zero $u \in (0, 1)$ of D again has $M(u) = -1$. Since $M(1) > 0$, let $r \in (u, 1)$ be the largest zero of M. Then $M > 0$ on $(r, 1]$, while (69.9) and $D(1) > 0$ force a zero of D there. As before that zero makes $M = -1$, a contradiction. Finally $D(0) = 0$ implies $A(0) = 0$ and $M(0) = -1$, already excluded. □

Theorem 31.4 (no forward-ray polynomial witness family; proved; source Theorem 69.4)**.** *Let*

$$(s, m_0) \in \{(2, 12), (21, 4), (505, 3)\}, \qquad K \ge 1. \tag{69.10}$$

There is no forward-ray polynomial datum (69.1)–(69.2). More strongly, there is none whose conditions are required only for all sufficiently large integer t*. There is no degree, coefficient, or cylinder-period restriction in this statement.*

Proof. Lemma 31.1 turns the tail pointwise conditions into (69.5) and makes all specializations integral. It also gives $B = (M-3)/4 \in \mathrm{Int}(\mathbb{Z})$, so

$$M(j) = 4B(j) + 3 \neq 0 \tag{69.10a}$$

for *every* integer j, not merely on the eligible tail.

Choose an integer $T \geq 1$ from which all conditions are eligible. Start with $M(T) > 0, D(T) > 0$. For $j = T, T-1, \ldots, 1$, apply Lemma 31.3 to the translated polynomials $P_j(x) = P(x+j-1)$. Its value at $x = 1$ is the already-positive pair at j, and (69.10a) supplies its required $M_j(0) = M(j-1) \neq 0$. Hence it gives $M(j-1) > 0, D(j-1) > 0$. This is a literal finite induction back to the *original* parameter value zero; translation is used only inside each one-step application and does not move the anchor.

The identities and their integer-valued cofactors then make every recovered positive-integer position eligible as well. In particular $M(0) > 0, D(0) > 0$, while (69.10a) gives $M(0) \equiv 3 \pmod 4$ and hence $M(0) \geq 3$. Lemma 31.2 makes the zero specialization a witness for sm_0^2. In each of the three rows this contradicts Theorem 22.3. The argument applies unchanged when M is constant. □

Thus the two candidate escape mechanisms in (69.7) are algebraically possible at an isolated specialization but impossible in a witness identity: a real zero of M forces $D < 0$, while a sign-restoring zero of D forces $M = -1$. This is the step missing from the wave-28 attestation.

31.3. **Constant divisors and the bounded hunt.** The constant-divisor stratum admits an elementary complete description. Put $M(t) = \alpha t + \beta = 4(at+b) - 1$ with integer coefficients. For $d = \prod p^{v_p(d)} > 0$, set

$$q(d) = \prod_{p \mid d} p^{\lceil v_p(d)/2 \rceil}. \tag{69.11}$$

Then

$$d \mid A(t)^2 \ (t \geq 1) \quad \Longleftrightarrow \quad q(d) \mid a, \quad q(d) \mid b. \tag{69.12}$$

Indeed $d \mid x^2$ is equivalent to $q(d) \mid x$, and two consecutive values of $at+b$ give necessity; sufficiency is immediate. In particular eligibility always transfers to $t = 0$. For a general integer-valued polynomial the same conclusion follows either from Lemma 31.1 or directly: if a denominator c clears its coefficients, translation by cd preserves its value modulo d, so a congruence true at every positive integer is true at zero.

There are no constant-D, linear-M anchored rays. If $\alpha \neq 0$, polynomial divisibility by M would make $s(m_0 + Kt)^2 + 4d$ vanish at the real root $-\beta/\alpha$, impossible. If $\alpha = 0$, finite differences show that validity is equivalent to

$$\begin{aligned} &\beta \mid sm_0^2 + 4d, \\ &\beta \mid sK^2 + 2sm_0K, \\ &\beta \mid 2sK^2, \end{aligned} \tag{69.13}$$

besides $\beta \equiv 3 \pmod 4$, positivity, and $d \mid ((\beta+1)/4)^2$. The first congruence in (69.13) would be an anchor witness. Thus (69.11)–(69.13) completely classify every tuple $(d, \alpha, \beta, m_0, K)$, including the degree-zero edge, and the valid anchored list is empty.

Computational 69.1 (proof-aware coefficient and cylinder replay; exact stated boxes). Block `(bp)` used the period panel

$$\mathcal{K} = \{12, 24, 36, 44, 48, 72, 84, 88, 132, 168, 231, 252, \\ 264, 396, 462, 627, 693, 1254, 1881\}, \tag{69.14}$$

the Section 28 periods $12, 24, 44, 84, 132, 231$, the projected modulus 627, and their two- and three-fold multiples. For all three anchors it counted every integer monomial-coefficient array with $\deg M \le 2$, $\deg D \le 4$, and coefficient absolute value at most 500. Exactly 31,375,250 of the M-arrays induce the polynomial-function law $M(t) \equiv 3 \pmod 4$; with all D-arrays and all 57 anchor-period rows this is

$$1{,}797{,}349{,}098{,}035{,}336{,}234{,}639{,}250 \tag{69.15}$$

syntactic candidates. Theorem 31.4 certifies zero identities in this box; the count is a proof-aware exhaustive ledger, not independent numerical evidence obtained by iterating that enormous set.

The direct constant-D, linear-M reducer used (69.11). At bound 500 there are 56,226 triples (d, α, β) passing mod 4, positivity, and eligibility (1,240 constant and 54,986 nonconstant M); across (69.14) and the three anchors this gives 3,204,882 rows. The root test disposes of all nonconstant rows, and the exact three coefficient congruences (69.13) dispose of every constant row. Under `ES_FULL_SCAN=1`, the same reducer runs to coefficient bound 5,000: the corresponding counts are 5,792,112 (21,390 constant and 5,770,722 nonconstant), or 330,150,384 anchor-period rows, again with no hit. The full flag also pins the larger syntactic ledger 178,285,380,203,533,476,742,414,087,642,500.

Finally the block used all 65 classes in the exact projection (66.16), not only the anchor class 12. For $m(t) = r + 627t$, the two nonconstant finite differences show that a fixed modulus must divide both $4 \cdot 627^2$ and $2 \cdot 627^2 + 4r \cdot 627$. Exhausting their common divisors gave 445 moduli $M \equiv 3 \pmod 4$, and exhausting all 1,709 divisors of the corresponding $((M+1)/4)^2$ gave no fixed-(M, D) ray. This last zero is complete for constant data on those 65 projected classes, but says nothing about their nonconstant polynomial rays; only the class $r = 12$ is covered by Theorem 31.4.

Block `(bp)` also performs deterministic resultant pins, 128 constructed integer-wise examples, and a complete small-coefficient collapse ledger. For $\deg G \le 2$, $\deg F \le 3$, all monomial coefficients in $[-2, 2]$, and $20 \le k \le 40$, exactly 3,472 pairs have $G(k) \ne 0$ and $G(k) \mid F(k)$ throughout; every one has zero exact polynomial remainder and an integer-valued quotient, including 280 quotients outside $\mathbb{Z}[t]$. It separately pins the $t^2 + t + 2$ adversarial box and the exact resultant-4 Bezout identity above. An exhaustive anchor-shadow check for $|A(0)|, |D(0)| \le 100$ finds 652 signed endpoint pairs, all blocked by (69.7), and no positive datum. These are regression tests for the proofs, not substitutes for them. The default block has an executable 15-second budget. After the wave-29 pins, it took 0.60 seconds by default and 2.06 seconds for its isolated full scan on the review host.

31.4. **Consequence and honest walls.** The forward-ray escape hatch in Section 28 is closed for polynomial data in the maximal integer-valued sense: pointwise divisibility already supplies polynomial integer-valued cofactors, and even tail positivity propagates back to the infinite anchor. Hence a polynomial witness family cannot settle an anchored escape cylinder by covering its whole forward ray. This hardens, rather than resolves, the restricted prediction C'_{SQ}: a proof must use non-polynomial or nonuniform witness structure.

Theorem 31.4 does not obstruct an arbitrary function $t \mapsto (M_t, D_t)$, a different polynomial on each finite piece, or a ray whose zero specialization is not one of the three proved infinite anchors. The 64 unanchored projected classes in (66.16) therefore remain open to nonconstant identities. Nor does the theorem prove that every fixed $m \neq 12$ has a witness, produce another $W = +\infty$ value, or exchange the quantifiers in (66.19). Higher-degree polynomial strata need no further hunt on the three anchored rays, but all genuinely non-polynomial supply and the pointwise classification of $W(2m^2)$ remain beyond this obstruction.

31.5. **Review attestation (wave 29). Verdict: sound-after-repairs.** Refutation-class review rederived the full quantifier chain and found no counterexample to the polynomial-ray obstruction. The repairs expose the integer-valued quotient issue, make the $t_0 > 1$ induction and its nonzero premise explicit, add the Section 28 forward pointer, and strengthen block `(bp)`. The theorem remains strictly about one polynomial datum on an anchored ray; no non-polynomial witness function or classification claim is covered.

- **Critical (backward positivity).** The proof uses two forced signs. At a real zero r of M, one has $A(r) = 1/4$, so $D(r)E(r) = 1/16$, while $sm(r)^2 + 4D(r) = 0$; hence $m(r) \neq 0$ and $D(r) < 0$. At a real zero u of D, the identity $A^2 = DE$ gives $A(u) = 0$, hence $M(u) = -1$. If M changes from negative at 0 to positive at 1, its last zero forces $D < 0$, and any restoration to $D(1) > 0$ forces $M = -1$ inside a subinterval where $M > 0$. If instead $M(0) > 0$ and $D(0) < 0$, a first D-zero gives $M = -1$; the last later M-zero repeats the same contradiction. Finally $D(0) = 0$ forces $M(0) = -1$. This case split is complete under the stated $M(0) \neq 0$ hypothesis; positivity of an arbitrary polynomial alone is never invoked.
- **Critical (tail quantifiers).** Tail congruence implies $(M-3)/4 \in \mathrm{Int}(\mathbb{Z})$, so every integer specialization of M is 3 (mod 4) and therefore nonzero. Starting at any eligible $T \geq t_0$, Lemma 31.3 applies successively to $P(x + j - 1)$ for $j = T, \ldots, 1$. Each step inherits the identities and uses the positivity just recovered at the preceding step. It ends at the original $t = 0$, not at a shifted anchor. Thus every degree, every integer $K \geq 1$, every tail start, and all three Theorem 22.3 anchors are covered uniformly.
- **High (integer-wise collapse).** Euclidean division gives a remainder ratio $R(k)/G(k) \to 0$ lying in the discrete group $c^{-1}\mathbb{Z}$, hence $R = 0$. Tail-integrality of the quotient extends to all integers in its Newton/binomial basis. This allows $t(t+1)/2$ and does not assert coefficient-wise divisibility in $\mathbb{Z}[t]$. Independently, the coprime residual satisfies $e_1(k) \mid \mathrm{Res}(e_1, b_1)$, so it is constant. The explicit $t^2 + t + 2$ test and resultant-4 Bezout identity found no content or periodic-residue loophole.
- **High (independent finite replay).** A fresh standard-library implementation reproduced the bound-500 constant-divisor ledger $(56{,}226; 1{,}240; 54{,}986; 0)$, the bound-5,000 ledger $(5{,}792{,}112; 21{,}390; 5{,}770{,}722; 0)$, both syntactic totals, and the 65-class fixed-data counts $(445, 1{,}709, 0)$. It also screened 50,000 seeded compatible coefficient rows with no 12-point survivor. The zeros remain bounded-hunt facts; Theorem 31.4, not sampling, supplies the all-degree anchored conclusion.
- **Register.** Section 28 now points forward to this theorem. Together the two sections show only that anchored forward cylinders resist a single polynomial witness family. This hardens the difficulty of C'_{SQ}; it is not evidence for that conjecture and leaves arbitrary non-polynomial $t \mapsto (M_t, D_t)$ untouched.

- **Verification.** Block `(bp)` now pins the collapse classification, integer-valued/non-integer-coefficient quotient cases, resultant bound, shadow blockage, all default ledgers, and a default runtime below 15 seconds. Its full-scan path pins the bound-5,000 ledgers. The prescribed full suite completed green in 210.01 seconds with 341,704 KiB peak resident memory; syntax parse and the control-byte sweep were also clean.

32. The a-frame failure law: confinement, the fixed-modulus scale, and congruence-wall freeness

Status and headline. The subgroup classification, the integer-frame Selberg–Delange asymptotics, the fixed-prime budget bound, the fixed-modulus upper bounds on shifted primes, the composite-modulus subgroup classification, and congruence-wall freeness below are **proved**. They use only elementary finite-group theory and the standard fixed-modulus Selberg–Delange and upper-bound-sieve facts stated explicitly below. They inherit **nothing** from the provisional machinery elsewhere in this paper. Block `(bq)` is **Computational**, exact only in its displayed finite ranges. All results are for each fixed modulus; no uniformity sufficient to stack a growing family of moduli is proved, and no pointwise Erdős–Straus result is claimed.

Fix for now a prime

$$a \equiv 3 \pmod 4, \qquad G = (\mathbb{Z}/a\mathbb{Z})^*, \qquad Q = G^2. \tag{70.1}$$

Thus $G \cong C_{a-1}$, Q is the subgroup of quadratic residues, and $|Q| = (a-1)/2$ is odd. For $(h, a) = 1$, write

$$F_a(h) = \{-1 \notin \mathrm{Rat}_a(h)\}, \qquad \sigma_a(h) = \#\{q : q \mid h,\ (q/a) = -1\}, \tag{70.2}$$

where (q/a) is the Legendre symbol and σ_a counts distinct prime divisors. The ratio spectrum is the one in (62.1). In the hard-prime application $(h, a) = 1$ is automatic by Theorem 24.1.

32.1. The unique involution and the exact failure decomposition.

Lemma 32.1 (unique-involution subgroup classification; proved; source Lemma 70.1)**.** *Let K be a finite group. Every even-order subgroup of K contains an element of order two. Consequently, if K has a unique element τ of order two, then*

$$\tau \notin H \quad \Longleftrightarrow \quad |H| \text{ is odd} \tag{70.3}$$

for every subgroup $H \le K$. In the particular cyclic group $G = C_{2n}$, where $n = (a-1)/2$ is odd, one furthermore has

$$-1 \notin H \quad \Longleftrightarrow \quad |H| \text{ is odd} \quad \Longleftrightarrow \quad H \subseteq Q. \tag{70.4}$$

Proof. Cauchy's theorem gives an element of order two in every finite even-order group. If the ambient group has the unique involution τ, that element must be τ. Conversely a subgroup containing τ has even order, proving (70.3). A cyclic even-order group has the unique involution represented here by -1. Its subgroup Q has odd order n. Every subgroup of odd order is contained in the unique subgroup of order n, while every subgroup of Q has odd order. This proves (70.4). □

Corollary 32.2 (quadratic confinement forces failure; proved; source Corollary 70.2)**.** *If $\sigma_a(h) = 0$, then $F_a(h)$ holds.*

Proof. Every prime factor of h, every inverse of such a factor, and hence every bounded product in $\mathrm{Rat}_a(h)$, lies in Q. Quadratic reciprocity gives $(-1/a) = -1$, so $-1 \notin Q$. □

Corollary 32.3 (the prime-modulus F2 class is empty; proved; source Corollary 70.3)**.** *If $\sigma_a(h) \geq 1$, then the subgroup generated by the residue classes of the prime factors of h contains -1. Thus a failure with $\sigma_a(h) \geq 1$ is caused solely by the bounded exponent budgets, not by a subgroup miss.*

Proof. A nonresidue g has nontrivial image in $G/Q \cong C_2$, so its order is even. The cyclic group $\langle g \rangle$ therefore contains an involution. The involution is the unique one, namely -1, by Lemma 32.1. Since $\langle g \rangle$ is contained in the subgroup generated by all prime-factor classes, the conclusion follows. □

In the taxonomy of source §19.2 this proves, rather than samples, that F2 is empty for every prime $a \equiv 3 \pmod 4$. The zero F2 column in the random part of that old-frame table is consistent with the theorem; the table included other moduli and targets, so it was not itself a proof of this statement. The criterion is the same signed-subset-product mechanism as source Lemma 12.3, now in the fixed-a ratio coordinate of Theorem 24.1.

Corollary 32.4 (F1/F3 decomposition; proved; source Corollary 70.4)**.** *Among units modulo a,*

$$\begin{aligned} F_a &= F_{1,a} \,\dot\cup\, F_{3,a}, \\ F_{1,a} &= \{h : \sigma_a(h) = 0\}, \\ F_{3,a} &= \{h : \sigma_a(h) \geq 1, -1 \notin \mathrm{Rat}_a(h)\}. \end{aligned} \tag{70.5}$$

Here F1 is all-quadratic-residue confinement and F3 is pure finite-budget failure.

Proof. Corollary 32.2 puts all of the first set in F_a. Every other failure has $\sigma_a(h) \geq 1$, and Corollary 32.3 identifies it as a budget failure. The alternatives are disjoint and exhaustive. □

The distinction is load-bearing: at $a = 7, h = 17$, (62.16) has a generated subgroup equal to all of G, but the valuation-one spectrum still misses -1.

32.2. **The confinement semigroup and its half-dimensional law.** Let $f_a(h)$ be the indicator that $(h, a) = 1$ and every prime divisor of h is a quadratic residue modulo a. Then f_a is multiplicative (in fact its value is determined prime by prime), and $F_{1,a}$ has indicator f_a.

We use the following standard fixed-modulus form of Selberg–Delange. It is stated here to make all analytic input visible.

Standard Fact 70.1 (fixed-class Selberg–Delange and fixed-mark corollary). Suppose a set $\mathcal{P}$ of primes is a union of reduced residue classes modulo a fixed integer k, of relative Dirichlet density δ, with finitely many primes added or removed. If f is the indicator of integers all of whose prime factors lie in $\mathcal{P}$, then

$$\sum_{n \leq x} f(n) = \frac{G(1)}{\Gamma(\delta)} x (\log x)^{\delta - 1} + o_k\big(x(\log x)^{\delta-1}\big), \tag{70.6}$$

where

$$\sum_{n \geq 1} \frac{f(n)}{n^s} = \zeta(s)^\delta G(s)$$

has G holomorphic and nonzero in a neighbourhood of $s = 1$. A fixed reduced progression may be selected by Dirichlet-character orthogonality. The expected principal-character main term applies when the integers counted are automatically coprime to the progression modulus and every nonprincipal twist has a strictly smaller real pole exponent. This is a hypothesis to check, not a blanket progression theorem: tied twists can cancel the principal term when a progression is incompatible with the prime semigroup.

A standard coefficient corollary will also be used. If disjoint fixed-class prime sets $\mathcal{P}_0, \mathcal{P}_1$ have densities δ_0, δ_1, then, for each fixed K,

$$\#\{n \le x : p \mid n \Rightarrow p \in \mathcal{P}_0 \cup \mathcal{P}_1,\ \Omega_{\mathcal{P}_1}(n) \le K\} \ll_{k,K} x(\log x)^{\delta_0 - 1}(\log\log(3x))^K. \tag{70.7}$$

Here $\Omega_{\mathcal{P}_1}$ counts with multiplicity. Formula (70.7) follows by inserting a marker z in the Euler factors for $\mathcal{P}_1$, applying (70.6) uniformly on a small circle about $z = 0$, and extracting the first $K + 1$ coefficients. This is the customary Selberg–Delange theorem and its fixed-order Landau–Sathe corollary (for example the Selberg–Delange theorem in Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Chapter III.5, Theorem 5.2). Only fixed k, δ, K are asserted here.

Theorem 32.5 (integer confinement law; proved; source Theorem 70.5)**.** *For each fixed prime $a \equiv 3 \pmod 4$,*

$$\#\{h \le H : F_{1,a}(h)\} \sim C_a \frac{H}{\sqrt{\log H}}, \tag{70.8}$$

where the positive constant is

$$C_a = \frac{1}{\sqrt{\pi}} \left\{ \left(1 - \frac{1}{a}\right) L(1, \chi_a) \prod_{\substack{q \text{ prime} \\ \chi_a(q) = -1}} (1 - q^{-2}) \right\}^{1/2}, \tag{70.9}$$

and $\chi_a = (\cdot/a)$. Put $b = (a + 1)/4$. In the progression forced by the hard class,

$$\#\{h \le H : h \equiv b \pmod 6,\ F_{1,a}(h)\} \sim C_a^{\mathrm{hard}} \frac{H}{\sqrt{\log H}}, \tag{70.10}$$

with

$$\frac{C_a^{\mathrm{hard}}}{C_a} = \begin{cases} 1, & a = 3, \\ 1/4, & a \equiv 7 \pmod{24}, \\ 1/3, & a \equiv 11 \pmod{24}, \\ 1/2, & a \equiv 19 \pmod{24}, \\ 1/6, & a \equiv 23 \pmod{24}. \end{cases} \tag{70.11}$$

In particular, both constants are strictly positive. The exponent $1/2 = 1 - 1/2$ is one minus the density of quadratic-residue prime classes.

Proof. The Euler series is

$$\sum_{h \ge 1} \frac{f_a(h)}{h^s} = \prod_{\chi_a(q) = 1} (1 - q^{-s})^{-1}. \tag{70.12}$$

Comparison of Euler factors gives

$$\left(\sum \frac{f_a(h)}{h^s}\right)^2 = \zeta(s) L(s, \chi_a)(1 - a^{-s}) \prod_{\chi_a(q) = -1} (1 - q^{-2s}). \tag{70.13}$$

The product on the right converges absolutely at $s = 1$, and $L(1, \chi_a) > 0$. Taking the local analytic square root writes (70.12) as $\zeta(s)^{1/2} G_a(s)$, with the value at one displayed in (70.9). Standard Fact 70.1 proves (70.8).

For (70.10), take the least nonnegative residue $r \equiv b \pmod 6$, put $d = (r, 6)$, and write $h = dn$. The possibilities for a prime a are

$a \bmod 24$	3	7	11	19	23
r	1	2	3	5	0
d	1	2	3	1	6
$6/d$	6	3	2	6	1.

(70.14)

For $a > 3$, every prime dividing d is a residue modulo a, while every prime dividing $6/d$ is a nonresidue; these four assertions follow immediately from the supplementary laws for $(2/a)$ and $(3/a)$. Thus $f_a(dn) = f_a(n)$. Put $M = 6/d$. Every prime dividing M is a nonresidue modulo a, so an integer counted by f_a is automatically coprime to M. Character orthogonality may therefore be applied to the reduced condition $n \equiv r/d \pmod M$. Since $(a, M) = 1$, the Chinese remainder theorem gives prime-pole exponent $1/2$ for the principal character and exponent zero for every nonprincipal character modulo M: the sum of that character over $(\mathbb{Z}/M\mathbb{Z})^*$ vanishes independently of the quadratic- residue condition modulo a. Hence all nonprincipal twists are lower order and the main term acquires the factor $1/(d\varphi(M))$. These factors are the last four entries of (70.11). When $a = 3$, all allowed primes are $1 \pmod 3$ and odd, so every integer counted by f_3 is already $1 \pmod 6$; the first entry follows. □

This proves, in particular, the asymptotic that was only contextual in Assessment 62.1 at $a = 3$.

32.3. **Fixed-a budget failures are lower order.** For an inversion orbit $\mathcal{O} = \{g, g^{-1}\}$ of nonresidues in G, put

$$k_{\mathcal{O}} = \frac{\operatorname{ord}(g)}{2}, \qquad K_a = \sum_{\mathcal{O} \subseteq G \setminus Q} (k_{\mathcal{O}} - 1), \qquad \beta_a = \frac{1}{2} + \frac{1}{a-1}. \tag{70.15}$$

Because $G = C_{2n}$ with n odd, every nonresidue has order twice an odd number. Hence $k_{\mathcal{O}}$ is an odd positive integer and $g^{k_{\mathcal{O}}} = -1$. The self-inverse orbit $\{-1\}$ contributes zero to K_a.

Lemma 32.6 (budget rigidity; proved; source Lemma 70.6)**.** *If $h \in F_{3,a}$, then*

$$\Omega_{G \setminus Q}(h) \le K_a. \tag{70.16}$$

Moreover, if a nonresidue class g contains a prime divisor of h, then no prime divisor of h lies in the quadratic-residue class

$$c_g = -g^{-1}. \tag{70.17}$$

Proof. Let $V_{\mathcal{O}}$ be the sum of the valuations in h of primes whose classes lie in $\mathcal{O}$. Using only those primes, their signed exponent intervals realize every power g^j with $|j| \le V_{\mathcal{O}}$: changing from g to g^{-1} only reverses one interval, and sums of integer intervals remain complete intervals. If $V_{\mathcal{O}} \ge k_{\mathcal{O}}$, the spectrum therefore contains $g^{k_{\mathcal{O}}} = -1$, contrary to failure. Hence $V_{\mathcal{O}} \le k_{\mathcal{O}} - 1$ for every orbit. Summing proves (70.16).

Now choose a prime factor $q \mid h$ in class g. The class c_g is a residue, since both -1 and g^{-1} are nonresidues. If a prime factor $r \mid h$ had class c_g, choosing exponent

one on q, r and zero on every other factor would give $qr \equiv -1 \pmod{a}$. This is forbidden in $F_{3,a}$, proving (70.17). □

Theorem 32.7 (fixed-modulus budget bound; proved; source Theorem 70.7)**.** *For every fixed prime $a \equiv 3 \pmod 4$,*

$$\#\{h \le H : F_{3,a}(h)\} \ll_a \frac{H(\log\log(3H))^{K_a}}{(\log H)^{\beta_a}}. \tag{70.18}$$

Consequently, for every fixed $\beta < \beta_a$,

$$\#\{h \le H : F_{3,a}(h)\} \ll_{a,\beta} H(\log H)^{-\beta}, \tag{70.19}$$

and

$$\frac{\#\{h \le H : F_{3,a}(h)\}}{\#\{h \le H : F_a(h)\}} \longrightarrow 0. \tag{70.20}$$

For $a = 3$, $K_a = 0$ and $F_{3,3}$ is empty. For $a > 3$, the proved logarithmic exponent is $\beta_a = 1/2 + 1/(a-1) > 1/2$; its infimum as a varies is only $1/2$, so this gain is not uniform in a.

Proof. By Lemma 32.6, every $h \in F_{3,a}$ has at most K_a nonresidue prime factors with multiplicity. It also has at least one such factor. Choose one of its at most $(a-1)/2$ nonresidue classes g. Lemma 32.6 excludes the one residue class c_g from all prime factors. The unrestricted residue classes other than c_g have prime density

$$\delta_0 = \frac{1}{2} - \frac{1}{a-1}; \tag{70.21}$$

the nonresidue classes are the marked set. Standard Fact 70.1, (70.7), therefore bounds this one choice of g by

$$H(\log H)^{\delta_0 - 1}(\log\log(3H))^{K_a} = \frac{H(\log\log(3H))^{K_a}}{(\log H)^{1/2+1/(a-1)}}.$$

The finite union over g proves (70.18), and a fixed power of $\log\log H$ is absorbed by every positive loss in the exponent, proving (70.19). Finally (70.8) is a positive lower bound of order $H/\sqrt{\log H}$ for the denominator in (70.20); division of (70.18) by it tends to zero. □

This proof does not use the quadratic-character moment alone. That moment only bounds the number of nonresidue factors and leaves the larger $(\log H)^{-1/2}(\log\log H)^{O_a(1)}$ scale. The extra $1/(a-1)$ comes from the deterministically forbidden correction class (70.17). The Fourier large-deviation criterion of source Lemma 12.4 remains a useful more general reduction, but no unproved transfer step from source §12.5 is being imported here.

Corollary 32.8 (integer per-modulus failure law; proved; source Corollary 70.8)**.** *For every fixed prime $a \equiv 3 \pmod 4$,*

$$\#\{h \le H : F_a(h)\} = (C_a + o_a(1))\frac{H}{\sqrt{\log H}}, \tag{70.22}$$

and, in the hard-class progression,

$$\#\{h \le H : h \equiv (a+1)/4 \pmod 6,\ F_a(h)\} = (C_a^{\mathrm{hard}} + o_a(1))\frac{H}{\sqrt{\log H}}. \tag{70.23}$$

Proof. Use the disjoint decomposition (70.5), Theorem 32.5 for F1, and Theorem 32.7 for F3. Restricting F3 to a progression can only decrease its global count, so it remains lower order in (70.23). □

32.4. **Transfer to shifted primes: an upper law.** For clarity, the only sieve input needed here is recorded separately.

Standard Fact 70.2 (fixed two-form upper sieve). Let $L_0(n), L_1(n)$ be fixed primitive affine linear forms with nonzero determinant. Let $\mathcal{P}$ be a fixed union of prime residue classes with

$$\sum_{q<z,\ q\in\mathcal{P}} \frac{1}{q} = \theta \log\log z + O(1). \tag{70.24}$$

Outside the finitely many primes dividing the coefficients or determinant, the roots $L_0(n) \equiv 0 \pmod q$ and $L_1(n) \equiv 0 \pmod q$ are distinct. The fixed-dimensional upper-bound fundamental lemma then gives

$$\#\{n \le X : L_0(n) \text{ is prime},\ q \nmid L_1(n)\ (q \in \mathcal{P})\} \ll \frac{X}{(\log X)^{1+\theta}}. \tag{70.25}$$

If at most a fixed K prime factors from $\mathcal{P}_1$ are permitted and all factors from a disjoint set $\mathcal{P}_0$ are excluded, then the right side is

$$\frac{X(\log\log(3X))^K}{(\log X)^{1+\theta_0+\theta_1}}. \tag{70.26}$$

Here each θ_i is the coefficient in (70.24).

This is the standard upper-bound beta-sieve fundamental lemma (for example Friedlander–Iwaniec, *Opera de Cribro*, Theorem 6.9) plus its fixed-K almost-prime variant. To see (70.26), select the at most K exceptional small prime divisors, apply the same upper sieve after those roots are fixed by the Chinese remainder theorem, and sum their reciprocal products; the latter sum is $O((\log\log z)^K)$. Prime factors above the sifting limit need no exception. Taking a fixed power $z = X^\eta$, and covering $L_0(n) \le z$ separately, proves the displayed bounds. This is exactly the elementary exact-remainder technology used in Theorem 25.7, not a level of distribution assertion about primes.

Theorem 32.9 (fixed-a prime-frame upper law; proved, effective; source Theorem 70.9)**.** *Let $a \equiv 3 \pmod 4$ be a fixed prime. Then*

$$\#\left\{p \le N : p \equiv 1 \pmod{24},\ p \text{ prime},\ F_a\left(\frac{p+a}{4}\right)\right\} \ll_a \frac{N}{(\log N)^{3/2}} \ll_a \frac{\pi(N)}{\sqrt{\log N}}. \tag{70.27}$$

More precisely, the F3 part satisfies

$$\#\left\{p \le N : p \equiv 1 \pmod{24},\ p \text{ prime},\ F_{3,a}\left(\frac{p+a}{4}\right)\right\} \ll_a \frac{N(\log\log(3N))^{K_a}}{(\log N)^{3/2+1/(a-1)}}. \tag{70.28}$$

The same bounds hold, with constants depending also on the modulus, after intersecting with any fixed compatible reduced progression of hard primes.

Proof. Write $p = 24m+1$ and $h = 6m + (a+1)/4$. The second form need not itself be primitive: with $d = ((a+1)/4, 6)$, as tabulated in (70.14), put $L_1(m) = h/d$. This is a primitive integral affine form. For $a > 3$, every prime factor of the fixed content d is a residue modulo a; for $a = 3$, $d = 1$. Thus removing the content changes neither F1 nor the nonresidue budget. Sieve the zero class of the prime form $L_0(m) = 24m+1$ for every prime, which contributes dimension one. For F1, also sieve the zero class of L_1 for primes q with $(q/a) = -1$. These have density

one half by Dirichlet's theorem. The two roots are distinct outside the finite set of primes dividing $4da$, by $4dL_1 - L_0 = a$. Standard Fact 70.2 with $\theta = 1/2$ gives the first bound for F1.

For F3, Lemma 32.6 permits at most K_a nonresidue prime factors and, after one of finitely many classes g is chosen, excludes every prime in the additional class c_g. If the fixed content d contains such a prime, that choice of g gives the empty event; otherwise it contributes only a finite local factor. In (70.26) the two second-form prime sets therefore have densities $1/2$ and $1/(a-1)$. This proves (70.28). Adding F1 and using (70.5) proves (70.27).

After imposing a further fixed compatible progression, the prime form is primitive. Divide the second form by its fixed content as above. If that content contains a nonresidue, F1 is empty; in F3 its fixed multiplicity uses part of the at-most-K_a budget. A chosen correction class already present in the content likewise makes that subcase empty. In every remaining subcase the same two primitive forms and root calculation apply, with only finitely many local factors changed. All Mertens estimates are in fixed progressions and the upper sieve is effective. □

No matching prime-frame lower bound is proved. Such a lower bound would require shifted primes p for which $(p+a)/4$ has its entire factorization in prescribed residue classes; the upper sieve does not cross that parity-type obstacle. Thus (70.27) is an upper law, not a prime-frame asymptotic.

32.5. **Composite coefficients: the correct confinement kernels.** The tempting statement that every maximal -1-avoiding subgroup is the kernel of an odd quadratic character is false for composite moduli. The following gives the exact replacement.

Theorem 32.10 (composite confinement classification; proved; source Theorem 70.10)**.** *Let $a \equiv 3 \pmod 4$ be any odd integer, put $G_a = (\mathbb{Z}/a\mathbb{Z})^*$, and let $\tau = -1$. A subgroup $H \le G_a$ is maximal, by inclusion, subject to $\tau \notin H$ if and only if*

$$G_a/H \cong C_{2^k} \quad (k \ge 1), \qquad \tau H \text{ is its unique involution.} \tag{70.29}$$

Equivalently, the maximal avoiding subgroups are the kernels of surjective characters

$$\psi : G_a \longrightarrow \mu_{2^k}, \qquad \psi(-1) = -1, \tag{70.30}$$

where characters differing by an automorphism of μ_{2^k} have the same kernel. Their exact number is therefore

$$\sum_{k \ge 1} \frac{1}{\varphi(2^k)} \#\{\psi : G_a \to \mu_{2^k} : \operatorname{ord}\psi = 2^k,\ \psi(-1) = -1\}. \tag{70.31}$$

The sum is finite. If $\omega(a)$ is the number of distinct prime factors of a, the $k = 1$ term consists of exactly

$$2^{\omega(a)-1} \tag{70.32}$$

odd quadratic characters. Higher-k kernels can occur.

For $(h, a) = 1$, let $B_a(h)$ denote the budget event in which -1 lies in the subgroup generated by the prime-factor classes but not in $\mathrm{Rat}_a(h)$. Then

$$F_a(h) \subseteq \bigcup_{\substack{\psi \text{ as in (70.30)} \\ \ker\psi \text{ maximal}}} \{h : q \mid h \Rightarrow q \bmod a \in \ker\psi\} \ \cup\ B_a(h). \tag{70.33}$$

Thus subgroup confinement remains a bounded union of multiplicative semigroup conditions.

Proof. Suppose first that H is maximal avoiding τ, and set $A = G_a/H$. Every nonzero subgroup of A contains τH, since otherwise its inverse image would be a strictly larger avoiding subgroup. There can be no subgroup of odd prime order. Hence A is a 2-group. It has only one subgroup of order two, generated by τH; a finite abelian 2-group with a unique subgroup of order two is cyclic, by its invariant-factor decomposition. This proves the forward implication in (70.29). Conversely every nonzero subgroup of a cyclic 2^k-group contains its unique order-two subgroup. Therefore any subgroup strictly above the kernel in (70.29) contains τ, proving maximality. Characters (70.30) are exactly the quotient maps, and the $\varphi(2^k)$ automorphisms of the cyclic quotient act freely on them, which proves (70.31).

By the Chinese remainder theorem, $G_a/G_a^2 \cong C_2^{\omega(a)}$. Since $a \equiv 3 \pmod 4$, at least one prime divisor is $3 \pmod 4$, so -1 is not a square modulo that divisor. Evaluation at -1 is therefore a nonzero linear functional on the $2^{\omega(a)}$ quadratic characters; exactly half have value -1, proving (70.32).

Finally, if the subgroup generated by the prime factors of h misses -1, enlarge it to a maximal avoiding subgroup. The classification puts all prime-factor classes in one kernel in (70.33). If the generated subgroup contains -1 but the bounded spectrum misses it, the event is $B_a(h)$. These are the only two possibilities. □

For example $G_{15} \cong C_2 \times C_4$, with $-1 = (1, 2)$. The subgroup $C_2 \times \{0\}$ is maximal avoiding and has cyclic quotient of order four; it is not the kernel of any quadratic character. This explicit counterexample is why replacing (70.30) by only odd quadratic characters would make (70.33) false.

32.6. **Congruence-wall freeness.** A progression $h \equiv r \pmod m$ will be called unit-compatible if it contains infinitely many integers coprime to a, equivalently it is not the case that $a \mid m$ and $a \mid r$.

Theorem 32.11 (fixed-a congruence-wall freeness; proved; source Theorem 70.11)**.** *For every fixed prime $a \equiv 3 \pmod 4$ and every fixed unit-compatible progression,*

$$\frac{\#\{h \le H : h \equiv r \pmod m,\ (h,a) = 1,\ F_a(h)\}}{\#\{h \le H : h \equiv r \pmod m,\ (h,a) = 1\}} \ll_{a,m} \frac{1}{\sqrt{\log H}}. \tag{70.34}$$

In particular success has conditional density one, and every such progression contains successful h.

Let $p \equiv c \pmod Q)_0$ be any fixed progression whose intersection with $p \equiv 1 \pmod{24}$ is a reduced class and hence contains infinitely many primes. Then

$$\frac{\#\{p \le N : p \text{ in that hard-prime class},\ F_a((p+a)/4)\}}{\#\{p \le N : p \text{ in that hard-prime class}\}} \ll_{a,Q_0} \frac{1}{\sqrt{\log N}}. \tag{70.35}$$

Thus no fixed congruence class of hard primes forces failure at this a. For every fixed $Z \ge 3$, the hard primes with $a_1(p) > Z$ contain no complete compatible hard-prime progression; indeed they have relative density zero in every such progression.

Proof. Corollary 32.8 gives the global upper bound $O_a(H/\sqrt{\log H})$ for failures. Its intersection with a progression is no larger. Unit compatibility makes the denominator in (70.34) asymptotic to a positive constant times H, proving the estimate. This argument also handles a fixed prime divisor of m which is a nonresidue and divides every member: it may annihilate F1, but it changes only a fixed local condition and cannot force the budget failure F3.

For primes, Theorem 32.9 in the additional fixed progression gives a numerator $O_{a,Q_0}(N/(\log N)^{3/2})$. The prime number theorem in the fixed reduced progression

gives a denominator asymptotic to a positive constant times $N/\log N$, proving (70.35). Finally, for all sufficiently large hard primes the coefficient $a = 3$ is admissible. The condition $a_1(p) > Z$ implies failure at $a = 3$, so (70.35) with $a = 3$ proves the last assertion. □

This is the precise contrast with Theorem 17.1. In the M-frame the residue-one trick puts **every** prime in explicit progressions inside $\{W(p) > T\}$. In the fixed-a frame every compatible progression has asymptotically full-density success, both for integers and, by an upper sieve plus the fixed-progression prime number theorem, for hard primes. The result concerns fixed coefficients and fixed progressions; it does not exclude a moving congruence modulus or a correlated growing family.

Computational 70.1 (finite structural replay). Block `(bq)` enumerates every subgroup of $(\mathbb{Z}/a\mathbb{Z})^*$ for primes $a \equiv 3 \pmod 4$, $a \le 199$, and checks all three sets in (70.4). It also enumerates all eight subgroups of G_{15}, finds exactly the three maximal -1-avoiding subgroups, and pins the higher-order kernel $\{1, 11\}$ with cyclic quotient of order four. For $a = 3, 7, 11, 19, 23$, it computes every ratio spectrum for unit $h \le 20,000$, with exact counts

a	3	7	11	19	23
# units	13334	17143	18182	18948	19131
$\#F1$	1896	3350	2861	2427	4888
$\#F3$	0	2149	4532	7666	6722
$\#F_a$	1896	5499	7393	10093	11610.

(70.36)

It verifies F2 emptiness on every row, replays (62.16), and prints the finite ratios normalized by $H/\sqrt{\log H}$ as information only. For $a = 7$, every one of the twelve progressions modulo 12 has a success by 20,000; their finite success proportions range from 44.37% to 100%, not yet near the asymptotic density one. Finally, for the first 200 hard primes it compares an exact bounded-spectrum computation of a_1 with an independent literal divisor scan $D \mid h^2$ from Theorem 23.4. All agree; the maximum is 23. With `ES_FULL_SCAN=1`, the spectrum census extends to $h \le 200,000$ and the independent comparison to the first 2,000 hard primes, where the maximum is 63. These are bounded regression checks, not proofs of any asymptotic.

32.7. **Honest walls and heuristic outlook.** All constants, error terms, the budget K_a, and the character/sieve unions above may depend on the **fixed** modulus a. A stacking argument would need, at minimum, uniform Selberg–Delange constants and errors for $a \le (\log H)^\theta$, uniform control of the number and conductors of the characters in (70.31), and a budget argument whose $(\log\log H)^{K_a}$ cost does not explode. None is supplied here. The 22% F3 share among the old-frame record-prime failures in source §19.2 is an empirical warning that conditioning on deep records can strongly enrich the nominally lower-order budget mechanism.

This does not contradict Assessment 63.2. Its mass $\sum 1/(a-1) = \frac{1}{2}\log\log Y + O(1)$ stacks the low-mass D1 class-exclusion slices. The event isolated here is different: confinement loses about half the prime-factor classes at each fixed prime modulus. A larger marginal deficit does not create independence. The same moving prime p controls every shift $(p+a)/4$, exactly the correlation warning in Assessment 63.2. The stacking analysis of Section 33 addresses separately what current joint technology yields.

Heuristic 70.1 (displayed independence model; no evidence claim). If, in addition to the proved prime-modulus laws, one *postulates* that all admissible

coefficients $a \equiv 3 \pmod 4$ have comparable failure scale $c_a(\log p)^{-1/2}$, that their geometric-mean constants stay bounded, and that distinct coefficients are quasi-independent, then there are $Z/4 + O(1)$ such coefficients up to Z, and

$$\begin{aligned}\Pr[a_1(p) > Z] &\approx \prod_{\substack{a \le Z \\ a \equiv 3(4)}} c_a(\log p)^{-1/2} \\ &= \exp\left\{-\frac{Z}{8}(1+o(1))\log\log p + O(Z)\right\}.\end{aligned} \tag{70.37}$$

On this model, making $N \Pr[a_1 > Z] < 1$ asks for

$$Z \asymp \frac{\log N}{\log\log N}, \tag{70.38}$$

inside the conceivable depth $a \le (\log N)^{1+o(1)}$. The extrapolation to composite coefficients in (70.37) is part of the heuristic, not a consequence of Theorem 32.10. If one multiplies only the prime moduli actually covered by (70.22), their number is instead $(1+o(1))Z/(2\log Z)$, giving the different prime-only model

$$\exp\left\{-(1+o(1))\frac{Z\log\log p}{4\log Z}\right\}; \tag{70.39}$$

this arithmetic distinction cannot honestly be suppressed.

The attraction of (70.37) is structural only. The M-frame crossing is blocked by the genuine progressions of Theorem 17.1, whereas Theorem 32.11 shows no analogous fixed-congruence wall in this frame. This is not a count-below-one theorem. Proposition 19.5 excludes only its named absolute-remainder architectures and explicitly leaves room for inputs that see actual locations; the present fixed-a laws do see those locations, but no joint growing-a transfer has been proved.

What would falsify the outlook; what this section does not do. Persistent large positive correlations between different shifted failures, rapidly decaying constants, or budget costs overwhelming the confinement term in the growing-a range would invalidate (70.37) as a useful model. The finite census cannot decide those alternatives. The proved statements do not give uniformity in a, a lower bound along shifted primes, independence, a tail estimate for growing Z, a bound for every $a_1(p)$, or a proof that $a_1(p) < \infty$. Since Theorems 23.4 and 24.1 give $W(p) = +\infty \iff a_1(p) = +\infty$ for hard primes, any one of those missing joint steps is load-bearing for the exceptional set. No such step is claimed here.

33. Fixed-J stacking in the first-witness frame: proved finite dimension and the uniformity frontier

Status and headline. The group-theoretic separation of confinement from the bounded-multiplicity obstruction, the finite-local factor, the count-below-one reduction, and the fixed-J mean-value bounds below are **proved**. The mean-value input in the last item is quoted from Nair–Tenenbaum and Henriot with its exact fixed-dimension hypotheses. Shiu's one-function theorem is also quoted. None of those papers gives a quantitative growth rate for its constant as the number of forms grows. The moving-J tail is therefore **conditional** on the two named hypotheses below, and the local-product discussion is explicitly a **Heuristic**. The unconditional first-witness bound here has only fixed-J, one-excluded-class strength. It is weaker than source §12.3, Vaughan's cited 2/3 exceptional-set shape, Pomerance–Weingartner's

modern 2/3 version, and the CLAIMED/PROVISIONAL 3/4 campaign bound (the exponent 3/4 was obtained earlier by [16]). No unconditional Erdős–Straus record or pointwise conclusion is claimed.

Put $L = \log N$. For $Z \geq 3$ let

$$\mathcal{P}_Z = \{a \leq Z : a \equiv 3 \pmod 4,\ a \text{ prime}\}, \qquad h_a(n) = \frac{n+a}{4}. \tag{71.1}$$

For $n \equiv 1 \pmod{24}$, every $h_a(n)$ in (71.1) is an integer. Restricting to prime a is deliberate: it gives a cyclic unit group and a clean quadratic-residue subgroup. Omitting composite admissible a can only weaken an upper bound obtained from necessary conditions.

33.1. **Confinement, budget failure, and the integer frame.**

Lemma 33.1 (prime-a separation; proved; source Lemma 71.1)**.** *Let $a \equiv 3 \pmod 4$ be prime and let $(h, a) = 1$. Write $F1_a(h)$ for the assertion that every prime divisor of h is a quadratic residue modulo a. Then:*

(1) $F1_a(h)$ implies failure at a.

(2) If $F1_a(h)$ is false, the subgroup generated by the prime-divisor classes of h contains -1. Any remaining failure is solely a failure of the bounded ranges $|f_q| \leq v_q(h)$ to reach that subgroup element. Call this disjoint event $F3_a(h)$. Thus

$$\{\textit{failure at } a\} = F1_a \,\dot\cup\, F3_a. \tag{71.2}$$

(3) Failure at a implies the completely multiplicative necessary condition

$$G_a(h): \quad q \not\equiv -1 \pmod a \quad (q \mid h \text{ prime}). \tag{71.3}$$

Proof. The group $(\mathbb{Z}/a\mathbb{Z})^*$ is cyclic of order $a - 1 = 2m$ with m odd. Its quadratic-residue subgroup has order m and therefore does not contain the unique element of order two, -1. This proves (1) from Theorem 24.1. If a prime class is a non-residue, its discrete logarithm to a primitive root is odd, so its order is even; the midpoint power in its cyclic subgroup is the unique involution -1. This proves the subgroup assertion and makes (71.2) a definition of exactly the residual bounded-budget event. Finally, if some $q \mid h$ satisfied $q \equiv -1 \pmod a$, the permitted choice $f_q = 1$ and $f_r = 0$ for the other primes would put -1 in $\mathrm{Rat}_a(h)$. Theorem 24.1 would give success, proving (3). □

The first implication is one-way: confinement is sufficient for failure, not necessary. Consequently an intersection of $F1_a$ events does **not** majorize the all-failure event. The exact pointwise majorant is

$$\prod_{a \in \mathcal{P}_Z} \mathbf{1}_{\mathrm{fail,a}} \leq \prod_{a \in \mathcal{P}_Z} (\mathbf{1}_{F1_a} + \mathbf{1}_{F3_a}), \tag{71.4}$$

whose expansion has one term for each choice of the $F3$ coordinates. Without an $F3$ estimate, (71.3), not $F1$, is the unconditional multiplicative majorant.

For hard primes, Theorems 23.4 and 24.1 and definition (63.25) give, for every Z,

$$\{p \leq N : p \text{ is an Erdős–Straus counterexample}\} \subseteq \{p \leq N : a_1(p) > Z\}, \tag{71.5}$$

and $a_1(p) > Z$ means failure at every admissible $a \leq Z$. For $p > 3Z/2$, every $a \in \mathcal{P}_Z$ is admissible. Therefore

$$\#\{3Z/2 < p \leq N : a_1(p) > Z\} \leq \#\{n \leq N : n \equiv 1 \pmod{24},\ G_a(h_a(n))\ (a \in \mathcal{P}_Z)\}. \tag{71.6}$$

Including composite n and values with $(a,n)>1$ only enlarges the right side. The omitted primes contribute $O(Z)$ to a global tail and are absent in a dyadic range $(N/2,N]$ once $Z=o(N)$. This is the promised integer frame.

Lemma 33.2 (the a_1 count-below-one reduction; proved; source Lemma 71.2)**.** *Suppose a proved estimate, in its stated validity window, gives*

$$\#\{p\le N : a_1(p) > Z(N)\} < 1 \tag{71.7}$$

for every member of an unbounded covering sequence of endpoints N. Then every sufficiently large prime satisfies the Erdős–Straus conjecture. The conclusion is effective if the estimate, its validity threshold, and $Z(N)$ are effective.

Proof. The count in (71.7) is a nonnegative integer, hence zero. By (71.5) no counterexample is at most such an N. A sequence of endpoints whose intervals cover the tail, for example all sufficiently large dyadic endpoints, proves the assertion. □

This is the direct a_1 analogue of Theorem 19.3. Unlike the absolute-remainder architectures in Proposition 19.5, the route below asks for a mean value of products of multiplicative functions on shifted linear forms. Assessment 57.3 explicitly did not exclude such a new mean-value input. Theorem 17.1's residue-one construction forces the different statistic W to be large; it does not force a_1 to be large (compare Section 32). Thus the M-frame congruence wall is not a wall for (71.6).

33.2. **What the three cited mean-value theorems actually say.** The quotations in this subsection preserve the hypotheses and dependency sentences of the archived papers. Notation is converted only to the present LaTeX typography.

Fact 71.1 (cited; Shiu 1980, Theorem 1, pp. 162–163, verbatim statement). Shiu defines $\mathcal{M}$ to consist of nonnegative multiplicative f satisfying

$$f(p^l)\le A_1^l \quad (p \text{ prime}, l\ge 1), \qquad (\forall \epsilon>0)(\exists A_2>0)\ f(n)\le A_2 n^{\epsilon}.$$

His theorem reads:

> Let $f\in\mathcal{M}$, $0<\alpha<1/2$, $0<\beta<1/2$ and let a,k be integers satisfying $0<a<k$, $(a,k)=1$. Then, as $x\to\infty$,
>
> $$\sum_{\substack{x-y<n\le x\\ n\equiv a \pmod k}} f(n) \ll \frac{y}{\phi(k)}\frac{1}{\log x}\exp\left\{\sum_{\substack{p\le x\\ p\nmid k}}\frac{f(p)}{p}\right\}, \tag{71.8}$$
>
> uniformly in a,k,y provided that
>
> $$k<y^{1-\alpha}, \qquad x^{\beta}<y\le x.$$

The paper then states: "All the constants implied by the $\ll$ and O notations depend at most on α,β,f." This is a one-function theorem; it contains no number-of-forms parameter.

Fact 71.2 (cited; Nair–Tenenbaum 1998, Theorem 1, pp. 123–126, verbatim statement with its defined notation). For fixed k,A,B,ϵ the paper defines $\mathcal{M}_k(A,B,\epsilon)$ by

$$F(m_1n_1,\dots,m_kn_k)\le \min(A^{\Omega(m)},Bm^{\epsilon})F(n_1,\dots,n_k), \quad m=m_1\cdots m_k, \tag{71.9}$$

when $(m_j, n_j) = 1$ for every j. It factors $Q = \prod_j Q_j = \prod_{h=1}^{r} R_h^{\gamma_h}$, writes $Q_j = \prod_h R_h^{\gamma_{jh}}$, lets ρ, respectively ρ_h, count roots of Q, respectively R_h, and defines

$$v(n; F, \boldsymbol{\rho}) = \sideset{}{^\dagger}\sum_{n_1 \cdots n_r = n} F\left(\prod_h n_h^{\gamma_{1h}}, \ldots, \prod_h n_h^{\gamma_{kh}}\right) \frac{\rho_1(n_1) \cdots \rho_r(n_r)}{n_1 \cdots n_r}, \tag{71.10}$$

where the dagger requires the n_h to be pairwise coprime and coprime to the discriminant D^* of $\prod_h R_h$. The theorem reads:

> Let k be an arbitrary positive integer and let $Q_j \in \mathbb{Z}[X]$ $(1 \le j \le k)$ be such that $Q = \prod_{j=1}^{k} Q_j$ has no fixed prime divisor. Denote by g the degree of Q, by r the number of irreducible factors of Q and put $\rho = \rho_Q$. Then for any $A \ge 1$, $B \ge 1$, $0 < \epsilon < 1/(8g^2)$, $0 < \delta < 1$ and $F \in \mathcal{M}_k(A, B, \epsilon)$, we have
>
> $$\sum_{x < n \le x+y} F(|Q_1(n)|, \ldots, |Q_k(n)|) \ll y \prod_{p \le x} \left(1 - \frac{\rho(p)}{p}\right) \sum_{n \le x} v(n; F, \boldsymbol{\rho}) \tag{71.11}$$
>
> uniformly for $x \ge c_0 \|Q\|^\delta$ and $x^{4g^2\epsilon} \le y \le x$. The implicit constant in the $\ll$-sign depends at most on $A, B, \epsilon, \delta, k, r, g, D$ and the constant c_0 depends at most on $A, B, \epsilon, \delta, k, r, g$.

Here D is the paper's finite discriminant-supported integer defined in §2. In particular, the bound is a fixed-k theorem and its implicit constant is allowed to depend on both k and D. No formula or growth rate in k is stated.

Fact 71.3 (cited; Henriot 2011, Theorem 5, Corollary 2, and Theorem 7, pp. 4–8, verbatim statements). Henriot writes $Q = \prod_{j=1}^{k} Q_j = \prod_{h=1}^{r} R_h^{\gamma_h}$, $Q^* = \prod_h R_h$, lets D^* be the nonzero discriminant of Q^*, and defines

$$\widehat{\rho}_{\mathbf{R}}(n_1, \ldots, n_r) = \#\{n \bmod [n_1\kappa(n_1), \ldots, n_r\kappa(n_r)] : n_h \parallel R_h(n) \ (1 \le h \le r)\}. \tag{71.12}$$

His main theorem reads:

> Let k be a positive integer and let $Q_j \in \mathbb{Z}[X]$ $(1 \le j \le k)$. Let $Q = \prod_{j=1}^{k} Q_j$ and assume that Q is primitive. Let (2.1) be the decomposition of Q in irreducible factors and define $g, \rho, \widehat{\rho}_{\mathbf{R}}$ as in the previous section. Let $0 < \alpha < 1$, $0 < \delta < 1$, $A \ge 1$ and $B \ge 1$. Let also
>
> $$0 < \epsilon < \frac{\alpha}{50g(g + 1/\delta)} \quad \text{and} \quad F \in \mathcal{M}_k(A, B, \epsilon).$$
>
> Then we have, uniformly in $x \ge c_0 \|Q\|^\delta$ and $x^\alpha \le y \le x$,
>
> $$\begin{aligned} &\sum_{x < n \le x+y} F(|Q_1(n)|, \ldots, |Q_k(n)|) \\ &\ll y \prod_{g < p \le x} \left(1 - \frac{\rho(p)}{p}\right) \sum_{n_1 \cdots n_r \le x} \widetilde{F}(n_1, \ldots, n_r) \frac{\widehat{\rho}_{\mathbf{R}}(n_1, \ldots, n_r)}{[n_1\kappa(n_1), \ldots, n_r\kappa(n_r)]}, \end{aligned} \tag{71.13}$$
>
> where c_0 and the implicit constant depend at most on g, α, δ, A, B.

Corollary 2 gives the more legible upper bound

$$\Delta_{D^*} y \prod_{g < p \le x} \left(1 - \frac{\rho(p)}{p}\right) \prod_{\substack{p \le x \\ p \nmid D^*}} \prod_{h=1}^{r} \left(1 + \widetilde{G}^{(h)}(p) \frac{\rho_{R_h}(p)}{p}\right), \tag{71.14}$$

with

$$\Delta_{D^*}=\prod_{p\mid D^*}\left(1+\sum_{\nu_h\le \deg R_h\ \ (1\le h\le r)}\widetilde{G}(p^{\nu_1},\dots,p^{\nu_r})\frac{\widehat{\rho}_{\mathbf{R}}(p^{\nu_1},\dots,p^{\nu_r})}{p^{\max_h \nu_h+1}}\right). \tag{71.15}$$

The corollary says its dependencies are those of Theorem 5. Under the same assumptions and $Q(0)\neq 0$, Theorem 7 also states the prime-input variant

$$\sum_{x<p\le x+y}F(|Q_1(p)|,\dots,|Q_k(p)|)\ll \frac{|Q(0)|}{\phi(|Q(0)|)}\frac{\Delta_{D^*}y}{\log x}\prod_{g<q\le x}\left(1-\frac{\rho(q)}{q}\right)\sum_{n_1\cdots n_r\le x}\widetilde{F}(\mathbf{n})\frac{\widehat{\rho}_{\mathbf{R}}(\mathbf{n})}{[n_1\kappa(n_1),\dots,n_r\kappa(n_r)]}. \tag{71.15a}$$

Its dependencies are again those of Theorem 5. Thus the implicit constant in Henriot is uniform in D^*; the exact finite factor (71.15) carries the discriminant dependence. Theorem 7 can supply the usual prime-density factor $1/\log x$ in the application below, but does not improve dimension uniformity. For J linear forms, $g=J$. The paper says only that the constants "depend at most on g,α,δ,A,B," not $\exp(O(g))$, $\exp(O(g\log g))$, or any other rate.

Assessment 71.1 (the actual $C(J)$ intelligence). Shiu has no multi-form parameter. Nair–Tenenbaum has an unspecified constant depending on k,r,g,D. Henriot removes D from that implicit constant but leaves an unspecified dependence on total degree g, which equals J here; its displayed Δ_{D^*} is not a bound for that hidden dependence. Therefore none of the three archived papers proves $C(J)\le\exp(O(J))$, $\exp(O(J\log J))$, or even a stated computable asymptotic rate as $J\to\infty$. They prove exactly fixed J. This is the unresolved Phase-1 $C(k)$ question in `PROJECT.md`, not a technicality that can be absorbed into $\ll$.

33.3. **The best unconditional fixed-J consequences.** For $a\in\mathcal{P}_Z$ put $b_a=(a+1)/4$, $d_a=(6,b_a)$, and

$$R_a(t)=\frac{6t+b_a}{d_a}\in\mathbb{Z}[t]. \tag{71.16}$$

The R_a are primitive, pairwise nonassociate linear polynomials. The discarded constant d_a causes no problem. If $2\mid d_a$ then $a\equiv 7 \pmod 8$ and 2 is a quadratic residue modulo a; if $3\mid d_a$ then $a\equiv 11\pmod{12}$ and 3 is a quadratic residue modulo a. Also neither 2 nor 3 is $-1 \pmod a$ in the cases where it occurs. Thus both indicators below take value one on d_a.

Define completely multiplicative 0/1 functions by their prime-power values

$$\begin{aligned} g_a(q^e)&=\mathbf{1}_{q\not\equiv -1 \pmod a},\\ f_a(q^e)&=\mathbf{1}_{q\neq a}\,\mathbf{1}_{(q/a)=1}\qquad (e\ge 1),\end{aligned} \tag{71.17}$$

and give both value one at 1. Then $G_a(h_a(24t+1))$ is $g_a(R_a(t))=1$, while $F1_a(h_a(24t+1))$ is $f_a(R_a(t))=1$.

Theorem 33.3 (fixed-J shifted-indicator means; proved from the cited Henriot theorem; source Theorem 71.3)**.** *Let $\mathcal{A}$ be any fixed finite set of J primes $a\equiv 3 \pmod 4$. As $X\to\infty$,*

$$\#\{t\le X: g_a(R_a(t))=1\ (a\in\mathcal{A})\}\ll_{\mathcal{A}} X(\log X)^{-\delta(\mathcal{A})},\quad \delta(\mathcal{A})=\sum_{a\in\mathcal{A}}\frac{1}{a-1}, \tag{71.18}$$

and

$$\#\{t\le X: f_a(R_a(t))=1\ (a\in\mathcal{A})\}\ll_{\mathcal{A}} X(\log X)^{-J/2}. \tag{71.19}$$

The corresponding prime-input estimates retain the prime-density saving: for $u_a = g_a$, respectively $u_a = f_a$,

$$\#\{p \le X : p \equiv 1 \pmod{24},\ u_a(h_a(p)) = 1\ (a \in \mathcal{A})\} \ll_{\mathcal{A}} \frac{X}{\log X} \begin{cases} (\log X)^{-\delta(\mathcal{A})}, & u_a = g_a, \\ (\log X)^{-J/2}, & u_a = f_a. \end{cases} \tag{71.19a}$$

Proof. Apply Henriot's Theorem 5 and Corollary 2 with $Q_j = R_{a_j}$ and $F(n_1, \dots, n_J) = \prod_j g_{a_j}(n_j)$, respectively $\prod_j f_{a_j}(n_j)$. These products lie in $\mathcal{M}_J(1, 1, \epsilon)$ for every $\epsilon > 0$. The product of the primitive R_a is primitive. Choose fixed $\alpha, \delta \in (0, 1)$ and a sufficiently small ϵ; all hypotheses of Fact 71.3 then hold.

Outside the finite set of primes dividing the leading coefficients or the discriminant, the J roots are distinct, so the logarithm of the Euler product in (71.14) is

$$-\sum_{q \le X} \frac{1}{q} \sum_{a \in \mathcal{A}} (1 - u_a(q)) + O_{\mathcal{A}}(1), \tag{71.20}$$

where $u_a = g_a$ or f_a. The fixed-modulus Mertens theorem for primes in arithmetic progressions gives

$$\sum_{q \le X} \frac{1 - g_a(q)}{q} = \frac{1}{a - 1} \log\log X + O_a(1), \qquad \sum_{q \le X} \frac{1 - f_a(q)}{q} = \frac{1}{2} \log\log X + O_a(1). \tag{71.21}$$

(The single prime $q = a$ changes only the constant.) Equations (71.20)–(71.21), and the fixed finite Δ_{D^*}, prove each estimate on a dyadic interval. Summing the dyadic intervals proves (71.18)–(71.19).

For (71.19a), put $S_a(T) = T + a$ and let $\bar{u}_a$ agree with u_a at odd prime powers but have $\bar{u}_a(2^e) = 1$. If $p \equiv 1 \pmod{24}$, then $\bar{u}_a(p + a) = u_a((p + a)/4)$: when $2 \mid (p + a)/4$ one has $a \equiv 7 \pmod 8$, so $u_a(2) = 1$ for both indicators. The monic linear forms S_a have primitive product and nonzero constant term $\prod_{a \in \mathcal{A}} a$. Apply Henriot's Theorem 7 to the $\bar{u}_a(S_a(p))$, dropping the progression restriction. Extend its nonnegative truncated mean-value sum to the full Euler product. Away from the fixed discriminant primes, distinct linear roots allow at most one nonzero valuation coordinate; summing the exact-valuation terms over $\nu \ge 1$ gives $\bar{u}_a(q)/q$. Thus the local factor is $1 + \sum_a \bar{u}_a(q)/q$, and after multiplication by $1 - J/q$ it has exactly the logarithmic expansion (71.20)–(71.21). The factor $1/\log X$ in (71.15a) proves (71.19a). □

The discriminant accounting is fully explicit here. If $R_{a_i}(t) = u_i t + v_i$, then

$$D^*_{\mathcal{A}} = \prod_{i<j} (u_i v_j - u_j v_i)^2 = \prod_{i<j} \left(\frac{6(b_{a_j} - b_{a_i})}{d_{a_i} d_{a_j}} \right)^2. \tag{71.22}$$

For the weak indicators g_a and the first six moduli

$$3, 7, 11, 19, 23, 31, \tag{71.23}$$

(71.15) evaluates exactly as follows; this is the discriminant penalty, not the hidden fixed-degree constant.

J	$\mathrm{factor}(D^*)$	Δ_{D^*}
1	1	1
2	3^2	$4/3$
3	$2^4 3^2$	$49/18$
4	$2^{14} 3^8$	$49/18$
5	$2^{16} 3^8 5^2$	$1472/225$
6	$2^{16} 3^{18} 5^4 7^2$	$20564/1225.$

(71.24)

Each entry is the finite valuation count (71.12) substituted in (71.15); no estimate for D^* is being hidden.

Corollary 33.4 (fixed-J first-witness tails; proved; source Corollary 71.4)**.** *For the first J moduli in (71.23),*

$$\#\{p \le N : p \equiv 1 \pmod{24},\ a_1(p) > a_J\} \ll_J \frac{N}{\log N} (\log N)^{-\delta_J}, \tag{71.25}$$

where

J	a_J	$\delta_J = \sum_{j \le J} 1/(a_j - 1)$	$J/2$ ($F1$ *stacking*)
1	3	1/2	1/2
2	7	2/3	1
3	11	23/30	3/2
4	19	37/45	2
5	23	859/990	5/2
6	31	446/495	3.

(71.26)

Proof. Apart from the fixed range $p \le 3a_J/2$, a prime counted on the left fails at each selected a_j. Lemma 33.1(3) and the g_a case of (71.19a) prove (71.25); the fixed omitted range is absorbed in the implied constant. The last column is the additional stacking exponent in the f_a case of (71.19a), so the F1-intersection bound itself is $N(\log N)^{-1-J/2}$. By (71.4) it is not an unconditional first-witness exponent. The fractions in the third column are direct sums. □

For every fixed Z, the same proof gives exponent

$$\delta(\mathcal{P}_Z) = \sum_{a \in \mathcal{P}_Z} \frac{1}{a-1} = \frac{1}{2} \log\log Z + O(1). \tag{71.27}$$

This is precisely the class-exclusion mass diagnosed in Assessment 63.2. Even if one were allowed to insert $Z \asymp \log N$ into the fixed-J formula (one is not, without uniform constants), it would yield only

$$\frac{N}{\log N}\exp\{-(1/2+o(1))\log\log N\ \log\log\log N\}. \tag{71.28}$$

Source §12.3 instead obtains $N\exp\{-c(\log\log N)^2\}$ by using all composite $w \equiv 3 \pmod 4$ and further exclusion classes; its mass is of order $\log W$ rather than (71.27)'s $\log\log W$. Thus the a-frame is new bookkeeping, not a stronger unconditional estimate.

33.4. **Exact finite-prime correlations.** The shifted values are not locally independent. The exact collision can be written without an asymptotic theorem.

Lemma 33.5 (one-prime local confinement factor; proved; source Lemma 71.5)**.** *Let $\mathcal{A}$ be a finite set of primes $a \equiv 3 \pmod 4$, let C be a squarefree product of any subset of $\mathcal{A}$, and let $\ell \geq 5$ be prime. Parameterize $n = 24t+1$ and put*

$$r_a(\ell) \equiv -6^{-1}b_a \pmod \ell, \qquad B_\ell = \{r_a(\ell) : a \in \mathcal{A},\ (\ell/a) \neq 1\}. \tag{71.29}$$

Here $(\ell/a) = 0$ when $\ell = a$, so that placement is bad. Let

$$E_\ell = \begin{cases} \{-24^{-1} \pmod \ell\}, & \ell \mid C, \\ \varnothing, & \ell \nmid C. \end{cases} \tag{71.30}$$

Among residues n modulo $\operatorname{lcm}(24, 4\ell C)$ satisfying $n \equiv 1 \pmod{24}$ and $(n, C) = 1$, the exact proportion for which every forced divisibility $\ell \mid h_a(n)$ is compatible with all the confinement conditions $F1_a$ is

$$\boxed{\lambda_\ell(\mathcal{A}; C) = \frac{\ell - |E_\ell| - |B_\ell \setminus E_\ell|}{\ell - |E_\ell|}.} \tag{71.31}$$

Proof. Since $h_a(24t+1) = 6t + b_a$, $\ell \mid h_a$ holds in exactly the residue $t = r_a(\ell)$. If several a give the same residue, all are forced there; compatibility fails exactly when at least one has $(\ell/a) \neq 1$. Hence the union of bad t-residues is the set B_ℓ, with collisions counted once. The coprimality condition $(n, C) = 1$ is independent at primes other than ℓ by the Chinese remainder theorem. When $\ell \mid C$ it removes the single residue $24t + 1 \equiv 0 \pmod \ell$, namely E_ℓ; when $\ell \nmid C$ it removes none. Counting the remaining good residues gives (71.31). □

For example, with $\mathcal{A} = \{3, 7, 11, 19, 23, 31\}$ and $C = 3 \cdot 7 \cdot 11$, (71.31) gives

ℓ	5	7	11	13
λ_ℓ	3/5	2/3	7/10	9/13.

(71.32)

Block `(br)` enumerates the complete residue frames and checks these exact fractions.

Heuristic 71.1 (displayed model, no evidential force). Let $J = |\mathcal{P}_Z| \sim Z/(2\log Z)$. In a balls-in-residue-classes model, a given modulus a contributes a bad placement at ℓ with probability about $1/2$, and its root is spread over ℓ bins. A bin is then safe with probability

$$(1 - 1/(2\ell))^J \approx e^{-J/(2\ell)}, \tag{71.33}$$

so the model suggested by the exact collision formula is

$$\prod_{\ell\le Z}\lambda_\ell \approx \exp\left\{-\frac{J}{2}\sum_{\ell\le Z}\frac{1}{\ell}\right\} = \exp\left\{-(1+o(1))\frac{Z\log\log Z}{4\log Z}\right\}. \tag{71.34}$$

This is only a heuristic: for fixed small ℓ the bad roots may even cover every available residue, making the exact factor zero.

A $2^{-\pi(Z)+o(\pi(Z))}$ factor would arise from the different sparse toy assumption of one forced, random-character placement per rational prime:

$$2^{-\pi(Z)+o(\pi(Z))} = \exp\{-(\log 2+o(1))Z/\log Z\}. \tag{71.35}$$

The actual prime-modulus system has about J/ℓ placements, not one, so (71.35) is not its precise aggregate. Both (71.34) and the sparser (71.35) are subdominant, when $Z = L^\theta$, to the putative independent one-form saving

$$\exp\{-(J/2)\log L\} = \exp\{-(1+o(1))Z/(4\theta)\}. \tag{71.36}$$

This correction matters: with prime a only, the main exponent is order Z, not $Z\log\log N$.

33.5. **A named conditional frontier and its exact thresholds.** The first hypothesis isolates the bounded-budget problem. Its normalization is relative to the size of the progression; omitting that normalization would not express a $(\log x)^{-1/2}$ failure probability.

Hypothesis 33.6 ($H_{\mathrm{FAIL}}(\theta_0)$; open; source Hypothesis H_{FAIL})**.** *There are effective $C_0, N_0 > 0$ such that, uniformly for $x \ge N_0$, primes $a \equiv 3 \pmod 4$ with $a \le (\log x)^{\theta_0}$, and compatible reduced classes $r \pmod{24a}$,*

$$\#\{n\le x : n\equiv r \pmod{24a},\ -1\notin \mathrm{Rat}_a(h_a(n))\} \le \frac{C_0 x}{24a}(\log x)^{-1/2}. \tag{71.37}$$

Replacing $24a$ by $\phi(24a)$ under a reduced-residue normalization is equivalent up to an absolute constant. The hypothesis includes F3. Section 32 proves the corresponding fixed-a statement; it does not provide the displayed uniformity as a moves with x.

Hypothesis 33.7 ($H_{\mathrm{STACK}}(\theta,\gamma)$; open; source Hypothesis H_{STACK})**.** *For $Z = L^\theta$, $\mathcal{A} = \mathcal{P}_Z$ and $J = |\mathcal{A}|$, the one-form estimates (71.37) tensorize on the hard frame with*

$$\#\{n\le N : n\equiv 1 \pmod{24},\ (n,\textstyle\prod_{a\in\mathcal{A}} a)=1,\ n \text{ fails at every } a\in\mathcal{A}\} \le C_* N\exp\{KJ(\log(2J))^\gamma\}\{C_0L^{-1/2}\}^J, \tag{71.38}$$

for fixed effective C_*, K. Local factors such as (71.31) may be retained if they are at most one. This is required only for $0<\theta\le\theta_0$. The exponent γ is a parameter, not a literature claim: Facts 71.1–71.3 provide no admissible value for growing J.

Theorem 33.8 (conditional moving-J tail; proved implication; source Theorem 71.6)**.** *Assume $H_{\mathrm{FAIL}}(\theta_0)$ and $H_{\mathrm{STACK}}(\theta,\gamma)$ with $0<\theta\le\theta_0$. In the dyadic range $N/2 < p \le N$,*

$$\log\frac{\#\{N/2<p\le N : a_1(p) > L^\theta\}}{N} \le -\frac{J}{2}\log L + KJ(\log(2J))^\gamma + O(J), \tag{71.39}$$

where

$$J = |\mathcal{P}_{L^\theta}| = (1+o(1))\frac{L^\theta}{2\theta\log L}. \tag{71.40}$$

Consequently:

- *if $\gamma < 1$, then*

$$\#\{N/2 < p \le N : a_1(p) > L^\theta\} \le N \exp\{-(1+o(1))L^\theta/(4\theta)\}; \tag{71.41}$$

- *if $\gamma = 1$, (71.41) follows only when the actual constant is favorable, specifically $K < 1/(2\theta)$ in the normalization (71.38), with a correspondingly reduced positive exponent;*
- *if $\gamma > 1$, (71.38) by itself gives no saving, because its stated prefactor is larger than the negative main term.*

Proof. Here $Z = L^\theta = o(N)$, so every selected modulus is admissible for all primes in the dyadic range. A prime with $a_1(p) > Z$ fails at every selected prime modulus, and (71.38) applies. Taking logarithms and absorbing C_0^J and C_* gives (71.39). The prime number theorem in the progression 3 (mod 4) gives (71.40). Hence

$$\frac{J}{2}\log L = (1+o(1))\frac{L^\theta}{4\theta}, \qquad J(\log J)^\gamma = (1+o(1))\frac{\theta^{\gamma-1}}{2}L^\theta(\log L)^{\gamma-1}. \tag{71.42}$$

For $\gamma < 1$ the second expression is $o(L^\theta)$. For $\gamma = 1$ it is $(1/2 + o(1))L^\theta$, giving the displayed constant condition. For $\gamma > 1$ it dominates. □

For each fixed $\theta < 1$, summing dyadic ranges gives the same exponent shape for counterexample primes up to N: a counterexample has $a_1 = +\infty$, and the initial range $p \le N^{1/2}$ is negligible compared with $N\exp\{-cL^\theta\}$. At the count-below-one endpoint one must instead use (71.39) directly on successive dyadic ranges; this also avoids any small-p admissibility issue.

Under the favorable $\gamma < 1$ branch, (71.41) beats the cited Vaughan/Pomerance–Weingartner shape $\exp(-cL^{2/3})$ when $\theta > 2/3$, and beats the CLAIMED/PROVISIONAL Theorem 10.7 shape $\exp(-cL^{3/4})$ when $\theta > 3/4$. Equality of powers does not by itself beat either bound because the constants are not comparable.

Fact 71.4 (cited; Pomerance–Weingartner 2025, Theorem 1.3, p. 2). There is an absolute positive constant C such that, for $4 \le m \le (\log N)^2$, the number of $n \le N$ for which m/n is not a sum of three unit fractions is at most

$$\frac{N}{\exp\{C(\log^2 N/\phi(m))^{1/3}\}}. \tag{71.43}$$

At $m = 4$ this has power $2/3$. The paper says its proof largely follows Vaughan. Theorem 10.7's $3/4$ statement remains CLAIMED/PROVISIONAL as recorded in source §39.7. Theorem 14.2 and Corollary 14.3 also give much stronger witness-modulus tails in their stated windows, with their exact source §51 review qualifications; although W and a_1 are different statistics, either tail already bounds counterexamples. Thus every unconditional result in the fixed-J subsection above is weaker than the standing literature bound and the campaign's existing record shapes.

For count below one, suppose the two hypotheses are strengthened only enough to remain uniform when Z is a fixed positive multiple of L (the present $Z = L^\theta$ notation does not itself provide this endpoint freedom). Then (71.39) has the exact requirement

$$\frac{J}{2}\log L - \log C(J) > L + O(J). \tag{71.44}$$

If $\log C(J) = o(J\log J)$, this requires

$$J > (2+o(1))\frac{L}{\log L}. \tag{71.45}$$

Since the selected moduli are primes $3 \pmod 4$, $J \sim Z/(2\log Z)$. Therefore (71.45) corresponds to

$$Z > (4 + o(1))L, \tag{71.46}$$

not $Z \asymp L/\log L$. At $Z = L/\log L$ one has only $J \sim L/(2(\log L)^2)$ and saving $L/(4\log L) = o(L)$. Equivalently, if J denotes the number of prime forms, the needed $J \asymp L/\log L = \log N/\log\log N$ has largest prime modulus of order L. Using every admissible (including composite) a would put the largest modulus near $L/\log L$, but that is a different system and is not covered by Lemma 33.1 or the hypotheses above. This is a necessary logarithmic-bookkeeping correction to the tempting prime-only formulation.

Assessment 71.2 (budget-transition comparison). A typical integer $h \asymp N$ has $\omega(h) \approx \log\log N = \log L$, so the informal bounded-product transition

$$3^{\omega(h)} \approx L^{\log 3} = L^{1.0986\ldots} \tag{71.47}$$

lies above the corrected prime-modulus crossing depth $Z \asymp L$. This comparison is strictly asymptotic: against the coefficient 4 in (71.46), $L^{\log 3} > 4L$ only for $L > 4^{1/(\log 3 - 1)} = 1.274\ldots \times 10^6$, so it gives no favorable finite-range evidence at ordinary sizes. Thus count below one is not *asymptotically* ruled out by that budget transition. This is only a scale assessment: neither uniform $F3$ control up to $a \asymp L$ nor growing-dimensional tensorization is known.

33.6. **Honest walls and falsification register.**

(1) **The hidden constant is the main analytic wall.** Facts 71.1–71.3 justify fixed J only. No cited theorem supplies even $\exp(O(J\log J))$, much less the $\gamma < 1$ regime required for (71.41) without a favorable leading constant. Discriminant uniformity and J uniformity are different questions.

(2) **The budget wall is independent.** Confinement has the natural $L^{-1/2}$ scale (Assessment 62.1 gives the $a = 3$ case), but failure also contains $F3$. Hypothesis H_{FAIL} is open beyond fixed a; the subgroup argument and Section 32's fixed-a result do not supply its moving-modulus uniformity.

(3) **Even the dream chain nearly reaches logarithmic depth.** Any fixed $\theta < 1$ leaves the exponent in (71.41) $o(L)$. Count below one needs (71.45)–(71.46), including its endpoint constants. Theorem 19.3's moral therefore survives: a strong exceptional set is not yet a pointwise proof.

(4) **Correlation can help or hurt locally, but does not prove tensorization.** Lemma 33.5 computes one rational prime exactly. Multiplying its factors, exchanging the a- and ℓ-limits, or treating Legendre symbols as random are the unproved steps isolated in Heuristic 71.1.

The value of the fixed-J result is therefore an independent shifted-form route, an elementary exact treatment of its local collisions, and a precise map of the conditional frontier—not an improved numerical tail. The cited implicit constants are not numerically printed, so no explicit numerical constant is claimed.

A violation of (71.37) for an unbounded sequence of allowed (a, x) falsifies H_{FAIL}. A family for which the left side of (71.38) exceeds its right side for every proposed fixed C_*, K falsifies the corresponding $H_{\mathrm{STACK}}(\theta, \gamma)$. One unfavorable finite sample does neither. Conversely, the exact checks in block `(br)` establish only the displayed finite local factors, exponent fractions, and bounded counts. They do not validate either hypothesis, an independence model, or an Erdős–Straus conclusion. This is the same scope discipline as the closing register of Section 19.

34. The a-frame failure census: per-modulus frequencies, dependence, and the third decade of a_1

Status and headline. Every census and count in this section is **Computational** and exact only in its displayed finite, half-open range. The normalized ratios, shares, and comparisons are **INFO** summaries of those counts. In particular, no finite-window stability, dependence ratio, transition pattern, or record stall below is an asymptotic claim.

34.1. **Method and four finite windows.** The committed runner `scripts/afr_census.py` uses a segmented NumPy Boolean bitmap only to enumerate primes, converts every prime immediately to a Python integer, and keeps all factorization, exponent, and residue paths in Python integers. It factors

$$h_a(p) = \frac{p+a}{4} \tag{72.1}$$

with `sympy.factorint`. For each prime $a \equiv 3 \pmod 4$, it starts with $R = \{1\}$ and replaces

$$R \quad \text{by} \quad \{r(q \bmod a)^e \bmod a : r \in R,\ -v_q(h_a) \le e \le v_q(h_a)\}. \tag{72.2}$$

Thus failure means exactly $a-1 \notin R$, as in Theorem 24.1. On every scanned failure the runner computes $\sigma = \#\{q \mid h_a : (q/a) = -1\}$: F1 means $\sigma = 0$, and F3 means $\sigma > 0$. It asserts F1-or-F3 exhaustiveness and, independently of the budget, asserts -1 belongs to the subgroup generated by the factor classes on every F3 row. There was no counterexample.

The four windows are exactly

$$[10^5, 2\cdot 10^5), \quad [10^6, 1.2\cdot 10^6), \quad [10^7, 1.02\cdot 10^7), \quad [10^8, 1.002\cdot 10^8), \tag{72.3}$$

with respectively 1,031, 1,817, 1,561, and 1,355 hard primes. The next table shows the required compact panel: each cell is $F = F1 + F3; I; s$, where the first three numbers are exact counts, $I = (F/n)\sqrt{\log \bar{p}}$ is INFO, $\bar{p}$ is the arithmetic mean of hard primes in that window, and $s = F3/F$ is an INFO percentage. The runner also records the same data for every prime $a \equiv 3 \pmod 4$, $a \le 199$.

a	$[10^5, 2\cdot 10^5)$	$[10^6, 1.2\cdot 10^6)$	$[10^7, 1.02\cdot 10^7)$	$[10^8, 1.002\cdot 10^8)$
3	$500 = 500 + 0; 1.6744; 0.0$	$821 = 821 + 0; 1.6852; 0.0$	$676 = 676 + 0; 1.7391; 0.0$	$522 = 522 + 0; 1.6535; 0.0$
7	$257 = 257 + 0; 0.8606; 0.0$	$433 = 433 + 0; 0.8888; 0.0$	$340 = 340 + 0; 0.8747; 0.0$	$275 = 275 + 0; 0.8711; 0.0$
11	$372 = 302 + 70; 1.2457; 18.8$	$598 = 492 + 106; 1.2275; 17.7$	$474 = 393 + 81; 1.2195; 17.1$	$392 = 327 + 65; 1.2417; 16.6$
19	$771 = 373 + 398; 2.5819; 51.6$	$1302 = 595 + 707; 2.6725; 54.3$	$1030 = 478 + 552; 2.6499; 53.6$	$846 = 369 + 477; 2.6798; 56.4$
23	$247 = 227 + 20; 0.8271; 8.1$	$416 = 390 + 26; 0.8539; 6.2$	$315 = 296 + 19; 0.8104; 6.0$	$255 = 240 + 15; 0.8077; 5.9$
31	$585 = 325 + 260; 1.9590; 44.4$	$977 = 543 + 434; 2.0054; 44.4$	$734 = 399 + 335; 1.8884; 45.6$	$608 = 333 + 275; 1.9259; 45.2$
43	$909 = 224 + 685; 3.0440; 75.4$	$1544 = 412 + 1132; 3.1692; 73.3$	$1277 = 342 + 935; 3.2853; 73.2$	$1057 = 280 + 777; 3.3481; 73.5$
59	$788 = 335 + 453; 2.6388; 57.5$	$1301 = 559 + 742; 2.6705; 57.0$	$1044 = 473 + 571; 2.6859; 54.7$	$859 = 370 + 489; 2.7209; 56.9$
103	$857 = 275 + 582; 2.8698; 67.9$	$1403 = 473 + 930; 2.8798; 66.3$	$1148 = 353 + 795; 2.9535; 69.3$	$971 = 305 + 666; 3.0757; 68.6$
199	$933 = 369 + 564; 3.1243; 60.5$	$1607 = 575 + 1032; 3.2986; 64.2$	$1321 = 479 + 842; 3.3985; 63.7$	$1109 = 379 + 730; 3.5128; 65.8$

(72.4)

At fixed a, the four INFO values of I are fairly close in these windows: among all 24 scanned prime moduli, the largest max/min spread is 19.3%, and among the displayed moduli it is 12.4%. This is consistent with the proposed $(\log p)^{-1/2}$ scale in these four windows, but it does not establish that scale. The F3 share is not monotone in a: for example, in the third window it is 53.6% at 19, 6.0% at 23, 73.2% at 43, 54.7% at 59, 69.3% at 103, and 63.7% at 199. Larger coefficients often have a larger F3 share in this panel, but modulus-specific oscillation is substantial.

34.2. **Pairwise dependence on different shifts.** For the two middle windows, the exact marginal failure counts for $a = (3, 7, 11, 19, 23, 31)$ are respectively

$$(821, 433, 598, 1302, 416, 977) \quad (n = 1817), \qquad (676, 340, 474, 1030, 315, 734) \quad (n = 1561). \tag{72.5}$$

In each cell below, J is the exact joint count, $E = f_a f_b/n$ is the displayed INFO product-of-marginals count, and $r = J/E$ is INFO.

(a, a')	$[10^6, 1.2 \cdot 10^6) : J/E/r$	$[10^7, 1.02 \cdot 10^7) : J/E/r$
$3, 7$	$177/195.6/0.905$	$112/147.2/0.761$
$3, 11$	$291/270.2/1.077$	$214/205.3/1.043$
$3, 19$	$570/588.3/0.969$	$433/446.0/0.971$
$3, 23$	$251/188.0/1.335$	$180/136.4/1.320$
$3, 31$	$431/441.5/0.976$	$283/317.9/0.890$
$7, 11$	$163/142.5/1.144$	$119/103.2/1.153$
$7, 19$	$274/310.3/0.883$	$203/224.3/0.905$
$7, 23$	$92/99.1/0.928$	$71/68.6/1.035$
$7, 31$	$214/232.8/0.919$	$136/159.9/0.851$
$11, 19$	$417/428.5/0.973$	$321/312.8/1.026$
$11, 23$	$141/136.9/1.030$	$93/95.7/0.972$
$11, 31$	$296/321.5/0.921$	$229/222.9/1.027$
$19, 23$	$305/298.1/1.023$	$213/207.8/1.025$
$19, 31$	$679/700.1/0.970$	$458/484.3/0.946$
$23, 31$	$219/223.7/0.979$	$140/148.1/0.945$

(72.6)

Most displayed ratios are near 1 on this finite panel, but two repeatable departures are visible: $(3, 23)$ is 1.335 and 1.320, while $(7, 11)$ is 1.144 and 1.153. The lowest value is 0.761 for $(3, 7)$ in the second window. These are not same-integer events: they concern $(p + a)/4$ and $(p + a')/4$, respectively. Near-independence across those different shifts is the question, and (72.6) is only finite evidence; it is not an independence claim.

34.3. **Complete third-decade a_1 census.** For every hard prime $p < 10^8$, the runner tests all admissible coefficients $a = 3, 7, 11, 15, \ldots$, including composites, until the first exact hit. It streams one million integers at a time and writes an atomically replaced per-chunk partial histogram when `-checkpoint` is supplied. The complete histogram is

a_1	3	7	11	15	19	23	27	31	35	39	43	47	51	55	59	63	71	79	91	107
$\#p$	430409	235146	35036	7736	5323	4040	437	1059	143	227	33	120	16	20	20	6	7	1	1	1.

(72.7)

The count is exactly 719,781, agreeing with Section 27. Restricting the same run to $p < 10^6$ and $p < 10^7$ reproduces every entry of (63.26), respectively 9,732 and 82,887 hard primes. The maximum remains

$$\max_{p<10^8} a_1(p) = 107 \quad \text{only at } p = 8803369. \tag{72.8}$$

Thus the new decade does not move the old maximum. It does add the values $a_1 = 79$ at 66,222,601 and the composite $a_1 = 91$ at 22,605,361. The complete upper tail is

63	87481, 34103161, 37767409, 57399241, 67824409, 90108841
71	1430641, 2451289, 12426481, 20322481, 36740929, 54932329, 77599729
79	66222601
91	22605361
107	8803369.

(72.9)

34.4. **Deep-failure anatomy.** For a prime failed modulus, write $a : Fj_\sigma$. Composite failed moduli are marked `C` by the runner but have no F1/F3 label, because this taxonomy was defined only for prime a. The complete strict-record anatomy on the prime moduli is

p	a_1	failed prime moduli in order
73	7	$3 : F1_0$
1129	11	$3 : F1_0, 7 : F1_0$
1201	23	$3 : F1_0, 7 : F1_0, 11 : F3_1, 19 : F1_0$
21169	31	$3 : F1_0, 7 : F1_0, 11 : F1_0, 19 : F3_1, 23 : F1_0$
67369	35	$3 : F1_0, 7 : F1_0, 11 : F1_0, 19 : F3_1, 23 : F1_0, 31 : F3_1$
87481	63	$3 : F1_0, 7 : F1_0, 11 : F1_0, 19 : F1_0, 23 : F1_0, 31 : F3_1, 43 : F3_1, 47 : F1_0, 59 : F3_1$
1430641	71	$3 : F1_0, 7 : F1_0, 11 : F3_2, 19 : F1_0, 23 : F1_0, 31 : F3_1, 43 : F1_0, 47 : F1_0, 59 : F1_0, 67 : F3_1$
8803369	107	$3 : F1_0, 7 : F1_0, 11 : F1_0, 19 : F1_0, 23 : F1_0, 31 : F1_0, 43 : F3_1, 47 : F1_0, 59 : F3_1, 67 : F3_1, 71 : F1_0, 79 : F3_2, 83 : F3_1, 103 : F3_1.$

(72.10)

There are exactly 225 record-class/near-record primes with $a_1 \geq 43$. Their a_1-counts are $(43, 47, 51, 55, 59, 63, 71, 79, 91, 107) \mapsto (33, 120, 16, 20, 20, 6, 7, 1, 1, 1)$. The next table aggregates their complete prime-modulus failure ledger; the last entry gives the exact histogram $\sigma : \#$. It therefore records every F1/F3 and sigma count along all 225 paths, while the runner emits the individual per-prime rows.

a	$F1$	$F3$	$\sigma : \#$
3	225	0	$0 : 225$
7	225	0	$0 : 225$
11	204	21	$0 : 204, 1 : 12, 2 : 9$
19	148	77	$0 : 148, 1 : 38, 2 : 39$
23	214	11	$0 : 214, 1 : 11$
31	166	59	$0 : 166, 1 : 42, 2 : 17$
43	74	118	$0 : 74, 1 : 83, 2 : 31, 3 : 4$
47	60	12	$0 : 60, 1 : 9, 2 : 3$
59	8	8	$0 : 8, 1 : 6, 2 : 2$
67	0	10	$1 : 9, 3 : 1$
71	2	1	$0 : 2, 1 : 1$
79	1	1	$0 : 1, 2 : 1$
83	0	2	$1 : 2$
103	0	1	$1 : 1.$

(72.11)

For auditability, the canonical 225-row tuple $(p, a_1, [(a, \text{kind}, \sigma)])$, including composite `C` rows, has SHA-256 `5d1f08c0054b6ff858259c8dd59c8a5c8d60c0cde83b37d1d19d77af9e2bf1e8`. Among all 371,903 failed prime-modulus pairs encountered before the first hit for all 719,781 primes, 5,314 are F3 (1.429% INFO). Among the corresponding 1,648 pairs for the 225 deep primes, 321 are F3 (19.478% INFO), a 13.63-fold finite enrichment. On the eight strict-record primes the split is 35 F1 and 16 F3 (31.373% INFO). This is the a-frame analogue of the source §19.2 record-conditioning observation, not a selection-free distribution law.

34.5. **Budget transition by $\omega(h)$.** In the third window, the exact entries below are `failures/F3/F3-percent` for each number $\omega(h)$ of distinct prime factors. A dash means that there was no failure in that cell.

a	$\omega = 1$	2	3	4	5+
31	–	290/109/37.6	364/199/54.7	78/27/34.6	2/0/0.0
43	287/138/48.1	622/470/75.6	328/289/88.1	40/38/95.0	–
59	–	306/137/44.8	539/319/59.2	186/108/58.1	13/7/53.8
79	–	297/129/43.4	545/327/60.0	212/162/76.4	17/11/64.7
103	–	261/124/47.5	594/440/74.1	267/208/77.9	26/23/88.5
127	–	304/143/47.0	591/459/77.7	280/233/83.2	26/26/100.0
151	–	284/136/47.9	601/403/67.1	312/243/77.9	39/31/79.5
199	–	322/157/48.8	596/385/64.6	361/266/73.7	42/34/81.0

(72.12)

The proposed transition near $a \sim 3^{\omega(h)}$ is visible only irregularly. At fixed $\omega = 3$, for example, the share moves from 54.7% at $a = 31$ to 77.7% at 127, but is already 88.1% at 43 and falls to 64.6% at 199. The $\omega = 4$ row likewise rises broadly after 31 but is not monotone. These cells show that budget blocking is numerically material in the predicted coefficient range; they do not identify a transition law. The empty $\omega = 1$ cells are genuine exact zeros in this window, except at 43; congruence restrictions on $h_a(p)$ differ with a.

34.6. **Replay scope, timings, and honest wall.** A 9,732-prime pilot through 10^6 took 0.374 seconds inside the scanner (2.55 seconds including environment startup), so the full extrapolation was well inside the fallback threshold. The complete $p < 10^8$ run took 40.74 seconds inside and 41.36 seconds wall, with 72,328 KiB peak resident memory. Its 100 one-million-integer chunk histograms were checkpointed. The four failure/dependence windows took 33.54 seconds inside and 35.91 seconds wall, with 79,372 KiB peak resident memory. Block `(bs)` recomputes the full $p < 10^6$ histogram and the three smallest-modulus first window by default, pins the full 10^8 histogram and three deep values, and structurally audits a seeded sample. With `ES_FULL_SCAN=1` it recomputes (63.26) through 10^7, all 24 prime-modulus rows in all four windows, both dependence panels, and the budget-conditioning table. Isolated final runs took 1.90 seconds on the default path and 56.78 seconds on the full path. The prescribed default suite completed green in 206.20 seconds wall with 345,272 KiB peak resident memory.

The evidence is limited to four short windows and one initial census segment. Different windows, larger coefficients, or deeper record conditioning could change every INFO comparison. In particular, (72.4) supplies no uniform failure estimate, (72.6) supplies no decorrelation theorem, (72.12) supplies no threshold theorem, and the unchanged maximum in (72.8) supplies no bound on $a_1(p)$. None of these finite computations proves that a_1 is finite for the next hard prime, let alone for every hard prime.

35. General-numerator tails and the effectivity perimeter

This section truncates the general-numerator machinery; it does not extend its range. The fixed-polylogarithmic row has the internal review qualification of Lemma 14.1 and Section 10, but uses neither Theorem 9.4 nor Theorem 11.6. The cubic row and record recovery inherit them and remain CLAIMED/PROVISIONAL. Effectivity below means computability of constants conditional on those correctness

labels. Put

$$\lambda_m = \frac{\eta_1(m)}{\varphi(m)}, \qquad \theta_m = \frac{\eta_2(m)}{\varphi(m)}; \tag{55.1}$$

both are $\asymp 1/m$ up to absolute constants.

35.1. **The minimal general-numerator witness modulus.** For $m \geq 3$ and positive p, define

$$W_m(p) = \min k\ell, \tag{55.2}$$

where the minimum is over positive k, ℓ, u, v, w satisfying

$$k\ell \equiv -1 \pmod m, \qquad \frac{k\ell+1}{m} = uvw, \qquad pv \equiv -u \pmod{k\ell}; \tag{55.3}$$

put $W_m(p) = +\infty$ if no datum exists. The coordinate ℓ need not be prime. Lemma 11.1 turns $W_m(p) \leq T$ into a representation of m/p, and at $m = 4$ this is exactly (51.1)–(51.2).

Let $t = \log X$, $KX \leq T$, and use the canonical boxes from the proof of Theorem 11.6. Their two masses are

$$M_1 = \lambda_m t^2 \log K, \qquad M_2 = \theta_m t^3 \quad (K = X^\kappa). \tag{55.4}$$

Lemma 35.1 (m-uniform truncated assembly; proved internally with the Section 10 review qualification; source Lemma 55.1)**.** *Fix $B > 0$. Uniformly for $m \leq t^B$, if K is a sufficiently large fixed power of t, $KX \leq T$, and*

$$\log N \geq C_B t(1 + M_1), \tag{55.6}$$

then

$$\#\{p \leq N : p \text{ prime},\ W_m(p) > T\} \ll_B N \exp\{-c_B M_1\}. \tag{55.5}$$

For fixed $0 < \kappa < 1/240$, if $K = X^\kappa$, $KX \leq T$, and

$$\log N \geq C_{B,\kappa} t(1 + M_2), \tag{55.8}$$

then

$$\#\{p \leq N : p \text{ prime},\ W_m(p) > T\} \ll_{B,\kappa} N \exp\{-c_{B,\kappa} M_2\}. \tag{55.7}$$

The second assertion is CLAIMED/PROVISIONAL *through Theorem 11.6. Every constant is effective in the precise sense of Subsection 35.2.*

Proof. The full coprime multiplier family has mass comparable to M_1; replacing η_1 by η_2 in the upper profile and moments changes only an absolute factor. The cubic family has mass M_2 by (43.20), (43.27), and the relative-mass argument (43.36)–(43.38). Choose the small-prime quarantine and even Bonferroni degree of order M_i when it grows, and fixed degree otherwise. The exact ledger is

$$\deg = O(1 + M_i), \qquad \log d_{\text{term}},\ \log \sum |c_{\text{term}}| = O_B\{t(1 + M_i)\}. \tag{55.9}$$

Conditions (55.6) and (55.8) pay this ledger. Every retained atom has modulus at most $KX \leq T$; primes dividing m and fixed initial ranges are absorbed effectively. □

Theorem 35.2 (general-m master tails; status as in Lemma 35.1; source Theorem 55.2)**.** *Fix $B > 0$. Uniformly for $m \leq (\log T)^B$, put*

$$\mathcal{M}_1(m, T) = \lambda_m (\log T)^2 \log(2 + \log T), \qquad \mathcal{M}_2(m, T) = \theta_m (\log T)^3. \tag{55.10}$$

Set $\mathcal{P}_{m,T}(N) = \#\{p \le N : p \text{ prime}, W_m(p) > T\}$. *With effective constants depending only on the displayed fixed parameters,*

$$\begin{aligned} &\mathcal{P}_{m,T}(N) \ll_B N e^{-c_B \mathcal{M}_1(m,T)}, && \log N \ge C_B\{\log T + (\log T)\mathcal{M}_1(m,T)\},\\ &\mathcal{P}_{m,T}(N) \ll_{B,\kappa} N e^{-c_{B,\kappa} \mathcal{M}_2(m,T)}, && \log N \ge C_{B,\kappa}\{\log T + (\log T)\mathcal{M}_2(m,T)\}. \end{aligned} \tag{55.11}$$

The first line uses only the Layer-1 supply. The second is CLAIMED/PROVISIONAL. *No uniformity is asserted when* m *exceeds every fixed power of* $\log T$.

Proof. Take $X = T^{1/2}$. In the first row choose $K = t^{D(B)}$ large enough for the sign selection below; then $KX \le T$ and $\log K \asymp_B \log(2+t)$. In the second, rescale X by a harmless fixed power so $K = X^\kappa$ and $KX \le T$. Lemma 35.1 gives both rows; bounded T is absorbed in the constants. □

At the top of the cubic window, $\log N \asymp \theta_m (\log T)^4$, so the saving is $\theta_m^{1/4}(\log N)^{3/4}$, exactly (43.35). In the fixed-gap range $m \le (\log N)^{3-\varepsilon}$ the extra $\log T$ term is smaller and the $m \le t^4$ ledger closes. The first row similarly recovers the scale $\lambda_m^{1/3}(\log N)^{2/3}(\log\log N)^{1/3}$ and the fixed-gap range $m \le (\log N)^{2-\varepsilon}$. These are consistency checks, not range or status upgrades.

Corollary 35.3 (general-m polylogarithmic tails; source Corollary 55.3)**.** *Fix* $A, B > 0$ *and* $m \le (\log\log N)^B$. *With* p *restricted to primes,*

$$\begin{aligned} &\#\{p \le N : W_m(p) > (\log N)^A\} \ll_{A,B} N \exp\{-c_{A,B}\lambda_m (\log\log N)^2 \log\log\log N\},\\ &\#\{p \le N : W_m(p) > (\log N)^A\} \ll_{A,B} N \exp\{-c_{A,B}\theta_m (\log\log N)^3\}. \end{aligned} \tag{55.12}$$

The second line is CLAIMED/PROVISIONAL. *The almost-all result does not prove the pointwise hypothesis*

$$H_{\mathrm{MOD},m}(A): \quad W_m(p) \le (\log p)^A \quad \text{for every sufficiently large prime } p. \tag{55.13}$$

35.2. **The conductor coupled to** m**.** The prime progression in the general-m supply is

$$q = muv, \qquad \ell \equiv -k^{-1} \pmod q, \tag{55.14}$$

not modulo uv or $\mathrm{lcm}(m, uv)$. Thus if a Page conductor r divides m, every modulus is divisible by r and deletion would erase the entire supply. Retaining the exceptional term gives the effective form

$$\begin{aligned} \mathcal{E}_r(y;q,a) &= \psi(y;q,a) - \frac{y}{\varphi(q)} + \frac{\mathbf{1}_{r\mid q}\chi_r(a) y^\beta}{\beta\varphi(q)},\\ \sum_{q\le Q} \max_{(a,q)=1} \max_{x\le y\le 2x} |\mathcal{E}_r(y;q,a)| &\ll_A \frac{x}{(\log x)^A}. \end{aligned} \tag{55.15}$$

Here $Q \le x^{1/2}/(\log x)^{B_A}$; constants and thresholds are effective, and the exceptional term is absent if there is no Page character. This is the displayed-exceptional-character variant obtained by retaining the one primitive real character in the proof of Lenstra–Pomerance [17, Lemma 11.2]; their lemma itself prints the excluded-conductor form.

Lemma 35.4 (m-uniform Page repair; proved; source Lemma 55.4)**.** *Uniformly for* $m \le t^B$, *effective aggregate lower class mass holds for exactly the full coprime multiplier family used by the Layer-1 chain and the structured fibres* $\mathcal{J}_c$ *used by the*

c-free chain. This is not a per-multiplier lower profile and not an arbitrary-sparse-subfamily claim. The conductor cases are:

1. *With no exceptional conductor, retain every box.*
2. *If $r \nmid m$, choose a prime a with $v_a(r) > v_a(m)$ and impose $a \nmid uv$. Then $r \nmid muv$, including when r and m share other prime-power factors.*
3. *If $r \mid m$, retain exactly the multipliers satisfying*

$$\chi_r(-k^{-1}) = -1. \tag{55.16}$$

For every progression actually retained, the exceptional term in (55.15) then adds to the ordinary main term. This family retains a fixed positive fraction of the aggregate mass needed downstream.

The low-ω deletion and low-congestion pruning may be imposed simultaneously.

Proof. In case 2, omitting a from uv leaves the a-adic exponent of muv below that of r; Lemma 14.5's local proportion and large-a estimate apply pointwise in k.

Suppose $r \mid m$. The lifted nonprincipal real character takes each sign on half the units modulo m. Uniformly in reduced $b \pmod m$, Möbius expansion gives

$$\sum_{\substack{Y<k\le 2Y\\ k\equiv b\ (m)}} \frac{\varphi(k)}{k} = \frac{Y}{m}\prod_{p\nmid m}(1-p^{-2}) + O(\log(2Y)). \tag{55.17}$$

After summing favorable classes and dividing by $k \asymp Y$, intervals with $Y \ge Cm^2$ contribute $\gg \eta_1(m)$. If F_r is the set in (55.16), $H_m(K) = \sum_{k\le K,(k,m)=1} \varphi(k)/k^2$, and $K \ge m^D$ for a fixed large D, then effectively

$$h(F_r) \ge c_0\eta_1(m)\log K \ge c_1 H_m(K). \tag{55.17a}$$

The cubic family has $K = e^{\kappa t} \gg m^D$; the Layer-1 tail uses $K = t^{D(B)}$.

For the structured fibre $\mathcal{J}_c = \{k : (k,m) = (k,c) = 1\}$, (43.36) gives the adversarially safe subtraction

$$h(F_r \cap \mathcal{J}_c) \ge h(F_r) - H_m(K)\sum_{p\mid c,\ p\mid L_{m,K}} \frac{1}{p} \ge (c_1 - Z(c))H_m(K). \tag{55.17b}$$

Thus the same Chernoff event as (43.37), now with $Z(c) \le c_1/2$, leaves a fixed positive favorable mass without asserting character balance in an arbitrary sparse family. If an incidence statement formally requires $1 \in \mathcal{J}$, adjoin it only for bookkeeping and omit its triples from the BV lower sum when its sign is unfavorable. Increasing the low-ω cutoff and applying the existing low-congestion pruning preserve the order of mass. □

This quantifier boundary is load-bearing: an arbitrary sparse $\mathcal{J}$ could contain only the unfavorable sign. The full Layer-1 family and the fibres $\mathcal{J}_c$ are exactly the two downstream families for which Lemma 35.4 proves enough positive mass.

Theorem 35.5 (effectivity perimeter; proved internally, status-preserving; source Theorem 55.5)**.** *"Effective" means that constants and thresholds are computable uniformly in the theorem's displayed fixed-gap range. The exact perimeter is*

statement	*effective?*	*inheritance*
Layer-1 tails	*yes*	*Section 39 review no 34.8 or 43.7*
thinned-cubic tails	*yes, conditional on correctness*	*CLAIMED/PROVISIONAL 34.8 and 43.7*
Theorem 43.12	*yes, conditional on correctness*	*CLAIMED/PROVISIONAL 34.8, 43.7, and Section 39.7*

(55.18)

In every row, $r \nmid m$ uses deletion in uv, while $r \mid m$ uses the favorable k-half. The cubic rows effectivize only the surviving class mass, followed in Theorem 43.12 by the existing effective ledger and semigroup transfer.

Proof. Lemma 35.4 repairs the only Siegel-sensitive lower prime-progression input. Shiu, Brun–Titchmarsh, the box and incidence bounds, exact CRT moments, Chernoff, Bonferroni, integer class counting, and the Rankin semigroup transfer are effective. At the cubic top, (55.8) is the existing (43.40) ledger, so no parameter or range changes. □

The table does not effectivize the inherited ancillary absolute-error assertion in Theorem 11.6 over *all* low-congestion triples. Restoring deleted unfavorable triples restores the unresolved exceptional term; no row of (55.18) uses that assertion.

35.3. **Comparison and finite companion.** Truncating Pomerance–Weingartner's general-m, $k = 1$ supply gives mass $\asymp (\log T)^2/\varphi(m)$. The ratios of the two masses in (55.10) to that benchmark are

$$\eta_1(m) \log\log T, \qquad \eta_2(m) \log T. \tag{55.19}$$

Thus PW's pure $\varphi(m)^{-1}$ dependence is formally stronger below the crossovers where either ratio is one, and the multiplier exponents are formally stronger above. For every fixed B, the below-crossover regions eventually lie outside the proved window $m \le (\log T)^B$; this separates the unrestricted formal comparison from actual uniformity. The cubic side is provisional, PW is proved, and unknown constants preclude a finite crossover. At the optimized N-window the standing comparison remains (43.33)–(43.34), including PW's $\varphi(m)^{1/3}$ uniformity. No pointwise theorem or range extension follows.

Computational 55.6 (exact finite regression; informational). Verifier block `(bb)` harvests all classes (55.3) for $m \in \{3, 5, 6, 7\}$ and $k\ell \le 1500$. For primes $p \le 10^5$ prime to m, the counts with $W_m(p) > 25, 100, 400$ and those still unharvested are

m	$\#p$	three tail counts	unharvested at 1500
3	9591	$(0, 0, 0)$	0
5	9591	$(875, 61, 5)$	1
6	9590	$(3136, 663, 77)$	17
7	9591	$(2222, 438, 35)$	10

(55.20)

The block independently checks the intrinsic class description, reproduces the $m = 4$ classes from `(ax)`, and verifies

$$\eta_2(3) = \frac{9}{11}, \quad \eta_2(5) = \frac{25}{29}, \quad \eta_2(6) = \frac{36}{55}, \quad \eta_2(7) = \frac{49}{55}. \tag{55.21}$$

For the Page toys $(m, r) = (5, 5), (6, 3)$, both signs occur and the favorable side retains between one third and three quarters of the finite harmonic weight. This is a finite shadow of (55.17a), not evidence for its asymptotic uniform assertion.

36. Residue-dispersion frontier

Put

$$L = \log X, \qquad z = \frac{L^3 \log\log L}{\log L}, \qquad \Lambda = \frac{L^3}{\log L}.$$

For a composite z-rough $M \le X$, $M \equiv 3 \pmod 4$, choose the least positive divisor $D \mid ((M+1)/4)^2$ in each fibre of $D \mapsto -4D \pmod M$, and delete a representative

when its class is implied by a prime-modulus atom. Call the remaining set $\mathcal{D}^\circ(M)$. With the exact conditional weight

$$\kappa(M)=\prod_{\substack{q\mid M,\ z<q\le Y\\ q\equiv 3(4)}}\frac{q}{q-F(q)},\qquad n_{M,p}(a)=\#\{D\in\mathcal{D}^\circ(M):-4D\equiv a\pmod p\},$$

the pair-level dispersion quantity is exactly

$$\mathcal{V}_X=\sum_{z<p\le X}p\sum_{a\bmod p}\left(\sum_{\substack{M\le X,\ M\text{ composite},\ P^-(M)>z\\ M\equiv 3(4),\ p\mid M}}\frac{\kappa(M)}{M}n_{M,p}(a)\right)^2. \tag{40.5}$$

This is the residue-dispersion reformulation; canonical deduplication, prime powers, and the prime-implied deletion are part of the formula.

Write $D=R^2/s$, with $s\mid \mathrm{rad}(R)$ and $M=4Rk-1$. For $p\nmid 4R$, let $c_p(R)\in\{1,\dots,p-1\}$ be the inverse of $4R$ modulo p. The condition $p\mid M$ gives $k=c_p(R)+jp$. Splitting into the endpoint $j=0$ and regular tail $j\ge 1$ gives vectors $u^{\mathrm{end}}_{p,a}$ and $u^{\mathrm{reg}}_{p,a}$. The regular-tail theorem is proved, with the full canonical and retention restrictions allowed:

$$\mathcal{V}_X^{\mathrm{reg}}:=\sum_{z<p\le X}p\sum_a(u^{\mathrm{reg}}_{p,a})^2\ll L^2\log L+\frac{L^7}{z\log z}=o(\Lambda^2). \tag{40.14}$$

Indeed the contribution of fixed D is at most $CL/(pR_0(D))$; after expansion, $p\mid D-E$, while $\sum_D L/R_0(D)\ll L^3$ and $\sum_D(L/R_0(D))^2\ll L^2$. For $D\neq E$ there are at most $2L/\log z$ prime divisors above z, and for $D=E$ Mertens' estimate costs $O(\log L)$, giving (40.14).

The exact remaining pair target is therefore

$$\boxed{\mathcal{V}_X^{\mathrm{end}}=\sum_{z<p\le X}p\sum_a(u^{\mathrm{end}}_{p,a})^2=O(\Lambda^2).} \tag{40.18}$$

Equivalently, in divisor variables, it is the explicit short-cofactor estimate

$$\sum_{z<p\le X}p\sum_{a\bmod p}\left(\sum_{\substack{R,s,c,q:\ s\mid\mathrm{rad}(R),\ 1\le c<p\\ pq=4Rc-1\le X,\ P^-(pq)>z,\ pq\equiv 3(4)\\ R^2/s\equiv -4^{-1}a\ (p),\ q>1\ (\text{so } pq\text{ composite})\\ R^2/s\text{ is the retained representative at }pq}}\frac{\kappa(pq)}{pq}\right)^2\ll\Lambda^2. \tag{40.19}$$

Every term has $q>z$ and $q<4R$.

36.1. **Endpoint sparsity and proved ranges.** For $(p,4R)=1$, let $q_p(R)$ be the unique integer in $[1,4R)$ with

$$pq_p(R)\equiv -1\pmod{4R}. \tag{42.1}$$

Let $I_p(R,s)$ assert all the conditions in (40.19), including $s\mid\mathrm{rad}(R)$, $q_p(R)>z$, $pq_p(R)\le X$, roughness, and canonical retention. Then endpoint incidences are exactly the pairs with $I_p(R,s)=1$, and

$$q=q_p(R),\quad z<q<4R,\quad 1\le c=\frac{pq+1}{4R}<p,\quad \frac{z}{4}<R\le\frac{X+1}{4}. \tag{42.2}$$

Thus there is one cofactor for each (p, R), not one for each p; the moving multiplicity $2^{\omega(R)}$ in s remains.

Put $A = (pq+1)/4 = Rc$. The complementary divisor to $D = R^2/s$ is $A^2/D = c^2 s$, and hence

$$D \equiv \overline{16c^2 s} \pmod{p}, \qquad -4D \equiv -\overline{4c^2 s} \pmod{p}. \tag{42.3}$$

Two endpoint incidences over p collide exactly when $c^2 s \equiv c'^2 s' \pmod{p}$. Distinct incidences cannot have equal integer shifts: $R^2/s = R'^2/s'$ forces the common value's radical root $R = R_0(D) = R'$, then $s = s'$, and (42.1) gives the same q. The endpoint energy therefore has only genuine nonzero divisibility coincidences off the diagonal.

Theorem 36.1 (fixed endpoint fibres)**.** *For every fixed squarefree s, uniformly in s,*

$$\sum_{z<p\le X} p \sum_{a \bmod p} (u_{p,a}^{\mathrm{end},(s)})^2 \ll L^3 + L^2 \log L = o(\Lambda^2). \tag{42.4}$$

A prescribed set of K_0 squarefree fibres has energy $O(K_0^2 L^3)$ and therefore closes when $K_0 = O(L^{3/2}/\log L)$.

Proof. Drop canonical restrictions by positivity and use $\kappa \le 2$. For $1 \le c < p$, put

$$B_{p,c}^{(s)} = \sum_{\substack{z<q\le X/p \\ pq\equiv -1 \ (4c) \\ s \mid \mathrm{rad}((pq+1)/(4c))}} \frac{1}{q}.$$

If $p \nmid s$, the map $c \mapsto -\overline{4c^2 s}$ has fibres of size at most two, and therefore

$$p \sum_a (u_{p,a}^{\mathrm{end},(s)})^2 \le \frac{8}{p} \sum_{c<p} (B_{p,c}^{(s)})^2.$$

The q's lie in one progression of spacing $4c$. Separating its first member and writing $\ell_p = 1 + \log^+(X/(pz))$ gives

$$(B_{p,c}^{(s)})^2 \le \sum_q \frac{1}{q^2} + O\left(\frac{\ell_p}{cz} + \frac{\ell_p^2}{c^2}\right).$$

After multiplication by $1/p$ and summation, the first terms cost $O(L^3)$: with $M = pq$ and $A = (M+1)/4$, their sum is at most $\sum_{M\le X} \tau(A)/M \ll L^3$ by Theorem 4.2. Mertens' estimates bound the errors by $O(L^2/z + L^2 \log L)$, proving (42.4). Cauchy over the prescribed fibres gives the last assertion. □

For every fixed $B > 3$, the endpoint also satisfies

$$\sum_{z<p\le L^B} p \sum_a (u_{p,a}^{\mathrm{end}})^2 = O_B(\Lambda^2), \tag{42.5}$$

and there are no endpoint incidences for $p > X/z$. Indeed Theorem 7.2 gives $t_p^{\mathrm{end}} \le 2A(b_p + u_p)$; uniformly for $p \le L^B$, $pb_p = O(\Lambda)$ and $pu_p = L^{o(1)} = o(\Lambda)$. Residue Cauchy and Mertens prove (42.5). The super-polylogarithmic moving-s range remains nonempty and has harmonic length $\Theta(\log L)$.

36.2. **Kloosterman-fraction audit.** For an endpoint incidence put $m = 4c^2 s$ and aggregate its cofactors by

$$G_{m,p} = \sum_{q} \frac{\kappa(pq)}{q}, \tag{45.1}$$

where the sum includes every condition in (40.19); set it to zero unless $m/4 = c^2 s$ with s squarefree. This decomposition is unique. Additive Parseval and (42.3) give the exact matrix identity

$$\mathcal{V}_X^{\mathrm{end}} = \sum_{z<p\le X/z} \sum_{h \bmod p} \left| \frac{1}{p} \sum_{m\ge 1} G_{m,p} e_p(-h\bar{m}) \right|^2 . \tag{45.2}$$

The support has $(m, pq) = 1$, $z < q \le X/p$, and $m \le c(pq+1) < p(X+1)$. The only valid reciprocity orientations are

$$e_p(-h\bar{m}) = e_{pq}(-hq\,\bar{m}) = e_m(h\bar{p}) e\left(-\frac{h}{mp}\right). \tag{45.3}$$

In particular, q is never the phase modulus.

The DFI theorem, checked directly against the obtained 1997 article [5], allows arbitrary *separate* sequences α_m, β_n and gives

$$\left| \sum_{\substack{m\sim M, n\sim N \\ (m,n)=1}} \alpha_m \beta_n e(a\bar{m}/n) \right| \ll_\varepsilon \|\alpha\|_2 \|\beta\|_2 (|a| + MN)^{3/8} (M+N)^{11/48+\varepsilon} \tag{45.4}$$

for every nonzero integer a, with no range hypothesis. Its balanced power-saving window is $N^{5/6+\varepsilon} \ll M \ll N^{6/5-\varepsilon}$. Bettin–Chandee [1] similarly allow three separate sequences and prove, for nonzero real ϑ,

$$\begin{aligned} \left| \sum_{\substack{a\sim A, m\sim M, n\sim N \\ (m,n)=1}} \nu_a \alpha_m \beta_n e(\vartheta a\bar{m}/n) \right| &\ll_\varepsilon \|\nu\|_2 \|\alpha\|_2 \|\beta\|_2 \\ &\quad \times \left(1 + \frac{|\vartheta| A}{MN}\right)^{1/2} \\ &\quad \times \{(AMN)^{7/20+\varepsilon} (M+N)^{1/4} + (AMN)^{3/8+\varepsilon} (AN+AM)^{1/8}\}. \end{aligned} \tag{45.5}$$

Neither statement allows the joint coefficient matrix $G_{m,p}$. An exact singular-value decomposition introduces its nuclear norm. Worse, (45.2) asks for the complete row-by-row h-norm: summing DFI over h loses the full number of frequencies, while dualizing for Bettin–Chandee introduces an arbitrary (h,p) matrix. Their scalar savings do not pay these projective-norm losses. A long interval for q does not help because q stays inside $G_{m,p}$, and near $p = X/z$ that interval can be arbitrarily short. Thus the actual DFI PDF and the Bettin–Chandee theorem do not prove (40.19); the missing input is an operator/large-sieve estimate for the joint matrix, not a stronger scalar bilinear form.

Theorem 36.2 (small-fibre moving-s closure)**.** *Let $\mathcal{E}(W_0)$ contain all endpoint incidences with $m = 4c^2 s \le W_0$. If $4 \le W_0 \le z/2$, then*

$$\mathcal{V}(\mathcal{E}(W_0)) = \sum_{z<p\le X/z} \frac{1}{p} \sum_{4\le m\le W_0} G_{m,p}^2 \ll W_0 (L^3 + L^2 \log L). \tag{45.6}$$

Consequently $W_0 = \lfloor L^3/(\log L)^2 \rfloor$ *gives* $\mathcal{V}(\mathcal{E}(W_0)) \ll \Lambda^2$ *over the entire endpoint prime range. This is the injective prefix used by the wider closure below.*

Proof. Here $m, m' < p$ and $|m - m'| < p$, so distinct supported cells are distinct modulo p; inversion preserves distinctness. Orthogonality of the full h-sum in (45.2) therefore leaves no cross-cell terms and proves the equality in (45.6). For each unique decomposition $m = 4c^2 s$, the cells with common s lie in the fixed-s family of Theorem 36.1. Its positive diagonal gives

$$\sum_p \frac{1}{p} \sum_m |G^{(s)}_{m,p}|^2 \ll L^3 + L^2 \log L.$$

There are at most $W_0/4$ possible squarefree s, which proves the upper bound. The specialization uses $W_0 = o(z)$ and $W_0(L^3 + L^2 \log L) \ll L^6/(\log L)^2 = \Lambda^2$. □

The injective prefix is joined by a supported-cell argument. If $\mathscr{G}_C$ denotes cells $m = 4c^2 s$ with $c \geq C$, progression occupancy of their cofactors gives $G_{4c^2 s,p} \leq 2/z + L/(2c)$ and hence

$$\mathcal{D}(\mathscr{G}_C) \ll \frac{L^4}{C \log z} + \frac{L^4}{z(\log z)^2}.$$

A residue bucket modulo p contains at most $T/p + 1$ cells with $m \leq T$; thus Cauchy gives $\mathcal{V}(\mathscr{A}(T, C)) \leq (1 + T/z)\mathcal{D}(\mathscr{G}_C)$, where $\mathscr{A}(T, C) = \{m \leq T, c \geq C\}$.

Theorem 36.3 (prefix/wedge consequences of source Theorem 46.2)**.** *Put*

$$W_1 = \left\lfloor \frac{zL^2}{\log L} \right\rfloor \asymp \frac{L^5 \log\log L}{(\log L)^2}, \qquad W_2 = \lfloor zW_1 \rfloor. \tag{46.8}$$

Then

$$\mathcal{V}(\mathscr{A}(W_1, 1)) = O(\Lambda^2), \qquad \mathcal{V}(\mathscr{A}(W_2, z)) = O(\Lambda^2). \tag{46.10}$$

Thus the full prefix through W_1 *and the wedge* $c \geq z$, $m \leq zW_1$ *close over the complete endpoint-prime range.*

Indeed the two substitutions in the preceding diagonal bound give (46.10). Set

$$\mathscr{A} = \{m \leq W_1\} \cup \{c \geq z,\ m \leq W_2\}, \qquad \mathscr{B} = \{m > W_1\} \cap (\{c < z\} \cup \{m > W_2\}).$$

Then $\mathcal{V}(\mathscr{A}) = O(\Lambda^2)$, while the complete cell diagonal is $o(\Lambda^2)$. Positivity and the two-vector inequality give:

Corollary 36.4 (exact endpoint remainder; source Corollary 46.3)**.** *The endpoint target (40.19) is equivalent to*

$$\boxed{\sum_{z < p \leq X/z} \frac{1}{p} \sum_{\substack{m \neq m' \in \mathscr{B}_p \\ m \equiv m' \ (p)}} G_{m,p} G_{m',p} = O(\Lambda^2).} \tag{46.13}$$

Equation (46.13) is the current pair-level open core, not all $m > W_0$. Theorem 8.8 refutes the literal raw hierarchy (40.28) and raw-H target (37.19), while Theorem 8.12 refutes the antichain hierarchy (47.16) and the ordinary reduced-count moment target. Neither refutation reaches (40.19), (37.27), H'_{PF}, or the conjecture.

Status. This is not a submission. Both 3/4 headlines remain CLAIMED/PROVISIONAL, especially through Theorem 9.4; the internal proofs and refutations above do not close the open critical-window ledger, (46.13), H'_{PF}, either H_{WIN} window, or either

supercritical crossing hypothesis. The count-below-one theorem is conditional, and the presently realized window is below its thresholds. The square-escape theorem and complete linear/quadratic polynomial classification are proved, but they separate the all-integer and polynomial-family escape mechanisms from the still-open prime-survivor problem rather than proving a pointwise prime representation. The all-degree cyclotomic obstruction is qualitative and ineffective; C_{POLY} and repaired C'_{SQ} remain open. The original C_{SQ} is refuted: a computer-assisted theorem proves the three former constant candidates have $W = +\infty$. They are composite statistic-escapes with explicit Egyptian-fraction representations, not Erdős–Straus counterexamples. The complete census through 200,000, and the optional replay through 10^6, finds exactly the squares and those three values, but this proves no unbounded classification. The proved hard-prime criterion and its exact ratio-spectrum reformulation are terminating equivalences; the six small-a laws, dual fixed-D taxonomy, and gcd cancellation cover are exact local structure. The fixed-a confinement, budget, shifted-prime, composite-kernel, and congruence-wall laws are proved only with fixed-modulus constants and errors. The shifted-form mean values are unconditional only for each fixed J. Every moving-J tail and count-below-one consequence in that frame remains conditional on both open hypotheses H_{FAIL} and H_{STACK}; no unconditional result there beats the standing literature. The effective square-root law is weaker than the Section 51 tails on both axes and is only almost-all coverage of the explicit $D = 1$ layer. None of these proves the finite divisor list nonempty for every hard prime, a pointwise bound, a fair-share estimate, or a new Erdős–Straus theorem. The enlarged twisted-square hunt covers only its exact coordinate boxes and proves no unbounded classification or finiteness; only the right-to-left inclusion in C'_{SQ} is proved. The two-decade witness census, third-decade a_1 census, and record-stall anatomy are Computational/INFO finite-range evidence. Their maxima, dependence ratios, F3 shares, budget histograms, absent post-10^7 witness record, parity percentages, purity depths, and normalized suprema imply no growth law; the spacing model supplies no evidence, and every $A \geq 1$ remains open. The twisted-family cylinders prove three unbounded limsups but no new fixed infinite escape; the family certificate ceiling covers only finite harvested laws. The forward-ray theorem rules out a single polynomial witness family on the three anchored rays in both divisibility regimes, for every degree, period, and tail start. It leaves arbitrary non-polynomial and nonuniform witness supply untouched, does not classify $W(2m^2)$, and is not evidence for C'_{SQ}. The Pomerance–Weingartner audit is source-bounded and changes no provisional priority status. The raw and antichain factorial-moment formulations are refuted and that axis is dormant; the pair endpoint and other possible pointwise minorants retain exactly the narrower open scope stated above.

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